

# EU IVDR for Computational and Molecular Diagnostics

*A case-driven assessor and consultant apprenticeship in product route, performance evidence, software, AI, and lifecycle*

BOOK 2 • v7

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## INTRODUCTION

# From a Diagnostic Result to a European Decision

## Begin with the work people actually do

A laboratory receives a tissue sample from a person with cancer. Deoxyribonucleic acid (DNA) is the genetic material in its cells. A sequencing system reads many short pieces of that DNA, and software compares the readings with a reference sequence. This is the next-generation sequencing, or NGS, route used throughout the book. It may identify a *variant*: a difference from the reference that needs interpretation. A laboratory professional checks the proposed finding before releasing a report. A clinician may then use the report alongside other information when discussing treatment. Finding a variant, explaining its clinical meaning and identifying a person likely to benefit from a treatment are different jobs. A manufacturer must say which job its test is meant to do.

In digital pathology (DP), tissue is made into a slide and a scanner turns it into a digital image of the whole slide. Software may put the image earlier in a pathologist's worklist or point to a region for closer examination. The pathologist still makes the diagnosis in this assistive workflow. A study of how often the software highlights a region answers one question; a study of whether the pathologist can use that help during real work answers another.

Now imagine that each manufacturer wants to offer its product in the European Union (EU). The central question of this book is: **Which exact product and clinical purpose are being proposed, and what would justify a European conformity decision for that purpose?** We follow the route from the manufacturer's description, through the applicable law and evidence, to the people who make particular decisions. We also follow products after they enter use. In a separate fictional deployed NGS service, an operator records more invalid initial reports after a change in materials; the cause remains under investigation. In a separate fictional deployed DP service, supported images reach the worklist later than the local target after a service change; a diagnostic error is not established. Those observations raise focused questions about the earlier evidence without turning every field signal into a proven defect.

The In Vitro Diagnostic Medical Devices Regulation (IVDR), Regulation (EU) 2017/746 is the principal EU rulebook for the diagnostic examinations this volume studies. It is law, with definitions, duties and procedures. This book teaches you to connect that law

to a particular product and its evidence. The scientific-evidence and software volumes offer deeper methods for judging the studies and system design; here we ask which EU decision each piece of work must serve. The essential starting ideas are taught locally.

What would the European reviewer need to know before opening a successful study? First, the manufacturer has to describe the job it offers: whose tissue or image is used, who operates the product, what result reaches whom, and what the recipient is meant to do with it. The IVDR calls this the **intended purpose**; Article 2(12) ties it to the manufacturer's labeling, instructions, promotional statements and performance-evaluation material. For the NGS example, a tissue-variant report is a smaller job than identifying who is likely to benefit from a particular medicine. For the DP example, help with prioritizing images is different from making a diagnosis. The words used to describe the job decide which later questions the evidence must answer.

A clear job description raises the next question: what kind of product does that job make it under EU law? **Qualification** is the name for deciding whether a function belongs under the IVDR, the Medical Device Regulation (MDR), another framework or outside device law. A DP program that only moves images needs a different analysis from one that interprets specimen-derived images for a medical purpose. If a product qualifies as an in vitro diagnostic (IVD), the manufacturer then applies the IVDR's risk rules to its stated purpose; that is **classification**. We cannot classify the DP program merely because it uses an algorithm, or carry an NGS class across to a claim about a different specimen.

Knowing the class still leaves a practical question: which assessment must actually happen? Once the purpose and class are justified, the manufacturer identifies the required **conformity-assessment route**: the prescribed process for showing that the product meets applicable EU requirements. Some routes require an independent assessor to examine specified parts of the product or quality system. That assessor is a **notified body**. Its work is one part of the applicable route. The manufacturer still owns the product, its evidence and the decision to apply. Completing the applicable process supports the manufacturer's EU declaration of conformity and **CE marking**, the conformity mark used for EU market placement. A favorable study or an early notified-body conversation does not itself complete that process. Chapter 1 explains who can answer each preliminary question; Chapters 2–4 show how purpose, qualification, class and route are actually established.

## Follow the families without mixing their histories

The book returns to three kinds of diagnostic work. Oncology NGS lets us examine how specimen type, variant finding, interpretation, treatment-related claims, software and evidence versions interact. DP lets us test whether the proposed software function is an in vitro diagnostic, another kind of medical-device software or something else, and whether its evidence covers the scanner-to-reader workflow. Near-patient infectious-disease testing puts users outside a central laboratory, connectivity, invalid results and practical follow-up in view. These are recurring *families of questions*, not three secretly continuous products.

The source exercises often use anonymous fictional manufacturers and local cases. The U.S. oncology package in Chapter 1 belongs to that chapter; its later exhibit does not become the history of every NGS example. The separate deployed NGS and DP observations above are bounded teaching scenes, with no supplied certification history, injury or established cause. They are not HelixBridge Dx, PathSight Cloud, LumaCell Review, Asterion, or the anonymous dossiers elsewhere in this volume. A familiar modality, number or software label does not establish identity across cases. A worked example states its own premises, and a later conclusion must use only evidence actually connected to it.

## The route from promise to continuing evidence

Once the manufacturer has named the product's job and legal route, a reviewer needs to follow a result backward through the product. For the NGS report, that means the tissue, laboratory steps, software and professional release. For the pathologist's image, it means the slide, scanner, image-processing steps, worklist and display. A good study about one step is useful, but it may leave another step unexamined. The evidence file must let someone see which result-producing path was studied and which path will be offered.

The IVDR has **general safety and performance requirements (GSPRs)** in Annex I. Instead of simply writing “complies” beside one, the manufacturer links the requirement to a design choice, risk control, test or justified rationale. For example, if an invalid NGS result must be prevented from appearing as a reportable finding, the file should show the rule and evidence that it works. A DP worklist needs its own evidence about getting supported images to the reader. These are different controls for different product jobs.

The company also needs a way to make those decisions consistently. A **quality management system (QMS)** governs development, manufacture, release, investigation and controlled change. **Technical documentation** under IVDR Annexes II and III is the inspectable account of the product and evidence those processes produced. A folder

with the right headings is useful navigation; it cannot make a missing test result appear. Chapters 5 and 6 follow a requirement through these company processes into that file.

An NGS test can accurately report a variant even if the proposed clinical meaning of that variant has not been established. The first evidence question is therefore whether the marker has a sound association with the clinical condition or purpose. The IVDR calls this **scientific validity**. Without that relationship, an accurate measurement may have no established meaning for the offered medical job.

A supported relationship leaves the next question open: can this assay actually detect or measure the marker in the intended specimens and conditions? That is its **analytical performance**. A paper about the marker cannot show how this product handles the specimen.

Even that measurement evidence leaves a further question: how do this product's results relate to the clinical state in the people and use being claimed? That is **clinical performance**. The three questions build on one another because each leaves something the next must establish. For a DP function that qualifies under the IVDR, the same discipline reaches from the information the image is meant to provide, through the configured image examination, to the result used in the clinical workflow. Its evidence must fit that function; copying a molecular study would not explain the pathologist's task.

Someone must then judge how the strands fit the whole claim, and keep that judgment current as evidence changes. The IVDR calls this continuing work **performance evaluation**. Before the results arrive, the manufacturer specifies the questions and methods in a **performance evaluation plan (PEP)**. Later, a **performance evaluation report (PER)** appraises what the accumulated evidence supports and where it stops. A plan is valuable because it shows what the team intended to test; it is not evidence that the planned study succeeded. Chapters 7–11 move through that sequence in detail.

Evidence work continues after release. **Post-market surveillance (PMS)** is how the manufacturer gathers and acts on information from use. In our separate fictional deployed NGS service, the team needs to count invalid initial reports, locate affected materials and compare periods before claiming a cause. In the separate deployed DP service, a late worklist entry calls for checking the image-to-reader path, even if the model itself has not changed. Each observation tests a particular earlier assumption; neither proves a patient injury.

The planned performance-evidence part of that work is **post-market performance follow-up (PMPF)**. It can answer a specific question that remains after a justified initial decision; it cannot substitute for a missing basis for the central claim before market placement. **Vigilance** is different again: it is the formal reporting and response route when the legal conditions for serious incidents or field safety corrective actions are met.

The NGS and DP observations need investigation and an assessment of any timely protective or reporting duties while the cause is uncertain. Later information may confirm an earlier conclusion, call for a focused bridge or require stronger action. Chapter 15 teaches those choices.

Some NGS products make an additional promise involving a medicine. A **companion diagnostic (CDx)**, in the IVDR's defined sense, is an IVD essential for safe and effective use of a corresponding medicinal product to identify, before or during treatment, patients most likely to benefit or at increased risk of serious adverse reactions. A variant finding alone does not establish that relationship. If the claim does, the notified body seeks a scientific opinion from the appropriate medicinal-products authority or the European Medicines Agency (EMA) as part of the device assessment. Chapters 3, 4 and 13 return to the exact test–medicine relationship and the distinct institutional roles.

## Who makes which decision?

The **manufacturer** chooses the NGS or DP product it wants to offer. It defines the claim, develops and documents the evidence, controls the product and decides whether to apply with the present purpose or change it. A consultant can help prepare that decision, but cannot make it for the manufacturer. When the device passes through the EU supply chain, an authorized representative, importer or distributor has duties tied to that role. The IVDR calls these parties **economic operators**. Their participation does not transfer the manufacturer's responsibility for the product's underlying conformity.

The **notified body** is an independent assessor for devices whose legal route requires one. It may assess the manufacturer's quality system and evidence and later perform certificate surveillance, but only for work within its designation. For the proposed NGS assay, a body can discuss whether it has the right designated scope and later judge the submitted evidence. For the prospective DP function, the manufacturer first needs to explain what its software does; calling it “artificial intelligence (AI) pathology” does not itself tell a body which regulatory route applies. A conversation with either manufacturer does not produce a certificate.

Now consider an NGS manufacturer planning a study involving people or their specimens. Before it starts, the sponsor must identify the study route under the IVDR and the relevant Member State processes. A national **competent authority** performs the public regulatory functions assigned to it, including performance-study oversight and, later, vigilance and market surveillance. An **ethics committee** examines ethical acceptability through the applicable national process. The notified body may explain how the study could help a future evidence assessment; its email does not establish the national start conditions. The same distinction applies if a DP manufacturer studies a pathologist's interaction with software.

If a proposed NGS test is a CDx, another question arises: is it suitable for use with the corresponding medicine? The relevant medicinal-products authority or EMA gives a scientific opinion through the applicable consultation. The notified body considers that opinion in its device assessment. Each actor contributes a different judgment; none is a general EU approval desk.

When the separate deployed NGS service reports more invalid results, or the separate DP service reports late worklist entries, the manufacturer investigates and considers whether protective steps or a legal report are due. A notified body's surveillance role and a competent authority's vigilance role may then matter, according to the product and event facts. The observation alone does not assign every institution a task.

The European database on medical devices (**EUDAMED**) holds different actor, device, certificate and other regulatory information through its modules. Registration is a distinct operating duty, not a substitute for performance evidence or a certificate. Its module obligations and national alternatives have changed over time, so Chapter 15 checks their current status rather than treating a remembered date as timeless.

## How to use the book

Readers may come from laboratories, pathology, clinical work, engineering, regulatory assessment, quality management or consulting. You need to be willing to follow a specimen or image through a workflow and to distinguish a proposed claim from a study observation. The book supplies local explanations of the legal and technical steps. You should leave able to identify an EU decision and its owner, find the controlling source, trace evidence to the exact purpose and product state, preserve a supported narrow conclusion, and specify what would close a material uncertainty. This is a reasoning apprenticeship, not authorization to make a statutory decision or to perform a specialty for which separate competence is needed.

Read Chapters 1–6 in sequence if you are new to the field. They establish the product, legal route and evidence file that later chapters assume. Chapters 7–11 turn the claim into an evidence plan and conclusion. Chapters 12–16 address studies, linked medicines, software and field work, and cross-jurisdictional choices. Chapter 17 integrates the route. If you return for reference, start with the decision you face and then follow its dependency backward: for a changed intended user, return to purpose and qualification before changing a study plan; for a postmarket performance signal, revisit the exact evidence and configuration before revising the PER.

Most chapters contain a decision pause followed by feedback. The pause gives a moment to form a judgment from the evidence available so far. The feedback then shows how an experienced reviewer reaches a bounded conclusion and which fact would change it. A changed-case transfer alters one premise and asks what must be reopened. An independent assessment in Chapter 17 uses an unfamiliar packet; its



instructor feedback is separate from this learner manuscript. Neither the introduction nor earlier examples give away its answer.

The **Reviewer Lens** asks what a submitted record establishes now, for which purpose, product and conditions, and what focused question would change the judgment. The **Consultant Lens** uses the same facts to help a manufacturer define a claim, compare options, sequence evidence work, assign owners and propose a closure condition. Advice, evidence authorship, implementation, manufacturer decisions and institutional decisions remain distinct. A consultant may identify a needed national or clinical specialist without pretending to exercise that specialist's authority.

The short working states in the chapter checks are **A** (adequately explained for the bounded item), **U** (present but unclear), **M** (missing or inconsistent), **D** (deeper analysis required), **C** (consultation with the relevant specialist or institution indicated), and **N/A** (not applicable, with a reason). They are descriptions of an evidence item at a particular moment. They are not grades, a vote tally, a device-level pass or an approval. A useful letter is accompanied by the evidence location or absence, the reason for that state and the next action. Without those, it hides the reasoning it is meant to retrieve. Existing checklist IDs remain useful retrieval keys; they are not a scoring system.

Technical names are introduced as the work needs them. The source glossary, historical chapter crosswalk, bibliography and cross-volume route map remain reference aids. A regulation, an implementing act or Common Specification, a harmonized standard, nonbinding guidance and a body's practice have different authority. The chapter should show what each source can establish locally; a glossary entry is not the explanation.

## The route through all 17 chapters

**Chapters 1–4 choose the question before trying to answer it.** Chapter 1 starts with a product someone wants assessed and asks who owns each European decision and which source can support it. That actor-and-source record lets Chapter 2 ask what the manufacturer actually intends the NGS or DP function to do and whether it belongs under the IVDR, MDR or another category. Only after the function is bounded can Chapter 3 apply the IVDR classification rules; it shows why an added treatment, specimen or user claim can change the risk context, without assuming that a changed user always changes class. Chapter 4 takes the justified class and asks which conformity procedure is available, what a notified body's designation covers, and how any special consultation fits. A class name alone cannot select or complete that procedure.

**Chapters 5–6 turn the route into a traceable product record.** Chapter 5 takes an applicable general safety and performance requirement and follows it to a design decision, test, risk control or rationale. The result is a thread one can inspect, rather than a checked box. Chapter 6 gathers these threads into technical documentation and a quality system: what was built, which configuration was tested, who supplied

consequential parts, what changed, and whether the file still describes the product being offered. A later NGS knowledgebase change or DP scanner-interface change can send the reviewer back to this record even when the product name has not changed.

**Chapters 7–11 construct and judge performance evidence.** Chapter 7 develops the performance evaluation plan before the final result is known, linking each claim to a question, method and decision rule. Chapter 8 asks whether the biological or clinical association itself is supported and how literature and the current state of the art are appraised; a cited article is not automatically evidence for this product. Chapter 9 tests the actual measurement system, including invalid or incomplete outputs and the configuration that produced them. Chapter 10 then asks whether results work for the proposed clinical population and for the human workflow, especially a pathologist using assistive software rather than an algorithm working alone. Chapter 11 brings the strands into a performance evaluation report: what is adequate now, what remains uncertain, and which specific question post-market performance follow-up can investigate after a justified market decision. It does not use future follow-up to excuse a missing premarket basis for a central claim.

**Chapters 12–13 handle evidence that crosses institutional and product boundaries.** If new performance evidence requires a study, Chapter 12 asks what the whole protocol does to people and specimens, which legal route applies, and what the sponsor must resolve with each Member State and ethics process before starting. Chapter 13 examines the extra relation between a companion diagnostic and a medicine: who gives the scientific opinion, what the trial test shows, and how the proposed commercial test can or cannot be bridged to it. The study decision cannot be inferred from a notified-body email, and an assay comparison cannot alone settle the medicinal-product relationship.

**Chapters 14–16 follow the product into change and practical choice.** Chapter 14 applies the earlier product and evidence discipline to medical-device software (MDSW), AI, cybersecurity and controlled updates; its discussion of the EU Artificial Intelligence Act identifies the source and applicability question rather than using an AI label as a shortcut. Chapter 15 asks what an actually operating postmarket system observes and does: performance follow-up, periodic safety reporting, vigilance, registration and certificate maintenance each have different triggers and owners. It distinguishes a released configuration from a device merely described in a final technical report. Chapter 16 returns to the familiar U.S. evidence package and asks what may be reused for the *same* EU proposition, what needs bridging, and which claim and work sequence management should choose. It makes the choice reviewable without predicting a certification date or business outcome.

**Chapter 17 makes the route repeatable.** It revisits the decisions in passes, shows how a later fact can reopen an earlier state, and asks for a concise actor-and-evidence memorandum. Then an unfamiliar independent packet tests whether you can make,

revise and defend a bounded conclusion without the familiar NGS or DP case cues. Its six requested outputs precede the separate instructor feedback, so the unfamiliar evidence can be judged on its own terms.

The thread running through the book is simple to remember: define the exact product and purpose; identify the actor and source; ask what evidence would answer that actor's question; give credit for what the evidence supports; and reopen only the conclusions touched by a new fact. The work is careful because one product may have several valid evidence statements but still leave one proposed claim undecided.

## CHAPTER 1

# The European Decision System: Product, Actors, and Sources

A laboratory professional is about to release a report about a tissue specimen. Software has proposed findings, but the professional still needs to review them. What would a European assessor need to know about the product behind that report? A convincing study is part of the answer. The assessor also needs to know exactly what the manufacturer promises, which product produced the evidence, and who has authority to decide the next question.

In a next-generation sequencing examination, often shortened to NGS, the laboratory reads many short pieces of DNA, the genetic material in cells. Software compares those readings with a reference and proposes **variants**—differences that need interpretation. The professional reviews the proposed result before a report is released. Detecting a difference and explaining its clinical meaning are connected jobs, but evidence about one does not automatically answer the other.

In **digital pathology**, tissue becomes a slide and a scanner turns the slide into an image. Assistive software may prioritize that image or highlight a region. The pathologist remains responsible for the diagnosis, while the software influences what the pathologist encounters. Both workflows let us follow the same practical question: which exact job is being proposed for use in the European Union (EU), and what supports that proposition?

## SECTION 01

### Which product crossed the Atlantic?

A fictional U.S. oncology-diagnostics company sends a successful submission package to a prospective EU notified body, an independent organization that assesses specified conformity work. The package includes extensive studies of test performance, a clinical study, software documentation and labeling. The company expects to place its documents under new headings. Instead, the body asks which use the company will claim in Europe, which assay, software and report-rule combination produced the results, why its proposed class and review route apply, and what each study establishes. What is missing from the idea of simply moving the documents?

A study keeps its observations when it crosses a border. Its specimen, population, tested output and limitations remain the same. A successful package under a U.S. Food and Drug Administration (FDA) pathway does not, however, establish conformity under the EU In Vitro Diagnostic Medical Devices Regulation (IVDR). That conclusion concerns a particular European proposition and its applicable requirements. The opening packet supplies neither the actual U.S. decision nor the proposed EU product state. The body's questions seek that missing connection; they are not findings that the studies are bad.

The connection starts at the report. Whose specimen enters, which information comes out, who uses it, and what action is it meant to support? Together these define the **intended purpose**. Article 2(12) ties that purpose to the manufacturer's labeling, instructions, promotional statements and performance-evaluation material. For the sequencing workflow, the tissue, finding and proposed clinical action belong in the same account. For pathology, the account connects the slide and scanner to what the software displays and what the pathologist does next. A general product name cannot carry those details. Prioritizing images for a pathologist is a different promised job from diagnosing without one; these are alternatives to analyze, not additional claims assigned to the fictional product.

Once the job is clear, the next question is which law covers it. Deciding whether the function is an in vitro diagnostic (IVD) under the IVDR, a medical device under the Medical Device Regulation (MDR), or something else is **qualification**. Pathology makes the need for this step visible. A program storing images, a program providing specimen-derived information for an IVD purpose, and a program with another medical purpose require different analyses. Knowing that all three handle pathology images would leave the legal question unanswered.

If the function qualifies as an IVD, the manufacturer can then apply the IVDR's Annex VIII risk rules to its purpose. This is **classification**. The opening company proposes class C; the proposal needs a rule and the facts showing why that rule fits. A changed specimen or user can send the analysis back to those facts without automatically determining a new class. A pathology function governed by another regulation cannot simply inherit the sequencing product's class.

The justified class then makes it possible to select the applicable Article 48 **conformity-assessment route** and examine whether a notified body is designated for the required work. Class, procedure and body eligibility answer successive questions. A class name completes none of the later procedures. The package therefore needs an explanation of its product and claim before its document headings can become useful navigation.

## SECTION 02

### Which setup produced the result?

Suppose the sequencing instrument and main program stay the same, but the information used to interpret a variant changes. Could the report change? Following the specimen reveals why the opening body's version question extends beyond the name on the software box.

The laboratory prepares the tissue and generates sequence readings. An ordered set of computational steps checks the readings and proposes variants; this is the work often described as the software pipeline. The proposed variant then needs interpretation. Controlled information in a **knowledgebase** helps supply that interpretation, and **report rules** determine which information belongs in the draft. A professional reviews the draft. Changed knowledgebase content can therefore change how the same detected variant is described without changing either the instrument or the main program.

For pathology, the corresponding path starts with a slide and scanner. Image preparation makes the scanner data suitable for the software. An assistive model may calculate a score, a number used by a rule to prioritize the image or highlight a region. The worklist and display determine what the pathologist actually sees. Changing a scanner profile may change the input data. Changing a worklist rule may change when the image is encountered even if the model calculates the same score. Neither the number nor the highlight is the pathologist's diagnosis.

The study must identify this result-producing combination. Its **configuration** includes relevant materials, equipment, software, controlled information, settings and behavior presented to people. For sequencing, that can reach from assay materials through the knowledgebase to the report rules. For pathology, it can reach from scanner and image preparation through the model to the worklist. The shared scientific and engineering work is to identify the problem, control this product, test the relevant behavior and learn from use. The jurisdictions' different procedures do not remove that common foundation.

A study directly describes the configuration it examined. Even a final technical report is a record of evidence about a state. A manufacturer's release is a separate decision accepting a state for a specified purpose. Lawful market availability is another matter, and actual clinical deployment is another again. The report's completion alone proves neither market availability nor deployment. This distinction becomes essential when later evidence appears to describe a familiar product but the actual installed state is unknown.

How can the company locate the particular devices involved? The EU **unique device identification (UDI)** system provides linked identities at different levels. Consider a

purely illustrative tissue sequencing kit, called model T, with units made in lot L. The letters are teaching labels, not assigned EU identifiers. They supply no evidence that several models belong in one regulatory group.

The group question comes first: which models share the same intended purpose, risk class and essential design and manufacturing characteristics? An eligible group's **Basic UDI device identifier (Basic UDI-DI)** connects its regulatory documentation and database record. Grouping requires justification; the opening company has supplied none here. This documentation key does not appear as a third part of the UDI on a device or its packaging.

That UDI has two parts, subject to the IVDR's application details. The **UDI device identifier (UDI-DI)** identifies the particular device model or version—model T in the teaching trace. The **UDI production identifier (UDI-PI)** carries production information—lot L here. If an investigation concerns that lot, its production identifier helps locate the units, while the device identifier tells the investigator which model those units belong to. A trade name or internal software build number replaces neither role.

A pathology software change raises the corresponding question about affected versions and distributed units. That does not mean every software update requires the same identifier change. Article 24 and Annex VI, together with the Commission's UDI explanation, supply the identification framework; Chapter 15 treats assignment and registration in context. Identification makes an evidence comparison possible. It does not establish that the identified product performs adequately.

## SECTION 03

### What does each study actually answer?

A tissue assay can reliably detect a variant while leaving open what the variant means for the proposed clinical use. Which evidence would close that gap? The question shows why the opening company needs to explain the job of each study instead of offering “a clinical study” as a complete answer.

First, there must be a reason to examine this marker in relation to the condition or purpose. Research supporting that association addresses **scientific validity**. The association makes the marker worth measuring, but says little about whether the configured assay can measure it reliably. That next question belongs to **analytical performance**: what happens when this assay is challenged with relevant specimens and conditions? A strong biological association cannot rescue an assay that misses the variant.

Successful detection then permits another question: how do this assay's reported results relate to the clinical state in the proposed population and use? This is **clinical**



**performance.** It concerns the product's result in that clinical setting. Association, measurement and clinical use can each have good evidence, yet remain different conclusions. Evidence does useful work when its actual conclusion is carried forward with its limits.

For a pathology function that qualifies under the IVDR, the same logic follows the image's actual job. Evidence that a specimen image carries relevant information differs from evidence that the configured software produces its claimed output on relevant images. Both differ from evidence about that output in the proposed clinical workflow. A model-only highlight rate cannot describe how a pathologist uses a changed worklist. This transfer is an evidence relationship, not an assertion that an image has the sequencing assay's variant-detection limit.

The manufacturer needs to settle those questions and methods before collecting and combining results. A **performance evaluation plan (PEP)** specifies what will be examined, under which conditions and how. A well-designed plan is valuable, but its planned evidence has not thereby been obtained. After the results are appraised, the **performance evaluation report (PER)** explains what their combined support permits the manufacturer to conclude and where it stops. The continuing appraisal joining these strands is **performance evaluation**.

The performance argument sits within the product's wider conformity record. The IVDR's Annex I **general safety and performance requirements (GSPRs)** state applicable requirements, while Annexes II and III organize the technical documentation. A heading gives a reviewer a place to find the argument; the linked evidence must still support it. The first route can now be read as a chain of questions rather than a list of documents.

The table helps locate where an application question belongs before a study or institution is asked to answer it.

| Question                         | What an answer settles                                      | Where to look in these workflows                                        |
|----------------------------------|-------------------------------------------------------------|-------------------------------------------------------------------------|
| What job is promised?            | The purpose being proposed.                                 | Specimen or image, user, setting, output and intended action.           |
| Which rulebook covers it?        | The function's qualification.                               | What the pathology software actually does beyond handling an image.     |
| Which risk class follows?        | The Annex VIII conclusion for a qualified IVD.              | The opening company's proposed class C rule and supporting claim facts. |
| How will conformity be assessed? | The applicable Article 48 procedure and institutional work. | The route and designated body for that defined product.                 |



| Question                      | What an answer settles                                          | Where to look in these workflows                                                 |
|-------------------------------|-----------------------------------------------------------------|----------------------------------------------------------------------------------|
| What supports it through use? | The connection between product, evidence and later information. | The studied assay or image-to-reader configuration and the scope of each result. |

The pathology row about qualification must be answered before an IVDR class can be assigned. In the final row, a sequencing detection study can be credited for its technical result while a different clinical proposition still needs its own evidence. None of the rows supplies a certificate or study-start permission. A later study or field observation can send the team back to an earlier row when it changes a premise on which that answer depended.

**Reviewer Lens.** The most useful first request often names the missing connection rather than another document. For the opening package, the purpose and configuration let a reviewer ask which study answers which claim element. Model T and lot L show how identity could locate tested units; they establish neither group eligibility nor clinical adequacy. A pathology score needs its scanner and reader context before it can support a claim about professional performance. Enough orientation means those connections can be inspected. It does not mean every question is closed, and an adequate study should not lose its credit because a different question is still open.

**Consultant Lens.** Early scope choices determine which work will later be needed. The company's alternative EU statements can be compared through specimen or image path, output, user, configuration and evidence question before a report is assembled. It is often cheaper to expose those dependencies then than after a polished file obscures them. A narrower defensible proposition may reduce preparation work, but promises neither certification nor launch. Management owns the product choice. The consultant makes the alternatives and required laboratory, clinical, software or regulatory expertise visible, including the factual basis for any proposed regulatory grouping.

## SECTION 04

### A new claim arrives

The opening package has so far left its actual European proposition unresolved. The following exhibit supplies a change to that local case. It belongs to this anonymous company, not to HelixBridge or another sequencing dossier.

The clinical study used the correct commercial assay. The proposed European intended purpose adds **liquid biopsy** and a **predictive treatment-selection role**, both absent from the U.S. claim. Liquid biopsy means testing material obtained from a body fluid rather than the earlier tissue specimen; neither the fluid nor its collection path is specified. The proposed predictive role uses the result to help identify who is likely to

benefit from a corresponding treatment. The technical file is complete only for a narrower product.

## Decision

**Should the company simply map each U.S. document into the closest IVDR Annex II heading? What can be credited, what question remains, and who should choose the proposed scope?**

### How a reviewer reads the changed proposition

The correct-commercial-assay fact matters. The clinical study remains useful for the assay and conditions it examined. The new problem is the distance between those conditions and the added liquid-specimen path and predictive role. A complete file for the narrower product cannot provide an unobserved result for either addition.

The next comparison therefore connects each European claim element with the study's material, population, output, clinical use and configuration. The manufacturer must first stabilize the proposition it wants assessed. A staged tissue application is an option if its own purpose, evidence and applicable route can be defended; it is not a promised certificate. A broader proposal needs evidence for its added claims or a justified bridge showing why existing evidence applies across the difference. The notified body assesses the resulting application within its designation. It does not choose the company's product scope.

A short **actor–source–route record** keeps this inference visible to someone who did not attend the discussions: the question, supplied fact, responsible actor, controlling source, present conclusion and what could close it. The table below uses three questions to show how different kinds of incompleteness can coexist in the same application.

The letters are **working states** for individual items. **A** means adequately explained for the bounded item; **U**, present but unclear; **M**, missing or inconsistent; **D**, deeper analysis required; **C**, relevant specialist or institution consultation indicated; and **N/A**, inapplicable with a reason. An evidence location or its absence and the reasoning come before the letter. Letters neither add up to a grade nor authorize a product.

| Question and present fact                                                                   | Actor and source                                                                     | Present conclusion and closure                                                                                                                                              |
|---------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Does the study support the wider purpose? The draft adds specimen and clinical-role claims. | Manufacturer; IVDR Articles 2(12), 10 and 56.                                        | <b>D.</b> Wider applicability is unsupported by this packet. Regulatory and clinical owners compare each claim with study/configuration evidence; management chooses scope. |
| Can this body assess it? Its exact designation is not supplied.                             | Manufacturer checks official designation and written application review; Article 49. | <b>U.</b> Eligibility is unclear. Dated scope and exclusions must be matched to the defined product before eligible bodies' practical capacity is compared.                 |
| May a planned study begin? Complete protocol and national start basis are not supplied.     | Sponsor; Articles 57, 58 and 66–70 and concerned Member State processes.             | <b>M.</b> The start basis is missing. The full protocol, countries, submissions, ethics process and lawful commencement conditions need an appropriate decision record.     |

The first row concerns the reach of evidence, so D does not say the original study failed. The second concerns institutional scope, so U is not a finding against the body. The third concerns an absent start basis, so M predicts neither refusal nor a particular national procedure. For a prospective pathology application, the first question would instead begin with its actual software function: specimen-derived diagnostic information, another medical purpose or only data handling. The record must name the product question before assigning a state.

## Decision

**A reviewer has checked the exact professional-use claim, current Annex VIII class rationale, and a body's applicable designation and exclusions. The body's written application review confirms that the product is within its scope. Evidence assessment remains pending. Can the eligibility item be A, and what remains open?**

## How a reviewer closes a positive narrow finding

Yes. The dated designation and written application review support A for eligibility. They leave capacity, evidence sufficiency, certification and launch as separate questions. Reopening eligibility merely because the whole product remains undecided would discard a conclusion the record already supports.

The same discipline applies to a later investigation: a documented comparison can establish the affected configuration even while cause or clinical effect remains unknown. Positive closure is useful precisely because it tells the next reviewer which question has already been answered.

**Reviewer Lens.** “Unsupported” becomes useful only when its object is explicit. Here the wider claim is unsupported by this packet, while the correct assay study remains creditable. The focused review follows purpose and configuration, the study's actual observation, the claim difference, and the adequacy of what sits under the documentation heading. That sequence avoids both an indiscriminate rejection and a premature acceptance.

**Consultant Lens.** The practical company comparison places unchanged, staged and expanded purposes beside the evidence each would require, an owner and the event that would permit a decision. A narrower option can make preparation dependencies visible without being marketed internally as an assured certificate. The reviewer judges the support now supplied; the consultant must also help management understand the work behind its alternatives. Neither role can convert a scope preference into an institutional decision.

### **Changed-case transfer**

#### **A distinct near-patient test.**

A fictional infectious-disease test has strong FDA evidence for laboratory staff. Only the intended user changes, to trained pharmacy personnel. Which user tasks, setting, risk controls, instructions and postmarket signals need reconsideration? What would be needed before drawing a class conclusion?

### **How the changed user alters the question**

The evidence still describes laboratory staff. Its applicability to pharmacy personnel depends on their actual tasks and setting, the controls and instructions they can use, and the signals that would reveal problems during use. Treating the change as distribution alone would hide those differences. Yet a changed user does not by itself establish a new class: the exact purpose and applicable rule still need analysis. Chapter 3 develops that step. This test is an independent transfer case, not a new version of the oncology assay.

## SECTION 05

### Who can decide the next question?

A separate fictional manufacturer asks its notified body whether a three-country leftover-sample study needs a notification or application in each country. In the same week, it asks a national competent authority to approve its class C technical documentation. Both requests concern the same broad ambition of bringing a product into use. Why might they still have reached the wrong decision desks?

Start with the decision that belongs inside the company. The **manufacturer** chooses its design and intended purpose, assembles the technical documentation and decides which application or change to pursue. Its **quality management system (QMS)** supplies the controlled processes through which those product and evidence decisions are made. A consultant can compare a staged claim with a broader one or develop a bounded analysis. That assistance leaves the manufacturer's responsibility in place.

The application then meets an independent question: does the product conform through the applicable procedure? A **notified body** is designated to carry out specified assessment work. Its remit can include the quality system, technical documentation and later certificate surveillance, within its designation. The Commission's notified-body page and NANDO listing identify bodies and designated scopes. The positive example in §04 supplied enough to establish eligibility; the opening oncology packet did not. A useful preliminary conversation and a formal scope record therefore retain different roles.

A study-start question concerns another kind of authority. A Member State's **competent authority** performs public regulatory functions, including relevant performance-study administration, market surveillance and vigilance. An **ethics committee** addresses ethical acceptability and participant protection through the applicable national process. The notified body can discuss how study evidence may contribute to conformity assessment. Its informal view cannot perform the national study or ethics process. Conversely, the national authority does not ordinarily perform the notified body's class C technical-documentation assessment.

The sponsor still needs enough protocol information to determine which national questions arise. Do results affect care? Is there extra specimen collection or another procedure? Which activities occur in each country? What is the device's status and intended purpose in the study? “Leftover-sample” answers none of those questions completely. The practice case supplies neither the full protocol nor country procedures or authority responses, so there is no final route to infer. The IVDR study provisions and the Medical Device Coordination Group's study guidance, MDCG 2025-5, help frame the whole-protocol inquiry; the guidance remains nonbinding.

A medicine-linked purpose adds a different decision again. Article 2(7) defines a **companion diagnostic (CDx)** as a device essential to the safe and effective use of a corresponding medicine. Before or during treatment, it identifies patients most likely to benefit, or patients likely to face increased risk of serious adverse reactions from that treatment. Where that relationship applies, the notified body seeks a scientific opinion on the test's suitability in relation to the medicine. The opinion comes from a medicinal-products authority or, where applicable, the European Medicines Agency (EMA). The EMA's CDx overview distinguishes when EMA rather than a national medicines authority is consulted. The notified body retains the device conformity decision. An oncology variant result alone establishes neither the defined test–medicine relationship nor the need for this consultation.

Finally, reaching the EU supply chain brings role-specific duties for an authorized representative, importer or distributor. These **economic operators** do not become substitute manufacturers. This is the same discipline at a different handoff: participation in one part of the route does not transfer every underlying product or evidence responsibility.

The actor table is useful when two requests sound similar but need different decisions. Its last column identifies the separate judgment that would remain after the actor had done its own work.

| Who is involved?                                               | What that party owns here                                                                      | Which question still belongs elsewhere?                                 |
|----------------------------------------------------------------|------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| Manufacturer and project owners                                | Product, purpose, evidence, quality system, application/change choice and field investigation. | The independent regulatory or conformity decision.                      |
| Notified body within designation                               | Applicable conformity assessment and justified certification or certificate maintenance.       | National study start, ethics review and the medicinal-products opinion. |
| National competent authority                                   | Statutory study, vigilance or market-surveillance work within its remit.                       | Routine notified-body technical-documentation certification.            |
| Ethics committee                                               | Ethical acceptability and participant protection under the national process.                   | Device certification and scientific performance conclusions.            |
| Medicinal-products authority or EMA, when consultation applies | Scientific opinion on suitability with the corresponding medicine.                             | The notified body's device conformity decision.                         |

| Who is involved?                                   | What that party owns here                                 | Which question still belongs elsewhere?                            |
|----------------------------------------------------|-----------------------------------------------------------|--------------------------------------------------------------------|
| Authorized representative, importer or distributor | Duties attached to its particular economic-operator role. | The manufacturer's underlying product and evidence responsibility. |

In the practice case, the notified-body row preserves the value of advice about future evidence assessment. The national-authority and ethics rows show why that advice cannot finish the start question. The manufacturer row also prevents the reverse substitution: asking an authority to approve the class C file does not discharge the company's work or select the correct assessment process. For the prospective pathology product, the function must first be qualified before the company assumes which conformity body, if any, is needed.

**Reviewer Lens.** The verb in a proposed decision often reveals the mismatch more quickly than the institution's name. Defining a claim, assessing conformity, authorizing an activity and giving a scientific opinion require different records. For the three-country case, a focused request for the protocol, country activities and national-route record is more useful than assuming three identical applications. A medicine-linked claim needs the actual medicinal-product relationship and consultation actor. Enough means the response comes from the process entitled to answer that particular question; a helpful response from another process remains useful advice.

**Consultant Lens.** An early responsibility record can prevent the company preparing a polished answer for the wrong recipient. Each question needs an internal owner, the institution or specialist to approach, facts to stabilize before contact and the response that would close it. Advisory meetings and formal submissions have different jobs. A regulatory lead can coordinate claim and body scope, a study specialist can resolve protocol and national dependencies, and pathology and software specialists can explain the image function. The consultant's sequencing advice makes those dependencies manageable while keeping the resulting decisions visibly owned by the right party.

## SECTION 06

### Which source supports the answer?

The opening company now knows why its proposed purpose and the body's scope matter. Could a body application guide supply the legal answer? It might explain where to put the documents, yet the reason a particular requirement applies must come from a source capable of establishing it. A useful source hierarchy distinguishes authority, scope and date together. Higher authority does not make every lower source dispensable.



For the claim itself, binding law is the starting point. The IVDR defines IVDs and intended purpose, assigns manufacturer duties and provides classification, conformity, performance-evidence and postmarket provisions. Article 2(12), applied to the manufacturer's actual statements, can therefore support a purpose analysis in a way the phrase “diagnostic software” cannot. The MDR may govern a pathology function with a medical purpose that does not qualify as an IVD. The EU Artificial Intelligence Act, Regulation (EU) 2024/1689, may add duties for a qualifying AI system when the relevant provisions apply. An AI label chooses neither obligations nor an effective date. The relevant definition, Article, Annex, exception, amendment and transition must fit the actual decision date.

The main regulation may itself authorize a later instrument. That raises the question of what the Commission was empowered to do. An **implementing act** sets conditions for applying existing law uniformly across Member States; it does not amend or supplement that law. A **delegated act** may supplement or amend specified nonessential elements when the law grants that power. The Commission's explanation connects these different roles to Articles 290 and 291 of the Treaty on the Functioning of the European Union. The parent provision and adopted instrument's scope and date therefore belong together in the analysis.

Article 9 supplies one relevant example. When its conditions are met, an implementing act may adopt a **Common Specification (CS)** for defined requirements or evidence topics. Within its applicable scope, the manufacturer must comply with the Common Specification unless it duly justifies another solution ensuring at least an equivalent level of safety and performance. This is a compliance-or-justified-equivalent threshold; it differs from the voluntary use of a harmonized standard discussed next. Locating a Commission information page is useful navigation; the page is not the adopted instrument. A material or software change needs a comparison with the particular covered requirement before the words “Common Specification” can support a conclusion.

A technical standard answers another question: what support does conformity with its provisions provide? Under IVDR Article 8, a relevant **harmonized standard** or part whose reference is published in the Official Journal can provide a presumption of conformity for the requirements it covers. The effect depends on the officially cited edition and scope, the clauses applied, deviations and the actual product. The Commission's harmonised-standards explanation confirms that use remains voluntary; technical value and the particular legal effect must not be confused. No specific standard is declared harmonized for a product in this chapter.

That scope limit also protects positive evidence. If the cited edition and applied clauses adequately address a product requirement, the reviewer should credit that support rather than automatically demand duplicate testing. If materials or software later move outside the assessed conditions, the affected applicability needs reconsideration. The

earlier conclusion about a still-covered process need not vanish. The standard's title alone can establish neither of those two findings.

Difficult boundaries may still benefit from interpretation. **MDCG guidance** offers common interpretation and practical application, including for IVD qualification, software and studies. The Commission's guidance register identifies the documents as nonbinding and subject to revision. For example, MDCG 2024-11 can structure the qualification reasoning for the sequencing or pathology function while the IVDR definition remains the governing source. A changed interpretation may warrant revisiting that reasoning; it is not evidence of a field defect.

Once eligibility is established, a body procedure can help the manufacturer prepare the interaction. Instructions from a particular body and documents from the European Association of Medical Devices Notified Bodies, Team-NB, describe operational expectations. They can prevent a missed administrative step without converting a body preference into a universal legal duty. The Commission's designation information answers who may assess a device type. A body's written procedure answers how it asks eligible applicants to present the file. For the opening company, format advice cannot establish the added claim's adequacy. A later surveillance request likewise needs its particular body and certificate context.

Someone must finally connect the product facts to these sources. That **professional synthesis** is essential work: no regulation supplies the company's completed comparison of claim and study. Its credibility depends on visible assumptions, evidence locations, alternatives and decision ownership. “A narrower tissue application may be worth evaluating” is a management option. “This study probably needs notification” is a provisional view until the full protocol and national processes are examined. Neither acquires binding force because a consultant wrote it confidently.

The table helps distinguish what a cited source could establish before relying on it in a conclusion.

| Source                                                  | What it can support                                            | What makes it applicable here?                                                                 |
|---------------------------------------------------------|----------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| IVDR, MDR or applicable AI Act provisions               | Binding duties and procedures.                                 | Exact product definition, provision and applicable date, including exceptions and transitions. |
| Implementing act, delegated act or Common Specification | The effect permitted by the parent law and adopted instrument. | Parent power, effective version, covered requirement and any permitted alternative.            |
| Officially cited harmonized standard or part            | Presumption for the requirements covered.                      | Official Journal citation, edition, applied clauses, deviations and product scope.             |

| Source                             | What it can support                                  | What makes it applicable here?                                              |
|------------------------------------|------------------------------------------------------|-----------------------------------------------------------------------------|
| MDCG guidance                      | Nonbinding interpretation and application help.      | Revision/date, legal basis and fit of the example to the product.           |
| Team-NB document or body procedure | Association or body operating practice.              | Issuer, designation, product context and any written body response.         |
| Professional synthesis             | A reasoned recommendation joining facts and sources. | Assumptions, alternatives, evidence location, owner and decision authority. |

For the opening purpose question, the law row supplies Article 2(12); a guidance example may help interpret the actual function; a body procedure may help present the resulting file. Those sources contribute different things to one decision. In the standards row, a valid cited scope can close a covered requirement without deciding every other requirement. Reading down the table is therefore not a sequence in which later rows become progressively worthless.

A compact source ledger makes this use reviewable. Its entry identifies the title, version or edition, publication and verification dates, status, scope, controlling clause, official location and the conclusion supported. One dated line beside a consequential conclusion is often more useful than an undifferentiated bibliography. Revision-sensitive guidance, body designations and exclusions, European database modules and national procedures particularly need verification at the decision point.

**Reviewer Lens.** A precise reference can still be attached to an overbroad sentence. The useful reading joins the sentence to the scope of the source and then to the product evidence. “Every GSPR conforms because standard X was used” invites a question about edition, cited coverage and the requirement-to-evidence link. A correctly bounded statement may already be adequate. During a change assessment, comparing the affected material or worklist behavior with the earlier scope keeps a narrow applicability question from becoming a demand to reopen the whole file.

**Consultant Lens.** Source maintenance is cheapest when the decision record already says which conclusion each source supports. Then an owner can verify the volatile item before management relies on it, and a changed revision has an identifiable consequence to examine. A standard may supply useful presumption while a particular product gap still needs direct evidence. A national study, clinical or legal question may need the relevant specialist; documenting that input preserves its value without silently elevating advice into law. The consultant can recommend a route, while management and the relevant institutions retain their decisions.

## SECTION 07

## When does an earlier conclusion reopen?

Two separate fictional deployed services now supply observations from use. One sequencing service records more invalid initial reports after a materials change. One pathology service records supported images reaching its worklist later than a local target after a service change. Neither service is the opening company, the prospective pathology illustration or a named series product. No counts, denominator, established cause, patient effects or regulatory history are supplied. What can these observations change while so much remains unknown?

Each identifies a comparison worth making. The sequencing service can reconstruct materials and configurations, count initial reports and compare the affected conditions with those in its earlier analytical evidence. Timing alone does not prove that the material caused the invalid result. The pathology service can reconstruct the image-to-worklist path and time window, then compare them with the workflow represented in earlier human-use evidence. An unchanged model calculation leaves that timing question open. Neither observation automatically disproves an earlier clinical study.

This information belongs in **post-market surveillance (PMS)**, the manufacturer's process for gathering and assessing information from use. The investigation also has an immediate legal question. **Vigilance** is the distinct formal reporting and response route when applicable incident or field-action conditions are met. The scenes do not establish those triggers, but the manufacturer must promptly assess protective and reporting duties on facts already known. Investigation of seriousness, affected configuration, denominator and possible mechanism can continue alongside that assessment; causal certainty is not a prerequisite for considering a proportionate precaution.

The evidence response may be narrower than a complete repeat of earlier work. A justified comparison or targeted evidence bridge can show which prior conclusions remain applicable and which need further examination. The planned performance-evidence part of ongoing work, **post-market performance follow-up (PMPF)**, can address a justified residual question after a defensible initial market decision. It cannot supply a missing initial basis for the central claim. New field information may prompt a bridge or investigation without erasing unrelated adequate evidence.

These decisions need somewhere to remain connected. A **route control sheet** is the small working record that holds product, intended purpose, configuration and lifecycle state beside the next question, source, actor, evidence location, present conclusion and event that would reopen it. It is not an additional legal form. The book calls this first orientation **Pass 0**. Later review passes revisit actors and sources; purpose and

qualification; class and route; documentation and risk; performance evidence; studies and companion-diagnostic consultation; software and change; and postmarket capability. These eight passes organize review and return questions, rather than creating a new legal sequence. Chapter 17 joins them after the intervening chapters develop their methods.

The following record compares three different kinds of unresolved question. It is useful when a broad phrase such as “regulatory issue” conceals what actually needs to change.

| <b>Local case and question</b>                                                 | <b>What the packet supports now</b>                                                                                                                                                              | <b>Who can close or reopen it?</b>                                                                                                                                                                            |
|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Opening oncology company: wider purpose supported?                             | Correct-commercial-assay study remains useful; added liquid and predictive claims remain unresolved here.                                                                                        | Clinical/regulatory owners compare specimen, population, outcome and configuration; management chooses purpose. New claim or evidence reopens the match.                                                      |
| Prospective pathology product: which law covers its function?                  | Storage, specimen-derived IVD information and another medical purpose still need distinguishing. Scanner/display conditions, pathologist action and proposed evidence belong in the description. | Manufacturer reasons from the actual function using Article 2 and applicable nonbinding software guidance. Changed function or purpose reopens qualification.                                                 |
| Separate deployed sequencing and pathology services: what do the signals mean? | Invalid initial reports and late supported-image entries are observed in their respective scenes; cause and harm remain unknown.                                                                 | Product owners assess immediate duties, reconstruct configurations and denominators, then investigate mechanisms and prior evidence applicability. Cause, severity or affected scope can change the decision. |

In the first row, a study result already exists but does not support every added use. In the second, the function needed for qualification is still undefined. In the last, the observation is supplied but its explanation is not. The table does not turn the deployed-service observations into measured evidence for either prospective product. Nor does it collapse the two services into one event.

The actor map must change with the actual question. The sequencing signal starts with the manufacturer and laboratory establishing counts, configuration and possible harm. The pathology signal starts with timing and workflow evidence. Manufacturer investigation, notified-body certificate surveillance and competent-authority vigilance may all become relevant, but the facts determine their duties. A field note alone does

not trigger all three. The source map is equally specific: a possible mechanism is a hypothesis, and the legal significance of a delay requires the applicable definition and event facts.

**Reviewer Lens.** A useful row is a complete inference: this product and state are described by this evidence; this source gives this actor the relevant role; and this closure condition tests the missing assumption. The opening study can retain credit while the added claim is examined. The pathology function can remain unqualified without being assigned a speculative IVDR class. In either deployed service, an adequate configuration comparison can close “what changed?” while cause and reportability remain open. A narrow closure should be preserved until new evidence actually touches its premise.

**Consultant Lens.** The source of cost differs across these rows. The opening company can compare evidence work for a tissue application and a wider proposal before selecting scope. The pathology team can settle the intended function before commissioning studies for a particular route. The deployed services need owners for preservation of observations, focused investigation and timely protective/reporting assessment now. Later options may include a targeted bridge, revised labeling or stronger change control as cause, scope and risk become clearer. Management chooses the product action; the consultant makes its dependencies visible and arranges reassessment when new facts alter them.

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## Decision

**A notified body informally says a planned study is likely noninterventional. Can that view be the Member State study-start decision? Which missing product and protocol facts, relevant actor and closure record would make this particular decision reviewable?**

### How a reviewer turns useful advice into the next question

The view can help anticipate future conformity assessment, but cannot itself provide the national start decision. The sponsor needs the whole protocol under the IVDR and the processes of the concerned Member States, including ethics requirements. Activities may extend beyond leftover samples. Countries and commencement conditions are not supplied in this prompt.

A country-by-country decision record from the proper processes would connect actual device and intended-purpose status, result use, collection and procedures to the relevant submissions and lawful start conditions. The email remains dated advisory input, and facts that could change the route stay visible. This is the same discipline that keeps a worklist investigation or invalid-report trend from being recorded as an authority decision before the legal trigger facts are known.

The resulting record lets each professional ask a distinct question. The notified-body reviewer considers designation and support for conformity. A competent-authority reviewer addresses the statutory matter within that authority's remit. A medicinal-products reviewer gives the scientific opinion where the companion-diagnostic route requires it. The consultant translates these dependencies into work the manufacturer can execute while keeping decision ownership visible.

**Transfer check**

For a class question, performance study, companion-diagnostic consultation, product change or field signal, how would the question differ for the notified body, competent authority, medicinal-products authority where relevant, and company management? Which facts would make an actor's involvement necessary?

**How the actor changes the question**

For either separate field service, management can ask what investigation it should fund and who owns it. A competent-authority question arises when legal facts make that authority's statutory role relevant. A body's certificate-surveillance question would concern its own remit; a medicinal-products opinion requires the applicable test–medicine relationship. If every actor receives the same question, the decision and its owner probably have not yet been distinguished.

**SECTION 08****Quick discrimination lab**

Each record below is independent. The question is what its evidence establishes now, what remains open, and which source or actor could change that judgment.

**A study email**

A notified-body contact says a proposed study appears noninterventional. The team records the email as authorization to begin in three Member States.

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**Decision**

**Can the email close the study-route question? What exact record must replace it, and what value can the email retain?**



**How a reviewer distinguishes advice from a start basis**

The email remains useful conformity-assessment advice with a date and limited scope. It cannot close the national study route. The sponsor needs the complete protocol and each country's applicable competent-authority and ethics processes, including lawful commencement conditions.

Those facts have not been supplied. The number three cannot establish three identical applications, or whether any particular notification or application is required. Closure is a country decision record tied to the actual activities and processes. Retaining the email alongside it preserves useful advice without attributing an authority it does not have.

**A standards citation**

A technical file cites a familiar standard and marks every related general safety and performance requirement conforming. The cited edition differs from the edition used in design. The cloud service controlling a clinical output was outside the assessed scope.

**Decision**

**Does the citation provide a presumption that the complete product conforms? What support may still be credited, and which checks would resolve the remaining gap?**

**How a reviewer follows the citation into the product**

The reference alone cannot support the complete-product conclusion. The actual edition or relevant part must be checked against the official citation, covered requirements and what the manufacturer applied. The assessment must also show whether the clinical-output service and its risk are covered. Here the packet explicitly places the cloud service outside the assessed scope.

Correctly applied clauses can still supply useful support. The uncovered product question then needs direct evidence or a justified rationale; dismissing all standards evidence would be as unhelpful as treating the title as a blanket result. A later cloud-workflow change in a separate deployed pathology service would reopen the corresponding scope question even with unchanged model calculations. A materials change in a separate sequencing service would require comparison with the standards and tests that actually covered those materials.

**SECTION 09****Leave with an actor–source–route record**

Another professional should be able to follow the exact product question, actor with authority, source of the right status and scope, evidence location or absence, present bounded conclusion and closure condition. These seven checks are usable on their own. Workbook Part A2, if available, is an optional companion containing expanded cards with the same retrieval identifiers. Its letters describe evidence items, never a numerical score or overall authorization.

The historical checklist numbers identify source-era topics rather than this edition's chapter numbers. For example, current Chapter 6 contains cards from source topic 07 because earlier topics were integrated there. The original identifiers remain at the ends of the headings; the original-to-integrated crosswalk locates every topic.

**Exact purpose and configuration • card IVDR-CH01-A01**

Which purpose and product state does the evidence describe? Controlled wording, component records and release, market and deployment records make the answer inspectable. A trade name alone identifies neither the claim nor the state.

**Evidence and its EU documentary role • card IVDR-CH01-A02**

What did the study observe, and where does that observation fit in the performance evaluation plan and report? Protocol, report, tested configuration and claim-to-evidence mapping preserve the study's original limits when it enters an EU evidence strand.

**Changed claim and downstream route • card IVDR-CH01-A03**

Which qualification, class, Article 48, consultation or study question does a changed claim reopen? Claim alternatives and the Annex VIII rationale show the dependency without predetermining the new answer.

**Adequacy behind the headings • card IVDR-CH01-A04**

Does the Annex II/III index lead to evidence and sufficiency reasoning for the actual purpose? A present document cannot support a claim it did not examine merely because it occupies the expected heading.

**U.S.-EU reuse and claim difference • card IVDR-CH01-C01**

Which proposition can reuse the existing evidence? A comparison of claim, specimen, user, configuration and study applicability establishes what transfers; the submission's origin alone does not.

**The manufacturer's scope choice • card IVDR-CH01-C02**

What work would unchanged, staged and broader purposes each require? Management can compare those options and choose its proposition. A consultant's comparison cannot promise certification.

**The relevant specialist or institution • card IVDR-CH01-C03**

Which exact technical or legal question requires which response? Body designation, national-procedure records and focused specialist input each resolve their own item; none substitutes for all the others.

The separate eligibility example supports A for documented scope eligibility. The opening company's wider evidence applicability remains D until the claim-by-study analysis is done; its unshown study-start basis is M. N/A requires a reason tied to the actual purpose. The evidence location and explanation make those states useful, and a new fact reopens only the conclusions it affects.

**Source trail and next reading**

The governing legal source for this chapter is the IVDR, Regulation (EU) 2017/746, consolidated text of 10 January 2025, especially Articles 2, 8–10, 17–18, 24, 48–49, 56–58 and 66–70, and Annexes I–III, VI, VIII and XIII. The Commission's explanation of implementing and delegated acts explains their distinct roles under Treaty Articles 290 and 291. The Commission's notified-body/NANDO page, UDI page, and MDCG guidance register give current operating and guidance locations. MDCG 2024-11 addresses IVD qualification, MDCG 2019-11 rev.1 (June 2025) addresses software, and MDCG 2025-5 addresses performance studies; all are nonbinding. The EMA CDx overview identifies the medicinal-products consultation role. Check current versions, Member State rules, designation and product facts at the actual decision date. Book 1A Chapter 1 offers optional deeper practice reconstructing a product-to-result system; the present chapter is self-contained.

Chapter 2 now takes the first field in the route record—intended purpose—and asks what product functions it actually describes. A sound actor and source map cannot rescue an undefined purpose; a precise purpose can, in turn, make the qualification question answerable.

## CHAPTER 2

# Intended Purpose, Scope, and Qualification

**Retrieval check — before reading.** Chapter 1 distinguished the manufacturer's product decision, a notified body's conformity assessment and a national competent authority's statutory role. Who must first make and document the argument about what kind of product is being offered, and what product facts would let another person assess it?

A platform can give a pathologist an image, a score and a place in a queue. Those outputs arrive together, but they need not do the same job. **What does each function do with specimen-derived information, for whom, and which European device definition fits that job?** A class or a conformity route becomes useful only after the object of that decision is clear.

## Follow what happens to the image or specimen

Follow the image first. In **digital pathology**, a scanner turns a human tissue slide into an image that a pathologist can examine. Copying that image into an archive preserves it for retrieval. Putting it in a **worklist**, the queue of cases awaiting review, affects when the pathologist encounters it. Examining its content to produce a patient-specific score or highlight may supply information for diagnosis. The same screen can sit at the end of these different contributions.

The visible report can hide an equally consequential contribution in sequencing. In a tissue **next-generation sequencing** examination, the laboratory extracts DNA, the genetic material in cells, and reads many short pieces of it. Software checks the readings, compares them with a reference and proposes **variants**—differences for professional interpretation. A quality classifier might decide whether the specimen enters that analysis at all. A rejected usable specimen or an accepted unsuitable one could change whether, when or how a report appears, even if the gate contributes no field to the final report.

**Opening case — a cloud pathology application.** This fictional application receives scanned whole-slide images and returns a risk score and highlighted regions. It also stores images and manages a worklist. The manufacturer calls the whole offering a “laboratory information service” and points out that the pathologist makes the final

diagnosis. The team wants to know whether it belongs under the **In Vitro Diagnostic Medical Devices Regulation (IVDR)**, the **Medical Device Regulation (MDR)** or outside device law.

The packet does not explain precisely what the score means, which finding the highlights represent or how the clinical user is meant to act. Those are genuine gaps. This is a local application, separate from PathSight Cloud, Asterion, LumaCell Review, Chapter 1's deployed pathology service and the independent examples below.

The manufacturer cannot get a sound answer by replacing “information service” with another platform label. It first needs an account of what enters each function, what leaves and what the person does with it. Chapter 1's actor distinction matters here: the manufacturer owns the purpose and qualification argument. A consultant can clarify options and evidence; a notified body may later assess the application within its role; a competent authority exercises its public duties. Neither institution supplies the company's product proposition for it.

## Describe the promised job before naming a class

For the cloud application, “the pathologist uses it” leaves the central question unanswered: uses *which output*, *to do what*? The manufacturer's **intended purpose** is the use for which it offers the product. As Chapter 1 explained, the relevant statement extends beyond a short label. IVDR Article 2(12) includes labeling, instructions, promotional and sales statements, and the performance evaluation. These sources need to describe the same offered job.

That is why the image's path belongs beside the claim. Which tissue preparation and scanner supply it? Does a function copy, order, reject, score, highlight or interpret it? What reaches the pathologist, and for which diagnostic decision? Low-scored, unscored and failed images also need a described path. The opening packet leaves answers missing. A purpose precise enough to test would expose the proposed clinical role of the score and highlight; it would not infer autonomous diagnosis merely because a score is computed.

Reading design and workflow alongside the claims protects against two different mistakes. “Informational” wording cannot explain away a feature presented and configured to change clinical work. Yet a hospital's use of a general archive does not automatically give the manufacturer an independent medical claim. An unexpected customer action requires its own examination before it is attributed to the manufacturer's purpose.

Now consider the proposed sequencing classifier. A visible flag for a technologist and removal from the only analysis path would leave different things for the professional to do. So would a gate that changes a reported variant. The product description needs the intended specimen, users and laboratory setting, and whether the output protects

technical quality, supplies patient-specific clinical information or controls a diagnostic instrument. “Professional use” cannot choose among those actions. Deciding early who may override a rejection also tells the study team which errors and recovery paths must be examined.

The purpose questions below help expose such an undecided choice before a study is designed. They do not fill in the missing answers for either proposal.

| <b>Question about the offered job</b> | <b>Cloud pathology application</b>                                                | <b>Proposed sequencing gate</b>                                               |
|---------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| What enters, and from where?          | Which tissue slide, preparation and scanner supply the image?                     | Which tissue specimen and sequence-quality data reach the gate?               |
| What changes, and what comes out?     | An archived image, queue position, score or highlight?                            | A review flag, exclusion or changed result?                                   |
| Who uses the output, where, and how?  | Does the pathologist see every image, and what action is expected from the score? | Can a trained operator inspect or reverse an exclusion before the report?     |
| What clinical role is claimed?        | Which condition or decision does the score assist, if any?                        | Technical quality control, medical information or control of the examination? |

Take the second row through the cloud application. Evidence that the image was copied says something useful about its stored output. It does not describe what the score means or how a highlight directs attention. For the sequencing gate, the same row distinguishes a flag someone can examine from an exclusion that stops analysis. Those outputs determine the next user-action question. Evidence for one input path may not cover another, and the professional's final responsibility does not describe the changes already made upstream.

A narrower, accurate purpose is often easier to test than vague “decision support.” Accuracy is the condition: deleting medical words while retaining the function and clinical presentation does not create a non-device conclusion. Adding a treatment or autonomous-diagnosis promise for the sake of precision is equally unhelpful. Claim, behavior and evidence must meet at the same product.

### Reviewer Lens – purpose and output

A review can go wrong by treating one well-described output as a description of the whole platform. The stored image may be clear while the clinical role of the score remains unexplained. An experienced reviewer follows that difference into the instructions, demonstration and study configuration. For the sequencing gate, a demonstration of the visible flag would be useful only if a flag is the offered behavior. A focused request asks what happens to the specimen after that output; it need not reopen an adequately described neighboring function.

### Consultant Lens – purpose and output

An early flag-versus-exclusion choice is inexpensive to discuss and expensive to leave implicit until the evidence is collected. It changes who must act, what can be recovered and which errors a study must examine. The analogous cloud choice is which combination of storage, ordering and diagnostic scoring the company will offer. A consultant can show the user actions and evidence consequences of each alternative. Management chooses the function and claim; the consultant's tidy purpose sentence cannot make that decision disappear.

## Decide what kind of function the purpose makes it

Suppose the cloud team establishes what the score is intended to tell the pathologist. Does being embedded in the platform decide its legal status? First comes the question Chapter 1 called **qualification**: whether the function meets a legal product definition and which framework covers it. **Classification** then assigns a risk class to the qualified, bounded purpose. Cloud hosting, a scanner or an algorithm supplies no fixed answer to either question. The program can transfer data in one function and supply medical information in another.

For the score, the first possible definition concerns information obtained from a human specimen examined outside the body. An **in vitro diagnostic (IVD)** is intended by its manufacturer for that examination solely or principally to provide one of the kinds of medical information specified in IVDR Article 2(2); software used alone or with other products is included. Relevant purposes here include information about a physiological or pathological state, predisposition to disease, predicted treatment response or the definition or monitoring of treatment. A tissue-image finding about disease or a sequencing result intended to guide treatment may answer that information question; simply moving the same specimen data may not. Specimen origin alone, however, does not tell us that this software function is intended to provide information within the definition. The score's promised meaning still matters.

If a function has a medical purpose outside that IVD definition, the MDR medical-device definition may be the relevant one. Article 2(1) includes software



intended for a specified human medical purpose, such as diagnosing, monitoring or treating disease, whose principal intended action is not achieved by pharmacological, immunological or metabolic means. MDR Article 1(6)(a) excludes IVDs covered by the IVDR. The decisive change would therefore concern what medical information or action the function is intended to provide and whether it meets the specimen-examination definition—not whether it runs in a cloud. The opening score’s missing meaning prevents that choice from being completed. The binding definitions are in the IVDR text and MDR text. The **Medical Device Coordination Group (MDCG)** gives nonbinding interpretive help: MDCG 2024-11 addresses IVD qualification, and MDCG 2019-11 rev.1 addresses software. The guidance helps frame the product question; the law supplies its definition.

A function that creates a patient-specific finding from an image or interprets a specimen-derived result illustrates why software can have a medical purpose of its own. That is the defining feature of **medical device software**. Its neighboring copy-and-retrieve function need not acquire the same independent purpose. The archive’s availability and interface can nevertheless affect the regulated analysis. The manufacturer therefore needs intelligible module boundaries and interfaces, with the complete offered configuration assessed. A worklist that merely offers convenience in one product may select clinically relevant inputs in another.

What if the software creates no medical information of its own but operates the instrument that does? Software controlling an IVD instrument or supplying output about its functioning may **drive or influence** the device. Its intended purpose and supply may place it within the device as a part or component, or support an accessory analysis.

An **accessory** under IVDR Article 2(4) is not itself an IVD. It is intended for use with one or more particular IVDs to enable their intended use, or specifically and directly assist their medical functionality. The word *particular* makes a practical difference: installation somewhere in a laboratory is insufficient. A separate code module also does not establish separate supply, accessory status or an independent medical purpose.

This leaves three connected questions. Does the software have its own medical purpose? Does it drive or influence another device, or have the particular-device relationship needed for an accessory? How is it supplied—as a product in its own right or an integral component? These dimensions cannot be reduced to exclusive labels.

Consider two hypothetical arrangements. An image interpreter can create a diagnostic highlight while integrated into a larger device; integration does not erase its own medical purpose. A separately offered sequencing instrument controller can operate the instrument without producing its own medical information; separate supply does not create that information. Each needs the appropriate product analysis. Neither arrangement adds a fact to our local packets. The software guidance preserves

independence of medical purpose under either supply form; Chapter 3 applies the class rules after these relationships are described.

Return to the cloud score. Its score and highlights warrant the inquiry into medical information, while storage and the worklist need their own functions and interfaces mapped. The packet still cannot settle IVDR versus MDR for the whole application. The proposed sequencing gate likewise needs its claimed output, interaction with the assay and supply established: an accept/reject rule may belong to the controlled examination, and may have a distinct claimed medical output. The existence of code that performs the gate cannot answer all those questions.

A useful conclusion can therefore close one function while leaving another open. A carefully bounded finding about an archive's lack of independent medical purpose would not settle a connected scorer. The reasoning moves from the observed job to intended medical information or device use, then to the legal definition and supplied boundary. It gives a later classification decision something definite to classify.

### **Reviewer Lens – function and supply**

The difficult review judgment is often whether two apparently different descriptions answer the same question. “Inside the platform” answers where the scorer is supplied; “diagnostic information” describes what it is meant to contribute. Neither substitutes for the other. For the sequencing gate, the corresponding inspection asks whether it controls the offered assay, creates a distinct medical result or does both. A reviewer can credit a clear supply description while requesting the missing output claim, or close a plain-copy archive item while leaving the scorer open.

### **Consultant Lens – function and supply**

A company is naturally tempted to settle the route from the module name before product management has settled the offer. Separating medical output, device-control action, particular-device relation and supply in the product map exposes that shortcut. The image interpreter and its archive/worklist dependencies may need different findings; the gate and assay may need another set. These maps also make later investigation cheaper. In Chapter 1's separate deployed pathology service, late worklist entries after a service change point to a dependency even if the model is unchanged. In its separate sequencing service, more invalid first reports after a materials change call for comparison of material, gate and analysis states. Neither observation proves a new purpose or makes an integrated module a separately supplied device.

## Manage a borderline by naming the fact that would decide it

The cloud team could commission a study of highlight accuracy while still being unable to say what the risk score estimates. What would a successful study then establish? A useful borderline inquiry names the fact that would distinguish the possible purposes: is the score intended to inform diagnosis, direct attention to a possible lesion or arrange technical quality review? The claims and actual worklist should show what the pathologist sees and can do. The name “laboratory information service” supplies a setting, while “pathologist responsible” supplies a final actor. Neither describes the score's contribution.

For the sequencing classifier, the corresponding uncertainty is observable in a specimen's path. Is rejection visible and recoverable? Is it part of the assay's intended quality control? Can accepted input proceed despite inadequate data? False rejection could withhold or delay a usable specimen; false acceptance could admit unsuitable input. These are mechanisms to investigate, with no supplied error rates or demonstrated clinical harms.

A claim label still deserves its proper weight. **Research use only** means that the manufacturer intends research use rather than clinical diagnostic examination. IVDR Article 1(3)(a) excludes research-use-only and general-laboratory products unless, in view of their characteristics, the manufacturer specifically intends them for in vitro diagnostic examination. That makes the label relevant evidence of purpose, to be read with the offered function, presentation and integration. If a customer uses a product without authorization, that fact needs its own examination; it is not automatically the manufacturer's assigned purpose. The inquiry concerns the actual product proposition, rather than which short label sounds most reassuring.

Once that proposition is defined, a study can ask a meaningful question about it. An investigation to establish or confirm an IVD's analytical or clinical performance is a **performance study**. For the cloud proposal, the question might concern how the highlights help pathologists with the claimed task. For the gate, it might concern false acceptance or rejection. A protocol about a different output would leave the original purpose question unanswered.

This also explains the distinction between manufacturer and **sponsor**. Under IVDR Article 2(57), the sponsor initiates and manages the performance study and arranges its financing. The manufacturer owns device purpose and the conformity argument. One organization may occupy both roles, but a sponsor cannot make an undefined purpose testable merely by producing a protocol. The full study route and national processes belong to Chapter 12; defining a study question here supplies no permission to begin it.

The uncertainty can now be expressed as a decision another person could resolve: here is the present bounded claim; here is the alternative if a specified function or claim differs; here is the item that will distinguish them. For the cloud score, the items include instructions, marketing description, score meaning, image-to-worklist logic and user display. For the sequencing gate, they include assay configuration, gating rule, override and outcome path. A later change in materials or workflow in Chapter 1's separate deployed services would reopen the match between the observed product state, original purpose and earlier evidence. The field observation itself would still supply no new diagnosis, harm or class.

### **Reviewer Lens – borderline and study question**

An impressive study result can make a poorly defined question less visible. Highlight accuracy would answer a calculation question; it would leave a distinct question about how images reach the pathologist. Similarly, variant-call accuracy among accepted specimens says nothing about usable specimens rejected before analysis. A reviewer can recognize the value of the tested output and still request the missing path. The useful purpose alternatives are concrete—diagnostic attention or technical quality ordering, for example—and stay alternatives until claims and display establish the offered role. The manufacturer's argument and the sponsor's later study-start record remain separate.

### **Consultant Lens – borderline and study question**

Before studies are planned, the consultant can show the company what each unresolved choice costs in evidence: define the score's role and study the relevant reader/worklist task, or change the actual function and presentation so that a narrower purpose is truthful. The sequencing choice similarly needs accepted and rejected paths considered together. When a performance study follows, named responsibility for initiating, managing and financing it prevents a manufacturer–sponsor overlap from becoming an omission. Advice clarifies the choices and responsibilities; it does not select the offered product or authorize the study.

## **See how an early scope error travels**

Suppose the whole cloud platform is described as storage. The evidence team may then study a product boundary that leaves out the score and highlights. A later class or conformity discussion inherits that incomplete object. Calling a sequencing gate an administrative filter can similarly omit false acceptance, false rejection and recovery from the evidence plan. An early description matters because other people build their work from it.

The affected questions may extend to classification, technical documentation, risk controls, performance evidence, registration, postmarket duties and any applicable

artificial intelligence (AI) obligations. Their applicability depends on the actual product and rules. Neither “cloud” nor “classifier” activates them all. The next exhibit gives new facts about the opening application, so its current conclusion can be reconsidered on evidence.

**Additional exhibit — opening cloud case.** Time-to-review data show that low-score positive cases are reviewed several hours later during peak workload. The interface hides the secondary queue by default. The packet supplies no count, denominator or established patient injury and does not isolate how much of the delay is attributable to any one feature or workload factor. These facts belong to this local cloud case, not Chapter 1's separate deployed pathology service.

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### **Decision**

**The company proposes keeping a non-device conclusion and calling the score “informational.” What would you examine before accepting or rejecting that conclusion? What does the exhibit establish, what remains open, and which record needs reconsideration first?**

### **How a reviewer follows the new timing evidence**

The exhibit changes the factual account before it settles the legal one. Low-score positive cases were reviewed several hours later during peak workload, and the secondary queue was hidden by default. Both observations deserve a place in the record. They warrant examination of purpose, risk and workflow evidence even though the pathologist still makes the diagnosis.

Their relationship remains partly unknown. The packet does not isolate the score's, queue's, workload's or another factor's causal share. With no count or denominator, it establishes no overall delay rate; it establishes no patient harm. Calling the score “informational” resolves none of those factual questions.

The intended-purpose and function record is the first place to return. The useful evidence concerns the score's claimed meaning, entry into each queue, what users see and can recover, and how the manufacturer presents the combined workflow. Qualification still requires the product-specific medical-information and supply analysis, even if one module eventually proves to lack its own medical purpose.

## Reviewer Lens

The quickest way to test the revised account is often to follow one low-score case from image intake to review beside a demonstration of the default and secondary queues. The purpose statement, function map and screens should describe the same route. This keeps two supplied findings—later review and a hidden queue—distinct from a causal conclusion the packet cannot support. The precise medical purpose and product relationship remain open as well. For a proposed sequencing classifier, following a rejected usable specimen and an accepted unsuitable one serves the same purpose: “quality only” cannot show the consequences of either path.

## Consultant Lens

After this exhibit, advice needs to compare real product choices. Visible prioritization and queue recovery with evidence of the full workflow is one possibility; removal or redesign of the consequential selection function is another. A narrower claim is useful only if the offered function and presentation match it. Pursuing the applicable device route with evidence is also a management option. Design and clinical owners need to understand what demonstration would settle each remaining uncertainty and how the options affect product value and work. For the sequencing gate, changing override, recovery or the gate's role within the assay would likewise require renewed function and evidence assessment. None of these options promises a non-device result, certification date or market outcome.

## Work three separate function boundaries

Can a review ever close the qualification question positively when neighboring questions remain difficult? Consider a **separate general archive**. It copies and retrieves images without interpreting, selecting or changing their content. Its manufacturer assigns no diagnostic purpose and no intended role specifically assisting a particular IVD. These stipulated facts support a bounded finding: no independent medical purpose and no supplied basis for IVD-accessory status. Hospital location does not reverse it. The exact behavior and claims make the finding defensible and identify what later changes would need comparison.

This archive and the next two examples are independent local products. They are not successive releases of the opening cloud application.

A **separate integrated image gate** shows why a positive finding can have a different scope. It rejects specimen images before analysis using a score intended to protect diagnostic-result quality. It may stop an unsuitable image but could also exclude a useful one. Its place in the medical system's function, risk and evidence map is therefore clear. Its precise legal form is not: the packet leaves open its own medical-information purpose, driving or influencing relationship, possible accessory

status and component or separate supply. Claim, output, control of another device, user visibility and supply facts must decide those questions. This gate is neither LumaCell Review nor the opening cloud application.

A **third, separately offered treatment-information module** interprets specimen-derived features to provide patient-specific treatment-selection information. First, that promised information and its specimen source must be tested against IVDR Article 2(2). Only if it qualifies as an IVD does the additional **companion diagnostic (CDx)** question arise.

Article 2(7) asks for more than treatment relevance. The device must be essential for the safe and effective use of a corresponding medicinal product, identifying before or during treatment patients most likely to benefit or at increased risk of serious adverse reactions from that treatment. The packet supplies no corresponding-medicine relationship. Output, intended medicine relationship and supplied boundary must be established before either qualification conclusion; class and conformity route follow afterward.

The following comparison is useful when deciding which part of a function record can close. Each row has its own supplied facts and its own stopping point.

| <b>Independent example</b>   | <b>What can be concluded now?</b>                                             | <b>What keeps that conclusion bounded?</b>                                                                 |
|------------------------------|-------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Copy-and-retrieve archive    | No independent medical purpose or supplied IVD-accessory basis.               | No interpretation, selection, content change, diagnostic claim or particular-IVD assistance is stipulated. |
| Integrated image gate        | Its role belongs in the medical system's function, risk and evidence account. | Own medical purpose, device relation and supply still require facts.                                       |
| Treatment-information module | IVD qualification is the first legal test; CDx is an additional test.         | Specimen source, medical output, intended medicine relationship, clinical use and supply must be known.    |

The archive row can close at its stated scope. The gate row closes a different question—whether its role must be included—while leaving own medical purpose and supply open. Its evidence needs the input, accept/reject/unavailable paths, visibility, recovery and medical claim. For the treatment module, a named corresponding medicine, if claimed, is part of the additional inquiry. The table is a way of retaining these different conclusions; it is not a ladder of increasing device status.



**Decision**

**Change only the general archive's function. It now uses image content to remove suspected nondiagnostic slides from the *only* worklist, under a claim that this ensures reliable diagnostic analysis. Which record should be reopened first, and which evidence would resolve the changed boundary?**

**How a reviewer handles the changed archive**

The original positive finding was adequately bounded to copying and retrieval. The new behavior is image-content selection, so the earlier purpose and function qualification needs reconsideration before a class table can help. This does not make the earlier finding wrong on its original facts.

Selection logic, claims and instructions need to be connected to the image-to-worklist path. The useful evidence includes the number and nature of accepted, rejected and unavailable paths in the intended use, and how a rejected slide can be found and recovered. These facts establish a medically consequential function to assess; they do not automatically establish software with its own medical purpose. Supply and particular-device relationships still need facts.

An archive newly excluding sequencing specimens would raise the same new-function question. Describing the change as a metadata update would leave the actual exclusion unexplained.

**A qualification record a reviewer can use**

A record that says “pathologist final” leaves the opening application's route to that person unexplained. A useful qualification record follows each function that can change an image's or specimen's path: input, transformation, output, user and action. Alongside them belong the manufacturer's claim sources, the applicable IVDR or MDR definition or possible nonmedical boundary, and the evidence location or missing fact. This is how another reviewer can understand why the unchanged archive's finding closes while its new selection function needs analysis.

For the sequencing gate, “no report field” is similarly incomplete. The accept/reject decision can change what reaches the report. The record needs to explain that contribution even while its precise legal status remains open.

Chapter 1's **working states** attach to those bounded items: **A**, adequately explained; **U**, present but unclear; **M**, missing or inconsistent; **D**, deeper analysis required; **C**, specialist or institution consultation indicated; and **N/A**, inapplicable with a reason. They never add up to a device approval.

The unchanged general archive's medical/accessory boundary can be A on its stipulated facts. The cloud record can retain the supplied later-review observation and hidden-by-default queue while marking their causal relationship and exact IVD-versus-MDR boundary D or U. An evidence location, a reason and the fact that would move the state make the letter useful. Without them, it merely hides the uncertainty in a shorter form.

## **A qualification memo a consultant can deliver**

If the cloud team has not decided whether low-scored images will remain hidden, a fixed “non-device launch” estimate rests on a product the team has yet to define. A useful consultant memo makes that dependency visible. It starts with the version and purpose, then connects the function map to the claim sources. The software's own medical-information purpose receives one finding; device relationship and supply receive another. Integration can coexist with its own medical purpose, so these cannot be competing boxes.

The memo can then present plausible branches, including a genuinely nonmedical archive or an accessory/component relationship, with the unresolved fact, its owner and the decision depending on it. Management can change the offered function, change a claim to match that function, or resource the applicable device route. Conditional advice is precise when it says which choice changes the work. A sequencing team undecided between an alert and exclusion likewise needs both branches understood before it selects the product. The consultant explains their consequences; management decides.

**Changed-case transfer — a sequencing gate.** A quality classifier rejects specimens before variant analysis. No error rate, report consequence or supply arrangement is supplied.

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### **Decision**

**Where could a false rejection and a false acceptance change the input-to-result path? Which further fact would answer each question about the classifier's own medical-information purpose, its control of the assay or assistance to a particular device, and whether it is supplied within the system or separately?**

### How the mechanism and product questions stay separate

A false rejection removes a usable specimen before analysis; a false acceptance admits unsuitable input. Those mechanisms identify paths to examine, with no supplied rate or demonstrated report outcome. The claimed information, interaction with the assay and supply arrangement remain separate facts to obtain. One presumed label cannot stand in for all three.

Chapter 1's *separate* deployed sequencing service has more invalid first reports after a materials change. That signal would prompt a comparison of material, gate and variant-analysis states; it does not prove this classifier caused those reports. Its *separate* deployed pathology service has later worklist entries after a service change, prompting an interface inquiry. Neither service becomes the opening cloud application's low-score exhibit.

## Quick discrimination lab

The two snapshots below are independent fictional products, separate from the opening cloud case and the three-function comparison. Each invites a present bounded finding, the missing fact and the next action by the appropriate actor.

**Case 1 — an image platform.** A pathology platform is marketed as image storage. Before images reach a diagnostic model, it automatically rejects slides below a proprietary quality score and never exposes them to the user.

### Decision

**Which described functions fit “storage,” and what needs analysis about the quality score? What remains unknown about its independent legal status?**

### How a reviewer separates the stored image from the rejected image

Copying and retrieval may still be storage. The score adds another operation: content-based rejection determines which specimen images reach the diagnostic workflow. Purpose, output, false rejection and acceptance, user visibility and recovery need to be connected to how the platform is supplied with the model. The function belongs in qualification, risk and evidence.

That is a consequential but bounded finding. The snapshot leaves the gate's own medical purpose, its particular-device relationship and its supply as part of the platform or separately undecided. A safer interface or narrower wording would require the actual function to be reassessed; neither gives an automatic non-device outcome. The manufacturer's next contribution is a controlled function and supply account that can resolve those questions.

**Case 2 — a genomic ranking module.** The manufacturer labels a module “research use only,” bundles it with a commercial assay and supplies a shortlist that a tumor board uses to select therapy. The snapshot supplies neither a corresponding medicinal product nor all the manufacturer's claim material.

### Decision

**How much can the research-use label establish alongside the supplied function and integration? Which product facts would decide IVD qualification, and which additional facts would be needed for a companion-diagnostic conclusion?**

#### How a reviewer reads the label beside the offered function

The label is one claim source, while the commercial-assay integration and therapy shortlist are facts about the offered work. What did the manufacturer intend and present? What specimen-derived information does the module create, and what clinical action is it meant to support? These facts may contradict a research-only designation and support an IVD-software argument, but this packet does not complete the determination.

The tumor board's shortlist leaves a further question open. It supplies neither the Article 2(7) relationship to a corresponding medicine nor the condition that the test is essential for that medicine's safe and effective use. Treatment relevance is insufficient to make that additional finding. The manufacturer must stabilize the product and claim; a consultant can show alternatives and evidence requests without promising a classification or certificate.

## Check the qualification record

A usable qualification record lets another person follow the specimen or image to the user's action and see which legal definition was tested. Copying, creating information, controlling another device and selecting inputs need enough distinction for the conclusion to be intelligible. A changed claim, function or interface reopens the affected finding. It does not join independent cases into a product history.

The seven checks below return to that record. Their original identifiers retrieve the expanded Workbook Part A2 cards; the original-to-integrated crosswalk preserves the lookup when source-era topic numbers differ from current chapters. Each working state belongs beside the evidence location or absence and the limited inference it supports.

**Purpose and user action • card IVDR-CH02-A01**

Do the controlled claims and actual image/specimen path agree on purpose, input, output, user, setting and clinical role? The cloud score's intended use and the sequencing operator's action need to be visible behind the word “support.”

**Storage and medical information • card IVDR-CH02-A02**

Which transformation, selection and worklist rules distinguish copying/retrieval from the other offered work? Pure storage can receive a bounded positive finding; content-based selection of clinical inputs needs its own analysis.

**Functions and their interfaces • card IVDR-CH02-A03**

Do module and supply boundaries show the failure, recovery and dependency paths? An archive without its own medical purpose can still have an interface that matters to regulated analysis.

**The fact that could change qualification • card IVDR-CH02-A04**

Which different score purpose, user action, device relationship or changed claim would move the boundary? A useful request identifies that fact more precisely than “more information needed.”

**Alternatives in the qualification memo • card IVDR-CH02-C01**

Does the memo connect a bounded purpose to controlling definitions, plausible branches, missing facts and owners? The adequate archive finding can remain beside the gate's open boundary.

**Management's product choice • card IVDR-CH02-C02**

What behavior and evidence would each claim, function or route option require? A wording change cannot by itself remove the offered medical function.

**Advice while the product remains undecided • card IVDR-CH02-C03**

Which unresolved product choice makes advice conditional, and who at the manufacturer owns it? A fixed route or launch estimate needs a defined function beneath it.

A closure can be short and still preserve its basis: “For this unchanged archive, the copy-and-retrieve behavior and claims support no independent medical or IVD-accessory purpose; a changed function or particular-IVD assistance would require comparison.” The cloud conclusion ends elsewhere: “Low-score positive cases were reviewed later and the secondary queue was hidden by default; the queue's causal share, score's exact medical purpose and supplied boundary remain to be documented.” Neither is a certificate or a conclusion about a named series product. Chapter 3 asks

which class rule applies to a qualified purpose; the archive does not acquire the scorer's class by association.

## **Source trail and next reading**

Binding sources are the IVDR, Regulation (EU) 2017/746, consolidated 10 January 2025, especially Articles 1(3), 2(2), 2(4), 2(7), 2(12) and 2(57), and Annex VIII 1.4 for the later software classification question; and the MDR, Regulation (EU) 2017/745, consolidated 19 July 2026, especially Article 2(1)–(2). MDCG 2024-11, October 2024, §§1, 2.4–2.5 and 2.8, explains IVD qualification, general-laboratory and research-use boundaries. MDCG 2019-11 rev.1, June 2025, §§3.1–3.4, 6–7 and Annex I, explains software qualification, driving/influencing, modules, information systems and digital-pathology image management. These MDCG texts are nonbinding; their examples guide questions but do not determine a real product's status. Dates, claims, configuration and supplied relations must be checked at the actual decision. Book 1A Chapter 2 offers optional further practice decomposing a clinical promise; this chapter is self-contained.

## CHAPTER 3

# Classification: The Claim Sets the Risk Context

**Before reading.** A pathology platform is called “storage,” but one function selects which images reach a diagnostic reader. What would its input, user, output and clinical role tell us that the product name cannot?

That question carries Chapter 2's qualification work into the next decision: **which risk rule fits the exact medical job the manufacturer intends to offer?** A sound answer will identify the rule, a plausible competitor and the fact that would change the result. It will also leave room for a positive class conclusion even when some of the evidence needed for the claim is still missing.

## Why one instrument cannot supply the answer

An anonymous fictional oncology team proposes a panel with several jobs. It detects inherited-predisposition variants, tumor variants and a biomarker proposed for selecting a specific therapy. The team offers one class rationale based on the instrument platform. Its regulatory lead sees a different starting point: what medical question will each result answer, and which rule will govern the device containing those purposes?

To understand the disagreement, follow what an ordinary tissue-sequencing report contains. The laboratory reads many short pieces of DNA, the genetic material in the sample. Software compares the readings with a reference and proposes variants—differences for a professional to interpret. An inherited difference may inform future disease risk; a difference in a tumor may inform a decision about an existing cancer. The same examination architecture can therefore produce information with different clinical meanings.

The inherited differences are often called **germline** variants; **somatic** variants arise in body cells, such as tumor cells. A measured feature used to provide medical information is a **biomarker**. That last name does not tell us which information is promised. In this case the proposed therapy result needs its own clinical meaning, just as the inherited-risk and tumor results do. No particular variant, medicine, added specimen, validation result, certificate or complete companion-diagnostic relationship is supplied. This local panel is separate from HelixBridge, Asterion and Chapter 1's U.S. package.



Chapter 2's qualification question came first: does the product meet the relevant legal definition? An in vitro diagnostic, or IVD, examines material from the human body outside the body to provide specified medical information. Once that qualified product and its purpose are bounded, **classification** assigns its risk class under the In Vitro Diagnostic Medical Devices Regulation (IVDR). Classes A, B, C and D reflect increasing inherent risk; Annex VIII supplies the rules. A class describes the risk context of the intended purpose. It is not a mark awarded for a complete study.

The same distinction applies to digital pathology. Copying a scanned tissue image, rejecting it for quality and estimating a disease-related feature are different jobs even when they run on one server. Neither the server nor the sequencing instrument can stand in for the claim being classified.

## Which medical question does each result answer?

For the opening panel, “inherited predisposition” begins one chain: a result reaches a defined person or population and says something about the risk of developing a condition. The tumor branch begins another: is the finding used to diagnose cancer, establish its stage, or inform some other decision? If it predicts a likely future course after a condition is present, it is a **prognosis**. That promise needs its own outcome and clinical role; it cannot be silently treated as diagnosis.

Once that difference is visible, the rule comparison becomes concrete. Human genetic testing makes Rule 3(i) a candidate for the inherited-risk claim. Cancer screening, diagnosis or staging brings Rule 3(h) into the comparison. A prognosis statement needs its actual cancer or patient-management purpose compared with the relevant rules.

Under Annex VIII Rule 3(g), the question is whether the device is intended for disease staging and a wrong result could lead to a patient-management decision creating a life-threatening situation for the patient or the patient's offspring. Rule 3(k) concerns devices intended for management of patients suffering from a life-threatening disease or condition. Neither premise follows from the word “prognosis” alone; the actual use must establish it, with the specific cancer purpose in 3(h) still compared. The opening prognosis claim therefore remains unclassified on the supplied facts. For the therapy branch, a different definition must be settled before Rule 3(f) can be used. Each branch needs its input, population, user, output and clinical role, rather than a general promise to “support decisions.”

This table helps locate the missing premise when one instrument description has been offered for several purposes.

| <b>Branch of the opening panel</b> | <b>What the result is meant to tell someone</b>       | <b>What makes the rule reasoning reviewable?</b>                                                 | <b>What still needs evidence?</b>                                                        |
|------------------------------------|-------------------------------------------------------|--------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Inherited predisposition           | A person's risk of developing a condition             | The controlled human-genetic-testing claim under Rule 3(i), with other relevant rules considered | The variants, population and proposed risk interpretation                                |
| Tumor finding or prognosis         | Present cancer, its stage, or its likely later course | The actual use compared with Rule 3(h), 3(g), 3(k) or another applicable rule                    | Each exact output and the clinical use promised for it                                   |
| Therapy-selection biomarker        | Which patient should receive a corresponding medicine | Full Article 2(7) definition before Rule 3(f)                                                    | The medicine relationship, patient-identification purpose and claim-specific performance |

The middle row is where the instrument rationale fails most visibly. The same measurement could be offered for a present diagnosis or a forecast of later course, yet those words raise different rule and study questions. The table is a map of questions, not three certified indications. A separately intended IVD instrument may itself have an Annex VIII rule; that rule does not classify every assay or interpretation function using it. A U.S. product code likewise cannot replace this European purpose analysis.

What if all these purposes belong to one offered device? Annex VIII 1.1 makes intended purpose control rule application. Implementing rules 1.7–1.9 require consideration of the relevant rules and the higher resulting class when multiple purposes or rules apply to that device. The reasoning for each branch remains useful even when the resulting class is the same. A justified C for a genetic or cancer purpose can coexist with an unevicenced prognosis claim. Adding a purpose that activates a higher rule could change the device's class; adding a purpose that also leads to C could still require substantial evidence.

That is why a proposed narrower configuration needs a real boundary: which claims and functions would it contain, and which would it exclude? A stage described only in a presentation cannot establish that boundary or promise a certificate. Nor can an event from another product fill it. Chapter 1's separate deployed sequencing service had more invalid first reports after a materials change; its own specimen and result path needs review. The separate deployed pathology service had later worklist entries after a service change; its own interface and risk need review. Neither observation establishes a class change or belongs to this oncology panel's history.

**Reviewer Lens — mixed purpose.** The useful first comparison is between the wording of each promised result and the rule premise it is supposed to satisfy. It is reasonable to close an adequately supported human-genetic-testing rule application while keeping medicine essentiality open. Treating every unresolved branch as a reason to discard the genetic finding loses information. The corresponding software trap is starting Annex VIII before knowing what the image score means, or deciding a specimen gate's status from its absence on the final report. The focused request concerns the actual information or control and the offered product state.

**Consultant Lens — mixed purpose.** Early separation of the claims makes management's choice visible while it can still influence study scope and release design. A broad proposal and a controlled narrower one may share a class yet require different evidence, consultation or software report rules. The tempting shortcut is to present “C” as cover for an immature prognosis claim. It leaves the study team with an undefined promise and the later file with a scope mismatch. A useful option comparison identifies exactly what would be withheld and what work would make a future expansion reviewable. The same reasoning applies to a changed image score or specimen-rejection rule even if an existing class remains plausible.

## What must the medicine relationship establish?

The therapy branch names a clinical destination, but that alone does not tell us which rule fits. A result can be discussed before treatment without being essential to the medicine's use. Article 2(7) reserves **companion diagnostic (CDx)** for a device essential for the safe and effective use of a corresponding medicinal product, intended before or during treatment to identify either patients most likely to benefit or patients likely to face increased risk of serious adverse reactions from that treatment.

There are therefore two linked questions behind the medicine name: why is the test essential to safe and effective use, and which of those patient-identification jobs does it perform? A biological association with response cannot answer both. Neither can a named therapy in a sales presentation or an established commercial partnership. The opening panel has not supplied the complete definition, so its therapy branch cannot yet close Rule 3(f).

Now consider a *separate stipulated example* that does satisfy every Article 2(7) element. Rule 3(f) assigns C. That is a genuine positive class conclusion. The Article 48 conformity procedure then adds the applicable consultation with a medicinal-products authority or the **European Medicines Agency (EMA)**. Consultation is an institutional step in the route; it is not the classification rule. Scientific, analytical and clinical support for the stated test–medicine relationship remains necessary. Chapter 4 develops the route and Chapter 13 the bridging and consultation work. The stipulated example supplies no missing status to the opening panel or another dossier.

**Reviewer Lens — the definition threshold.** A request for the corresponding medicine, the label-level patient-identification proposition and the reason for essentiality is more useful than a request to “confirm class C.” Another Rule 3 branch might already support C, concealing the unresolved therapy premise if only the class letter is recorded. In sequencing, the variant call and the treatment-selection decision need different explanations. In pathology, a highlighted feature raises this medicine inquiry only when that role is actually claimed; a highlight by itself supplies no companion-diagnostic conclusion.

**Consultant Lens — the definition threshold.** The company has a product choice: substantiate the full relationship or offer a narrower truthful purpose. The tissue assay, software report rules and medicine relationship need to tell the same story. A commercial agreement cannot replace performance evidence, and deleting “companion” from the label cannot remove an unchanged treatment function. If a separately deployed sequencing service changes a knowledgebase or report rule, the relevant comparison is its actual treatment-related output and approved claim. Its materials-change invalid-report signal supplies no medicine relationship and cannot be repurposed as this case.

## What does separate delivery actually tell us?

Imagine the two Chapter 2 comparisons again: an image interpreter supplies diagnostic information inside a larger device, while an instrument controller has its own installer but produces no independent diagnostic result. If we classified from packaging alone, we would miss the first function's medical purpose and invent one for the second.

The first question remains what the software does: does it give medical information of its own, control another device, or affect how that device is used? The next is how it is related and supplied—as a product in its own right, within a larger device, as an accessory, or in another documented combination. Own medical purpose and integration can coexist. A separate installer does not establish an independent medical-information output. These comparisons add no product facts to the oncology panel.

Those relationships explain Annex VIII 1.2–1.4. Devices used together and accessories receive their specified separate analyses. Software that drives or influences another device follows that device's class; software independent of another device is classified in its own right. “Independent” in this analysis concerns the device relationship, rather than the existence of a separate installation package.

For the prospective image-quality scorer, the facts still needed are what its score tells someone and whether it controls the images reaching diagnosis. Its particular-device relationship and supply also remain to be established before deciding IVDR, Medical Devices Regulation (MDR), or non-device qualification and the appropriate class. For

the sequencing specimen gate, a false acceptance can pass unsuitable input onward; a false rejection can withhold a usable specimen. Those mechanisms make the function worth understanding, but do not decide whether it has its own medical purpose, drives or influences the assay, or is supplied in a particular legal form. A separate deployed worklist delay or invalid-report signal can reopen configuration and risk work without resolving those unknown relationships.

**Reviewer Lens — software.** A useful finding has two parts: the medical-information or device-control function, and its relationship to the product being classified. Looking at the interface exposes errors that a module list can hide. A diagnostic highlight described as storage, or a gate classified only because it prints no report field, leaves out what changes for the pathologist or specimen. “Independent medical device software,” “component” and “accessory” should not be forced into exclusive boxes before the function and supply facts are known.

**Consultant Lens — software.** A versioned image-to-reader or specimen-to-report map is valuable while product owners can still decide what to offer. It joins engineering's module boundaries to the medical output and any intended particular-device assistance. That makes a conditional class analysis possible for management's actual configuration. If a service or material later changes, the map first locates the affected dependency. A revised interface or narrower purpose can then be compared with the necessary evidence; neither choice guarantees a cheaper route.

### Does the material set the measurement or check it?

A “quality control” label raises a similar problem for physical materials supplied alongside a fictional sequencing assay. Suppose the first material establishes how that particular assay responds to a reference so that it can make its measurement. This is a **calibrator**. Under Annex VIII 1.5 it takes the class of the device it is intended to calibrate. We have learned the classification relationship, but the assay's own class has not been supplied.

The second material has another job: it checks whether the assay gives an expected result. It is a **control material**. Its manufacturer supplies a “variant detected” target for checking a run. That is a qualitative assigned value; a specified numerical concentration used as the target would be quantitative. Under 1.6, a control with an assigned qualitative or quantitative value for one or more specified analytes takes the class of the device it is intended to control. The assigned target, specified analyte and intended assay make the relationship intelligible.

What if a third genuine control comes without a manufacturer-assigned target? The supplier may describe indicative content or a range while the laboratory establishes its own working target. That description alone is not an assigned value. For this stated control-only purpose, Rule 7 gives B, subject to any other applicable rule. These three materials are separate classification questions, with no performance result or class

supplied for the fictional assay. The software gate remains separate again: rejecting a specimen does not turn software into a calibrator or a physical control material.

**Reviewer Lens — materials.** The decisive evidence sits in the item's intended-use statement, its intended assay and the actual assignment of a target. “Control” on a box is not enough to distinguish 1.6 from Rule 7. The corresponding restraint for an image-quality algorithm or sequencing rejection gate is to retain the software-function inquiry; its quality-related name cannot turn it into a material.

**Consultant Lens — materials.** Separate product records for assay, calibrator, assigned-value control, genuine unassigned control and software gate prevent a supply change from disappearing under one broad label. Each record needs its device relationship, any assigned value, rule and evidence work. When management changes what is supplied, those relationships and documentation change with it; renaming the package “quality control” does not settle the class.

## What does the opening panel's new evidence support?

The difference between a purpose and its evidence becomes easier to see when the opening packet supplies more information. Similar chemistry can serve different users; similar sequencing can support different clinical promises. The next exhibit concerns only the original oncology panel, before the separate rule examples and practice platform that follow.

**Additional exhibit — opening panel.** Its proposed broad purpose combines predisposition, prognosis and selection of a named therapy. Validation and commercial partnerships are described as mature **only for the therapy claim**. “Mature” is qualitative: the packet supplies no performance results, medicine-essentiality finding, completed consultation, certificate or evidence for the other branches.

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### Decision

**Should all three purposes enter one initial claim because a higher class may govern the device? What class reasoning can be retained, what separate evidence question remains, and which choice belongs to the manufacturer?**

### How a reviewer reads the favorable description

The maturity statement is useful for deciding where to investigate readiness first. It is not a substitute for the studies themselves. For the actual proposed configuration, the manufacturer still needs every applicable rule and, separately, scientific association, measurement performance and clinical-use support for every intended-purpose element. The reported maturity does not complete Article 2(7) or establish certification readiness.

The reviewer can retain a justified bounded class finding while identifying the missing evidence for another claim. Management can compare a genuinely narrower configuration with the broad proposal, including what would have to be excluded from the initial offering. A gap in prognosis evidence is not a reason to pronounce every study inadequate; it is a reason to keep that proposed use open. The manufacturer's configuration choice remains distinct from the reviewer's assessment of the chosen proposal.

**Reviewer Lens — opening exhibit.** The partnership is easy to find in a submission and can make the therapy story feel complete. The harder, more useful comparison puts the actual rule premise beside the evidence for the exact use. The predisposition row needs the genetic-testing basis and support for its risk message. Prognosis needs a defined outcome and rule basis. The therapy row needs the Article 2(7) premise before Rule 3(f). The same separation would preserve a qualified pathology score's class finding while leaving reader-workflow performance open. Enough is a supported conclusion for each row, not equal confidence across them.

**Consultant Lens — opening exhibit.** A company can compare one broad configuration with controlled narrower configurations without promising a staged certificate. The comparison becomes practical when it includes labels, report rules, studies, consultation and documentation for each option. Durations stay conditional on that work and actual capacity. A later therapy-report change in a separate sequencing product, or clinical-prioritization change in a separate pathology product, needs the same comparison against the earlier purpose and evidence before an existing certificate scope is assumed to cover it.

## When can a class conclusion be positive and complete?

A specimen container is supplied sterile, so its assessment work differs from non-sterile supply. Has its class changed? This is a useful place to begin five separate rule examples because it forces the class question apart from the route question. Each product below is independent of the oncology panel and of the other examples. Each stipulates that no additional purpose activates a higher rule. As before, all applicable rules must be examined; another device merely used alongside a product does not automatically inherit its class.

The container is specifically intended for the primary collection and preservation of a human specimen before an IVD examination. Those words make it a **specimen receptacle** under Article 2(3), rather than any cup present in a laboratory. Rule 5(c) assigns A. Sterile supply changes the evidence and limited notified-body involvement under Article 48(10), while the class remains A. A reviewer can close the stipulated class and leave sterility evidence open. A consultant comparing sterile and non-sterile supply changes the route-work estimate, not the class cell.



What if the user also performs the examination? A separate sole-purpose urine pregnancy test is intended for a lay person to detect pregnancy. That is **self-testing**. Rule 4(a) generally assigns C to self-tests but expressly excepts pregnancy detection, giving **B** here. “Sole-purpose” matters because another claimed measurement could activate a higher rule. The exception answers class; it leaves sample handling, comprehension and invalid-result warnings to the actual lay-use evidence. The reviewer checks the full purpose before accepting the exception; the consultant plans for the home user rather than copying a professional laboratory protocol.

The next product has the full relationship the opening panel lacks. This *separate stipulated companion diagnostic* is essential to the corresponding medicine's safe and effective use and identifies likely beneficiaries before or during treatment. With all relevant Article 2(7) elements satisfied, Rule 3(f) gives **C**. Its reviewer can close that premise and class while reviewing performance and the distinct consultation route. Its consultant can organize those workstreams without treating a medicine association as an equivalent premise.

Does detecting a serious infectious agent necessarily give D? The fourth example supplies a more precise job: a test screens donated blood for a transmissible agent **to decide suitability for transfusion**. The first limb of Rule 1 gives **D** on that purpose. Detecting the same organism for an infected patient's care does not supply the donation-suitability premise. Other Rule 1 limbs could still apply to that diagnostic use, depending on its disease and purpose facts; they must be examined before a lower rule is considered. The reviewer preserves the donation condition for this positive example. The consultant cannot use D as a general synonym for serious infection.

The fifth example reaches **B** by exclusion. Rule 6 applies after the manufacturer has considered the other relevant rules and found no more specific result. The Medical Device Coordination Group classification guidance, revision 5, gives a professional near-patient urine dipstick for ordinary urinary-tract-infection assessment as a Rule-6 B example. That guidance example is not a decision on an actual product, whose pathogen, claims and error consequences still need checking. A reproducible rationale therefore records why Rules 1–5 do not apply and considers any distinct Rule 7 control-material question. The bounded B option belongs to the defined use, not to everything called a dipstick.

The table is useful after that reasoning: it locates what has closed and the next question that has not.

| Separate stipulated product purpose                                         | Supported rule and class          | Separate work still to assess                                               |
|-----------------------------------------------------------------------------|-----------------------------------|-----------------------------------------------------------------------------|
| Primary collection and preservation of a human specimen for IVD examination | Article 2(3); Rule 5(c): <b>A</b> | Collection and preservation evidence; limited notified-body work if sterile |

| Separate stipulated product purpose                                        | Supported rule and class      | Separate work still to assess                                                   |
|----------------------------------------------------------------------------|-------------------------------|---------------------------------------------------------------------------------|
| Lay urine test solely for pregnancy                                        | Rule 4(a) exception: <b>B</b> | Lay handling and interpretation; added claims if any                            |
| Full Article 2(7) companion diagnostic                                     | Rule 3(f): <b>C</b>           | Claim evidence and medicinal-products consultation                              |
| Donated-blood agent screen for transfusion suitability                     | Rule 1, first limb: <b>D</b>  | Assay performance and the route's detailed reference-laboratory or expert steps |
| Qualified IVD after other applicable rules have been examined and excluded | Rule 6: <b>B</b>              | Adequacy of the exclusions and evidence for the residual purpose                |

The first row resolves the container puzzle: extra sterile-supply work belongs in the last column while A stays in the middle. The last row has a different burden. Its B depends on the exclusion reasoning already having been done; “residual” is not permission to skip the specific rules. None of these rows supplies evidence for another fictional product.

One further distinction explains why the professional dipstick's location did not decide its class. **Near-patient testing** is performed outside a laboratory, near the patient, by a health professional. Training in an assay alone does not make someone a health professional for this definition. Rule 4(b) classifies such a test in its own right; location alone does not award C. Changing a purpose to lay home self-testing reopens Rule 4(a) alongside every other applicable rule, as well as the evidence for the user, environment and error detection. Article 48 procedure follows classification and may include self-test-specific documentation assessment.

### Decision pause – a changed specimen

A separate tissue companion diagnostic has a reasoned class-C classification. A proposed extension adds **liquid biopsy**—testing material derived from a body fluid rather than the earlier tissue—for the same marker. Assume Rule 3(f) still applies and no higher rule applies.

### Decision

**Does the unchanged class close the specimen extension? Which conclusion can remain closed, and what would the new path need?**

### How a reviewer holds the class steady while reopening evidence

The premise explicitly keeps the class at C. The new specimen still has to reach a justified result: collection, handling, analytical performance and clinical applicability need their own argument. The manufacturer must also assess the consultation and change implications under the applicable route. The packet supplies no fluid type, assay performance, commercial status or certificate scope for the extension.

A precise finding is “C on the stated rule assumptions; the broader specimen purpose is not yet evidenced.” That closes a real question without extending the conclusion into another. The consultant can sequence the evidence and management decision from that distinction, but cannot promise certification of the extension.

### Could another reviewer reproduce the result?

Suppose the class cell says B for the pregnancy test, C for the stipulated companion diagnostic and A for the sterile receptacle. The letters are correct on their premises, yet a later reader needs to know what those premises were. A reviewable record connects the controlled claim and configuration to the qualified product or function, every plausible Annex VIII rule, the wording that makes the selected rule fit and the fact that would favor a competing rule. A different field holds the evidence-sufficiency conclusion.

For the pregnancy test, that separation preserves the genuine B exception without inventing lay-user results. For the receptacle it preserves A without closing sterility work. For the opening panel it preserves a supported branch without granting the proposed breadth. If later evidence from the separate deployed pathology or sequencing services calls for renewed function or product-state analysis, it belongs to those services' records; it cannot be pasted into the panel's class history.

### What can management decide from that record?

The class record becomes useful advice when it compares actual product choices. A claim-option memo begins with a device boundary and the purposes for the *same configuration*, then shows the rule comparison and resulting class. Evidence, consultation and route work remain visible beside it. Management can choose between a unified broad proposal and narrower stages only if labeling, software, assay materials and evidence boundaries really control the exclusions.

That control is what makes a cost or timing estimate meaningful. The estimate depends on the actual work and notified-body availability; the class letter alone supplies neither. An image-scoring option cannot be priced as an IVDR route while its medical purpose and supply are undefined. A sequencing option cannot be priced as a companion

diagnostic while essentiality and the corresponding medicine remain unestablished. The consultant explains those dependencies; the manufacturer chooses the offering.

**Changed-case transfer — a different near-patient test.** A fictional infectious-disease test is intended for a trained professional near the patient. A new proposal changes only the user to a lay person at home; chemistry stays the same. How do the original and new purpose statements differ? What facts support the Rule 4(b) inquiry, what does Rule 4(a) now require considering, and which other rules could remain controlling? Which user, environment and route work reopens? No original or revised class is supplied, and neither “near-patient” nor “home” determines it. This is separate from the stipulated class-C cassette in the later lab.

## Practice studio — which product is the company proposing?

A second anonymous fictional company proposes one multiplex sequencing platform for hereditary cancer predisposition, tumor prognosis, a named therapy-selection relationship and general research reporting. It wants one broad intended purpose and assumes class C covers every use. Evidence, labeling and commercial partnerships are described as mature only for therapy selection. The packet gives no complete Article 2(7) analysis, study results or authorized claim.

This practice platform is independent of the opening panel, Chapter 1's U.S. package, HelixBridge and Asterion. The similar modality is a chance to apply the method again, not permission to borrow a product history. Its research output also needs its own account: it cannot be silently absorbed into a clinical purpose or used to erase a clinical function.

### What reaches the person making the decision?

For hereditary predisposition, the missing sentence connects a specimen and measured feature to a risk message for a defined population, user and setting. For prognosis, it connects a tumor result to a future outcome and the action that forecast informs. “Detects all clinically relevant variants” would leave those boundaries uncontrolled. For therapy selection, the sentence needs the corresponding medicine and the essentiality and patient-identification elements. General research reporting needs a truthful research purpose and an explanation of its separation from the offered medical functions. Each proposed use has an input, person, result and decision; vague support language obscures the differences.

The next comparison is between those promises and the actual studies, configuration, labeling and commercial plan. If a U.S. package exists, it can inform that comparison; it cannot transfer an IVDR purpose or certificate. A useful record of an added claim follows the added word into the changed output or user action, then into its rule question and missing evidence. For this practice platform, prognosis evidence is not

supplied. The maturity description for therapy selection provides no study facts that could bridge that gap.

**Reviewer Lens — practice purpose.** The same sequence readings can support two very different sentences. For therapy, the reviewer needs the medicine and intended patient-identification role; for prognosis, the projected course and population. That is why a general analytical result cannot make the two rows interchangeable. A separately qualified pathology prognostic score would need its own output and rule rationale. A neighboring quality gate does not acquire that purpose merely by sharing the image path.

**Consultant Lens — practice purpose.** The early decision is which of the four outputs management actually wants to offer. Comparing their purposes and evidence gaps exposes missing clinical, software or medicine facts and who can resolve them. A narrower first proposal only reduces scope if the product and labeling truly withhold the other functions. Research wording has to match actual supply and use. This is a way to scope work, not to predict certification or launch.

### **Which premise makes each rule fit?**

With those outputs in view, Rule 3(i) is a concrete candidate for hereditary human-genetic testing. Tumor prognosis still requires its exact cancer or patient-management use before the relevant Rule 3 clauses can be compared. The named therapy branch still lacks the complete Article 2(7) premise for Rule 3(f). “Oncology” cannot do the work of those separate explanations.

The same purpose discipline extends to any supplied materials. A calibrator for a particular assay follows 1.5; an assigned-value control for specified analytes follows 1.6; a genuine control without an assigned value raises Rule 7. The material's name or class cannot classify the whole platform. If all applicable purposes belong to one offered device, 1.8–1.9 give the higher resulting class, with each branch's reasoning retained. Scientific, analytical and clinical support must then be assessed for each claim.

Software adds the familiar two questions: own medical purpose and device relationship or supply. An integrated image scorer may create medical information; a specimen gate may drive the assay without independently diagnosing. Only the qualified, actually supplied product can receive the rule analysis. A familiar laboratory platform, U.S. product code or enthusiastic commercial partner supplies none of the missing product premises.

### **Which changed words would change the analysis?**

The following six contrasts are independent of the practice platform and of one another, except for the expressly compared laboratory and lay versions of the respiratory-test purpose. They reveal which words need attention before any class is closed. None supplies a study result or market history to the platform.

**A professional respiratory diagnosis.** A laboratory tests a patient's specimen for a respiratory infectious agent to help diagnose that person's illness. The purpose is patient care rather than deciding whether donated blood is suitable for transfusion. That removes the donation-suitability limb of Rule 1. Does it remove Rule 1 altogether?

No class can be selected without the remaining disease and intended-use facts. Another Rule 1 limb covers detection of the presence of, or exposure to, an agent causing a life-threatening disease with a high or suspected high risk of **propagation**—spread between people. A further limb concerns determining **infectious load**, the measured amount of infectious agent, when monitoring it is critical to managing a life-threatening disease. Detecting presence does not itself establish an infectious-load purpose.

If neither of those limbs fits, the next question is what infectious information the test provides and what a wrong result could do. Rule 3(c) gives C for detecting an infectious agent when an erroneous result carries a significant risk of death or severe disability to the tested individual, foetus or embryo, or the individual's offspring. Rule 3(e) gives C for determining infective disease or immune status when a wrong result could lead to a patient-management decision creating a life-threatening situation for the patient or offspring. Specimen also matters: Rule 3(b) covers detecting an infectious agent in blood or cerebrospinal fluid without high or suspected high propagation risk. These are distinct premises, not consequences supplied by the word “respiratory.” An established intended-information, specimen or error-consequence fact could select a relevant C branch. If no specific rule fits after the applicable rules are examined, Rule 6 remains the residual B inquiry. The contrast supplies neither the agent nor the complete purpose, so its class stays open. Its analytical and clinical studies must cover the professional laboratory path offered. A later reagent or specimen-handling change in a *separate deployed service* would call for an evidence-applicability comparison with that configuration; the change alone would not establish a new class.

**The chemistry moves into lay hands.** A lay person now performs the respiratory test and reads or acts on its output. That changes the self-testing purpose, rather than merely distributing the old professional use somewhere else. Rule 4(a) generally points to C, with every other applicable rule still considered: a higher Rule 1 purpose could govern, and the pregnancy exception cannot be borrowed for a respiratory test. Sample collection, instructions, interpretation, invalid-result warnings and realistic home conditions become part of the changed-use evidence. The reviewer reopens the user and rules; the consultant scopes the studies and route assessment before the company offers the home claim.

**Inherited cancer predisposition.** A separate germline panel looks for inherited differences associated with future cancer risk. Human genetic testing gives a positive C basis under Rule 3(i), unless a higher rule applies. That conclusion leaves the variant range, interpretation of uncertainty and target-population evidence to be established. It

does not transform future inherited risk into an acquired tumor finding for a present cancer decision. If a separately deployed product updates its variant knowledgebase, the interpretation may change despite an unchanged specimen or instrument. Its claim and evidence need comparison before the earlier rationale is assumed to cover the new output.

**A marker proposed for a named treatment.** A somatic tumor feature may be associated with medicine response. The association still leaves the Article 2(7) essentiality and patient-identification questions open. If the full relationship is stipulated, Rule 3(f) gives C; otherwise the actual cancer, genetic or other purpose must be compared with the other rules. Performance evidence may also need to connect the assay used in a study to the offered commercial assay. A marker association in the abstract does not supply that connection, and this contrast supplies no bridge.

**An image-quality score.** A pathology program scores scanned tissue images. Is it alerting the user to rescan, excluding images from diagnosis, or giving patient-specific diagnostic information? Those functions can have different qualification and product relationships. The score's purpose and its effect on what the pathologist sees come first. For an actually deployed, separate pathology service with late worklist entries after a service change, the inquiry concerns that interface and the applicability of earlier evidence. It imports neither Chapter 2's cloud-case delay nor an automatic class shift. The reviewer locates output and intended action; the consultant helps the manufacturer settle the offered behavior and claim.

**A sequencing specimen gate.** Its decision may never become a variant finding, yet a false rejection can withhold a usable specimen and a false acceptance can send unsuitable material into analysis. Own medical information, driving or influencing the assay, component supply and accessory relation need examination as related dimensions. They are not presumed exclusive labels. No error rate, independent supply or class is given. The separate deployed invalid-report scene after a materials change calls for comparison of material and gate states; it does not establish that this contrast's classifier caused those reports. The appropriate qualified product boundary remains the starting point for Annex VIII.

The table helps locate the controlling distinction after those explanations, especially when several products use similar laboratory technology.

| <b>Independent contrast</b>    | <b>What the wording must settle</b> | <b>Consequence for rule and evidence</b>                                                                       |
|--------------------------------|-------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Professional respiratory assay | Agent and patient-diagnosis purpose | Non-donation Rule 1 limbs, then relevant Rule 3 or residual Rule 6; studies of the offered laboratory workflow |



| <b>Independent contrast</b> | <b>What the wording must settle</b>                          | <b>Consequence for rule and evidence</b>                                                             |
|-----------------------------|--------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Respiratory lay self-test   | The lay person's performance and use of the result           | Rule 4(a) and any higher rule; collection, instructions, interpretation and home conditions          |
| Inherited predisposition    | Inherited future risk                                        | Rule 3(i) gives bounded C; variant and population support remain necessary                           |
| Named therapy marker        | Essentiality and identification of the Article 2(7) patients | Full definition before Rule 3(f); the study-to-commercial-assay connection may need evidence         |
| Pathology quality score     | Rescan alert, diagnostic exclusion or medical information    | Function and supply qualification first; reader and worklist effects need evidence                   |
| Sequencing rejection gate   | How specimens are allowed into analysis                      | Function and device relationship first; false acceptance, false rejection and recovery need evidence |

For the first row, “patient diagnosis” excludes only one Rule 1 purpose. It leaves the other limbs to be examined; the table therefore keeps the class open. The inherited-risk row can already credit C on its stated human-genetic-testing purpose. Those different outcomes reflect different supplied premises, not inconsistent caution.

The resulting workshop record ties the controlled product boundary, purpose alternatives, rule reasoning, evidence gaps and class-changing facts together. Technical documentation, device registration, labeling, the performance plan and an eventual certificate application must describe the same product state. A signed provisional memo makes the decision traceable; it does not show that an authority has approved the broader claim.

### **Decision**

**The practice company proposes one broad class-C purpose because third-party assessment would be needed anyway. Compared with controlled narrower configurations, what does a justified C establish? What remains to be shown for predisposition, prognosis, therapy selection and a possible later liquid-specimen purpose?**

### How a reviewer compares the proposed configurations

C establishes the risk-class context of the defined device. It does not establish the breadth of the proposed claims. The qualitative maturity of therapy work supplies neither full companion-diagnostic status nor the study results. Predisposition and prognosis need their own claim-specific evidence; a later liquid-specimen path needs an applicability argument even if its class stays C.

The comparison therefore follows the offered labels, software report rules and studies for the broad proposal and for genuinely narrower configurations. The manufacturer chooses what to offer. The notified body assesses an actual application through the applicable route. Neither the comparison nor a shared class predicts a staged certificate, an assessment date or launch.

### Who can settle each remaining question?

The manufacturer states the product boundary and intended purposes, owns the Annex VIII rationale and selects the configuration for submission. A notified-body reviewer assesses whether the rationale covers that purpose and complete product, and later whether the conformity evidence is adequate within the body's designation. That is different from choosing the manufacturer's claim.

A national competent authority has its statutory roles, including particular borderline or study questions where applicable. It is not a replacement for the manufacturer's ordinary rule analysis. If the product is a companion diagnostic, the appropriate medicinal-products authority or EMA supplies the scientific opinion through the applicable Article 48 consultation; the notified body retains its device-assessment role. A consultant can organize alternatives and engagement, but cannot assign the class by fiat or choose the claim for management. The same allocation keeps an image-score or specimen-gate team's provisional module label from becoming an invented institutional decision.

**Transfer check — change the user only.** A test initially used by a trained laboratory professional is proposed for a trained pharmacy **health professional** near the patient; a separate proposal would make it a home self-test. Chemistry is fixed. For each proposal, what changes in specimen handling, training, instructions, error detection, interpretation, risk controls, evidence and labeling? Which classification rules and Article 48 route questions need another look, and which fact could make a different rule controlling? Neither change necessarily changes the class.

## Quick discrimination lab

These two anonymous packets are independent of each other and of the oncology examples. Each provides its own premises; neither carries an unstated approval, study result or certificate.

### Case 1 – a cassette for another user

A fictional class-C near-patient assay changes from use by trained pharmacy health professionals to use by patients at home. Chemistry and cutoff are unchanged. The original class is stipulated; the pathogen, full rule rationale and new-use evidence are not supplied.

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#### Decision

Which part of the class rationale needs another look, and what could remain the same? Which evidence and route questions need their own answers?

#### How a reviewer separates the user change from the chemistry

The intended user and Rule 4(a) self-test analysis reopen alongside the other rules supporting the original C. The result may remain C. Unchanged chemistry neither guarantees that result nor automatically selects a higher or lower rule.

The changed person and environment make collection, error detection, usability, comprehension, labeling and risk controls new questions. Article 48's self-test documentation steps may also change while the class remains C. Analytical evidence applicable to the unchanged chemistry can be retained as the reviewer seeks the changed-use evidence. A consultant can scope that work without promising the final class, assessment timing or launch.

### Case 2 – a proposed certificate description

A separate company argues that its oncology assay's class C permits one certificate description covering tissue, liquid biopsy, prognosis and several drug relationships. Evidence for those additions and the actual certificate scope are not supplied.

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#### Decision

Which conclusion can the class support, and what would need to be established before that proposed description could be assessed?

**How a reviewer keeps certificate scope tied to the assessed product**

The class alone supports none of the added specimen, prognostic or medicine relationships. Each purpose first needs stable wording and an applicable-rule comparison, then scientific, analytical and clinical evidence for the exact purpose and configuration. A proposed companion-diagnostic relationship needs the full Article 2(7) definition and its consultation; a medicine name is insufficient.

Certificate scope comes from the applicable completed conformity assessment. It cannot be expanded by the team's class letter. The reviewer can preserve a bounded existing class finding while leaving breadth unsupported. The consultant can compare staged and broad options for management without guaranteeing either outcome.

**Could the class rationale survive a changed premise?**

A useful rationale begins with the offered function and purpose, confirms the IVDR product boundary and explains all plausible Annex VIII rules. For one device with several purposes it retains the higher applicable class. The separate evidence and Article 48 route questions then have a defined product to assess. A missing claim study belongs in the evidence finding; it should not be disguised as a different class.

The seven checks below retain their lookup identifiers for Workbook Part A2 and the crosswalk. The familiar item states remain **A**, adequately explained for a bounded item; **U**, present but unclear; **M**, missing or inconsistent; **D**, deeper analysis required; **C**, specialist or institution consultation indicated; and **N/A**, inapplicable with a reason. Each state needs an evidence location or absence and the reasoning for it. These letters are neither device grades nor approvals.

**Which purpose is being classified? • card IVDR-CH03-A01**

Do the controlled output, user, specimen and clinical action make the Annex VIII comparison reproducible? The instrument model cannot close the panel's claims.

**Which rule governs the complete offered device? • card IVDR-CH03-A02**

Does the rationale preserve the inherited, tumor, therapy and other purpose branches, every plausible rule, and the highest applicable class for that configuration?

**What is established about class, and what about evidence? • card IVDR-CH03-A03**

Does the rule memo sit beside the performance and risk records without borrowing their conclusions? A justified C can be credited while prognosis or liquid-specimen evidence remains open.

**Which fact could make another rule control? • card IVDR-CH03-A04**

What changed purpose, user, disease context, device relationship or specimen path would alter the comparison? Naming the competing rule makes the reopening point useful.

**Can management compare real claim options? • card IVDR-CH03-C01**

Are the broad and controlled narrower proposals each matched to reproducible purpose and rule reasoning, with evidence and route differences shown separately?

**Is a proposed stage actually controlled? • card IVDR-CH03-C02**

Do manufacturer-approved labels, configuration identifiers, product boundaries and release sequence make the exclusions real? Without them, a stage is only a presentation idea.

**What does the work depend on? • card IVDR-CH03-C03**

Do evidence, consultation and timing estimates state the actual work and capacity assumptions under Article 48, including any companion-diagnostic or self-test conditions? A class letter cannot supply a launch date.

For the stipulated receptacle, an item state of A can close the Rule 5(c) classification while sterility-route evidence remains open. A separate adequately explained item can close the stipulated full companion diagnostic's Rule 3(f) C while performance and consultation retain their own questions. The opening panel has no equivalent full-CDx premise. Its missing relationship belongs explicitly in its own record. These are compatible bounded findings about different products.

## Source trail and next reading

The binding text is the IVDR, Regulation (EU) 2017/746, consolidated 10 January 2025: Article 2(2)–(7) for relevant definitions, Articles 47–48 for class versus conformity procedure, and Annex VIII implementing rules 1.1–1.10 and Rules 1, 3–7 for the examples here. MDCG 2020-16 rev.5, September 2026, current on the Commission guidance register at this draft's check, gives nonbinding explanations and examples for IVDR classification; it supersedes the source chapter's rev.4 citation. MDCG 2019-11 rev.1, June 2025, §§3.2–3.4 and 6.1–6.2, informs the distinct software-purpose and supply questions. Those guidance documents do not decide any fictional product's class or legal status. Check product-specific Common Specifications, implementing acts and current guidance where they actually apply; none has been assumed to settle these anonymous cases. Book 1C Chapter 4 is optional deeper practice in turning a claim and risk into an evidence plan; Chapter 3's method stands on its own. Chapter 4 will turn an adequately justified class into a product-specific conformity and notified-body strategy, not an automatic certificate.

## CHAPTER 4

# Conformity Assessment and Notified-Body Strategy

### Retrieval pause

For one claim from Chapter 3, which Annex VIII rule is the leading candidate, which rule could compete with it, and what fact would decide between them? What evidence question would the resulting class still leave open?

### How the class finding carries forward

The purpose and controlling facts make the rule comparison reproducible. Even a justified class leaves the product's performance evidence to be assessed for its actual claim and configuration. The next step is to identify the procedure that will assess conformity, without turning the class letter into a certificate.

An audit opening can look like the date a company has been waiting for. Its value depends on what the body can assess, what the manufacturer will have ready and what work remains after the audit. The European decision now moves from a justified risk class to an assessable product, an eligible body and a sequence of distinct decisions.

## SECTION 01

### What did the finished file finish?

An anonymous fictional startup finishes what it calls a class-C technical file and approaches a well-known notified body. Although designated under the In Vitro Diagnostic Medical Devices Regulation (IVDR), the body either cannot accept the relevant scope or cannot start within the company's financing runway. Another body can consider the application but requests a preliminary packet that exposes unresolved classification, companion-diagnostic consultation and software-evidence questions. No body name, certificate or completed review is supplied. This startup is separate from the candidate exercise later in the chapter and from every named series product.

The first problem is therefore more specific than “the body is slow.” The file's title has not shown which product can enter which procedure. **Conformity assessment** examines whether a defined product meets applicable IVDR requirements through the



prescribed route. The manufacturer remains responsible for the device, its evidence and its declaration. When third-party work is required, a notified body performs specified tasks within its designation. A booked audit is one event within that work.

The product boundary gives those tasks an object. Specimen or image input, assay and software versions, instrument, user, result, intended clinical action and any related medicine need to describe the same proposed product as the Chapter 3 class rationale. Article 48 then identifies the available assessment procedure. For a sequencing panel reporting inherited risk, class C would tell the company which route to examine. It would not tell the reviewer which variants or populations were studied.

Consider first how the manufacturer controls that product. Its quality management system is the operating set of processes for designing, supplying, monitoring and changing it. An audit asks whether those processes work. For a qualified digital-pathology function that scores scanned tissue images, an auditor might trace how scanner and model versions were approved and released. For a separate sequencing manufacturer, the trace might concern a reagent supplier change. Those are possible audit inquiries, not new startup facts. They make the distinction concrete: a controlled approval process still leaves a question about the performance of what it approved.

That question leads into **technical documentation**, the device-specific account of configuration, design, risk, performance, labeling and postmarket evidence. The pathology reviewer needs the images, readers, scanners and outputs behind the claimed score. The sequencing reviewer needs the materials, report software and population behind the reported finding. An audit can support the operating process while this device evidence remains unresolved. Neither an audit booking nor a folder of tests can settle an undefined qualification or class.

A further question can require another expert's judgment. For a true companion diagnostic, the appropriate medicinal-products authority gives its scientific view on suitability in relation to the corresponding medicine. This **consultation** contributes to the route without replacing the notified body's device assessment. Successful completion of applicable work can then support **certificates** with identifiable scope and conditions. A quality-system certificate may cover an approved system and device range; a technical-documentation or type-examination certificate concerns its defined device scope. One is not shorthand for every other assessment.

**Reviewer Lens.** The useful reading separates what each record actually establishes. For the startup, the class-C rationale, possible companion-diagnostic purpose and software configuration are still questions despite the “finished” file. A completed audit deserves credit for its own scope, while scanner coverage or assay/report-rule evidence can remain open. This separation helps keep a request narrow: the missing device

comparison should be sought where it can be answered, rather than by asking for another general quality statement.

**Consultant Lens.** Different owners can prepare connected work at the same time, but dependencies must remain visible. Product leadership controls the proposed claim and configuration; quality leadership demonstrates the company's quality processes; analytical and clinical owners resolve evidence gaps; regulatory leadership maps Article 48 and any consultation. A schedule becomes credible when it shows which decisions must precede others. Calling the audit date the certificate date hides that work and makes a company commitment depend on an event that cannot fulfill it.

## SECTION 02

### Who is eligible before preferences matter?

The startup's familiar body may be unable to accept its product. Is that merely a disadvantage to weigh against reputation or an early opening? The answer depends on whether the obstacle is legal scope or practical capacity. They need different evidence and lead to different decisions.

A body's **designation** defines the product types and assessment activities it is authorized to assess. The Commission's New Approach Notified and Designated Organisations system, **NANDO**, publishes that legal scope. For the exact proposed product and Article 48 route, the relevant record includes the current IVDR entry, technical codes, procedures, conditions and restrictions. The body's written application-scope position then connects that record to the actual application. Missing designation cannot be offset by a helpful assessor, a low fee or a fast date. It places the body outside the eligible set for that product.

Among eligible bodies, the next question is who would do the actual work. A pathology application covering several scanners may need assessment of image acquisition, model behavior, reader workflow and cloud suppliers. A sequencing application may need assay, bioinformatics and variant-interpretation expertise, with medicine-related work if its purpose truly calls for it. A broad website list does not identify the people or assessment arrangement that will handle these dependencies.

That arrangement leads to capacity: whether the relevant team can accept the file, in what sequence, and under what assumptions. Language, sites, fees, response windows, consultation handling and continuity when an assessor changes also affect the practical choice. NANDO does not promise an intake slot or expertise in every configuration. Conversely, a capacity problem does not alter a body's legal designation. The company needs to know which kind of obstacle it faces before proposing a response.

**Reviewer Lens.** A dated designation and written scope response are most useful when kept beside the exact claim, configuration and route. A scanner addition requires

comparison with the resulting product and certificate scope. A treatment-selection addition may introduce a true companion-diagnostic question. If the legal scope fails, reviewing fee advantages adds nothing to eligibility. If management proposes a narrower claim, the actual remaining function and evidence must support that proposal; an edited title alone cannot establish the new boundary.

**Consultant Lens.** A shortlist has two stages. Legal eligibility identifies who can be considered; technical fit and commercial arrangements distinguish the eligible options. Keeping those stages separate prevents a persuasive sales conversation from quietly compensating for a legal gap. Genuine narrowing may change the eligible set and preparation work, but requires controlled exclusions of outputs, materials, report rules, scanners or users as relevant. Its effect on launch remains uncertain until the resulting product and work are known.

### SECTION 03

## What should the first contact make answerable?

The startup's second body asks for a preliminary packet. Sending the complete file may still leave the body unable to see which decision is being requested. The first pages need to identify what it would assess and how ready that exact product is.

The intended purpose and input-to-result path locate the product boundary. User, specimen, hardware, assay and software versions, and relevant manufacturing or cloud suppliers make that boundary concrete. The qualification and class rationale, followed by the proposed Article 48 route, let the body compare the requested tasks with its scope. These are product facts and proposed conclusions, not an invitation to infer a purpose from a trade name.

The next pages distinguish a document's existence from its status. Performance evaluation, studies, risk controls, quality-system implementation and standards or Common Specifications work may be planned, partly complete or ready for appraisal. That difference affects whether an application can proceed. If a companion-diagnostic relationship is proposed, the corresponding medicine and unresolved scientific relationship belong here explicitly. "Oncology" cannot supply them.

A short question log can join each issue to the record that would make an answer useful. Which class-C purpose is being offered, and where is the controlled label? Which configuration is proposed, and where is it identified? If Article 2(7) may apply, what establishes the medicine relationship? Which study or software work remains unfinished, and what is its dated status? The body can explain designation, intake criteria, process and likely assessment units. It cannot informally certify the unfinished design, replace the manufacturer's class rationale or authorize a national performance study.

The reply therefore needs a status as well as an owner. Contractual process information, a preliminary technical view and an unresolved manufacturer decision can all be useful, but they settle different matters. A controlled response log keeps the question, body statement, status, company owner, needed evidence and date to revisit together. Resolving a route-changing ambiguity before a major study or board commitment can avoid planning work around a premise nobody has established.

**Reviewer Lens.** The comparison that matters is between the packet and the proposed submission. In a separate pathology example, a multiple-scanner claim with only one scanner in the evidence index raises a coverage question. In a sequencing example, an added treatment-selection role raises consistency questions across purpose, class and any device-specific consultation. An absent fact is a readiness finding. A preliminary reply supplies neither a certificate nor a consultation outcome or national study permission.

**Consultant Lens.** The temptation is to ask for broad reassurance so that management can proceed. A more useful exchange asks a question the body can answer from supplied facts, then assigns the unresolved company work. That makes later reliance on the response traceable. The same discipline applies to a backup body: its actual scope and capacity need checking before it can serve as a contingency. A second familiar name that fails the same eligibility condition offers no alternative route.

## SECTION 04

### What happens when the proposed product will change?

A certificate concerns an assessed product and system. What happens if the manufacturer later changes a scanner, assay material or report rule? The company needs the earlier state to remain identifiable so that it can explain what the change affects.

Under the Annex IX quality-system route, periodic **surveillance** examines the operating quality system and post-market surveillance plan, including audits at least every 12 months. The manufacturer continues gathering field information, reassessing performance and risk and maintaining technical documentation. Post-market surveillance collects and analyzes experience; post-market performance follow-up supplies planned work to confirm and update performance conclusions. Neither turns an unsupported initial claim into an acceptable one after the fact.

The change comparison concerns the actual assessed state and route. A scanner, model, specimen, report rule, biomarker, medicine relationship, site or supplier change may need internal evidence and control. Depending on its effect on safety, performance, intended use or quality-system/product-range conditions, it may also require notification, a supplement or a new assessment. The word “update” cannot choose

among those responses. The certificate, procedure and actual change determine what needs to happen.

The two separate deployed teaching services from Chapter 1 make the need for that comparison visible. The sequencing service sees more invalid initial reports after a materials change; the pathology service sees supported images reach its worklist later than the local target after a service change. Neither has a supplied cause, harm, certificate history or notified-body reporting trigger. Both warrant reconstruction of the before-and-after state, investigation and proportionate assessment of applicable duties. These are not releases of the startup.

The opening startup now supplies another exhibit. Its multi-scanner claim depends on evidence remaining useful across the proposed scanner conditions. It has a cloud model and plans a later model update. The body's quoted schedule excludes resolving a possible Artificial Intelligence (AI) Act interface and treats scanner expansion as a future certificate change. The packet establishes neither AI Act applicability, the later model's behavior nor support for the scanner extension. Those items have no resolved timetable in the quote.

Evidence is said to have **generalizability** when it remains useful in the people, sites, scanners and conditions actually claimed beyond the original study. A scanner-to-scanner comparison asks whether the evidence travels across the changed image conditions. A sequencing specimen or reagent comparison asks the analogous question about that assay's input and measurement path. This is a property of the evidence's applicability, distinct from the body's legal designation.

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### Decision

**One body offers an earlier audit but has limited experience with the startup's AI-pathology and multi-scanner questions. Another may start later with stronger relevant staff. What must the manufacturer verify and compare before choosing? Which dependencies does an audit opening resolve, and which does it leave open?**

### How a reviewer reads the offered opening

The comparison begins with designation for the actual product and route. Once both options are eligible, their usefulness depends on the evidence-ready product and the team that will assess its scanner and model claims. Application acceptance, question rounds, any required consultation, audit and technical-review sequence, and treatment of planned changes remain part of the path.

A later start may be more efficient with the relevant team. An earlier body may also be suitable if its scope and staffing fit a genuinely narrower product. Neither outcome is

established merely by the offered date or reputation. The AI Act interface remains an unresolved scheduling question, not an assumed obligation or completed assessment. Its exclusion from a quotation needs to remain visible in the company decision.

**Reviewer Lens.** A quote becomes intelligible when its object is explicit: scanner set, cloud release, intended purpose, quality-system sites and any consultation. An opening, acceptance, audit completion, technical findings, their closure and certificate decision are distinct milestones. A planned model update belongs in the change strategy; it cannot silently join the initially assessed configuration. A sequencing quote needs the same precision about assay and software before its dates can be compared with another offer.

**Consultant Lens.** Management benefits from a dependency record showing prerequisite, owner, evidence state and uncertainty for each milestone. Scanner expansion and the possible AI Act interface have their own unresolved rows here. A narrower initial option is meaningful only if the supplied product and claims truly exclude the deferred function. Preparation work and opportunity cost can then be compared conditionally. They cannot be converted into a promised staged certificate.

## SECTION 05

### Which procedure fits the class and purpose?

Chapter 3's specimen receptacle illustrates why a class does not identify all the work. Supplying it sterile leaves the stipulated class A unchanged, yet adds a specific assessment task. To understand that difference, the annexes need to be read as procedures for different objects.

Article 48 concerns devices other than devices for performance study. Annex IX assesses the quality system and, as required, device technical documentation, with surveillance afterward. Annex X examines a defined representative device type and its documentation: this is European Union (EU) **type-examination**. Annex XI assesses whether production can consistently make the covered device, through **production quality assurance**; for applicable class D devices it includes manufactured-batch verification. These mechanisms provide different work. Class and any self-testing, near-patient or companion-diagnostic purpose determine the available combinations.

For the nonsterile class-A specimen receptacle, Article 48(10) requires Annex II and III technical documentation and the manufacturer's EU declaration of conformity. That branch has no notified-body assessment, while the other manufacturer duties and need for a compliant product remain. Supplying the same class-A purpose sterile invokes Annex IX or XI with notified-body involvement limited to establishing, securing and



maintaining sterility. The class stays A. The nonsterile declaration cannot stand in for evidence and the limited assessment of sterile supply.

Class B adds a broader independent examination. Article 48(9) uses Annex IX Chapters I and III, with Section 4 technical-documentation assessment of at least one representative device per **category**. Category is the statutory sampling unit in this B procedure; it must not be replaced by a sales family or the different class-C grouping. For Chapter 3's sole-purpose lay urine-pregnancy test, the class-B exception still leaves its self-testing purpose to be assessed. Annex IX 5.1 adds the applicable examination of handling, intended-user studies and information supplied with the device. Sampling does not establish every unreviewed claim's evidence.

Ordinary class C uses a different grouping. Article 48(7) provides Annex IX quality-system assessment and technical-documentation review of at least one representative device per **generic device group**. Article 2(8) describes shared or similar intended purposes, or technology in common, permitting a generic grouping. A brand name alone does not establish one. The sampled device gives the body something specific to examine; evidence must still cover the other offered variants. For a bounded genetic-testing panel, the record identifies the sampled assay and software report. A separately qualified class-C pathology function needs its own scanner and output evidence. Self-testing or near-patient use adds the applicable intended-user assessment.

A true companion diagnostic asks a device-specific medicine question within class C. Under the Annex IX route, Article 48(7) requires Section 5.2 and Sections 4.1–4.8 technical assessment for **every companion-diagnostic device**, followed by consultation with the applicable medicinal-products authority. Ordinary generic-group sampling cannot replace this work. First, however, the actual purpose must meet Article 2(7): the device is essential to the corresponding medicine's safe and effective use and identifies, before or during treatment, patients most likely to benefit or patients likely to face increased risk of serious adverse reactions from that treatment. A named medicine beside a marker does not settle that prerequisite.

Class C also has the Article 48(8) alternative: Annex X type-examination together with Annex XI, except Section 5. Section 5 is the manufactured-batch verification procedure for class D: its exclusion keeps that class-D task out of this class-C alternative. A change to a purpose that actually requires D would reopen the route and its batch work. A true companion diagnostic still receives the corresponding Annex X medicinal-products consultation. Selecting the alternative changes what must be produced and assessed. It does not bypass the device–medicine relationship or substantiate a broader claim.

For class D, Article 48(3) provides Annex IX Chapters I, II except Section 5, and III; Article 48(4) permits Annex X together with XI instead. Annex IX Section 5 contains special-purpose procedures; the default D path excludes it, while Article 48 separately adds Section 5.1 when the purpose is self-testing or near-patient testing. A



corresponding companion-diagnostic purpose also retains the applicable medicines consultation under either route. Chapter 3's donated-blood transmissible-agent screen illustrates a D classification. Its class alone does not establish every possible additional assessment condition.

One addition depends on a designated **EU reference laboratory (EURL)** for the device. Where one is designated, Article 48(5) requires the notified body to request laboratory testing to verify claimed performance and compliance with applicable Common Specifications or an equivalent chosen solution. Those specifications provide common technical or clinical requirements for meeting specified IVDR obligations. The laboratory addresses the relevant performance question; it does not certify the device. If no laboratory is designated for that device, this cannot be scheduled as a universal task.

Another addition depends on two conditions together. Article 48(6) requires **expert-panel consultation** on the manufacturer's performance evaluation report when no Common Specifications are available for the class-D device **and** this is the first certification for that device type. A manufacturer's first submission is not necessarily the first certification of a type. Nor does absence of a Common Specification remove the need for performance evidence. Laboratory testing and an expert opinion answer different questions and can both be relevant when their respective conditions are met. Neither replaces the notified body's final assessment.

The route table is a lookup after those product and special-purpose conditions have been established. It keeps the legal paragraphs beside the work they select.

| <b>Class and condition</b> | <b>Article 48 procedure</b>                                                                                                                 | <b>Scope that remains distinct</b>                                                               |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| A, nonsterile              | 48(10): Annex II–III file and manufacturer EU declaration.                                                                                  | No body under this branch; other duties remain.                                                  |
| A, sterile supply          | 48(10): Annex IX or XI.                                                                                                                     | Body involvement concerns sterility; class remains A.                                            |
| B                          | 48(9): Annex IX Chapters I and III, plus Section 4 review of at least one device per category.                                              | Self-test/near-patient purpose adds Annex IX 5.1. Sampling is not proof for every claim.         |
| C                          | 48(7): Annex IX Chapters I and III, Section 4 review of at least one device per generic device group; or 48(8): X plus XI except Section 5. | Every true companion diagnostic needs its device-specific assessment and medicines consultation. |

| Class and condition | Article 48 procedure                                                           | Scope that remains distinct                                                                                                      |
|---------------------|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| D                   | 48(3): Annex IX Chapters I, II except Section 5, and III; or 48(4): X plus XI. | Laboratory testing where designated; expert consultation only with no Common Specifications and first certification of the type. |

The sterile-A row shows a change in assessed work without a change in class. The ordinary-C row instead links the group sampling unit to device evidence, while its companion-diagnostic condition demands the separate device-specific work. Neither a sampled file nor a quality-system certificate can silently establish all other claims in a group.

Annex XII requires identifiable covered devices or groups, intended purposes and conditions in the certificates. A sequencing company proposing a liquid-specimen extension cannot use its quality-system certificate as proof that the added specimen was assessed. A pathology scanner list or cloud release outside the approved device description does not enter certificate scope because the product name stays the same. The actual certificates and procedure must be compared with the actual proposed state.

**Reviewer Lens.** The precise Article 48 paragraph and annex path are useful because they identify what completion would mean. For a class-C companion diagnostic, that includes the full definition, every-device technical review and applicable medicinal-products opinion. For a scanner extension, the device scope and evidence boundary come before a confident label such as “amendment.” A positive quality-system finding can remain closed while device or consultation findings stay open. Class D demands another distinction: laboratory designation and the two expert conditions need separate factual support.

**Consultant Lens.** The company needs to see which records cover its system and device range, which cover the defined product, and which arise from consultation. Comparing an available Annex IX route with an available X/XI alternative means comparing the actual work and maintenance under each, not asserting that one annex is faster. Planned scanner, supplier, reagent or software changes require capacity under the chosen route. A narrower first application only helps when the withheld use is genuinely absent and later evidence/change work has an owner.

## SECTION 06

## What does an available date establish?

The following engagement record is independent of the startup. Its two fictional candidates are teaching bodies, not actual NANDO entries. The designation facts are stipulated for this exercise. The table is useful because it puts an offered date beside the fact that determines whether the date can be used at all.

| Supplied candidate record                                                                                                                                      | What follows for this application?                                                                                                 | What the dates leave open                                                     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| First candidate (source E): week-6 opening; verified designation excludes this product.                                                                        | Excluded for this product. The earlier date cannot compensate for absent scope.                                                    | Whether a materially different, supported product could fit is a new inquiry. |
| Second candidate (source F): designation and application-scope review cover the exact product; evidence freeze planned for week 10, audit planned for week 12. | Eligibility is supported. The audit remains conditional on file readiness, application acceptance and continued slot availability. | Technical findings, consultation, closure and certificate date.               |
| No device-review or required-consultation completion dates are agreed.                                                                                         | The audit is a conditional milestone.                                                                                              | A defensible launch timetable.                                                |

The second candidate's scope finding deserves to remain positive. On the supplied facts, this body can assess the exact application. That does not establish evidence adequacy or assessment completion. The two weeks between the planned freeze and audit are simply the ordering supplied in this record; readiness, acceptance and continued availability still matter. The candidate without the required scope remains excluded even if its fee or calendar looks attractive. No fee is supplied here.

### Decision

**A project needs four weeks to freeze evidence and then six weeks for internal dossier review. An eligible body's intake opens at week 8, but it cannot accept the file before internal review ends. What is the earliest file-ready week? Does that establish a certificate date? These durations do not belong to either candidate above.**

### How a reviewer reads the sequential work

Evidence freezes at week 4. Internal review then runs for six weeks, from week 4 through week 10:  $4 + 6 = \text{week 10}$ . The week-8 intake occurs while that prerequisite is unfinished. An available opening cannot move the company's readiness backward by two weeks.

Nothing says intake remains available at week 10. Application acceptance, quality-system audit, technical assessment, questions, any consultation and closure also have no durations in this packet. The arithmetic can therefore close while the external schedule remains open. The consultant's next timing question concerns the next actual intake option, rather than a certificate date inferred from the earlier opening.

**Reviewer Lens — the route record.** A controlled purpose and class memo locate the product before a designation and route check. The selected Article 48 paragraph then gives each quality-system, device, consultation, certificate and surveillance record a specific job. The startup's "finished" file still has unresolved product questions; the second teaching candidate's eligibility is already supported. In a separate materials-change investigation, the connection to the assessed configuration determines which body communication, if any, needs consideration. Practical staffing risks deserve attention without rewriting a legal scope finding.

**Consultant Lens — the selection decision.** Management can choose a body and initial scope when the record shows eligibility first, then evidence-backed comparisons of team, capacity sequence, consultation, change procedure, fees, language, site logistics and continuity. The claim/configuration baseline, question log, owner, evidence-ready point and contingency trigger make the recommendation executable. A verified backup has its own scope and capacity record. It is not simply another name available for a presentation if the first schedule slips.

### Changed-case transfer

A separate class-C near-patient test has simple software but demanding field manufacturing and stability. After eligibility is established, which assessment capabilities become more important, and what evidence should a candidate show for those tasks? Its class and intended user are unchanged; it is not Chapter 3's pharmacy-to-home cassette.

### How the actual work changes the comparison

Production control, stability, intended-user performance and postmarket handling now carry more of the assessment burden than software depth. The useful evidence describes who will examine those areas and how that work fits the procedure and schedule. Reweighting the practical comparison does not relax the designation gate or imply a changed class or user.

**SECTION 07****A second company needs a selection decision**

A second anonymous manufacturer supplies a similar planning problem. Its packet is independent of the startup and the two teaching candidates; resemblance is a chance to reuse the method, not permission to carry over their scope findings or dates.

This manufacturer calls its class-C technical file complete. A prestigious IVDR-designated body lacks the needed scope, and its next application slot lies beyond the financing runway. Another body has capacity but requests a preliminary package, which exposes unresolved classification, companion-diagnostic consultation, supplier and software-change questions. The packet supplies no scanner list, assay result, named medicine, completed consultation or certificate. What would turn those facts into a defensible selection decision?

**Which product would each body actually assess?**

The first application needs a controlled claim and configuration, rather than a product-family slogan. That baseline carries the class and rule rationale into the exact Annex IX or permitted X/XI path, quality-system sites, device-documentation unit and expected certificate types. Any companion-diagnostic, self-test or near-patient condition belongs with the assessment it adds. If management wants stages, the first application must truly exclude deferred specimens, scanners or medicine relationships. Reusing a quality process or analytical method can reduce duplicated work; it cannot make an unexamined specimen or version evidenced by name alone.

The unresolved medicine relationship illustrates why this matters before contact. The company needs to establish whether it actually claims the corresponding medicine, patient-identification purpose and essentiality in Article 2(7). If so, the route includes device-specific review and consultation. If not, the different actual purpose still needs a reasoned route. “Oncology” supplies neither answer. A file may be internally complete for a narrower configuration while a broader board presentation remains unsupported. Broad and genuinely controlled narrow options can be compared, without promising that staging shortens certification.

**What would make the comparison actionable?**

The first practice body fails the supplied scope condition. Its late slot is therefore not the controlling reason to exclude it. The second body's capacity is favorable information, but its eligibility remains open until current IVDR codes, procedures, restrictions and written product-scope acceptance fit the chosen configuration. A decision aid must preserve that difference between known scope failure and scope not yet documented.

Once scope is supported, someone has to perform each assessment. For assay performance, sequencing bioinformatics, medical device software, cloud suppliers, cybersecurity, AI where relevant and any true companion-diagnostic work, a named assessment arrangement is more informative than a reputation adjective. A pathology file brings scanner and reader-workflow evidence; a sequencing file brings assay/software and variant-interpretation changes. The body need not solve the product's case in advance. It should be possible to understand how the necessary expertise enters the review.

That arrangement must also be available in the needed sequence. Application acceptance, quality-system audit, device review, specialist work, medicines consultation where required, responses and closure need separate assumptions. The practice company's second body has capacity, yet unresolved product questions may still move the filing date. An intake or audit slot therefore cannot become a revenue date by omitting the work between them.

The choice also needs to survive the planned product roadmap. A body can explain its communication and assessment process for scanner, model, specimen, database, reagent, supplier and software changes. The manufacturer must still assess the actual change against the certificate and legal thresholds. A frequently updated pathology cloud model and a sequencing knowledgebase can affect different outputs. Neither gains standing permission because its updates appeared on a roadmap.

Those assessment units make the commercial comparison meaningful. Published or proposed fees, language, site and travel requirements, document format, response windows and change-review terms need to cover comparable work. A lower entry fee that excludes specialist assessment or creates substantial rework is not automatically the lower total cost. A high price cannot cure missing designation. No candidate fee is supplied in this practice packet, so a numerical price score would be invented.

Even a suitable arrangement may change when an assessor leaves or capacity moves. The record should show how technical questions, handover and escalation are documented. A practical backup must itself be eligible, while moving bodies can require additional records and replanning. A later-start body with reliable specialist continuity might be preferable; the packet does not establish that outcome. Verified scope, informal impressions and commercial assumptions need to remain distinguishable.

The **selection scorecard** below is a comparison aid, not an arithmetic contest. It pairs each decision with the evidence that would make the comparison usable.

| Decision                         | Evidence that makes it answerable                                                       | Use in this practice case                                         |
|----------------------------------|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| Is the body legally eligible?    | Dated NANDO entry, relevant procedures/restrictions and written product-scope response. | First body fails; second body's capacity alone leaves scope open. |
| Can the team assess the product? | Named assessment arrangement for assay, software, workflow and any medicine task.       | Identifies specialist questions and dependencies.                 |
| Can the work be sequenced?       | Written intake, audit, device-review, consultation and closure assumptions.             | Connects capacity with actual readiness.                          |
| Can planned changes be handled?  | Contract and device/quality-system change procedure.                                    | Tests whether the roadmap has a workable review path.             |
| Are commercial terms comparable? | Fees, languages, sites, formats, response windows and change terms.                     | Budgets the same work; no fee facts are supplied here.            |
| Can the relationship continue?   | Handover, escalation and a verified backup option.                                      | Reduces dependence on an unexamined single-provider assumption.   |

The first row is a gate, so strength in the later rows cannot compensate for its failure. The third row is a conditional schedule, so it cannot yield a certificate date while specialist or consultation durations remain unknown. These are different forms of information, even when presented in one table.

**Reviewer Lens.** A credible comparison points to records rather than adjectives. The first body's stipulated absence of scope remains decisive. The second requires designation and application-scope evidence before its fee and slot comparison can support selection. A proposed narrower product deserves the same scrutiny of actual function and evidence as the original one. A title change cannot make an offered scanner, assay or medicine-related function disappear.

**Consultant Lens.** The recommendation becomes useful when product leadership accepts the included and excluded first-application scope, and the shortlist distinguishes eligible bodies from open scope questions. Timing ranges need assumptions; unknown specialist or consultation duration should stay unknown. The scorecard helps management choose an option and own its dependencies. It cannot perform the body's later evidence assessment on the company's behalf.



## What can the body answer from the packet?

A sendable packet needs enough product detail to orient the body and enough evidence status to expose gaps. The following is a worked structure, not a claim that this fictional company has completed its submission. Each page joins company content to one answer the body can give.

| Packet page               | Company facts and open items                                                                                                            | Body question and limit of its answer                                                                                       |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Claim and configuration   | Controlled purpose, user, specimen, output, assay/software versions and proposed release; broader board claims kept explicit.           | Can this described application be accepted within designation? Acceptance of scope does not approve the claim.              |
| Rule and route            | Annex VIII rationale, class-C premise and Article 48 choice; alternative analysis if companion-diagnostic status is unresolved.         | Which assessment units and materials belong to each stated route? Qualification/class reasoning remains the manufacturer's. |
| Evidence status           | Scientific, analytical and clinical work; performance plan, studies, software verification, risk and labeling, with missing work named. | Which gaps prevent acceptance or need early technical resolution? An informal reply does not establish sufficiency.         |
| Product operations        | Quality-system sites, critical suppliers, hosting, controlled changes and planned model/material updates.                               | Which sites, specialists and change communications are anticipated? Future updates remain unapproved.                       |
| Consultation and schedule | Actual medicine relationship if claimed; evidence freeze and internal review separate from available slots.                             | What consultation sequence and capacity assumptions can be recorded? Completion still depends on the unresolved work.       |

For the first page, a software release named in the claim must also be recognizable in verification and risk records. A sequencing specimen expansion should appear in the performance plan, and a pathology scanner in the evidence and technical boundary. These are consistency checks, not new supplied configurations for this practice company. The response log then connects the body's answer to its status, owner and due date. A binding procedure, preliminary view and unresolved manufacturer duty remain distinct. No informal discussion supplies national study authorization or a medicinal-products opinion.

## What would actually close a finding?

A persuasive response letter cannot resolve an unchanged underlying problem. Closure needs an owner, root-cause investigation, corrective action, an evidence cutoff and a map of affected documents. If a pathology review found no applicable evidence for a claimed scanner, the manufacturer could consider narrowing the offered scope or generating and appraising the missing evidence. Either response would affect the claim, file and potentially the assessment plan. If a sequencing supplier change affected specimen validity, configuration, risk, analytical evidence and labeling would need updating as warranted. These are conditional response paths, not findings established in either local packet.

The reviewer then asks whether the revised records agree and address the finding. The consultant coordinates owners and any body re-review without deciding adequacy for the body. This is why a response schedule must include the work needed to generate a defensible answer, rather than only a date for sending a letter.

After certification, a rolling **three-year planning calendar** can join surveillance audits, technical-documentation sampling, performance evaluation report updates, performance follow-up, applicable **periodic safety update report (PSUR)** obligations, suppliers, releases and planned changes. Three years is a planning horizon here, not a certificate promise or a universal legal update interval. The class and route determine the actual obligations and interactions.

For the separate deployed pathology service, the earlier late-worklist observation calls for comparison of release and workflow state while cause remains under investigation. For the separate sequencing service, the materials-associated invalid initial reports call for comparison of material, assay and gate states. Risk, evidence and any timely duties need assessment alongside that work. A resulting finding may warrant internal correction, body notification or reassessment according to its effect and approved scope. Neither scene supplies cause, injury, a reporting trigger or an identical change category.

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## Decision

**The company can seek the earliest audit from a body with limited AI-pathology and scanner-generalizability experience, or investigate a later body with stronger relevant staff. What must its selection record establish before preferring either one? How do legal scope, configuration, unfinished evidence and the path after the audit affect the choice?**

### How a reviewer compares the whole path

A body lacking designation for the actual product leaves the comparison before timing preferences begin. Among eligible bodies, an earlier option may work if its team can assess the controlled initial configuration and the company supplies adequate evidence. A genuinely narrower configuration may reduce some work only when the withheld function is absent from that application.

A later body may reduce question cycles through better relevant staffing, but the packet guarantees neither result. Readiness, audit, technical review, consultation, responses, closure and change handling all belong in the comparison. Uncertainty remains part of the decision record. The offered audit date cannot supply the certificate date.

### Who owns what remains?

The manufacturer chooses and controls claim, product state, application, evidence and corrective actions. The notified body assesses within designation, audits the quality system, examines required device documentation, manages its consultation steps and decides on its certificates. A national competent authority retains applicable study, vigilance, market-surveillance and dispute roles. For a true companion diagnostic, the relevant medicinal-products authority or European Medicines Agency gives the specified suitability opinion while the body retains device assessment. The consultant prepares the comparison, packet, response plan and conditional calendar. The reviewer tests these boundaries against the actual records; a preliminary discussion cannot transfer an institution's decision to another participant.

#### Transfer check

##### Change the work, keep the gate.

How would the eligible-body comparison change for a different class-C near-patient test with simple software and demanding field manufacturing/stability? How would it change for a different qualified cloud-pathology medical-device-software proposal with several scanners and frequent planned releases? Which specialist and change questions would matter, what evidence must the manufacturer bring, and which condition remains the legal gate? Any AI Act interface remains a product-specific question, not an assumed obligation or date.

**How the product determines the useful expertise**

The near-patient proposal puts production, stability, intended-user evidence and postmarket handling in view. The pathology proposal puts scanner applicability, software and reader-workflow evidence, suppliers and planned-release assessment in view. The manufacturer must bring a controlled claim/configuration and the corresponding evidence status for either. The body's legal designation for that actual product and procedure remains a prerequisite in both comparisons. No new class or user has been stipulated for the near-patient test, and no completed assessment has been supplied for either proposal.

**SECTION 08****Quick discrimination lab**

These two microcases are independent of the startup, practice manufacturer and teaching candidates. Each supplies its own facts before the decision question.

**An earlier application opening**

An IVDR-designated body offers a date three months earlier than competitors. Its NANDO entry is broad, but the relevant technical code does not cover the company's molecular companion-diagnostic configuration.

**Decision**

**Should the earlier date control selection? Which evidence belongs first in the review, and could stronger expertise or a lower fee repair the stated code gap?**

**How a reviewer keeps scope ahead of preference**

The described application falls outside the body's relevant designation. The NANDO/code mismatch therefore excludes this body for that product; expertise, fee and date cannot repair it. A manufacturer could analyze a genuinely different controlled product, but an edited heading cannot remove the same supplied companion-diagnostic function. The reviewer preserves the scope finding, and the consultant seeks an eligible route for the actual proposal.

**A configuration changes during review**

A company applies with Scanner A and cloud release 4.0, then adds Scanner B and changes image normalization while assessment is underway. No performance results for the added configuration are supplied.

**Decision**

**Is this merely extra documentation? Which product and evidence questions reopen, and what must the company agree with the body before treating the change as part of the application?**

**How a reviewer follows the changed input**

Scanner B can change the image input. Normalization changes how that image is prepared before the model receives it. Either may affect output and clinical workflow, so the object being assessed may have changed. Application scope, configuration identity, technical documentation, performance, risk, software verification and certificate strategy need reconsideration.

The manufacturer's change-impact and evidence proposal gives the body something specific to assess under the chosen procedure. The packet does not establish whether the change fits within the present review, needs additional assessment or belongs in a later application. The body decides how the proposal affects its assessment. Calling it clerical would skip that decision; the consultant can instead make the options and dependencies visible without promising approval.

**SECTION 09****Leave with a route and its actual dependencies**

A supported class, an eligible body, a completed quality-system audit and a technical certificate are different records. The route connects them without allowing one to stand for the others. These seven checks locate the evidence behind that connection; the original lookup identifiers remain at the ends of the headings and can retrieve the optional Workbook Part A2 cards.

The working states remain item descriptions: A, adequately explained for the bounded question; U, present but unclear; M, missing or inconsistent; D, deeper analysis needed; C, relevant specialist or institution consultation indicated; N/A, inapplicable with a reason. Evidence location or absence and reasoning make the letter useful. It is never a product grade, approval vote or certificate decision.

**Verify legal designation for the product • card IVDR-CH04-A01**

Do the dated IVDR entry, codes, activities, restrictions and written scope position cover this product and procedure? The candidate without that scope fails; the other teaching candidate's stipulated record supports eligibility only.

**Distinguish the assessment units • card IVDR-CH04-A02**

Which Article 48 paragraph and annex identify the quality-system, device-documentation, consultation, certificate and surveillance work? Each completed record must establish its own conclusion. An audit cannot complete the entire device decision.

**Establish the change process for this product • card IVDR-CH04-A03**

How does the body handle software, relevant AI, companion-diagnostic work and consequential changes? Its procedure needs to fit the actual product state and change-impact record; not every update follows the same notification path.

**Examine capacity and technical fit • card IVDR-CH04-A04**

Do staffing, evidence readiness and milestone assumptions describe the whole path, including unresolved specialist work? Intake and audit dates are development dependencies, not certificate forecasts.

**Make selection evidence-based • card IVDR-CH04-C01**

Does the decision record apply eligibility before comparing technical fit, capacity, change handling, cost and continuity? Favorable practical features cannot compensate for missing legal scope.

**Connect body choice to the initial application • card IVDR-CH04-C02**

Has the manufacturer accepted the claim/configuration baseline, evidence gaps, intended filing point and questions? Written preliminary replies need their status recorded so that process advice does not become assumed approval.

**Verify the contingency • card IVDR-CH04-C03**

What evidence supports backup eligibility, handover and the trigger for replanning if capacity, assessor continuity or scope changes? A nominal alternative without the necessary scope cannot provide the fallback it promises.

A can close the second teaching candidate's stipulated eligibility while M or D remains appropriate for absent technical or consultation facts. In the independent timing exercise, week 10 closes the file-ready arithmetic while later intake and certification stay unknown. Those bounded conclusions let the next person continue from a known point.

## Source trail and next reading

The binding route text is the IVDR, Regulation (EU) 2017/746, consolidated 10 January 2025: Article 2(8) for generic device groups; Articles 47–48; Annex IX Chapters I–III and Sections 4, 5.1 and 5.2; Annexes X–XII for type examination, production assurance and certificate content; and Annexes II–III for technical documentation. The Commission notified-body page links to NANDO for current designation and scope. The Commission EU reference-laboratory page explains designation-dependent class-D laboratory tasks; MDCG 2021-22 rev.1 is nonbinding guidance on the two-part Article 48(6) expert-consultation condition. Published procedures of a candidate body and Team-NB materials can inform engagement practice but are not universal law or a substitute for the actual contract. Book 2 Chapter 17 offers optional integrated route practice without changing this case's facts. Chapter 5 will trace the applicable safety and performance requirements to evidence; no certificate or launch is inferred here.



## CHAPTER 5

# From Safety Requirement to Product Evidence

**Retrieval check — before reading.** Chapter 4 separated the conformity route, evidence that a body is designated for the actual product, and changes that may reopen an application. For a product considered there, what would answer each of those three questions?

A pathologist needs the right image in the right patient's worklist. A manufacturer may have a page of safety requirements and several passing reports, but can another person follow the connection between them? **The useful thread runs from an applicable requirement to the product behavior meant to meet it, the evidence that behavior works, and the exact configuration tested.** It also tells us what can remain closed when another part changes.

## Why a green table may leave the product unanswered

Imagine a damaged image file, or one carrying a mismatched specimen identifier, arriving for review. If it enters the wrong patient's worklist, even a correct interpretation of its pixels could be attached to the wrong specimen. In sequencing, a perfectly measured **variant**—a difference from a genetic reference—could likewise appear in an invalid patient report if specimen, data-file and report identities became disconnected. The two products need different controls, but both raise a concrete question: how is the mismatch prevented or made visible, and how does the intended professional recover?

The **General Safety and Performance Requirements (GSPRs)** in Annex I of the **In Vitro Diagnostic Medical Devices Regulation (IVDR)** give that product question a legal basis. Applicable requirements call for more than a finished checklist. The manufacturer needs a behavior that meets the duty, an account of remaining risk, evidence of the behavior and a usable response to an exception. Annex II Section 4 asks the technical documentation to identify which requirements apply or do not apply, the methods used and the precise evidence locations. Those links let another person test the conclusion.

**Opening case — an apparently complete record.** An anonymous fictional company supplies a fully populated GSPR table. Every row says “compliant” and cites a validation

report or standards from the International Organization for Standardization (ISO): ISO 13485 for quality-system processes and ISO 14971 for risk-management processes. Yet the reader cannot tell which requirements apply, how this product meets them, where the proof sits or whether the report covers the scanner and model change being discussed.

An untraceable record establishes neither a failed changed product nor a tested one. This company is separate from the completed image-routing example later in the chapter and from Chapter 1's deployed pathology and sequencing services. Their evidence cannot fill its gaps.

### **Reviewer Lens – the opening record**

The first useful read is often one consequential row, such as an image whose specimen identity disagrees with the file. The intended behavior, implemented check, versioned test and operator's recovery step reveal more than the number of green cells. A specimen-to-report mismatch gives the sequencing reviewer the same starting point. The distinction to preserve is between a poorly indexed file and an unmet product requirement: the former leaves the evidentiary conclusion open until the underlying records can be followed. It is a reason for a focused request, not an invented failed test.

### **Consultant Lens – the opening record**

A table can be cosmetically repaired before anyone resolves which scanner, model and claim it describes. That is inexpensive work with little decision value. The useful sequence is to settle the offered boundary, locate the most consequential control and its reports with the software and risk owners, and connect those records to the controlled versions and release decisions with quality staff. If the changed configuration lacks support, characterization or testing belongs in the plan. Citation cleanup becomes useful once it points to the product actually being assessed.

## **Follow one requirement through the work it creates**

Why should the image be checked before entering the worklist? Annex I requires intended performance with controlled risks, and Section 13.2(f) specifically addresses reducing the risk of incorrect specimen identification. Its software provisions concern reliable operation, development, verification and validation, and the hardware, network and security conditions needed for intended operation. These duties leave the manufacturer to determine the product behavior that meets them. The resulting testable statement of what this product must do is a **design input**.

For the proposed image-routing function, “image integrity is safe” gives a developer little to implement and a tester little to challenge. A more usable input says that a file cannot enter a patient worklist until prescribed identity and integrity checks pass, and

that rejected files stay visible in a recoverable queue. Now there is an observable difference between acceptance and rejection, and a place where recovery can be examined.

The reason for those actions becomes clearer by following the possible consequence. Misleading diagnostic information associated with the wrong specimen is a potential source of harm—a **hazard**. A corrupted or mismatched file could carry that information into a patient worklist. If it is then used for clinical assessment, the patient is exposed to that source: this is the **hazardous situation**. A resulting wrong decision could delay or misdirect care and damage health. That damage is the possible **harm**. No such event is reported in this example; the path explains why the control deserves testing.

The identity check interrupts the path before worklist acceptance. The visible queue gives the operator a way to recover an excluded file. Annex I Section 4 gives the order for choosing controls: first reduce or eliminate risk through safe design and manufacture; then provide appropriate protection against risks that cannot be eliminated; then provide safety information and, where appropriate, training. Section 2 qualifies “as far as possible” by preserving the benefit–risk ratio. Here, a warning alone would shift detection onto the user even where the system could detect the mismatch first. If a risk remains after the available design measures, the protection and remaining user task need their own justification and evidence; that changed situation gives instructions a role without making them a substitute for feasible design. Risk analysis considers both each remaining risk and the overall residual risk. Copying this requirement sentence to another product would not establish that its particular control is adequate.

Once the behavior is specified, two different questions arise. Does the implemented release perform the specified check? **Verification** can challenge it with valid files and malformed, duplicate or mismatched identifiers, comparing the observed routing or rejection with the input requirement.

But a correctly rejected file still has to be dealt with. Can the intended operator recognize the rejection and recover it without creating another specimen error in the intended workflow? That contributes to **validation**, the question whether the resulting system meets its intended use. A code test does not show that a person can find a queue; a successful recovery task does not show that every file-integrity condition was implemented. Both reports need the tested build and inputs, acceptance criteria, results and limits so their different contributions remain visible.

The remaining user action then gives the **instructions for use (IFU)** their job. If the operator can see and act on the recovery queue, the instruction can explain that task. Assigning an action in a document is insufficient if the intended actor cannot perform it; the product and instructions need to agree.

After release, the manufacturer needs to know what would make this conclusion worth revisiting. A **post-market surveillance (PMS)** plan can identify rejected-file patterns, misrouting complaints or changes in supported input, with an owner to investigate them. That is a plan for future information, not a field result. When an actual signal arrives, investigation concerns the particular device state, while proportionate precautions and applicable timely duties are assessed on known facts without waiting for perfect causal proof.

The connected record now has a reason for every link: an applicable requirement motivates a testable input; the possible exposure path explains the control; software and intended-use evidence test different parts; usable instructions support the remaining task; planned and actual monitoring can reveal change. A later label cannot automatically compensate for an omitted feasible control. The **quality management system (QMS)** keeps decisions, tests, versions, release and subsequent changes controlled. Having that process does not itself supply a successful product test.

In the separate sequencing design illustration, the corresponding input would specify which specimen and report identifiers must match before release. The risk analysis would follow a possible wrong-patient report, verification would challenge the matching mechanism, and validation would examine how intended laboratory staff resolve a blocked report. This is an explanatory proposal, with no supplied assay test result. It shows why measurement and identity both need evidence: either can be wrong while the other is correct.

### **Reviewer Lens – one requirement, several records**

Following one identifier makes a large file manageable. In pathology it travels from image arrival to worklist acceptance, or through rejection and recovery; in sequencing it travels through extraction, analysis and report release. Each crossing raises a specific evidence question about the configured system and the person able to act. A passing implementation test can close the tested input behavior while operator recovery or a different scanner remains open. Keeping those findings separate prevents a single green row from quietly covering several unexamined relationships.

### **Consultant Lens – build the thread into the product plan**

The thread is cheaper to build while teams are making their decisions. Engineering knows the input and test; risk staff know the possible clinical path and control alternatives; clinical or usability staff know the intended task; labeling knows the proposed instruction; quality connects release and monitoring ownership. The useful deliverable joins those contributions with named gaps and decision dates. A planned sequencing identity check stays planned, and a tested pathology workflow keeps its scanner and worklist limits. This lets management see the actual next work without expanding the spreadsheet merely to make it look more complete.

## Know what each source can—and cannot—prove

The opening company has cited two respected process standards. Why do those citations leave the image-routing question unanswered? Start with what needs support: the binding IVDR duty in Annex I and its applicability and evidence account under Annex II Section 4. The product must have a way of meeting that duty. A standard can help organize the method; its title cannot extend a report to an untested scanner or make a changed sequencing report rule tested.

The quality-system standard ISO 13485 provides a method for organizing controlled processes. The risk-management standard ISO 14971 provides a method for identifying, evaluating and controlling risk. Standards are generally voluntary methods, and sound use of them can provide useful process evidence. That evidence has a different object from a test of this device: a quality procedure cannot report how often it misroutes a file, and a risk procedure cannot demonstrate that the implemented identity check rejects a mismatch.

A particular European citation can add a limited legal effect. Under Article 8, conformity with a relevant **harmonized European standard** or relevant part gives a presumption of conformity for the IVDR requirements covered when the reference to the exact European Standard (EN) edition has been published in the Official Journal of the European Union. The benefit is valuable within that coverage. It does not establish every feature or clinical claim.

The chapter's dated reference is Commission Implementing Decision (EU) 2021/1195, consolidated to 17 June 2026. Its annex lists specified EN editions of the two process standards. A file citing only their ISO numbers leaves the EN edition, amendment, covered requirements and any deviations unclear. Publication, restrictions and transition dates need checking at an actual assessment. Even a correctly supported process-standard citation still leaves the product's implemented control and result to examine.

The next question is whether a different legal mechanism supplies applicable common requirements. **Common Specifications (CS)** are technical or clinical requirements adopted by Commission implementing act for a defined scope under Article 9. Where applicable, the manufacturer must comply or duly justify another solution ensuring at least equivalent safety and performance. Regulation (EU) 2022/1107 supplies CS for certain class-D IVDs. It does not create a universal acceptance threshold for an artificial intelligence (AI) pathology model or every sequencing panel. “CS met” therefore needs the covered device and requirement, followed by product evidence or an equivalent-solution justification.

The distinction matters when reading the same “compliant” cell. A correctly cited harmonized standard can support a presumption within its covered requirements. An applicable CS calls for compliance or duly justified equivalent safety and performance

under Article 9. In both cases the technical file must still show the device, version, purpose and underlying results. Neither source category turns all product claims into passing results.

Other sources answer narrower practical questions. **Medical Device Coordination Group (MDCG)** guidance helps interpret the device rules without itself being law or a certificate. A body's intake format or a sector group's practice paper can help present evidence efficiently without setting universal device requirements.

Even a binding source must be addressed to the right question. Regulation (EU) 2026/977 sets uniform quality-system and procedural requirements, including process and timing duties, for notified bodies conducting conformity assessments. It does not set an assay's analytical-performance acceptance criterion. Using it to justify a product cutoff would move the source from the assessor's procedure to a task it does not govern.

The citation can now be read as a short sequence of checks: which edition and amendment was used; whether that exact EN reference has the claimed Official Journal status, timing and limitations; which legal requirement and product activity its relevant part covers; and where the device's implementation and result can be found. A risk record may adequately show that a pathology scanner change was evaluated while leaving image evidence for the changed configuration unresolved. A sequencing reagent change may have been properly reviewed under the quality process while still needing evidence of changed analytical performance. Those are compatible findings.

### **Reviewer Lens – source and evidence are separate checks**

The legal-status check and product-evidence check can each pass or remain open independently. A reviewer can credit correctly bounded process support while asking which images support a changed-scanner claim, or which verification and performance records support a revised sequencing report threshold. Marking the citation's status—law, applicable CS, cited harmonized standard, other voluntary method, guidance or body procedure—keeps its useful contribution attached to the conclusion it actually supports. Dismissing the whole source because it cannot answer the product question would discard valid evidence.

### **Consultant Lens – advise without laundering authority**

The most useful management comparison often has only two parts: what the source requires or helps demonstrate, and what this product still needs to show. A formatting request stays a body-process request. A possible class-D CS needs a scope analysis before anyone promises an equivalent solution; an inapplicable CS creates no fictional gap. Keeping the dated harmonization citation beside its covered requirement also makes a later edition change detectable. The consultant's value lies in making authority and remaining work visible together.

## One failure path can cross several teams

A scanner record, a model test and a worklist demonstration can each look sound. Were their parts tested together in the configuration now claimed? Follow one scanned tissue image: the scanner can affect its appearance, the model can affect the score, and a cloud queue can affect when or whether the result appears. An instruction can then affect recovery from an exception. Separate green cells for compatibility, software validation, usability, security and labeling may conceal the joins. The relevant question is how the claimed route keeps a correct-looking score from reaching the wrong person, arriving at the wrong time or being produced from an unsupported image.

Sequencing adds a reason to distinguish measurement from interpretation within that route. A specimen or reagent change can affect which genetic differences are detected. A computing step can change which **reads**—short stretches of sequenced DNA—are accepted. A **knowledgebase**, the controlled information used to interpret a detected variant, can change its reported meaning; a report template can change what the clinician sees. Following the result backward locates which links an edit can reach. It does not require every item to be retested after every edit: the impact assessment justifies which verification or validation evidence is affected.

The same logic prevents ordinary-input evidence from answering an untested security question. Malformed identifiers and a malicious substitution using properly authenticated access or valid-looking data can challenge different parts of the system. Rejection of the former does not establish resistance to the latter. Annex I 16.4's unauthorized-access question belongs in the security analysis. The ordinary identity check can receive credit while that distinct threat remains open.

### Reviewer Lens – test the joins

The useful comparison places the claimed release beside the scanner/model list, risk scenario, test inputs, pathologist workflow, instructions and monitoring plan. For sequencing, the corresponding comparison includes specimen and reagent state, computing pipeline, knowledgebase, analytical test, report label and invalid-result handling. The hardest gaps may be assumptions at the joins, where each team has supplied a valid document about its own part. A planned complaint trend is another such boundary: it says what will be watched, while an observed trend says what happened. Neither can substitute for the other.



### **Consultant Lens – prioritize the consequential join**

A change map is useful when it finds the shortest plausible path from an edit to a misleading result or lost recovery opportunity. The owner of that join can then identify the needed design or evidence work. A scanner extension may need image and preprocessing characterization; a sequencing rule change may need challenge data and report validation. Management can compare the supported scope with the broader scope and its required work. Narrower wording is a valid option only when the product actually offered fits it.

### **Reopen the affected thread when the product changes**

A report that matched yesterday's product may still match much of today's. What changed, and which parts of the requirement thread could that change reach? A specimen, scanner, model, supplier, cloud environment, report rule or intended user can alter the risk path, control, relevant standard or meaning of a test. A controlled before-and-after comparison establishes what changed, what stayed the same and why the chosen evidence still applies or needs supplementing. Neither automatic transfer nor automatic rejection of all earlier work answers that comparison.

**Additional exhibit — the opening company's scanner extension.** The Scanner A validation report is cited again for Scanner B. It contains no Scanner B images and no characterization of B's proprietary preprocessing: the scanner's internal alteration of acquired images before later software receives them. The exact alteration is not supplied. There is no adverse Scanner B result, numerical performance comparison or clinical outcome in the exhibit.

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### **Decision**

**The company's GSPR table is complete and cites broad procedures or standards throughout. What can now be concluded about the Scanner B claim? Which evidence would allow a decision about whether the Scanner A results transfer, and how does absent evidence differ from failed testing?**

### **How a reviewer keeps the tested result and the open extension separate**

The Scanner A report may remain useful for the configuration it tested. The supplied record does not show that Scanner B's images and proprietary preprocessing leave the relevant behavior unchanged. The gap is the bridge between the tested and proposed conditions, not an observed failed test on B.

The next evidence needs the exact intended B workflow, characterization of its processing, impact and risk analysis, and appropriately matched data or tests for the claimed output and users. That could support transfer, identify further work or limit the

offered scope; this packet decides none of those outcomes. A broader model update would need its own affected-thread analysis. Completing the index does not supply the evidence, and the independent completed router example below cannot supply it for this company.

### **Reviewer Lens – change disposition**

The review record should keep the A-supported and B-open findings readable at the same time. The focused question is whether the offered B images pass through processing absent from the old test, and whether the claim is limited to a supported scanner state. The parallel question for a separate sequencing materials change compares the earlier specimen and reagent conditions with those now offered. Chapter 1's separate deployed sequencing signal—more invalid initial reports—warrants investigation but supplies no cause, denominator or proof of harm. Protective action and applicable timely duties can be assessed while the cause remains under investigation.

### **Consultant Lens – an evidence-first change plan**

The change plan becomes actionable when it connects the current claim and versions, proposed change, affected requirement/risk/design/evidence links, owner and decision gate. For this company, Scanner B characterization and appropriate images come before citation polishing. For the separate deployed sequencing service, quality, analytical and supplier owners need the before-and-after materials and reporting path, an investigation and a reassessment point. In both settings, a plan remains a plan until evidence exists; a changed label cannot validate an altered feature. Keeping the unknown cause explicit helps management decide the next work without pretending the investigation is complete.

## **A bounded thread that really can be closed**

A different record contains the evidence needed to close a narrower question. The **tested image router** is an independent fictional application with an identified parser and routing rules. It is not a release of the Scanner A/B company. Its risk record follows misleading diagnostic information attributed to the wrong specimen, carried by a corrupted or mismatched file into a patient worklist and used in clinical assessment. Delayed or misdirected care could then damage health. This is a possible path to harm, with no such outcome supplied.

The worklist acceptance requirement makes that path interruptible. Drawing on Annex I 13.2(f) and software duties in 16.1–16.2, it permits entry only after prescribed identity and integrity checks. Rejected files go to a visible recovery queue with an audit record. Detection occurs before the point at which a user would otherwise need to heed

a warning. This requirement addresses ordinary file conditions; it leaves the separate unauthorized-access question under 16.4 for its own analysis.

The routing test report shows what the specified application, parser and rules actually did: valid files were routed, and the specified malformed, duplicate and mismatched identifier cases were rejected and recovered. The operator recovery report answers the next question: the specified intended operators successfully completed the tested rejected-file recovery tasks. Each report includes its methods, acceptance criteria, results and limitations, so the conclusions can be matched to the challenges and people involved.

The recovery instructions describe that task. The software lead owns the control, and the quality lead owns its release gate. Together, the controlled records support **A—adequately explained for this item** within the tested inputs and use conditions. They do not answer for Scanner B, an untested parser or malicious authenticated substitution.

This evidence table is useful when another reviewer needs to find each link. The original record identifiers are kept beside the plain names for exact lookup; they do not identify the opening company or any other case.

| Record to open                               | Original identity | What this record contributes                                                                                                           |
|----------------------------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Tested application, parser and routing rules | R2 / P1 / Q4      | The exact combination to which the routing and operator evidence applies.                                                              |
| Worklist acceptance requirement              | REQ-12            | Identity and integrity checks before entry; visible rejected-file recovery with an audit record.                                       |
| Routing test report                          | VER-12            | Valid-file routing and the specified malformed, duplicate and mismatched identifier challenges; methods, criteria, results and limits. |
| Operator recovery report                     | USE-12            | Successful tested recovery tasks by the specified intended operators; methods, criteria, results and limits.                           |
| Recovery instructions and release            | IFU-12            | The usable recovery task, with software ownership of the control and quality ownership of the release gate.                            |

Follow the acceptance requirement into the two report rows. The routing report shows implementation on the specified inputs; the operator report adds successful use of the recovery route. Neither alone supplies the other's evidence. The configuration row lets

their shared scope be checked against release and instructions. The resulting A belongs to that overlap. Changes to the parser, identifier mapping, queue visibility or supported input call for impact assessment and affected evidence; the record has no result for the separate scanner extension or the distinct malicious-substitution threat.

Can a requirement also close because it does not apply? A standalone software device with no patient-contacting component and no sterile-supply claim may have a reasoned N/A for a particular sterile-packaging requirement. That finding must name the actual boundary. It does not remove the entire infection-control or security domain from consideration; software, users and connected workflow still need their applicable duties examined. A sequencing kit containing physical collection or reagent materials has a different supplied boundary, so the app's rationale cannot simply be copied to it.

### **Reviewer Lens – credit a bounded positive**

The positive finding deserves the same precision as a gap. Matching the routing and operator reports to the acceptance requirement, configuration, instructions and release can justify closing this tested item. Asking how malicious authenticated substitution is addressed remains appropriate, but that is a separate question. A reviewer who turns the bounded A into closure of every Annex I software duty overstates the evidence; a reviewer who refuses to retain it because another threat remains open discards a useful result.

### **Consultant Lens – preserve and reuse valid evidence**

A linked requirement, risk rationale, tests, operator evidence, instruction and release decision make a later change cheaper to assess. Engineering can ask which specific relation a parser or queue change reaches, while quality can retain the evidence that still applies. Reuse has a clear limit: a new sequencing module may use the same recovery reasoning, but its specimen identifiers, staff and workflow need their own records. The router's passing result does not travel with the idea.

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### **Decision**

**The same tested router keeps its worklist behavior. A spelling correction changes only a noninstructional menu heading. A controlled diff and documented impact review establish that boundary, and the relevant display check passes. Can the identity-routing thread remain closed? Which records support that answer, and which wider conclusions remain unsupported?**

### How a reviewer preserves evidence for unchanged behavior

Yes. The controlled diff, impact review and passing display check establish the stated narrow change. Linked to the existing thread, they explain why the routing and operator recovery reports still apply to its unchanged behavior. A corrected menu word does not create a new unresolved identity gap merely because the file changed.

That conclusion remains local. It supplies no evidence for Scanner B, an altered parser, a different user task or a new medical claim. A later edit that reaches any of those links needs a fresh impact decision. The record preserves a justified closure without claiming that all possible changes are harmless.

### Reviewer field aid — read a row as a claim to inspect

A row can now be read as a small argument: this exact Annex I requirement applies—or does not—for this product boundary; this method and implemented control address it; these versioned documents at these locations show the result. The risk and tests belong beside the instructions and monitoring plan, because the conclusion depends on their agreement. A standard adds its exact edition, relevant part, scope and current Official Journal status where harmonization is claimed. An N/A needs an actual feature-based reason.

When scanner, assay material, model or user changes, comparison of the old and new states identifies which threads can be reached. The working state then belongs to each affected item, with reasons. A, U, M, D, C and N/A remain evidence states, never a device score, certificate or approval vote.

The three chapter findings can therefore stand together. The opening company's Scanner B bridge lacks image evidence and preprocessing characterization. The independent router has matched evidence for its tested behavior. Its documented spelling-only menu correction preserves that closure. Different evidence supports different endpoints; uniform coloring would make the record less informative.

### Consultant field aid — make the matrix a work plan

Which open item deserves the next day's work? A **GSPR evidence matrix** is useful when it connects potential consequence to the actual gap, evidence location and accountable owner. The three rows below compare independent records. They also show why “high priority” does not always mean “another test”: preserving a supported result under controlled change is a real disposition.

| Item and priority                                   | Evidence state                                                                                                                                              | Next work and owner                                                                                                          |
|-----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| High: tested router's identity path                 | The specified inputs and operators support bounded A. Recovery instructions match. Separate scanners and malicious substitution remain outside that result. | Software lead watches parser, mapping and queue changes; quality lead gates release. Preserve adequate evidence.             |
| High: opening company's Scanner B claim             | The cited report lacks B images and preprocessing characterization. No B failure is supplied.                                                               | Pathology engineering, risk and performance owners establish characterization and matched evidence before the broader claim. |
| Product-specific: proposed sequencing identity path | Identifier and invalid-result controls must fit the assay, software and staff. No test or field result is supplied.                                         | Assay, software and clinical owners define behavior, challenge tests and recovery validation before claiming closure.        |

The router row rests on the acceptance requirement, routing report, operator report and recovery instructions in the preceding table, with checks before entry and visible, auditable rejection. Its monitoring plan would identify rejected-file and misrouting signals; it asserts no complaint result. A mapping change reaches the requirement, risk, test, instruction and release links. By contrast, the documented spelling correction reached display and change-control records while leaving routing intact. For the opening company's second row, the next evidence is still about the unsupported scanner extension. The router's positive row cannot close it.

Priority follows the possible clinical and regulatory consequence of a missing control or unsupported claim, rather than the number of empty cells. That directs a choice among redesign, targeted testing, feasible instructions, specialist legal interpretation or justified inapplicability. A warning cannot carry an action the user cannot perform. Citation polishing cannot characterize an unknown scanner input. Equally, an adequate narrow finding need not be made newly active just to make the work plan look busy.

**Changed-case transfer — near-patient invalid handling.** A separate near-patient test, used by its intended health professional, may detect an invalid result and ask for a repeat. It has no supplied invalid rate or postmarket trend. It is neither Chapter 3's pharmacy-to-home cassette nor a release of the tested router.

**Decision**

**Which clinically relevant invalid state starts the thread? What must connect its detection and display, repeat logic, intended professional's action and planned monitoring before a valid-looking result can be trusted?**

**How the same thread fits a different clinical task**

Detection must make the invalid state visible, and the repeat logic must prevent a misleading valid-looking result. Intended-user evidence asks whether the professional can recognize and execute the instruction. “Repeat” on a label establishes neither the detection behavior nor successful user action. A laboratory-only usability study also does not automatically support a changed intended near-patient user.

The final link is a plan for complaint or invalid-result signals and who would investigate them after release. With no actual rate or trend supplied, this remains planned monitoring. The method transfers; none of the router's test results does.

## Quick discrimination lab

These independent miniature records each ask what is supported, what remains open and what focused action could resolve it.

**Micro-case 1 — a completed matrix.** A GSPR matrix marks every requirement conforming and cites the quality manual for most rows. No row identifies a risk control, verification, labeling or product version supporting the result.

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**Decision**

**Is the matrix adequate because every row is filled? What can the manual show, what can it not show alone, and which row-level record would be useful next?**

**How a reviewer reads past a completed index**

The manual can support a conclusion about the manufacturer's process for managing requirements and evidence. It cannot by itself show that this device's control works. A consequential row needs its applicability rationale, product version, implemented requirement and risk control, test or intended-use result, instruction and monitoring link.

Those records allow the conclusion to be assessed rather than guessed from a filled cell. Until their precise scope and result are checked, the product conclusion remains open. The process evidence keeps its value within that limit.



**Micro-case 2 — a cloud version instruction.** An IFU tells a community laboratory to verify the cloud knowledgebase version before every run. Only the manufacturer can see or change that version. The knowledgebase is the controlled information used to interpret the test output; its version can matter even if the sample measurement is unchanged.

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### **Decision**

**Can this warning alone close the version-control and risk thread? Who can currently perform the task, and what feasible repair direction could the manufacturer assess without assuming it already works?**

### **How a reviewer follows the instruction to the actor who can act**

Only the manufacturer can see or change the version in this packet. The laboratory cannot verify information it cannot observe, so the instruction does not implement the assigned safeguard. The manufacturer needs to retain and evidence effective version control, or supply a real user-visible mechanism and validate the user task if the user is to perform it.

Choosing between those directions depends on the system and risk analysis. Neither the inaccessible instruction nor a generic quality-system citation demonstrates a product-specific control. This is a finding about an infeasible assignment, with no evidence that a sequencing knowledgebase actually changed or caused an error.

## **What the evidence allows to close**

An adequately supported control can remain adequate for its tested configuration, users and conditions. A precisely inapplicable requirement can receive a reasoned N/A. An extension without matched evidence stays open. Those conclusions become defensible through the links behind the row: a generic citation, unexplained green cell or monitoring plan presented as a result cannot supply them. Chapter 6 uses these threads to organize technical documentation, the quality system, suppliers and changes for the actual offered product state.

### **Check what you just learned**

Each check below belongs beside an evidence location and reason. The working states remain **A**, adequately explained for this item; **U**, present but unclear; **M**, missing or inconsistent; **D**, deeper analysis needed; **C**, consultation indicated; and **N/A**, inapplicable with a reason. They do not grade the product. The original identifiers retrieve expanded Workbook Part A2 cards through the topic crosswalk.

**Applicability and its reason • card IVDR-CH05-A01**

Which exact Annex I requirement applies to this device boundary, and why? Controlled purpose and configuration support an applicable or inapplicable finding; a bare N/A cannot.

**Standards status and scope • card IVDR-CH05-A02**

What exact edition, current harmonization decision, covered requirement and relevant part support the claimed method? Deviations and limits remain visible. An ISO title alone supplies no Article 8 benefit.

**Requirement through use and monitoring • card IVDR-CH05-A03**

Does the consequential requirement connect to risk, design, implemented control, tests, intended-use evidence and feasible instructions? The final monitoring link distinguishes a plan from observed field data.

**What a change reaches • card IVDR-CH05-A04**

What do the change record, before-and-after configurations and evidence-applicability rationale show after a product or state-of-the-art change? An affected row may need work while an unaffected one remains adequate for a documented reason.

**Priority by consequence • card IVDR-CH05-C01**

Does the matrix connect possible clinical and regulatory consequence to claim impact, evidence location, residual gap and owner? The tested router is a model of bounded reasoning, not reusable evidence for another product.

**The management closure choice • card IVDR-CH05-C02**

Which gap needs design, testing, labeling or specialist interpretation, and why? An easier warning is insufficient reason to skip a feasible design or protective control.

**Product first, citation polishing second • card IVDR-CH05-C03**

Are the claim, configuration and evidence boundary stable enough for accurate citations to point to the right result? Formatting cannot repair evidence for the wrong scanner, specimen or software state.

**Further reading**

Book 1A Chapter 3 offers optional further practice connecting risk, product requirements and tests. The tested router, scanner extension and sequencing illustrations here keep their own facts and limits.

## Source trail and limits

- **Binding requirements:** IVDR consolidated text, 10 January 2025, Articles 5(2), 8–10; Annex I, especially Sections 3–4, 13.2(f) and 16.1–16.4; Annex II Section 4. These are the legal basis for the requirement/evidence thread and its documented applicability and location. The chapter's R2 results are fictional exercise facts, not evidence produced under a standard or for a real device.
- **Harmonization status:** Commission Implementing Decision (EU) 2021/1195, consolidated 17 June 2026, Article 1 and Annex entries 7 and 10, publishes specified EN ISO 13485 and EN ISO 14971 editions; the Commission harmonized-standards explanation describes the voluntary, scope-limited Article 8 effect. Exact cited edition, amendments, current status, restrictions and applicable requirements need checking for a real device; no standard here is claimed to validate Scanner B or an NGS result.
- **Different implementing acts:** Regulation (EU) 2022/1107 in the Official Journal, Article 1 and Annex, concerns CS for specified class-D in vitro diagnostic devices; Regulation (EU) 2026/977, title and Articles 1–4, concerns notified-body QMS and conformity-assessment procedures. Neither is a general DP/NGS product-performance cutoff. The Commission MDCG guidance register contains nonbinding interpretive material; selected-body operating preferences are not substituted for the binding acts.

## CHAPTER 6

# One Product Across the File, Quality System and Changes

**Before reading.** Chapter 5 followed a General Safety and Performance Requirement from legal duty to product design, possible harm and control, a test, intended-user evidence and the tested configuration. Which missing relationship would prevent a reviewer from applying that result to the product being offered?

A file can be easy to search and still leave that question unanswered. **Do its technical documentation, operating quality system and supplier records describe the same controlled product?** The answer matters to the person reviewing a report and to management deciding which work must happen next. Both need the same facts, even though they need different levels of detail.

## Easy to find, but which product is it?

An anonymous fictional sequencing manufacturer submits a meticulously indexed file. Its device description identifies **pipeline 7.0** but calls the knowledgebase “current.” Analytical work names **DB-18**; a final technical report concerns **DB-19**; risk work names **KB-2026Q1**. Software history calls the knowledgebase “external content.” Every record can be found, yet the packet supplies no single proposed commercial combination. The final technical assessment is not evidence that its database was commercially released or clinically deployed. Nor are the three content labels proved aliases. This is a local manufacturer, separate from the named series products.

Why does that uncertainty matter to the report? In an ordinary sequencing examination, the laboratory reads stretches of genetic information from a specimen. Software decides which readings are usable; a knowledgebase—the information used to interpret a detected variant—helps determine what the report says. The clinician receives the result after that work. The assay, instrument, pipeline, reference content, report rules and user information can each matter to what the tested result means. Sharing a trade name does not make different combinations interchangeable.

A digital-pathology report needs the corresponding account: which scanner and image conditions supplied the input, which processing and model produced the output, and how the pathologist used it. Finding the reader-study report is useful only after its tested combination can be distinguished from the proposed one.

An accountant faces a simpler version of the identity problem when an invoice, packing list and stock ledger each name a different bundle. A perfect index does not establish which shipment was sold. Reconciliation concerns the actual bundle, not consistent spelling. The diagnostic question goes further: a different knowledgebase can change a variant's interpretation and what a clinician sees. The office analogy explains why identity must be established; it supplies no analytical or clinical performance evidence.

**Reviewer Lens — the first question.** The specimen, assay, pipeline, content, report and intended use form a short product description with visible blanks. That is a more useful starting point than choosing the document with the most authoritative title. The database used in analytical work and the database in the final technical assessment each have a stated role; neither becomes “the marketed device” without release evidence. For a pathology proposal, the scanner, image processing, model and reader workflow must be equally visible before the reader-study result can be applied.

**Consultant Lens — stop rework at the root.** Renaming the contents page is easy, which makes it an attractive early response. The decision that actually unlocks work is the approved assembly and the record making it effective, if a release exists. A common provisional configuration sheet with unknowns exposed lets the analytical, software and risk teams locate affected evidence without pretending the content versions are equivalent. It also prevents a pathology team from commissioning every scanner test again before it knows which combination is being offered.

## Where should a claim lead inside the file?

Suppose the reviewer starts with a promised sequencing result. The file must connect its specimen and assay materials to the instrument and pipeline, the database and rules, the reported variant and its clinical meaning. That route then leads to analytical and clinical evidence, risk controls and instructions for use. The opening file stops at the uncertain assembly; adding another heading would not extend the route.

This is the purpose of the connected **technical documentation** required by the In Vitro Diagnostic Medical Devices Regulation (IVDR). Annex II explains the device and intended purpose, design and manufacture, information for users, applicable General Safety and Performance Requirements and their fulfilment, benefit-risk reasoning, and verification, validation and performance evidence. These are different questions about one product. A report title in the right folder does not establish that its evidence covers the proposed version.

Annex III follows that product into its post-market surveillance system: what information will be collected after market placement, how it will be assessed, and what resulting reports show when data exist. Both sets of documentation must be clear, organized, readily searchable and unambiguous. A surveillance plan describes future

work; it cannot stand in for an observed field result. Nor does the law require a separate physical file for every requirement if the controlled links can be followed.

For digital pathology the path runs from scanner and image conditions through model, display or worklist to pathologist use, then to performance, human-use evidence, risk and labeling. The questions differ from sequencing, but the stopping rule is the same: where does the evidence cease to identify the claimed product and use?

**Reviewer Lens — Annex II and III in use.** Matching the specimen, materials, pipeline and content across description, tests, risk and instructions gives the reviewer a defined object for the next question: what would its surveillance plan watch if it were deployed? The plan cannot repair missing version identity. For pathology, the supported scanner list should meet the scanners in model testing and the planned scanner or worklist signals. A mismatch is best described as the particular inference it prevents, rather than a formatting defect.

**Consultant Lens — two outputs from one finding.** An assessment map needs document identity, controlled version, owner, scope and evidence location. Management needs a shorter sequence: settle commercial identity, assess existing evidence, obtain what is missing, and update dependent labels and monitoring. The short view should still point to the detailed finding. Independently changing reference content or scanner image processing belongs in both views because either can affect output while a familiar software or model number stays fixed.

## What does the application version leave out?

The opening pipeline number identifies one component. “Current” identifies no content state that a later reviewer could reliably reconstruct. To recover what produced a report, the **controlled configuration** must identify the components assembled to deliver it: content version or digest, assay or reagent state, instrument, relevant report rules and labeling. Approval date and where that combination was supplied also matter if it was supplied. Unknown fields in this packet remain unknown; another exercise cannot fill them.

Once the assembly is visible, a change raises a more useful question: which conclusions depended on this part? A new knowledgebase could change variant classification or report wording without code changes. A changed scanner profile could change the image and model behavior while the model file stays fixed. The dependency map follows that possibility into design inputs, risk records, verification, performance evaluation, user information, certificate questions and surveillance. It gives the next supplier update somewhere specific to go.

There are two different ways records may “agree.” When identifiers refer to one controlled assembly, or their relationship is proved, the question is **configuration concordance**. When paired test results agree to a defined extent, it is **numerical**

**concordance.** Signed assembly records can establish the first without demonstrating sufficient assay-result agreement. High paired-result agreement can establish the second without identifying what was commercially deployed. The comparison and its unit need to be named before “concordant” becomes informative.

**Reviewer Lens — follow a dependency.** The content identifier in a report should connect backward to its approval and forward to the analytical, clinical and user-information evidence for that use. If the identifier is “current,” the missing item is a reproducible state. A pathology reviewer follows the scanner profile into the reader-study images in the same way. Identity and performance can both matter while needing different records; collapsing them into a general version concern makes the request harder to answer.

**Consultant Lens — make the map maintainable.** Each consequential component needs a steward, controlled identifier, approval record and known downstream documents. Alerts or gates become useful at the point where an outside update could change a delivered report or image. Their purpose is to route an impact decision, not to demand a full new study for every edit. The comparison may justify limited additional evidence or expose a need for broader work. A maintainable map makes that distinction before the next change arrives.

## Would another folder solve the problem?

Moving the analytical report under a preferred tab would make it easier to locate, but would leave its relationship to the proposed combination exactly where it was. Annex II and III specify information and presentation qualities, not one mandatory folder aesthetic. A notified body may accept another organization when the required information remains complete and navigable. Conversely, familiar titles do not turn a result on one version into evidence for another.

This gives a useful test for a format request: which assessment decision will the new organization make possible? An absent commercial manifest needs an identity record. A pathology scanner study with the correct configurations under a different clear index may already be assessable. The document's location and the evidence's reach deserve separate judgments.

**Reviewer Lens — ask for the missing relationship.** “The analytical result cannot yet be applied to the proposed configuration because the approved combination and content relationship are absent” tells the company what to resolve. “Use our template” does not. The pathology equivalent is the link between a study's scanner/model and the claimed release, even when the study itself is easy to find.

**Consultant Lens — repair substance before cosmetics.** The shortest useful response is the evidence path that permits the blocked decision: a manifest, a supplier-content comparison or targeted applicability work. Reindexing can follow if it improves



navigation. Rewriting old records to create apparent uniformity would destroy the very history needed to evaluate their original scope.

## What does the opening exhibit change?

The technical assessor and management can receive different-length accounts of the same uncertainty. One preserves every document version; the other identifies the product choice and owner needed next. Neither needs to infer a failed assay or a deployed final-assessment database in order to explain the gap.

**Additional exhibit — the opening record.** The managed knowledgebase can change report categories without changing the application version. No single release record names the commercial combination.

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### Decision

**What is the principal finding? Which relationship must be established before these reports support one product conclusion, and which existing work may remain useful within its own scope?**

### How a reviewer locates the root gap

The product and configuration identity remain unestablished. Different labels are clues to that missing relationship, rather than the whole problem. The approved combination and effective state must be identified before the analytical database, final-assessment database and risk-work knowledgebase can be mapped to it by records.

The analytical work and the final technical assessment may each remain useful for their documented subjects. Neither is shown to cover the proposed commercial assembly, and a final technical assessment is not itself deployment evidence. Risk, performance, software, supplier, label and postmarket records cannot converge on a product that has not yet been identified. The useful response preserves those report-specific positives and obtains the missing relationship.

**Reviewer Lens — the bounded opening finding.** The root finding is the missing release combination and its claim consequences. A date-like knowledgebase label cannot prove an alias, and the final report cannot create a field release. A definite tested configuration remains a definite tested configuration. A pathology record saying “scanner current” would similarly call for the scanner/profile list before it justified discarding every reader result.

**Consultant Lens — choose a decision, not a history.** The manufacturer's next step is one authoritative record for the product it proposes to assess, with an approver and effective date if it has actually been released. A defined content freeze, a version

comparison and bridge, a narrower offering or a redesigned update gate are possible choices after that. Each needs evidence and a feasible supply mechanism. The management sheet should keep the analytical and final-assessment roles visible while commercial identity remains open.

An approved reference configuration for a stated purpose is a **baseline**. It says what was assembled, who accepted it and what that acceptance meant. That makes a before-and-after comparison possible; it does not validate a later version. The database named in the final technical report is the report's subject, not an approved commercial baseline unless release records establish that status.

## Which blanks belong in the identity record?

The following filled map keeps each original identifier traceable without asking the reader to remember it. It describes evidence actually supplied and the question that stops the next inference. The independent monthly supplier case later in this chapter supplies no missing release date or version here.

| Supplied record                                                    | What is known                                                  | What would make the next inference possible?                                                                                   |
|--------------------------------------------------------------------|----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Device description:<br>pipeline 7.0;<br>knowledgebase<br>“current” | A pipeline label and a<br>moving content<br>description        | Exact content identifier, approved<br>assembly and effective state                                                             |
| Analytical report: DB-18                                           | Analytical evidence for<br>the named database as<br>documented | A justified comparison to another<br>content version or offered combination                                                    |
| Final technical report:<br>DB-19                                   | A final technical<br>assessment of the<br>named database       | Separate evidence of whether, when, to<br>whom and with which pipeline/label it<br>was released; deployment is not<br>supplied |
| Risk work: KB-2026Q1                                               | A third content label in<br>the risk record                    | A proved alias or an assessment of a<br>genuinely different baseline                                                           |
| Supplier record:<br>“external content”                             | Content source or<br>ownership is described                    | Manufacturer approval, delivery gate,<br>version capture, verification, recovery<br>and any field-state tracking               |

The final-technical-report row preserves a positive fact: its subject is identified. The next column asks a different question about supply and use. Management can respond

first by seeking the controlled release record; it need not commission every department to revise its reports before that identity question is settled.

An inventory helps only if it includes the route that can change the output. A **software bill of materials (SBOM)** lists components in delivered software, supporting dependency and vulnerability tracing. Managed knowledgebase content or a cloud annotation service can change through a channel outside that list. A complementary **service and update inventory** identifies the provider, content/version, access and publication path, notice and approval, monitored state and recovery options. Neither inventory replaces the other. Another party operating the service does not take over the manufacturer's responsibility for the supplied device's conformity.

**Reviewer Lens — use the filled map.** The analytical row leads to its method and tested configuration, then to the controlled relationship with the proposed report claim. An arriving alias map needs support in signed build or release records; a renamed PDF does not provide it. For pathology, the component list may identify a model library while the service inventory identifies an outside color-profile change. Both can be necessary when both can affect the output.

**Consultant Lens — sequence the work.** The first decision identifies the proposed product. The next proves or rejects the label relationships. Only then can the team judge which risk and performance work still applies and commission the remaining tests, labels or supplier controls. Each decision needs an owner and acceptance record. Calling a report final, or purchasing a supplier certificate, cannot substitute for the manufacturer's release decision.

### **Decision pause – the signed configuration-manifest example**

In a separate positive packet, signed build manifests show that **Pipeline 4.2, Clinical Release 5.0** and **Model Package M2** are component labels assembled in **CFG-8**. Its **performance evaluation report (PER)** and instructions for use explicitly name that configuration, and the supporting evidence reports identify its unchanged components. These records do not belong to the opening database case.

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#### **Decision**

**Can the naming discrepancy close? Which exact question can be closed, and which question can these manifests not answer?**

#### **How a reviewer credits the proved assembly**

Yes. The stipulated signed records establish the three labels' relationship to the configuration, with matching performance evaluation and user instructions. That adequately closes this packet's configuration-identity discrepancy.

The historical names should stay in their original reports, connected by the proved assembly or alias map. Rewriting them as though the older records used today's names would obscure the evidence. The manifests do not demonstrate new performance, erase a study limitation or establish the opening database's commercial state. Another packet using similar labels must be judged from its own records.

## What happens after the software blocks the report?

Consider an independent **treatment-report risk-control example**, identified in the source packet as RC-1. A stale interpretation database accepts a detected genetic variant under the wrong medicine-eligibility rule. The resulting apparently valid but misleading report reaches a clinician. If relied on, it could lead to inappropriate treatment and possible drug toxicity. This is a possible event chain, not an observed patient event, and it is separate from the opening pipeline and database file.

Following the chain keeps the risk words concrete. Stale content is the initiating condition. The misleading report is the potential source of harm. Clinical receipt and use expose the patient to it. Toxicity would be the harm, but no toxicity is reported here. These stages also reveal opportunities for a product or laboratory check before the information is used.

They do not all have the same probability. A stale-rule mismatch may occur; it may or may not escape checks and reach the clinician; toxicity may or may not follow that exposure. One way to organize the estimate is:

$$P(\text{harm}) = P(\text{hazardous situation}) \times P(\text{harm given that situation})$$

The first factor concerns reaching the hazardous situation; the second concerns the particular harm conditional on reaching it. Each segment needs justification for the intended workflow. A disagreement percentage describes agreement under its own denominator. It is not automatically either factor or the final probability of harm. This example supplies no rate, denominator, probability or severity estimate to insert into the equation.

The proposed control interrupts the chain by binding the report to an approved, compatible rule/database combination and blocking release when their identifiers do not match. Tests can challenge matching, missing and mismatched versions, inspect suppression and audit records, and confirm that compatible inputs still process. They establish whether the specified check was implemented for the tested inputs.

But the patient's route continues after a block. Intended staff must recognize it, recover within the clinical workflow and obtain or communicate a result in time. These questions concern the control's effectiveness. A block can prevent misleading

information while creating a delayed-result path. That additional possibility belongs in the risk review even when the software implements its requirement correctly.

The binding IVDR Annex I risk approach puts safe design and manufacture first, then adequate protective measures where appropriate for remaining risk, then safety information and suitable training. Its requirement to reduce risk “as far as possible” is qualified by not adversely affecting the benefit-risk ratio. The risk-management standard **ISO 14971:2019**, from the International Organization for Standardization, provides a method; it does not replace that legal duty. Clause 7 distinguishes selection, implementation and effectiveness checks, residual risk and risks introduced by a control; Clause 8 addresses overall residual risk. **ISO/TR 24971:2020**, an informative Technical Report, helps explain the in vitro diagnostic analytical-to-clinical event chain in Annex H. Here the version block is a design or protective response. Calling a warning an equally effective substitute would require evidence about the actual product and use.

**Reviewer Lens — separate findings.** The compatibility requirement and its tests can answer exactly which mismatches the software blocks. The intended-user task and timing evidence answer what happens next. A passing implementation deserves credit without silently crediting recovery, delay-risk acceptability or overall residual risk. A pathology worklist block raises the same distinction: rejection of an unsupported image may be technically correct while an invisible or unrecoverable queue leaves the clinical task unresolved.

**Consultant Lens — repair the clinical path.** Engineering owns the compatibility behavior; human-factors and clinical staff can define recovery and time-sensitive alternatives; risk leadership needs the resulting residual and overall assessment. That division allows the company to retain what passes and repair the actual gap. For sequencing, a recovery plan needs someone able to restore a valid report and communicate a clinically important delay. For pathology, someone must be able to locate and resolve the rejected image. These are design and management choices, not an asserted field incident or reporting trigger. Narrowing or delaying a feature still needs a controlled claim and supply decision.

**Additional record — the treatment-report control.** Verification shows correct blocking for every prespecified mismatch test and correct processing for every compatible control. A task simulation also shows that intended users **cannot recover from a blocked result in the allowed workflow**.

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## Decision

**Which finding can close? Which limitation is demonstrated, rather than merely missing evidence?**

**How a reviewer preserves the pass and names the failure**

The version-check implementation is adequate for the tested requirement and configuration. That positive result survives. The task simulation, however, demonstrates a recovery limitation: the intended users tested could not recover in the allowed workflow. It would be inaccurate to describe their recovery evidence as merely unsupplied.

Overall control effectiveness is therefore not established for that workflow, and possible delay needs assessment and control. A redesigned or otherwise feasible recovery route needs implementation and intended-use effectiveness evidence. Residual risk is the risk remaining after controls. The overall evaluation considers all remaining device risks together against the benefits of its intended use, using the method and acceptability criteria in the risk-management plan. A successful recovery redesign could resolve this task limitation; it would still need to enter that whole-device evaluation rather than automatically close it. The task finding supplies neither actual toxicity nor a measured delay rate.

**Where does the evidence stop?**

The opening file and the treatment-report example need different next records. For the opening, a product description with explicit unknowns must connect each consequential claim to its tested version, then compare description, risk, performance, software, user information and surveillance boundaries. Change history should explain departures. The analytical and final-assessment reports retain their known subjects while commercial identity remains open.

For the treatment-report example, the evidence stops at two different places: mismatch implementation passed, while intended-user recovery failed in the supplied task. A separate pathology study might likewise be adequate for its tested scanner while a scanner extension stays open. The useful finding identifies the boundary of the result rather than turning all incompleteness into the same state.

**Which work should happen first?**

The detailed assessment map and the executive repair sequence should point to the same root gap. In the opening file, identity comes before evidence applicability, and applicability before targeted repair. Simultaneous rewriting by every department would distribute the work without settling its common premise. A content freeze, version bridge, narrower scope or redesigned update route has different evidence and business consequences; none is an automatic fix.

In the independent control example, the roadmap instead separates working software implementation from recovery effectiveness and the new delay path. Management can then direct resources to the user route while preserving the tested block. This is a different response to a different finding, even though both appear in a technical file.

**Transfer — a separate pathology product.** Its intended scanner list, image-data summary, model description, reader-study report, instructions, security architecture and surveillance plan all assumed a stable scanner profile. A supplier firmware change challenges that assumption. Which profile and date would establish what changed, and which tests might still transfer? One changed premise can create several apparent document gaps. This supplies no successful bridge for Chapter 5's Scanner B case or for this chapter's sequencing file.

## Can the procedure handle the change that actually arrives?

A signed work instruction tells a factory how a unit should be assembled; the completed assembly traveler records what happened to a particular unit. That distinction explains why a quality manual and a dated change record do different jobs. A **quality management system (QMS)** is the manufacturer's controlled way of deciding, doing, checking, recording and correcting product work. Its procedure sets the expected route. A contemporaneous record shows who reviewed a particular proposed or deployed version, which evidence they examined, what they decided and when. Neither alone establishes the medical result's safety or effectiveness; product and use evidence still have their work to do.

In the separate **monthly supplier case**, a cloud annotation provider updates its medical knowledgebase each month. The manufacturer has a complete supplier file and a software-change procedure covering source-code releases. No process determines when supplier content changes need analytical assessment, clinical review, risk updates, labeling or notified-body communication. This company is independent of the opening database file. The monthly schedule gives that file no commercial date or version.

The gap becomes visible by following what the provider can alter. Annotation content can influence a report while the application binary remains the same. The relevant gate therefore concerns the delivered combination of code and content. A scanner supplier's firmware could similarly change color processing while the diagnostic model file stays fixed. Supplier ownership and unchanged code do not settle whether earlier evidence transfers; that requires an impact decision about behavior.

**Reviewer Lens — procedure versus transaction.** A dated example of the actual change is more revealing than another copy of the procedure: old and new content identity, clinical-output comparison, responsible review, release decision and field-state plan. Those records may show that the process worked; their existence cannot be inferred from the procedure alone. A scanner firmware notice belongs beside the



specific image profile and reader evidence, rather than disappearing into a maintenance folder.

**Consultant Lens — a gate the teams can use.** The proposed intake can begin with supplier notice and a content manifest, use automated comparisons for defined low-impact categories, and escalate changed therapy mapping, result meaning or supported workflow to assay, clinical, risk and regulatory owners. Someone must be able to accept, defer or reject the update and prevent publication or roll back an unapproved state. These are candidate controls for this packet. Their feasibility and operation still need to be established.

## What must the gate do, and what was built to do it?

“Accurate reports” would not tell engineering when to block this content. A candidate obligation for the monthly service is more precise: a therapy category cannot enter a report unless the delivered knowledgebase is approved and compatible with the report rules. This is a **design input**—the promised use and risk decisions translated into testable behavior. A version gate and recorded report-content manifest could be **design outputs**, the implemented rules and records intended to fulfil it. Outputs can also be an architecture, interface, manufacturing specification or user information. They are proposed design choices, not controls already demonstrated in this company.

The distinction makes the next tests meaningful. Verification compares those outputs with the input by challenging compatible and incompatible content. Validation asks whether the resulting system serves its intended users and clinical task, including the report and recovery from a block. The assessed configuration connects the evidence to the product. A meeting that approves a procedure without these links may leave the actual behavior undecided.

For a pathology product, a candidate input might specify supported scanner and color profiles and require images outside them to be rejected or clearly separated. The architecture is illustrative; actual limits come from the product and evidence. Once deployed, the same input/output comparison continues when a supplier changes content or firmware. A dated change record shows why old software, reader or clinical evidence still applies—or what must be added. It cannot retrospectively make an unevaluated version compatible.

**Reviewer Lens — trace a promise to a build.** One therapy-category promise should connect to a testable input, delivered content and rule identifiers, challenge tests and clinical-interpretation review. For pathology, the promised scanner leads to its color assumption, model-input check and pathologist-facing label. A phrase such as “cloud system” names an arrangement, not the output that meets the obligation. The request should expose measurable behavior and version, rather than solicit a longer slogan.

**Consultant Lens — assign the missing decision.** Product leadership knows what is promised, engineering what currently runs, and clinical staff what a changed category means. Quality needs a testable input and acceptance criteria before another update passes. Low- and high-impact branches become useful only after their boundary is defined. Scanner notices therefore need a route to model and reader-study owners as well as purchasing; organizational ownership alone cannot reveal an evidence dependency.

## What can a correct block leave the user unable to do?

The monthly service's proposed gate brings back the question exposed by the independent recovery example. Keeping an incompatible report from a clinician may prevent misleading treatment information, while leaving a time-sensitive result unavailable. Risk analysis follows the content update through rule and report behavior, laboratory or clinician checks, a decision and possible harm. A pathology change has its own possible path through altered color, model output or human interpretation; its worklist block can introduce delay. These are mechanisms to investigate, not patient events claimed for either case.

The binding Annex I duty applies through the device lifecycle. Safe design and manufacture come first, then appropriate protective measures for remaining risk, then safety information and suitable training. Its benefit-risk-qualified “as far as possible” obligation remains the legal test. ISO 14971:2019 Clause 7.1 provides a similar ordering as a method, with acceptability wording that cannot replace the IVDR requirement. Detection may be part of a design or protective measure; it is not a fourth equal tier allowing a feasible safer design to be skipped. Clause 7.2 separates implementation from effectiveness—the distinction the treatment-report simulation made observable.

For a changed state, the next question is what risk remains and whether the control creates another risk. A gate may protect report meaning and delay its delivery. A scanner check may protect the model and leave no accessible alternative for the pathologist. Safety information can help only when the assigned actor can observe the condition and perform the action. Software, assay, clinical and human-use records meet at that route; four isolated risk rows would hide their dependence.

**Reviewer Lens — two outcomes of one control.** The wrong information that might reach a clinician and the delay that might follow blocking it need separate evidence. The block test is one record; intended-user recovery is another. A scanner-profile detector needs both correct rejection on the tested images and an intended reader who can see and resolve the case. No software mismatch frequency becomes a patient-harm probability without the intervening clinical path.

**Consultant Lens — choose in order.** Engineering and clinical leads can first consider excluding unsafe content or images at source, then a feasible protective gate and recovery, then the residual instructions users need. Management can compare the remaining risks and evidence for each option, including a block's delay path. A warning's title cannot make an inaccessible cloud version observable or repair a failed recovery workflow.

## Can the manufacturer control the outside contribution?

The gate can operate only if the supplier relationship gives it something to inspect and a way to act. The contribution might be a reagent, scanner, reference data, model or annotation service, or a laboratory task. Control follows what that contribution can change in the clinical result, rather than whether purchasing calls it software, data or service. A content publication or scanner firmware event can matter while the application or model package stays fixed, and either event can sit outside the delivered-component inventory.

Supplier qualification and a contract provide a basis for control. The operating arrangement must specify technical requirements and accepted versions, notice before clinically relevant change, manufacturer approval or a controlled hold, enough detail for impact assessment, verification responsibilities, security and incident communication, continuity and recovery. Each actual change then needs its own record. If the provider cannot supply the information or control, management faces a product decision: whether the claimed configuration can still be supported, another technical control is feasible, or the supplier arrangement must change. Outsourcing the contribution does not outsource the manufacturer's IVDR conformity responsibility.

**Reviewer Lens — find the invisible dependency.** Comparing the provider's publication log, the manufacturer's accepted content manifest and the report versions that could reach users reveals whether notice arrived in time for a gate. For pathology, the comparison is between firmware/color processing and tested image characteristics. A generic supplier certificate may support the provider's system; it cannot demonstrate that this particular new mapping or profile preserves device performance.

**Consultant Lens — negotiate usable evidence.** Minimum deliverables include versioned content or firmware, a change summary and comparison inputs, a notice period, incident/security contacts, rollback or continuity arrangements, and records of which customers received which state. Manufacturer release then has a defined review and acceptance test. If a monthly feed cannot be paused, a proposed approval gate may be infeasible. That is an issue for management to solve, not a control the consultant can assume into existence.

## How far can the old evidence travel?

Calling an update “maintenance” tells us what activity the supplier performed. It does not tell us what changed in the result. A useful assessment begins with the old and proposed product states and the behavior that might differ, then follows that behavior into design inputs, risk controls, software tests, analytical and clinical studies, user information, certificates, registration and surveillance. Unchanged code can accompany changed medical meaning; many code edits may leave a defined output unchanged. File count and monthly cadence are poor substitutes for that comparison.

When evidence for one defined state remains applicable to another despite specified differences, the reasoned, evidenced argument is a **bridge**. It states which evidence transfers, between which versions, for which claim and with what limits. A comparison and limited regression testing may suffice, or new analytical, clinical or user work may be needed. “Minor change” is not the argument. The opening analytical database's results could be assessed this way for a later state, but no successful bridge is supplied; the monthly case cannot provide one for it.

Before an update is accepted, actual evidence supports a choice to release, hold, narrow the claim or redesign the update route. After deployment, the comparison also identifies affected field units or service tenants and any notified-body or authority communication required by the actual certificate, change and incident facts. Potential timely duties merit prompt assessment. None of these fictional packets establishes a particular notification trigger.

**Reviewer Lens — demand the difference.** The monthly content comparison should identify the changed therapy mapping and the old versus proposed report output against prespecified acceptance criteria. The next inquiry is whether analytical validity, clinical interpretation, risk, labeling or certificate scope is affected. A scanner firmware comparison similarly asks what changed in color and whether model and reader evidence cover it. An absent comparison leaves a bridge unsupported; it does not invent a failed study.

**Consultant Lens — real management branches.** A controlled update path can use defined comparison and regression criteria within a proven boundary and escalate changes in medical meaning to deeper review. Holds, rollback, affected customers, owners and timing belong in that plan. A frozen version or narrower claim is feasible only if the actual supply can enforce it and the remaining evidence supports it. No branch guarantees the old schedule or certification outcome.

## What can be done before the cause is known?

A supplier notice, complaint, discordant report, security issue or new scientific finding can raise a question before anyone knows its cause. **Post-market surveillance (PMS)** is the planned, ongoing collection and analysis of device quality, performance and safety after market placement. **Post-market performance follow-up (PMPF)** follows planned performance questions. Their plans identify sources, signals, methods and decision thresholds. An observation arrives only when actual information is collected.

Chapter 1's separate deployed sequencing service has more invalid initial reports after a materials change. Its separate deployed pathology service has supported images reaching the worklist later than the local target after a service change. Neither scene supplies a count, cause, harm, certificate or reporting trigger. Both explain why a signal must be tied to the configuration actually in use before earlier evidence can be reconsidered.

An **investigation** asks what happened, which versions and reports may be affected, and what caused it. For the monthly content service, that means comparing old and new therapy mapping, resulting report behavior and content actually supplied to users. Investigation is not itself a correction. A proportionate hold or other protection, and assessment of any applicable timely duty, may be needed while the cause remains open.

Suppose that investigation established an incorrect report. Its controlled replacement or withdrawal could be a **correction**, addressing the detected instance. This packet does not establish that report. If evidence instead or also established that an inadequate manufacturer gate let an incompatible mapping through, repairing that cause and testing the repair could be **corrective action**, intended to prevent recurrence. A check addressing a distinct, not-yet-observed failure path would have a different purpose: **preventive action** removes a cause of a potential nonconformity before it occurs. These are conditional examples of different jobs, not completed actions or determined causes in the monthly case.

The **corrective and preventive action (CAPA)** record connects cause, chosen action, implementation and effectiveness. A closed log alone cannot show that the field product now works. Any correction, design change or supplier change must be traced into affected configurations, risk, performance evaluation and user information, with field or other appropriate evidence of the intended effect. Article 10(8) expressly calls for CAPA management and verification of effectiveness; Articles 78–79 and Annex III connect surveillance, technical documentation and future action. The separate deployed examples likewise require attention to precautions and timely duties while causation is investigated. Causal closure and demonstrated effectiveness are different milestones.

**Additional exhibit — the monthly supplier case.** One update changes a therapy-associated classification and increases the frequency of an unresolved mapping anomaly. The direction of change is known. Counts, denominator, patient impact and commercial-deployment history are not supplied; the opening file supplies none of them for this company.

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### **Decision**

**Can every monthly content update be treated as routine maintenance because intended purpose and application code are unchanged? What should be held or examined for this update, and which conclusions would exceed the anomaly record?**

### **How a reviewer assesses this update's actual behavior**

The calendar and unchanged code do not establish low impact. The old and new content and report outputs need comparison, along with affected product states and users. That inquiry reaches risk, analytical and clinical meaning, performance evidence, user information, supplier gate, surveillance signals and any certificate or regulatory communication under the actual route.

The changed therapy classification and more frequent unresolved anomaly justify investigation and a controlled release or field-state decision. They supply no quantified error rate, established injury or demonstrated causal attribution to a particular party. Future changes might fit predefined, evidenced regression criteria. This update needs assessment against its own behavior and facts.

**Reviewer Lens — a change is a transaction.** The before/after manifest and comparison should connect rule and report behavior to approval, effective date, affected configurations and any field assessment. A supplier notice and a manufacturer acceptance decision are different records. For scanner firmware, image characteristics and model/reader evidence provide the corresponding comparison. The unquantified increase stays an observation; it does not become a harm probability or reporting conclusion by appearing in a trend discussion.

**Consultant Lens — useful change categories.** Prospective categories can distinguish metadata with verified no output effect, bounded content changes with predefined challenges and clinical sign-off, and changes to therapy categories or other medical meaning that need deeper review and possibly deferred release. The product still needs designed and validated thresholds, tests and stop conditions. Update authority, manufacturer acceptance, rollback and field-version tracing belong in both the supplier agreement and operating procedure. An effective arrangement could reduce repeated emergency reviews; this case has not demonstrated one.

## What would show that the quality system operated?

Starting with the intended purpose and latest approved state, a reviewer can follow one result through its design input, output and verification. The risk record should continue the software, assay, human and supplier events toward possible patient harm, then distinguish the control's implementation and effectiveness. That makes supplier proportionality concrete: what can the outside party change in the result, and which operating record shows how the manufacturer controlled it?

The change record then connects behavior to affected evidence. If field action was actually taken, its deployed configuration and effectiveness evidence matter; a filed procedure cannot answer either question. A scanner update may reach both model and reader evidence. A knowledgebase update may reach report meaning without a code release. The record is adequate when those relationships can be followed for the particular transaction.

## What belongs on the management decision sheet?

Management needs the old and proposed versions, potentially changed output, affected claim and risk, existing evidence and its boundary, and the remaining regression, analytical, clinical or user work. Those facts make release or hold authority meaningful. Field identification, recovery, certificate or authority questions, an owner and a decision date complete the operational choice.

The options may include automated acceptance inside proven criteria, gated release, manual expert review, freeze or rollback, or supplier replacement. Price and speed alone justify none of them. Prospective categories can make routine work efficient, but every actual update must satisfy its chosen category's conditions and exceptions. A replacement supplier brings its own product and evidence comparison.

**Transfer — a separate scanner.** After a reader study, its supplier calls a firmware/color-processing update an “image-quality improvement.” What if the model receives different color information? Which old/new profiles, image comparisons, model and reader conditions, and labeling claims would determine whether a bridge or repeated work is needed? The supplier's intention does not prove compatibility. This is neither a failed study nor another release of Chapter 5's Scanner B packet.



## Practice studio — return to the opening file

The opening record can now be read with two distinct questions in mind: can the product be identified, and can its claim be followed into evidence? A General Safety and Performance Requirement row connects intended claim, product obligation, risk control, test, performance evaluation, labeling and lifecycle records only if those records concern an identified configuration. The assessor needs the exact links. Management needs the first broken link and a decision that can repair it.

**The opening packet revisited.** All Annex II and III titles are present. The device description names the pipeline and calls its knowledgebase “current.” The analytical report names the database used in analytical work; the final technical report concerns a different named database; the risk file names the unresolved knowledgebase; the supplier procedure calls it external content. The exact identifiers remain in the earlier evidence map. This is the same opening record, with no added commercial release, alias, outcome or link to the independent monthly supplier case.

## What would make one reference record authoritative?

The manufacturer needs a controlled reference for each state it proposes to assess or supply—a **canonical configuration record**. For sequencing, it links purpose, specimen and user to assay materials, instrument, pipeline, interpretation content, rules and report. For pathology, the corresponding record connects scanner/profile, model, display or worklist and pathologist use. It also links accessories and cloud services, manufacturing and supplier sites, the applicable device identifier, user instructions, any certificate scope and effective dates.

That sounds extensive because it answers a consequential question: which pieces formed this offered product, and when? The record can point to controlled documents instead of duplicating their detail. One brand may legitimately contain several supported configurations, provided their tests and limitations are distinguishable. Deleting version letters from one test and another label would remove the evidence of a difference rather than prove agreement.

For the opening case, the analytical database has its documented test state, the final technical assessment names another state, and the risk-work knowledgebase remains unresolved. A proposed commercial assembly and its relationships must still be established. If no commercial version has been approved, the record should say that. A premarket proposal need not be rewritten as a deployment history.

**Reviewer Lens — authoritative does not mean invented.** An approved manifest, alias map, effective dates and any supplied-state record can connect the component identities to description, risk, performance and instructions. The final technical report remains a positive record of its own assessment while commercial status stays

unknown. For pathology, the needed scanner fact is the profile actually used in the reader study, not merely its manufacturer's model name.

**Consultant Lens — one naming authority.** A product owner can maintain the reference record while quality controls revisions and assay, software and clinical owners approve the parts they understand. Management then sees each offered configuration with its own claim and evidence. Externally updated content needs an effective-state capture and an enforceable block or rollback route. A central spreadsheet saying “current” would centralize the ambiguity rather than control it.

## Do several evidence paths meet at the same result?

Chapter 5 followed one legal requirement through its product-specific evidence. Here several paths must meet at one product. Consider a result-integrity promise: the specimen's sequencing report retains the right patient and approved interpretation content through calculation and transmission. Its design input specifies identifiers and version compatibility. Risk work examines wrong-patient or wrong-rule information and possible clinical consequences. Interface and rule tests challenge the path; intended-user work tests recovery from a block; instructions describe actions the intended person can perform.

The surveillance plan then identifies future mismatch or invalid-report patterns that could reopen the argument. Actual field data, if collected, would be another record. This distinction is essential in the opening file: the analytical report may support a measurement question, while the final technical assessment may support another question. Their titles do not establish the content in the proposed report or actual deployment. Until the accepted combination is known, the joined thread has a configuration gap.

The first question along that route is **performance under intended conditions**: what result, for which specimen or image, users and environment? A sequencing variant report needs its actual assay and interpretation state. A pathology score needs supported scanners, images and reader workflow. Analytical and clinical studies, software challenges and user tasks answer different parts. Later monitoring asks whether performance remains credible in supplied states; it cannot substitute for the initial basis.

Even a technically supported result must be understandable and usable. **Information and labeling** therefore concerns its meaning, limitations and the response to a blocked or invalid state. A laboratory cannot verify a hidden cloud version merely because instructions demand it; a pathologist cannot recover an image through an invisible route. The instructions and interface are design outputs whose clarity and usability need evidence. Later complaints or use errors can test whether that arrangement works in practice.

Those actions also depend on what was assembled and released. **Software and electronic systems** evidence identifies code, model, content, report rules, environment and security conditions, then shows how they were secured, released and maintained. The independent signed-manifest example established identity, not clinical performance. Challenge tests and release records establish their own behavior; vulnerability, supplier and patch monitoring may reopen it. A pathology model's intended use also depends on its scanner and display environment.

The software assembly is not the whole examination. **Specimen, biological and chemical safety** asks which collection, transport, reagent, contamination, interference and stability questions fall inside the actual kit or specimen pathway. A standalone pathology software function may supply neither reagents nor sterile packaging. That fact does not waive every infection or specimen question for a combined system. Applicability needs a reason for each requirement, followed by the relevant material, design, analytical or handling evidence. Lot, supplier or scientific signals can later change that reasoning.

The final connection is **measurement and result integrity**: does the right patient's specimen or image stay associated with the right data, calculation, content and report? Interface tests, reference comparisons, quality-control checks, audit records and recovery tasks each inspect a different part. A wrong identifier and a wrong interpretation can both make a plausible report misleading. The opening packet cannot join these records to one proposed combination until its versions are related; a pathology worklist likewise needs the tested image and scanner state before evidence transfers. Observed field discordance or corrections, if they occur, return to this thread.

The table is a retrieval aid after that walk, not five independent votes for conformity. Each final column identifies future monitoring or change questions unless actual observations are separately supplied.

| Question along the result path     | What needs to be settled for this product?                          | Evidence and the reason to return later                                                                                    |
|------------------------------------|---------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Performance in intended conditions | Specimen/image, range, user, environment and failure behavior       | Matched analytical, clinical, software and usability work; surveillance, performance follow-up and changed-state review    |
| Information and labeling           | Understandable limits, states and feasible recovery                 | Instructions/interface and comprehension or task results; complaints, use errors and controlled revisions                  |
| Software and electronic systems    | The assembled code, model, rules, content, environment and security | Architecture, traceability, challenge tests and security/release records; vulnerabilities, suppliers, patches and versions |

| Question along the result path           | What needs to be settled for this product?                                | Evidence and the reason to return later                                                                                                       |
|------------------------------------------|---------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Specimen, biological and chemical safety | Applicable materials, contamination, interference, stability and handling | Material/design, analytical, transport/stability and control evidence; lot, supplier and scientific signals                                   |
| Measurement and result integrity         | Preserved identity, calculation, transfer and meaning                     | Interface/reference/quality-control tests, audit and recovery for the identified configuration; discordance, CAPA and content/service changes |

The last row begins with the report's promise, points to distinct records testing it, then identifies later information that could reopen it. In the opening file those records are to be located or reconciled; the row does not assert they passed. A scanner/profile change may touch software, performance, labeling and integrity at once. A reagent or content change may touch specimen, measurement and report rows. The shared premise matters more than the number of rows affected.

**Reviewer Lens — one change, several questions.** The database used in analytical work has exact tested inputs that can be preserved. A proposed use of the final-assessment database needs an impact comparison across the relevant rows without an invented deployment. For a scanner extension, preprocessing leads into model input, display and user information. One outside change can unsettle several assumptions; unaffected evidence remains useful when a reasoned analysis supports that boundary.

**Consultant Lens — a common comparison for the teams.** Assay, software, clinical, labeling, risk, supplier and quality owners need the same product-state comparison and the document capable of answering each question. That exposes which decision is blocked by a missing record. Five formatting tasks cannot replace one missing content identity. A defined product can support a prospective surveillance or performance-follow-up plan, but the future data remain future data.

## What does each reader need from the same map?

The opening analytical report is located and names its tested database. A reviewer needs to know what makes that evidence applicable to the proposed product; management needs to know what decision must precede any additional study. Both questions can be answered from the same evidence map, provided its summary retains the unresolved commercial identity. A proposed owner in the table is a work assignment, not a claim that the company already holds the closure record.

| <b>Evidence link in the opening packet</b>     | <b>What is known?</b>                                               | <b>What blocks the next inference?</b>                                                                         | <b>Proposed owner and closure record</b>                                                            |
|------------------------------------------------|---------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Product description □<br>release identity      | Pipeline 7.0 and knowledgebase “current”                            | No approved commercial combination or effective content state is supplied.                                     | Product/quality: controlled assembly manifest, approval and effective-state history, if any.        |
| Analytical claim □<br>DB-18                    | Analytical report identifies its tested database.                   | Applicability to the proposed assembly needs the content and assay differences established.                    | Analytical/software: tested configuration and documented comparison or required new evidence.       |
| Final technical assessment □<br>DB-19          | Final technical report concerns this database.                      | Its scope establishes neither commercial release nor deployment.                                               | Quality/software: report scope and separate approved release/supply record, if one exists.          |
| Risk □<br>KB-2026Q1                            | Risk file identifies this knowledgebase label.                      | Its relationship to either report's database is unproved.                                                      | Risk/software: signed identifier map or assessment of a genuinely different baseline.               |
| Supplier and surveillance □<br>delivered state | Procedure calls content external; no field combination is supplied. | An outside update may change medical output without changing code; monitoring needs identified field versions. | Supplier/quality: update gate, service inventory, state trace and version-specific monitoring plan. |

Read the analytical row across. The report is useful evidence with a known subject. The missing relationship to the proposed assembly prevents a wider applicability claim. Its proposed closure can be a documented comparison; the table does not presume a new study is needed. The risk row may instead close through a proved alias. Closing that identity question would leave the analytical applicability and any other evidence questions with their own findings.

The executive roadmap can now be short without becoming vague. First, product and quality owners establish what is proposed or already supplied. Second, the relevant technical owners prove the identifier relationships and compare the evidence with that state. Third, management commissions the design, performance, labeling, supplier and monitoring work that the comparison shows is necessary. Each step has an owner and a

decision gate. Exact controlled references can connect the records; there is no need to manufacture one physical file for each requirement.

## Which outside party can change the result?

The last map row reaches beyond a delivered software component. A cloud interpretation service can change without shipping a new component; a reagent supplier can change material, a scanner supplier firmware, or a contract laboratory a process. Which technical requirements, version and release identity, change notice, test and approval responsibilities govern that contribution? Security and incident duties, continuity and exit arrangements matter because the manufacturer needs to keep producing or safely discontinue the promised result. The dated quality-system record then shows how those requirements operated for a particular change. The manufacturer's conformity responsibility remains attached to the resulting device.

For the opening packet, “external content” identifies a dependency but no commercial report's content state. In the separate monthly supplier case, a complete supplier file leaves the particular therapy-category update's assessment open. A pathology manufacturer similarly needs the scanner notice and tested image profile connected in an impact record. If the service will not disclose enough to assess its influence, management faces a product and supplier-control choice. A complete folder cannot resolve that lack of control.

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### Decision

**Could renaming the documents and putting the knowledgebase under “supplier maintenance” repair the finding? Which product and supplier records are needed before deciding whether the existing evidence applies? The monthly supplier case supplies no missing dates to this packet.**

### How a reviewer follows the proposed repair

Renaming may make documents easier to find after their relationship is established. First comes the proposed or actually supplied combination, its effective content state and controlled relationships among the labels. Release and manifest history, supplier delivery and approval records, and a difference and impact assessment would make that comparison possible.

The reviewer can then determine which analytical, technical, risk, performance, user-information and surveillance records apply and which require work. A new label cannot create an absent commercial baseline or successful bridge. The final-assessment database remains the subject of final technical work; the packet still does not establish deployment.

**Who uses the same records for a different decision?**

The notified body needs to reconstruct the device and assess whether its claim, safety requirements, risks and performance evidence fit through the applicable conformity-assessment route. A national competent authority may use configuration and field records for market surveillance or vigilance. That public role is distinct from the certificate decision. For the companion-diagnostic branch, the device must be essential to a corresponding medicine's safe and effective use, identifying before or during treatment patients most likely to benefit or at increased risk of serious adverse reactions. That relationship brings the separate medicinal-products scientific consultation into the applicable device route, developed in Chapter 4. An oncology word in the opening packet establishes no such relationship.

Within the company, the manufacturer maintains the product and evidence while a consultant can prepare both the detailed map and the management roadmap. The two outputs should preserve the same positives and unknowns. These records can support several institutional questions because they describe what happened, without turning preliminary advice into authorization.

**Transfer check — one separate change.** A pathology manufacturer proposes another scanner, or a sequencing manufacturer proposes another knowledgebase release. Beyond software tests, which device-description, input, risk, performance, user, supplier, instructions, certificate-scope and monitoring records are already sufficient, need comparison or are missing? The Chapter 5 Scanner B packet, this opening database record and the monthly supplier case retain their own facts. A software test can contribute to the answer while leaving other dependencies to examine.

**Quick discrimination lab**

These two packets are independent of each other and of the earlier examples. Similar words can help locate the right question; each finding still needs evidence from its own product.

**Another file with three labels**

A separate technical file uses Pipeline 4.2, Clinical Release 5.0 and Model Package M2. Its performance evaluation report and instructions for use do not show whether they identify the same commercial configuration. No signed assembly map is supplied.



**Decision**

**May the reviewer assess each document independently and import the earlier signed-manifest example's positive identity finding because the labels match? What would establish the relationship here?**

**How a reviewer treats matching labels**

Each report can be assessed within its stated version, preserving any support it actually supplies. Whole-product adequacy needs this product's approved configuration and release record, with documented concordance among the reports, performance evaluation report and instructions. Those relationships are missing from this packet.

The earlier signed manifests closed only their own example's naming discrepancy. Their identifiers looking familiar here supplies no shared evidence. The next request therefore concerns the missing assembly relationship, without treating every individual report as worthless or accepting a whole-product conclusion prematurely.

**A supplier edit after the reader study**

A scanner supplier updates firmware and color processing after a reader study. The manufacturer records supplier maintenance and does not revisit performance evaluation. No paired images, reader results, failure or deployment history are supplied.

**Decision**

**Does supplier ownership make the earlier reader evidence automatically applicable? Which comparison and downstream records are needed before deciding whether a bridge is enough?**

**How a reviewer compares the old and new input**

The maintenance description does not establish applicability. The old and new firmware and color states need identification, followed by a comparison of image characteristics and model behavior. The next question is whether the reader study represented the new input. Its original findings can retain their stated scope while that question remains open.

The affected risk, compatibility, analytical and clinical performance, instructions, surveillance and certificate-communication questions need review as applicable. A targeted bridge may be sufficient; further testing may be required. With no comparison

or results supplied, the packet establishes neither failure nor successful generalization. The manufacturer must document the decision even though the supplier made the edit.

## **Leave a product another reviewer can reconstruct**

A traversable file connects an identifiable product state to its claim, design, risk, verification, performance, labeling, suppliers and field controls. Its searchable titles help only when that relationship is established. A procedure organizes the work; dated records show whether it operated; product studies show what the device did. A change comparison can preserve a narrow positive result, identify a needed bridge or reveal a demonstrated limitation. The opening final technical assessment still supplies no commercial deployment record. Chapter 7 takes the defined claim and configuration into performance planning, before results are interpreted.

The fifteen checks below concern evidence items. **A** means adequately explained for that item; **U**, present but unclear; **M**, missing or inconsistent; **D**, deeper analysis needed; **C**, consultation indicated; and **N/A**, inapplicable with a reason. Evidence location, reasoning and the next decision give the state meaning. The letters never grade a device. The original identifiers also retrieve optional expanded cards in Workbook Part A2.

### **The exact product and purpose • card IVDR-CH06-A01**

Can the specimen-to-result or image-to-result path, user, claim and controlled assembly describe one product? The opening commercial content state stays explicitly unknown until a record establishes it.

### **Each claim's evidence and version • card IVDR-CH06-A02**

Which tested inputs, methods, limits and component identifiers support the claim? The opening analytical report and final technical report have different stated subjects; finding both titles does not join their evidence.

### **One boundary across the records • card IVDR-CH06-A03**

Do description, risk, performance, software, labeling and surveillance describe the proposed configuration? The risk-work knowledgebase label needs its own controlled relationship to the report identifiers before it is called an alias.

### **Change history and effective state • card IVDR-CH06-A04**

Do dated manifests, approvals, differences and evidence-impact decisions identify what actually became effective? A final report cannot supply a deployment event merely through its title.

**Document dependencies and assembly identity • card IVDR-CH06-C01**

Can another person follow both the evidence map and a proved component-to-assembly relationship? Agreement among identifiers answers a different question from numerical agreement between paired test results.

**Management's feasible options • card IVDR-CH06-C02**

What product state, supplier update path, evidence and decision gate make freezing, bridging, narrowing or redesign feasible? Freezing the word “current” leaves an uncontrolled publication channel unchanged.

**A detailed map and a short roadmap • card IVDR-CH06-C03**

Does each management action point back to a precise review gap? Report-specific positives and unresolved commercial identity belong in both views, however different their lengths.

The following identifiers retain source-era topic numbers. Both original topics now sit in this chapter; the original-to-integrated crosswalk preserves their location.

**Purpose, design and verification • card IVDR-CH07-A01**

Does a testable result or user requirement lead to an implemented rule or interface and its acceptance test in the same configuration? A requirement, an output and evidence of its behavior have distinct jobs.

**Risks through to patient consequence • card IVDR-CH07-A02**

How could an initiating condition lead through misleading or delayed information and clinical exposure to possible harm? Software, security, human and assay contributions meet in that chain. Implementation, effectiveness, any new delay path and residual risk need their separate findings.

**Supplier control proportional to clinical influence • card IVDR-CH07-A03**

Do the component inventory and service/update inventory expose the relevant provider, version access, obligations, notice and manufacturer acceptance? A managed service can influence a report while absent from the list of shipped components.

**Change through behavior and evidence • card IVDR-CH07-A04**

What differs in output, claims, risk, performance, security, instructions and certificate conditions? The before-and-after comparison answers an impact question that the label “maintenance” leaves open.

**Action effectiveness in the actual field product • card IVDR-CH07-A05**

Where corrective action is taken, do the root-cause finding, design or process action and affected deployed configurations connect to evidence of controlled recurrence? Closing an action log records an administrative step; effectiveness needs its own basis.

**Supplier and change controls working together • card IVDR-CH07-C01**

Which intake, comparison, escalation, acceptance, hold and recovery decisions govern each clinically influential dependency? Dated records of actual use show whether that framework operated for a particular update.

**A defensible update arrangement • card IVDR-CH07-C02**

What supports automated update, gated release, expert review or supplier replacement? Automation needs a proved boundary and effective stop conditions. A replacement needs its own product and evidence assessment.

**Prospective categories with real limits • card IVDR-CH07-C03**

Do change criteria, exceptions, evidence obligations and owners exist before the update? They reduce emergency decisions only when each actual update is checked against the chosen category's conditions.

**Further reading**

Book 1A Chapters 4–6 and Book 1B Chapter 5 offer optional practice with technical architecture, controlled evidence and supplier dependencies. This chapter stands on its own. Those readings add no facts to its separate opening database packet, pathology examples, signed-manifest packet or treatment-report control.

**Source trail and limits**

- **Binding law, checked 2026-09-25:** IVDR consolidated 10 January 2025, Article 10(2), (4), (8)–(9), Article 56, Articles 78–79; Annex I Sections 2–4; Annex II opening and Sections 1–6, especially 6.4; Annex III Section 1. These govern manufacturer risk/QMS, supplier and CAPA control, technical documentation, performance evaluation and PMS. IVDR Annex I Section 2 qualifies risk reduction “as far as possible” by benefit-risk; Sections 3–4 address risk management and control priorities. The chapter does not substitute a standard's acceptability wording for those legal requirements.
- **Method sources, licensed full text read-only:** ISO 14971:2019, relevant passages of Clauses 6–10, especially 7.1–7.5 and 8; ISO/TR 24971:2020, Annex H.2, especially H.2.3–H.2.7. The former is a risk-management method, and the latter is

informative IVD guidance on analytical-to-clinical event chains. Their concepts are explained in original prose; no source text, numerical hazard estimate or generic risk matrix is transplanted into the fictional packets. For any real product, exact standard edition, harmonized status, scope and actual device evidence need checking separately.

- **Packet limits:** The opening DB-19 final technical report is not an established commercial release. The positive CFG-8 manifests close only their packet's identity discrepancy; the later same-label micro-case lacks them. RC-1's passing mismatch tests and failed intended-user recovery task support different conclusions. The monthly supplier update reports a changed therapy classification and more frequent unresolved anomaly, without counts or proven patient harm. No figure or artwork from the source is used.

## CHAPTER 7

# Plan the Evidence Before Judging It

**Retrieval check — before reading.** Chapter 6's analytical report and final technical assessment named different databases, while the proposed commercial knowledgebase remained unidentified. Which assay, extraction or image input, software, content and release records would connect the tested product to the product now proposed? Why would a matching application number leave that connection incomplete?

A clinician receives a result; a study team must decide what evidence will make that result trustworthy for its offered job. **What needs to be established for each promised use, and what decision will show that the evidence is enough?** The answer begins with the claim and the configured product from Chapter 6. It leads to a sequence of questions, studies where needed, and decisions about what the manufacturer can support.

## Start with the result the user will rely on

Suppose a tissue sequencing assay reports information to inform treatment selection. It examines genetic material and identifies a **variant**, a difference from a reference sequence. A digital pathology system instead examines a scanned tissue image and presents information for a pathologist. Producing an output is only the beginning of either promise. Which specimen or image, people and users does the offered result cover, and what clinical job is it meant to do? A study of the wrong configuration could be well conducted and still leave that promise unanswered.

Before the next study begins, the team therefore needs to connect each part of the promise to an evidence question and a decision. That is the device-wide job of the **performance evaluation plan (PEP)** introduced in Chapter 1. It describes what must be established, how relevant existing evidence will be found and new evidence generated, how quality and sufficiency will be judged, and when to reconsider the conclusion. A controlled plan can evolve. What matters is preserving when the decision was made: writing a conclusion after seeing the result does not make it an earlier rule.

**Opening case — the plan arrived after the studies.** An anonymous fictional pathology company writes its PEP after its analytical and clinical studies are complete. The document accurately summarizes the work and declares each evidence area sufficient. It does not explain how the questions were chosen, what result would have led to more work or how missing conditions within the intended use will be handled.

This company is separate from PathSight and other named series products; no scanner evidence is inherited from Chapters 4–5.

The accurate summary has a useful job: it helps another person locate completed work. But what would let that person judge whether the work answers the offered claim? The manufacturer needs both an honest appraisal of the existing evidence and a plan for remaining decisions. Reconstructing a false planning history would obscure the very timing the reviewer needs to inspect.

## Which document controls which decision?

Imagine a separate pathology proposal intended for images from two scanners. Its PEP asks whether evidence can support that range across the claimed tissue and user conditions. It identifies sources, quality criteria, decision gates and a way to handle gaps. If the team needs a clinical performance study, someone must then specify exactly how that study will run. The **clinical performance study plan (CPSP)** does that narrower job: it describes who or what is studied, the device and scanner versions, image and reference acquisition, endpoints, statistical methods, bias controls, amendments and safeguards. An evaluation strategy that says “study both scanners” leaves these operating decisions to be made.

Once results exist, another question becomes possible: what do they support? The **performance evaluation report (PER)** appraises scientific-validity, analytical-performance and clinical-performance evidence together, including favourable and unfavourable findings, limits and the claim-level conclusion. One reader study might support its tested scanner and workflow while leaving the proposed range unresolved. For a sequencing assay, a study protocol might specify variant types, specimen conditions and handling of results that cannot be called; the PEP asks whether that and other work cover the treatment-information claim, and the PER judges the actual results. A change in the offered range can therefore reopen the evaluation even when the study report remains accurate.

These different jobs have a legal basis. Article 56 and Annex XIII Part A of the **In Vitro Diagnostic Medical Devices Regulation (IVDR)** require planned, continuing performance evaluation and clinical evidence sufficient for the device's characteristics and intended purpose. The manufacturer specifies and justifies the needed level. Article 56(4) requires clinical performance studies unless reliance on other clinical-performance data is duly justified. That justification must explain how the other data establish clinical performance for this device and intended use. Relevant existing data can therefore change what new work is needed, but a report title alone supplies no justification. The PEP governs generation and appraisal; a CPSP governs an actual clinical study; the PER records the evidence and assessment. Their distinct decisions and dates explain why none can simply replace another.



How will the team find the evidence worth appraising? A literature method identifies sources, search and selection rules, and how favourable and contradictory findings will be weighed. The plan also links the claim to relevant **General Safety and Performance Requirements (GSPRs)** and the possible consequence of a misleading result. A wrong sequencing treatment report could influence a treatment decision; a missed or misleading image output could influence a pathologist's review. These are reasons to choose evidence questions and justify their sufficiency, with no patient event asserted in either example.

### Reviewer Lens – find the decision before the result

The revealing comparison is between the dated claim, PEP and original study protocol. When did each input and population condition enter the offered purpose? When were endpoints and acceptance logic fixed, and which results were accessible then? A reviewer can credit a completed study's documented scope without assuming it covers everything now claimed. For sequencing, specimen and variant classes, pipeline and interpretation content make that comparison concrete; for pathology, scanner, image conditions and reader task do. The request becomes smaller and more useful when it identifies the particular mismatch.

### Consultant Lens – repair the sequence honestly

A useful repair starts with a dated chain: claim version, original protocol and criteria, study and result-access dates, current PEP questions, PER appraisal and next gate. Existing data keep their real history. A newly chosen criterion is marked as new, with its possible bias assessed. That gives management a prospective sequence for uncovered claim elements while preserving the work that remains useful. Backdating would remove the distinction on which that advice depends.

## Make one promise answer three different questions

What does “evidence for the claim” mean in practice? Begin with the ordinary output: from which specimen or image and configuration, for which people and users, to do which clinical job? A promise covering tissue and plasma creates a material question; one covering two scanners creates an input question. A move from trained laboratory use to community use can change handling and interpretation even when software stays the same. A treatment-selection promise also needs evidence about the decision the result informs. These claim elements tell us why several different questions must be asked about one result.

First, why should the marker or image output mean anything clinically? **Scientific validity** concerns its claimed relationship to a clinical condition or state. For tissue sequencing, relevant scientific and clinical sources must support the variant's claimed association, with contradictory findings and population limits retained. For an image

output, the plan must establish the clinical finding that the information represents. If that relationship is unsupported for the proposed people or use, a more repeatable scanner cannot establish it.

Given that relationship, can this device produce the intended result under its claimed input conditions? This is the **analytical performance** question. Extraction, sequencing, the pipeline and interpretation rules can affect whether a variant is found, missed, misclassified or left without a call. Repeating the same material tests stability of the call—**repeatability**. Other specimen or process factors that distort it raise **interference** questions. Near a **decision boundary**, a small signal change can alter the reported call, so failures and no-calls belong in the plan as well. In pathology, acquisition and processing change what the model sees. A changed input condition therefore needs an applicability argument or evidence; stable output elsewhere cannot establish performance there.

If the device can produce the result, how does that result relate to the relevant clinical condition or state in the intended population and use? **Clinical performance** asks this third question. A treatment-facing sequencing result needs an appropriate clinical reference and the proposed treatment-use relationship, including discordant results and specimens that yield no answer. A pathology output needs its relation to the clinical finding examined under the claimed scanner and reader conditions. A pathologist's ability to see and understand a display matters to safe use, but that successful task does not establish the output's clinical relationship. Nor does an established result relationship by itself demonstrate improved patient outcomes. Changing population or clinical use can reopen this question even when display and measurement stay the same.

The difference determines what a useful result can close. Literature may establish an association while leaving low-abundance detection untested. A technical study may show a stable assay or image output while leaving its meaning in another population unanswered. A favourable clinical result can coexist with an open analytical boundary. The table below keeps each piece attached to the question it can answer; it is useful when a single study is being asked to stand for the whole promise.

| Part of the promised result    | Question the plan must answer                                              | Limit of a narrower result                                         |
|--------------------------------|----------------------------------------------------------------------------|--------------------------------------------------------------------|
| Marker or image-output meaning | Does the claimed clinical relationship apply to these people and this use? | Repeatable measurement alone does not establish that relationship. |

| Part of the promised result    | Question the plan must answer                                                                           | Limit of a narrower result                                                                                                    |
|--------------------------------|---------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Result from the claimed inputs | Does the actual assay/pipeline or scanner/model work at relevant boundaries, including invalid results? | A result from another specimen or scanner needs an applicability argument before transfer.                                    |
| Result in clinical use         | Does the reported output relate to the relevant clinical state in the intended population and use?      | A usable display does not establish the relationship; evidence from studied people may leave another claimed population open. |

Follow the tissue treatment-information promise down the rows. Evidence that a variant is associated with a treatment relationship answers the first question. It still leaves the configured assay's ability to detect and report that variant to the second. The third asks about the commercial result in its intended adult use. If an answer concerns another assay or population, that difference marks a bridge to justify; it cannot silently disappear when the rows are combined.

### Reviewer Lens – walk a claim through the table

A reviewer can make the table concrete with the proposed phrase “marker-based treatment-selection information from tissue for adults.” Which marker–treatment relationship is supported? Which tissue assay and rules produce the output? How does that commercial result relate to the relevant clinical state in those adults? The pathology walk has the same structure: image-output meaning, scanner/model behavior, then the relationship to the finding in the reader workflow. Locating the first unsupported connection is more informative than labeling the entire claim “evidence present.”

### Consultant Lens – turn gaps into a work order

The order also prevents expensive work on an unsettled claim. Clinical and scientific staff can own the association question, the assay or image team the input/result behavior, and the clinical-study and user team the intended-use question. A plausible association is a reason to investigate the device, not evidence that development is finished. A poorly supported association is a reason to revisit the claim before commissioning a large performance study. Existing evidence may answer some questions; owners need a new protocol or analysis only for the questions that remain.

## Decide what would count before seeing the next answer

Once the question is clear, what answer would justify the next decision? A method describes how evidence will be collected; a sufficiency gate describes the decision that evidence must permit. Before outcomes are visible, the plan needs the population and input conditions, reference or comparator, endpoint, treatment of invalid and missing results, acceptable uncertainty, evidence quality and handling of conflict. Clinical consequence and present practice justify the criterion. An achieved result cannot justify itself by becoming its own threshold. The PEP can describe strategy and potential criteria, while a particular study protocol needs an executable endpoint, sample-size rationale and analysis rules before that study begins.

For a proposed scanner range, the actual conditions and denominators determine what a study's result can answer. A summary number does not disclose which boundary inputs or people were examined. For sequencing treatment information near a detection boundary, the gate needs to explain which missed or misleading reports matter, what uncertainty is acceptable and which evidence will support that judgment. A different clinical consequence or claimed input can therefore require a different justification. No numerical acceptance cutoff for that assay is supplied here.

What if a criterion changes after results have become visible? The original rule, new rule, reason, access dates and who knew what all matter because a later choice can favour the observed outcome. A justified amendment can be assessed honestly; it cannot be prospective for data already seen. Equally, a late PEP does not erase a completed study's methods or original criteria. Those, together with configuration, population and limits, allow appraisal of the existing work and a reasoned choice of further evidence or a narrower claim.

The reference for that judgment needs explanation too. **State of the art** means generally accepted current clinical and technical practice used to judge benefit, risk, methods and performance. It can make a criterion meaningful for the intended clinical job; it is not a single competitor result or a reason to copy an old threshold. The PEP identifies relevant practice and sources and explains their fit. The PER later compares the actual evidence against that reasoned reference. If accepted practice or the offered job changes, the reference may need reconsideration.

### Reviewer Lens – keep the time order visible

Dates reveal whether a rule directed the study or was selected after it. In pathology, the useful comparison joins the dated input and population claim to the protocol and report. In sequencing, the versioned protocol, no-call rules and raw accounting can show what a final accuracy number leaves out. A planned acceptance rule, an obtained result and a later reinterpretation deserve distinct findings even if they now appear on one page.

### Consultant Lens – set a useful stop rule

Clinical, statistical, laboratory or image, and regulatory staff need a shared decision about reference, precision and failure handling. If a quantitative protocol criterion or sample-size rationale remains undecided, the new study is not ready to execute. An explicit hold makes that dependency visible; assigning an unrelated worked-example number would only hide it. Management can still fund the work needed to settle the decision. A useful estimate names that work and its cost consequence without inventing a completion date.

### Which question can wait for use after market placement?

Some evidence questions concern how supported performance holds up during actual use. **Post-market performance follow-up (PMPF)** is planned, proactive collection and evaluation of performance and relevant scientific information from use after market placement or putting into service, within the supported intended purpose. Annex XIII Part B places it in the continuing performance evaluation and PER. A useful plan names the residual question, method, timing and trigger for reassessment. For a supported scanner, it might examine image or reader performance drift; for a supported sequencing configuration, whether new evidence or content changes a known interpretive limitation. These are illustrative future questions, with no observed result supplied.

The word “supported” determines which branch applies. An uncertainty essential to establishing the present claim needs a defensible premarket disposition; promising to watch it after launch supplies no missing initial evidence. A genuinely residual question within an adequately supported claim can have a specific follow-up method and reopening trigger. New evidence, a justified bridge or an accurately restricted offered use can change the first situation. Merely adding a PMPF heading cannot. The next exhibit lets us apply that distinction to the opening record.

**Additional exhibit — what the late plan did not cover.** The opening pathology company's studies omit Scanner B boundary conditions and an intended low-prevalence subgroup. Low prevalence means that the subgroup's condition is uncommon in the intended population. The pooled overall endpoint passed, and the late-written PEP declares no gaps. The packet supplies no subgroup count, separate performance result, documented bridge or reason those conditions can be left untested. These are the opening company's facts alone.

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### Decision

**The completed studies are accurately described. What can their passed overall endpoint contribute? What must the current PEP disclose or plan, and which questions cannot be assigned to PMPF merely by naming it?**

### **How a reviewer preserves the positive result and the open claim**

The passed overall endpoint is a positive result under the study's documented conditions. It remains available for appraisal. Scanner B boundary conditions and the intended subgroup are separate coverage questions: the pooled pass supplies neither their missing data nor proof of failure there.

The late PEP can organize the appraisal and plan the next work, but cannot establish that its original choices preceded result access. Original criteria, protocols, configurations, populations and result-access dates are needed to assess applicability and possible selection bias. The manufacturer can then consider new or bridging evidence, or a genuinely narrower claim supported by the device actually offered and its evidence.

PMPF becomes appropriate for a specified residual question after the premarket claim is adequately supported. “Watch the omitted scanner or population later” cannot establish the present broader purpose. No subgroup size, patient harm or certification outcome follows from this packet.

### **Reviewer Lens – a positive bounded result and an open claim**

A precise review retains two findings: what the completed study showed, and which part of the submitted claim remains unresolved. Original protocol, any scanner-specific data, subgroup definition and result-access history can explain the difference. Low prevalence does not disclose the missing subgroup's result or size. Nor does late planning alone show that the study was invalid. Both shortcuts replace an evidence question with an assumption.

### **Consultant Lens – choose an honest repair path**

The practical repair preserves the old protocol and final reports, then gives the current PEP a claim-by-claim evidence state, bias appraisal, candidate bridge or new work and explicit gates. A narrower purpose is a management option only if the supplied product, label and evidence fit it. Otherwise the missing premarket work needs a budget. Specific residual follow-up can be planned where justified; an undifferentiated promise to finish later would leave the central decision unresolved.

## Can a useful strategy still stop a study from starting?

A different record makes the distinction between planning and execution visible. The **candidate tissue assay** is an independent fictional proposal. Material sampled from solid tissue undergoes a specified extraction, assay and computational pipeline, then interpretation under a controlled rule set. The proposed output is marker-based treatment-selection information for intended adults: information meant to help identify who may benefit from a corresponding treatment. No particular medicine, marker, measured result or companion-diagnostic conclusion is supplied. The exact local identities—assay T1, extraction E1, pipeline P1 and rules K1—appear in its strategy record below. They have no identity link to Chapter 6's labels or the opening pathology company.

What can a filled row usefully establish before those results exist? It can define the boundary, the questions and who must settle the next decision. The first row below fixes the proposed tissue configuration; the next three carry the result from association through device behavior to clinical use. The protocol row then asks whether those strategic questions are executable. Reading in that order prevents a complete-looking final row from approving a claim whose earlier questions remain unanswered.

| <b>Candidate tissue strategy: PEP-T1</b> | <b>Planned question and decision gate</b>                                                                                                                                                                                                                                                                                                                                                        |
|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Claim and configuration                  | T1 tissue assay, E1 extraction, P1 pipeline and K1 interpretation; marker-based treatment-selection information for intended adults in a trained laboratory workflow. Liquid biopsy is outside this row. Record exact studied and proposed commercial configurations before transfer.                                                                                                            |
| Scientific validity                      | Appraise marker–condition and marker–treatment relationships, contradictory evidence, populations and source/cohort identity. The association must fit the stated clinical job; general marker biology is insufficient.                                                                                                                                                                          |
| Analytical performance                   | Plan detection, repeatability, interference and decision-boundary work for claimed tissue and variant conditions, retaining failures, no-calls and exclusions. The actual T1/E1/P1/K1 output must meet justified, prespecified criteria. No pass is asserted.                                                                                                                                    |
| Clinical performance                     | Compare the commercial treatment-facing result with a justified clinical reference and relevant trial or clinical relationship in intended adults. Include near-boundary discordance and unavailable specimens. The result must relate to the relevant clinical state in the intended use and population; agreement with another laboratory instrument alone cannot establish that relationship. |



| Candidate tissue strategy: PEP-T1 | Planned question and decision gate                                                                                                                                                                                                                                                                                              |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Criteria, protocol and timing     | Clinical/statistical leads justify endpoint, reference, sample-size rationale, quantitative acceptance and analysis rules from claim, consequences and state of the art, before access to new-study outcomes. <b>Execution HOLD:</b> these quantitative protocol decisions are missing. The strategy is not a study-ready CPSP. |
| Existing evidence and change      | Identify completed work, original protocol/criteria, configuration, population, dates and who could access which results when the new plan was written. Appraise it without backdating; retain amendments and possible bias. A claimed bridge needs an applicability argument.                                                  |
| Claim disposition and lifecycle   | Regulatory/clinical leads include a claim only after relevant association, device-performance, clinical-use and benefit–risk gates close. Quality assigns any justified residual uncertainty a PMPF objective, method, time and reopening trigger. No certificate, launch or completed follow-up is presumed.                   |

Follow the analytical row into the protocol row. “Test near the decision boundary” identifies a necessary kind of work; it does not specify the endpoint, quantity of evidence or acceptable uncertainty that will decide the result. The missing quantitative choices therefore create the stated execution hold. The strategy is useful because it makes those choices and their owners visible, but it is neither a completed study nor a favourable PER nor permission to recruit. If the proper protocol decisions and other prerequisites are resolved, the execution gate can be reconsidered. Chapter 9's number belongs to a different exercise and cannot supply this assay's medical criterion.

The first row creates another gate. A study using a different extraction or interpretation rule may still provide useful evidence, but its applicability to the stated assay requires assessment. In the final row, even technically successful work cannot alone close the association and clinical-use questions above it. These dependencies make the table a strategy record rather than a list of hoped-for results.

### Reviewer Lens – read a filled row without upgrading its state

A reviewer can positively recognize the defined tissue configuration and adult planning boundary, and the coherent scientific, analytical and clinical questions. The missing quantitative protocol remains a separate open item. Calling the whole record “no plan” would discard its actual contribution; calling it study-ready would invent the next decisions. The actual CPSP is needed before execution and the original study record before accepting an evidence bridge. A pathology counterpart would name scanner, processing, model, output and reader task, because copying assay measures would miss its input path.

## Consultant Lens – turn the hold into owned work

The hold is actionable once clinical and statistical leads own the endpoint, reference, sample-size rationale and acceptance logic, with laboratory and regulatory review of feasibility and consequence. Configuration control checks the stated assay, extraction, pipeline and rules before protocol freeze. Management may fund design and feasibility now and prepare a conditional study budget. Executing the study remains dependent on the missing criteria and other prerequisites. A tissue-only scope and a later plasma proposal are distinct budget choices; genuinely applicable existing evidence calls for appraisal before reflex repetition.

### Decision

**The manufacturer finds a well-conducted completed study matching the candidate assay, extraction, pipeline and rules, and its intended adult population. No earlier PEP existed. What should happen to the study, and what must the new planning record disclose about criteria and access to results?**

### How existing evidence enters an honest current plan

The matching configuration and population make the study relevant to appraise. Absence of an earlier PEP is not by itself a reason to discard it, and a backdated replacement would misrepresent its history. The original protocol, acceptance rules, amendments, data and full report establish what was actually done, with favourable and unfavourable findings and limits retained.

Any new criteria need a record of what planners knew or could access when setting them, together with appraisal of selection or post hoc bias. The current PEP can govern remaining evaluation and future work; the PER explains the completed study's contribution. Matching scope improves relevance without guaranteeing the whole claim, supplying missing new-study criteria or removing the execution hold automatically.

Before considering a specimen change, picture what would arrive at the laboratory. **Tissue** here is material sampled from a solid lesion or organ. **Plasma** is the fluid portion of blood after cells are separated. **Deoxyribonucleic acid (DNA)** carries genetic information; **cell-free DNA (cfDNA)** consists of DNA fragments circulating outside cells in that fluid. A plasma-based **liquid biopsy** therefore examines body-fluid-derived material instead of the tissue used in the strategy. The same marker might remain clinically relevant while the arriving material, and the meaning of a negative or unavailable result, changes.

## Decision

The manufacturer proposes adding plasma-derived cfDNA. Which collection and processing, analyte-abundance, interference, detection and clinical-applicability questions reopen? What tissue association evidence might remain relevant, and where would a bridge or new evidence be needed?

## Reviewer field aid — inspect the claim-to-question chain

The plasma question uses the same reasoning as a new scanner or population claim: which part of the result's justification changed? Four connected inspections make that question manageable. Each needs an evidence location and a reasoned working state rather than a judgment based on the number of studies.

First comes **purpose coverage**. The opening pathology record has separate Scanner B and intended low-prevalence subgroup questions; the candidate assay has tissue, trained laboratory use and a treatment-information role. A neighboring result cannot silently answer an unmentioned claim element.

Next comes **method and gate**. The dated search approach, analytical methods, clinical reference, quality criteria and acceptance logic show how a decision will be made. The candidate tissue strategy contains useful questions with a quantitative protocol hold. The opening pathology record needs original rules compared with the late plan before its pooled pass is treated as sufficient.

Those questions attach to an **exact product**. Assay, extraction, pipeline and interpretation content—or scanner, processing and model—must match the proposed state or have a justified bridge. Chapter 6's configuration map makes that comparison possible; a shared name cannot make it true.

Finally, the record distinguishes **premarket support from follow-up**. An unstudied condition inside the offered claim needs a defensible premarket disposition. Once a configuration is supported, a particular residual question can have a PMPF method and trigger to reopen the PER. The deciding fact is the evidence supporting the present claim, not whether a monitoring heading exists.

### How those inspections apply to the plasma proposal

Some marker-condition or marker-treatment evidence may remain relevant across specimens, subject to clinical applicability. That possibility deserves appraisal. It does not establish how blood collection and plasma processing affect fragmented, potentially scarce cfDNA, interference, detection or the meaning of a negative result.

Each changed link needs an explicit applicability argument and supporting evidence, or new work. The tissue results are neither inherited plasma performance nor

automatically worthless. Until the proposed extension is supported, the existing strategy retains its tissue-only scope.

## Consultant field aid — use the PEP to sequence choices

A consultant's claim-to-PEP matrix makes one promise visible at a time: output and user, specimen or scanner, controlled configuration, the three evidence questions, existing evidence and its quality and timing limits, decision gate, owner, next work and follow-up rationale. The candidate tissue table supplies one example. For the opening pathology company, the existing pooled study belongs in its own evidence entry; Scanner B boundary conditions and the intended low-prevalence subgroup remain distinct open entries. A blank denominator cannot be repaired by estimating one without data.

That record lets management compare real choices. It can generate or bridge evidence for the broader offered claim, restrict the offer to a genuinely supplied and supported use, or plan specific residual PMPF once the premarket claim is adequate. The dependencies also order funding: purpose and configuration, scientific applicability, executable study criteria before collection, then analysis and claim disposition. Later scanner behavior or changed sequencing knowledgebase content may reopen a row and the PER. A monitoring signal prompts reassessment; it does not itself supply a new performance result or proof of harm.

**Changed-case transfer — the funding decision.** The plasma cfDNA use remains a proposal. No plasma version, quantitative cutoff or result is supplied.

### Note

Which evidence about the scientific relationship could be appraised for relevance? Which questions now concern blood collection, processing and fragmented, potentially scarce DNA, and which concern the result or no-result in the intended population? What must remain tissue-only, and what may management fund while the protocol gate and possible bridge remain open?

## How the changed specimen changes the work sequence

The scientific relationship may supply a starting point if it fits the proposed clinical use. Analytical work must address the new material and processing path; clinical work must address the proposed result and unavailable-result interpretation. An explicit bridge can justify reuse where supported, while unanswered questions need new evidence. The tissue claim stays bounded until those extension questions close. Design, appraisal and feasibility funding can prepare that decision; it cannot convert undecided protocol criteria into an executable study.

## Quick discrimination lab

The two records below are independent of the opening pathology company and the candidate tissue assay. Each asks which contribution is real, which conclusion remains open and what would resolve it.

**Record A — document timing.** A team creates a polished PEP after all studies are complete and copies the acceptance criteria from the final reports. The resulting document accurately identifies the reports, but the packet supplies no earlier PEP approval.

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### Decision

**What does the late document help a reviewer find, and what fact about the original evidence decisions does it fail to establish? Which records would determine how the completed studies can be used?**

### How the document's date limits its contribution

The accurate PEP can index and organize the completed evidence. Copying criteria from final reports does not show when the original criteria were set. Original protocols, amendments, approvals, result-access dates, configurations and full reports can establish that history and support appraisal of actual methods, prior criteria and possible later selection bias.

Useful completed work need not be discarded. A current PEP can govern future gaps and updates with an honest date. Whether the earlier decisions truly preceded results depends on those records, not the polish of the new document.

**Record B — coverage of purpose.** Another PEP lists an accuracy study, a precision study and a clinical study. The proposed purpose also names liquid biopsy, a community-based intended user and treatment-selection information. The list never separates those claim elements or says what each study covers. The user's exact professional or lay status is not supplied.

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### Decision

**What would establish whether this plan is sufficient for the proposed purpose? Which specimen, user and clinical-use facts remain unanswered by the three study titles?**

### How the claim supplies the missing questions

Liquid-derived material raises collection, analyte and detection questions distinct from tissue. A community user raises handling, interpretation and environment questions, without this packet establishing professional or lay status. Treatment-selection information raises association and clinical-use questions.

For each promised element, the plan needs methods, quality and sufficiency logic, the product version, a disposition for gaps and an update route. “Accuracy” alone does not identify the specimen, user, clinical decision or failure states evaluated. Matching study details could establish coverage; absent details establish neither coverage nor failed studies.

## Keep the decision and its evidence in the same frame

A useful PEP connects a defined purpose to questions, evidence methods and future decisions. The study protocol makes a particular investigation executable; the PER appraises what the evidence actually shows. Association, reliable device output and the output's clinical relationship answer different parts of the promise. Usability and patient benefit remain connected questions with their own evidence. A pooled pass, a matching version or an accurate report list can each contribute without closing the whole claim. Original criteria and dates preserve that contribution; justified prospective work or an actually supported narrower use addresses the gaps. Specified residual follow-up then keeps the supported claim under review.

### Check what you just learned

The **working states** attach to each bounded item: **A**, adequately explained; **U**, present but unclear; **M**, missing or inconsistent; **D**, deeper analysis needed; **C**, consultation indicated; **N/A**, inapplicable with a reason. Evidence location and rationale explain why the state applies and what would change it. The seven source-era IVDR-CH08 identifiers below retrieve the expanded Workbook Part A2 cards through the original-to-integrated crosswalk; they are not device grades or authorization decisions.

### Every promise becomes a question • card IVDR-CH08-A01

Does the purpose-to-plan map account for each actual claim? Scanner B, an uncommon intended subgroup, plasma and community users require distinct questions only in the packets that claim them. No case can borrow its neighbor's evidence.

**Methods and sufficiency • card IVDR-CH08-A02**

Do method, reference, endpoint, failure handling, acceptance logic and original protocol make the next decision possible? The candidate tissue strategy has coherent questions and an explicit execution hold for missing quantitative criteria.

**The product behind each question • card IVDR-CH08-A03**

Do studied assay/extraction/pipeline/rules or scanner/model/input match the proposed device and change history? A difference needs an applicability bridge; a shared version name cannot supply it.

**Present support and residual follow-up • card IVDR-CH08-A04**

Does the premarket disposition support the offered claim, with a specific PMPF objective, method, timing and reopening trigger for residual questions? An unstudied intended condition cannot disappear into a monitoring heading.

**A usable claim-to-plan matrix • card IVDR-CH08-C01**

Can each promise be followed through scientific, analytical and clinical questions, evidence provenance, version, owner, gate and current state? A report list needs these connections before it can guide work.

**Management's scope and evidence choice • card IVDR-CH08-C02**

What would support the broader claim, and does a narrower supplied use have adequate evidence? Which residual question could follow market placement once present support is adequate? Neither a restriction nor follow-up promises approval or a date.

**Sequence and budget • card IVDR-CH08-C03**

Are prerequisites, protocol owners, stop rules and update points clear enough to govern funding? Design and feasibility can resolve the tissue strategy's missing decisions; study execution remains on hold while endpoint and acceptance criteria are undecided. A late summary remains a late summary.

**Further reading**

Book 1C Chapter 4 offers an optional route into evidence planning and acceptance questions. It does not add results or product identity to the independent cases here.



## Source trail and limits

- **Binding law, checked 2026-09-25:** IVDR consolidated 10 January 2025, Article 56(1)–(6); Annex XIII Part A §§1–1.3 and §2.3.2; Part B §§4–5. Article 56 connects sufficient clinical evidence to intended purpose, a planned continuous evaluation, scientific validity, analytical and clinical performance, and the PER. Annex XIII describes the PEP's device-specific methods and criteria, objective appraisal of favourable and unfavourable evidence, clinical study planning where used, and PMPF after market placement within the supported purpose. These legal provisions do not supply T1's missing quantitative criteria or a Scanner B/subgroup result.
- **Nonbinding interpretation:** MDCG 2022-2, §§6.3–6.4 describes claim-specific PEP factors and the evidence flow; it informs the original explanation but is not itself a new legal requirement or a substitute for study evidence. No licensed standard clause or numerical threshold is invoked in this chapter.
- **Packet boundaries:** The anonymous late-PEP DP record and the independent T1/E1/P1/K1 tissue strategy are not named-series products or one study. T1 has an execution hold, not a completed or authorized treatment-selection claim. The passed overall endpoint belongs only to the first record and supplies no Scanner B or low-prevalence subgroup result. Plasma cfDNA is a proposed extension, not an inherited tissue validation. No source figure or artwork is used.

## CHAPTER 8

# What Does the Marker Mean, and for Whom?

**Before reading.** A proposed tissue result will inform treatment selection. What evidence question would Chapter 7's plan need to ask, and what rule would determine whether an existing study answers it? Why does the date when its authors first saw the results matter to a criterion described as prospective?

## How the plan carries into appraisal

The question needs the proposed people, marker, treatment decision and device result. An existing study must be judged against those boundaries and a stated rule for relevance and quality. A criterion written after its authors could see results can still be explained and appraised, but cannot be represented as a rule set without that knowledge.

The next decision is **what the evidence establishes about the claimed clinical association, and what still has to be shown for this device**. A useful conclusion will tell a reviewer why a result matters and tell the manufacturer which claim or study decision can follow. The number of references cannot supply either answer.

## SECTION 01

## Which part of the claim does the trial answer?

A sequencing assay may accurately identify a variant—a difference from a reference in DNA, the genetic material in the specimen. That establishes a finding to interpret. The clinician's next question is why that difference matters to the condition or decision in front of them. Evidence for the marker's relationship with a particular clinical condition or physiological state answers that different question. This is scientific validity, the first of Chapter 7's three evidence strands.

The distinction is equally concrete in digital pathology. A model can consistently extract an image feature while its relationship to the claimed tissue finding remains uncertain. Conversely, a well-supported feature-finding relationship says nothing by itself about how reliably a particular scanner and model extract the feature, or what the complete

reader workflow produces. The association gives a device result a possible clinical meaning; the device still has to deliver an applicable result.

An anonymous fictional sequencing dossier describes its assay as a companion diagnostic, a medicine-linked test intended to help identify patients for the corresponding treatment. It contains strong clinical-trial evidence that patients with a named biomarker benefit from a therapy. The trial used a different assay and cutoff, with retrospective central testing. The manufacturer presents the relationship as proof of the commercial assay's clinical performance. Neither the therapy nor biomarker is named here, and no commercial decision-boundary bridge or authorization outcome is supplied. This dossier is independent of the marker Z library later in the chapter.

The strong treatment evidence deserves a place in the assessment. The question is which place. A cutoff separates results into categories that can put patients on different sides of a treatment-eligibility decision. If two assays assign those categories differently, the biological relationship cannot tell the reviewer how often that happens or which patients it affects. A useful assessment therefore retains the association while keeping the commercial-result question open.

Article 56 and Annex XIII of the In Vitro Diagnostic Medical Devices Regulation (IVDR) organize performance evaluation around scientific validity, analytical performance and clinical performance. The first concerns the marker–condition relationship just described. Trial and other evidence may support that relationship for the studied population and treatment role. It does not identify which individual specimens a different commercial assay will call positive.

To answer that question, the reviewer follows the specimen through the offered assay: extraction, measurement, software pipeline, reported variant classes and cutoff. Their ability to detect or measure what the device claims is its analytical performance. A change at one of those steps can change the reported call. In pathology, scanner image processing and model extraction can likewise change the output. Accurate measurement by the trial assay is useful information about that assay; it does not test another assay's behavior.

The next comparison joins the complete device result to the clinical condition or state in the intended population and user context. That is clinical performance. The commercial sequencing call, or the result reached through the reader-plus-software workflow, needs its own applicable evidence. Some observations may inform more than one strand, but the inference for each strand has to remain visible. The commercial result cannot inherit an untested mapping merely because the marker association is strong.

What happens when clinicians actually use that information? A changed decision could help, harm or make no useful difference compared with available care. The **clinical-utility lens** asks about that effect on justified decisions and patient outcomes,

including harms from wrong or delayed information. A change in management alone does not demonstrate improved benefit. This professional benefit question can matter to the treatment strategy and benefit-risk assessment, but it is not a fourth IVDR performance-evaluation pillar. The claim determines what comparative or other credible evidence is needed; no universal intervention-study design follows from these words.

That distinction also limits what a methods guideline can promise. The **Clinical and Laboratory Standards Institute (CLSI)** guideline **EP49, first edition**, provides a clinical-validity framework using **noninterventional** clinical-performance studies: the candidate result does not direct participants' care. It distinguishes clinical validity from utility, but does not supply a step-by-step interventional companion-diagnostic study or a method for developing clinical-utility evidence. The fictional examples here make no claim of EP49 conformity. The legal duties and the actual proposed use still determine the evidence decision.

**Reviewer Lens.** The opening trial can receive bounded credit for its biomarker, treatment role and population while the trial-to-commercial comparison remains open. The useful request identifies both assays, their cutoffs and cases near the eligibility decision. For an image feature, the corresponding first request concerns the clinical finding it is supposed to represent; scanner agreement comes afterward. Keeping those questions distinct prevents a good association study from being discarded for lacking device evidence, or from being credited with evidence it never collected. Multiple reports of its participants also remain one participant source.

**Consultant Lens.** The diagnostic and medicine teams need a shared claim that identifies the marker, population, treatment decision and assay version, with unknowns exposed. That allows literature appraisal, analytical comparison, clinical-performance work and any utility question to have separate owners and dependencies. If management changes the claim, the evidence question changes with it. A medicine partner's persuasive narrative, a guideline title or the word “companion” cannot select the study design on the team's behalf.

## SECTION 02

## What clinical job is the marker supposed to do?

Before judging whether a paper is relevant, the reviewer needs to know what the result promises. Is it helping identify a condition that is present, or estimating the risk of developing one? Those are diagnostic and predisposition roles. Once a condition is present, a forecast of its likely course or outcome is prognosis. That forecast concerns how people are likely to do, irrespective of which treatment is chosen. A predictive treatment role asks a further question: who is more or less likely to benefit from a particular treatment relative to a relevant alternative?

Suppose marker-positive patients have better responses than marker-negative patients while both groups receive one therapy. The observation connects marker status to response in treated people. Yet the marker-positive group might do better under other care as well. In that situation the response difference would not establish that this therapy gives them a greater advantage.

The treatment-selection question therefore begins with a comparison within marker-positive patients: how do outcomes under the treatment compare with outcomes under the relevant alternative? It then asks the corresponding question within marker-negative patients. Comparing those two treatment advantages reveals whether the advantage differs with marker status. This is the **treatment-by-marker interaction** question. The appraisal has to examine the actual comparisons, effect estimates and uncertainty, or other rigorous evidence suited to the claim. A response gap within one treated group cannot answer it, and naming the question does not mandate one universal trial design.

Population can change the promised job as much as treatment does. An association in advanced disease cannot simply become a screening claim for people without symptoms. Disease spectrum, prevalence, false-result consequences and clinical action differ. Nor does evidence for one variant establish a whole panel category. In pathology, a feature associated with tissue grade on expert-reviewed slides can have scientific relevance while automated triage in a community workflow remains untested. Feature meaning, model extraction and the complete clinical result still need their respective arguments.

**Reviewer Lens.** “Predicts outcome” leaves the critical comparison unstated. It could mean prognosis under any care, a response difference among treated patients or a difference in benefit versus another treatment. In the opening dossier, the precise relationship has to match the commercial eligibility claim. In pathology, an expert-slide finding needs its population and clinical role before it can be carried into a broader workflow. Naming the missing comparison produces an answerable finding; requesting “more literature” may produce more papers that answer the wrong question.

**Consultant Lens.** Agreement on a claim question before searching prevents a medicine or product team from changing the target after attractive results appear. Marker or feature, clinical state, population, role and treatment or workflow all belong in that question. Advanced-disease treatment and asymptomatic screening remain separate proposals. An added variant class or a wider pathology setting calls for an explicit applicability comparison: which association evidence stays useful, and which measurement or clinical-use question reopens? A narrower wording is a real option only if the offered device and evidence support that boundary.

### SECTION 03

## Can another reader recover the evidence judgment?

A marker's name in a paper does not make the paper evidence for every claim about it. The approved clinical question first becomes search concepts and synonyms: marker, condition, treatment and comparator, for example. The chosen sources, search dates, full source-specific strings, limits and reasons make that search recoverable. Before convenient papers determine the conclusion, the method also needs screening rules, a way to link reports about the same participants, extraction fields and a way to resolve disagreements. Annex XIII Part A requires systematic identification of relevant literature and gaps, followed by appraisal of all relevant data. Unfavorable findings belong in that body too.

Retrieval starts the judgment rather than finishing it. Does this report concern the claimed population, marker, clinical role and use? That establishes relevance. Do its design, reference, follow-up and missing observations let the result carry the proposed weight? That concerns trustworthiness. Finally, after allowing for overlapping participants and conflicting findings, what does it add to the body of evidence? That determines contribution. An unfavorable report can be essential to appraisal without supporting the desired claim. A large bibliography cannot be read as a vote count.

For a sequencing treatment claim, those questions become specific extraction fields: the assay and cutoff used, variant definition, treatment and comparator, endpoint, population and shared participants across publications. For pathology, the reviewer needs the feature and reference finding, scanners and images, and who read them. Those details expose whether expert grading evidence is being stretched into software-assisted triage. They matter because they determine the question the observation can answer. A later knowledgebase or scanner-workflow change can require a fresh applicability review; a new paper alone does not show that a deployed product has changed or failed.

**Reviewer Lens.** A saved query, source/date list, screening reasons, report-to-participant links, extraction sheet and appraisal rule let the reviewer follow

the inference. Tracing one favorable and one unfavorable sequencing report through the same rule tests whether convenience has shaped the evidence body. In pathology, a scanner/reader claim added only after expert-slide results appeared raises the same concern. A polished reference list cannot resolve a search that no one can recover.

**Consultant Lens.** The search ledger becomes useful at the next update when it records report identity, participant identity, exact claim relevance, strengths, limits, direction and decision impact. Two reviewers can screen independently and preserve a reconciled reason, with a named clinical owner resolving disagreements. A new contradictory treatment study, changed marker definition or new image-feature use then has a recognizable place to enter. The team can explain which conclusion changed in the performance evaluation report, instead of replacing it with an unexplained new narrative.

### **What does current practice make reasonable to expect?**

A new result enters an existing clinical decision. How is the condition identified now, what alternatives are available, and what do credible guidelines and studies say about their benefits and limits? This current generally accepted clinical and technical practice is the **state of the art** relevant to the proposed use. A competitor's advertised number cannot describe it by itself. The comparison needs to explain what useful information the new result adds and what risk it could introduce.

For a well-established marker–condition relationship, disciplined appraisal of high-quality existing evidence may answer much of the scientific-validity question. A new combined signature or algorithm-derived pathology feature may instead need more direct evidence for its particular relationship. Both situations leave the commercial measurement and clinical result to be assessed. Novelty changes what needs explaining; familiarity does not validate an untested assay.

The context can also change after deployment. Credible work may challenge a marker's treatment role or reveal a different meaning for an image feature in a particular population. That would prompt review of scientific validity, intended claim, risk and the performance evaluation report. No such field failure is asserted in these fictional packets. Post-market performance follow-up can observe a supported product's continuing performance, but cannot postpone the evidence required for a core premarket claim.



## SECTION 04

## What does the commercial boundary add?

**Additional exhibit — opening dossier.** The proposed commercial assay detects additional variant classes and uses a different cutoff from the trial assay. Overall concordance is described as high, but observations are sparse near the cutoff that changes treatment eligibility. No count, error rate, boundary estimate, successful bridge or outcome for the added classes is supplied.

### Decision

**What can the therapy trial support about the biomarker relationship? What remains open about the commercial result and its treatment-facing use, and why might high overall agreement answer the boundary question poorly?**

### How a reviewer separates the useful association from the missing comparison

The trial can retain its potentially strong support for the biomarker–treatment relationship in the studied population and role. It does not show that a commercial assay with a different cutoff and added variant classes puts the same patients on the same side of the treatment decision.

If most comparisons are far from the cutoff, they can dominate overall agreement while decision-changing disagreement near it remains poorly measured. The packet supplies neither the estimates needed for a trial-to-commercial bridge nor an appraisal establishing the added-class claims. That is an unresolved evidence question, not a demonstrated assay failure. Scientific validity and the commercial device's analytical and clinical performance keep separate findings.

**Reviewer Lens.** Narrow credit and a narrow request can coexist. The trial's relevant association is a bounded positive. Paired results near the decision-changing cutoff, their reference and clinical context, specimens without a reportable call and disagreements would address the commercial comparison. Each added variant-class claim also needs its evidence. Retrospective central testing raises an applicability question about the intended commercial workflow. Sparse boundary observations establish neither a pass nor a failure, and supply no particular disagreement rate.

**Consultant Lens.** A bridge plan becomes a joint diagnostic–medicine decision when it names both assays and cutoffs, the proposed eligible group, relevant archived specimens if available and a prespecified way to judge disagreements that change eligibility. Management can then compare additional testing, supported claim narrowing or a separately evidenced future expansion. Those remain conditional

options, with no promised post-approval path or launch date. The medicine partner must identify what trial evidence and labeling support; the diagnostic team must establish what its assay does. Calling the association strong cannot replace either comparison.

## SECTION 05

### What do four reports actually contribute?

The next packet is a separate, closed fictional library about **marker Z** and a proposed treatment-selection role for **assay C**. Neither the marker nor treatment has a real identity here. Its four reports let us examine search, participant overlap and appraisal. They are not a complete search of real literature and cannot validate the commercial assay. Nothing in the opening dossier supplies a missing result to this library.

The question comes before the search: does this marker distinguish benefit from treatment for the proposed people and decision, or only outcome or response? The saved teaching query is **marker Z AND (treatment response OR prognosis)** across all four supplied full-text records, with no date or language exclusion. Two reviewers screen independently and retain one reconciled reason per report. In a real search, database-specific syntax and synonyms, full strings and dates, wider retrieval and later updates would be needed. Here the query shows how the method is recorded within a closed set.

Why might four retrieved items yield fewer than four independent sources of participant evidence? A **record** is a search hit. A **report** is a document describing work. A **cohort** is the participant group whose observations produced results. The first response report and its secondary analysis use the same people; the prognosis and conflicting-response reports each use another independent group. Nothing is removed as a duplicate search record. The accounting is therefore four retrieved records, four distinct full-text reports appraised and three independent cohorts. Inclusion means the information matters to the question, not that it supports the desired claim.

Before weighing the reports, consider how an apparent marker effect could arise. If another factor is associated with both marker status and response, it can distort a simple marker-group comparison. This is **confounding**. For example, if one marker group had less advanced disease and disease stage also affected response, the observed difference might partly reflect stage. Depending on the pattern, this could exaggerate or mask the marker relationship. That is a possible mechanism to investigate, not a supplied fact about the first response report.

Missing outcomes can distort the comparison differently. **Attrition** is the loss of participants before their outcomes can be assessed. If people who would have responded poorly are more likely to disappear, the remaining participants may look too

favorable. Another loss pattern could bias the result the other way. The conflicting report supplies substantial follow-up loss but no account of who was lost or how their outcomes compared. Its uncertainty therefore affects the weight of its result; it does not authorize deleting the conflict.

The table follows those questions into each supplied report. Its first column preserves the observation, its second identifies useful appraisal work, and its last prevents an observation from being credited with a conclusion it cannot establish.

| Report and supplied observation                                                                                                                                 | What merits appraisal?                                                                                                                                                                                                                 | What does the observation leave open?                                                                                                                                                                                              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>L1 — first response report:</b> prospective trial in the claimed population, using assay A and a preset cutoff; treatment response differs by marker status. | The prospective design, preset cutoff and relevant population are useful features.<br>Treatment/comparison groups, marker definitions, endpoint, effect estimates, uncertainty, possible confounding and assay/cutoff need extraction. | No supplied comparison shows whether treatment benefit versus an alternative differs with marker status. Response association alone does not establish that predictive interaction. The trial also did not use commercial assay C. |
| <b>L2 — companion report:</b> secondary analysis of the first trial's participants and the same endpoint.                                                       | Link it to the first cohort. Appraise genuinely new analysis and whether its question and method were original or later.                                                                                                               | It can add detail without adding participants or independent replication.                                                                                                                                                          |
| <b>L3 — prognosis report:</b> independent cohort reporting prognosis.                                                                                           | Relevant disease-course context, with population and marker definition to compare with the claim.                                                                                                                                      | Prognosis does not establish different benefit from the proposed treatment.                                                                                                                                                        |
| <b>L4 — conflicting report:</b> independent treatment-response study conflicting with the first; substantial loss to follow-up.                                 | The conflict stays in view. Missingness, timing, possible bias, population, assay applicability and effect accounting determine its weight.                                                                                            | Attrition limits interpretation but does not erase unfavorable evidence. No supplied estimate resolves the disagreement or establishes the direction of missingness bias.                                                          |

Start with the first two rows. Their documents are different, but the same participants and endpoint cannot become two independent replications. Then compare the first and third rows: response and prognosis address related but different clinical questions. Finally, the last row prevents a favorable-only synthesis. Its missing follow-up needs scrutiny under the same appraisal discipline used for favorable work. Adding or removing a report changes the bibliography; a further independent participant source

would add a cohort, and only evidence answering the treatment comparison can close the predictive question.

The supported conclusion is therefore bounded. There is a relevant response signal, prognosis context and conflicting evidence. The packet does not establish that the marker is useless, yet the absent treatment-comparison details and estimates prevent it from establishing the proposed predictive role. A separate gap concerns the commercial assay: the first trial used assay A, while assay C's measurement and cutoff mapping have no supplied bridge. More literature about the marker cannot by itself show how C classifies patients.

The claim-boundary table records that result in words a product team could act on. Each proposed statement has a current support limit and a specific question that could change it.

| <b>Proposed claim</b>                                                                                 | <b>What this packet can contribute now</b>                                                                                                                                                            | <b>What could change the claim decision?</b>                                                                                                                                                              |
|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| “Marker Z is associated with response or disease course in studied people.”                           | A response-by-status observation and a prognostic observation are relevant, while the conflicting response and shared participants remain visible. Exact scope and strength require the full results. | Effect estimates, populations, comparator detail and follow-up accounting would allow favorable and unfavorable synthesis for each separate clinical role.                                                |
| “Marker Z identifies who benefits more from the proposed treatment than from a relevant alternative.” | The descriptions do not establish this. No treatment-by-marker effect comparison or justified equivalent evidence is given.                                                                           | Evidence answering the comparative treatment question, with its uncertainty, could support or alter this conclusion. Prognosis or a within-treatment response gap cannot be relabeled as that comparison. |
| “Commercial assay C can make that treatment-selection call.”                                          | The descriptions do not establish this either. The first trial used assay A; C's cutoff and result mapping lack a bridge.                                                                             | C's defined marker measurement and cutoff scope, decision-changing discordance and clinical applicability must be evaluated and joined to the marker-role conclusion.                                     |

The second row explains why commissioning only an assay-concordance experiment would leave a central gap: it could compare calls while the predictive relationship remained unproved. The third explains the reverse limit: even a stronger marker literature would leave the commercial mapping to be demonstrated. A study may legitimately inform several questions when its design permits, but citing it for another question does not create an independent cohort.

This is a scientific-validity report component and a claim-boundary table. It is not a completed performance evaluation report for the commercial assay. It gives the reviewer the useful evidence and the exact stopping points, while giving the consultant distinct work to put before management. The labels, tables and arithmetic have created no new participant observations or assay results.

### **Decision**

**A draft concludes, “Four independent studies support clinical performance because all four reports were included.” What should its conclusion say about records, reports, cohorts and clinical roles? What is the first commercial-assay applicability question?**

### **How a reviewer corrects the inference without discarding the library**

Four records yielded four reports from three independent cohorts. L2 reanalyzes L1's participants, so it is not another replication. Inclusion makes all four reports part of the appraisal, not four favorable results. L3 concerns prognosis. L4 conflicts and has substantial missing follow-up. L1's response-by-marker observation does not alone demonstrate different benefit from treatment versus an alternative.

The conclusion first states the association evidence being appraised and its uncertainty. It then asks how commercial assay C's result and cutoff map to the proposed treatment decision. Neither the predictive role nor C's clinical performance follows from the report count. A report with new participant data, an appropriate treatment comparison or a justified commercial bridge could change the respective finding; none is supplied here.

## **SECTION 06**

### **Who can act on the conclusion?**

A reviewer can reconstruct the library's inference through four connected questions. First, which clinical role does the intended statement promise: present-condition diagnosis, future predisposition, prognosis or prediction of treatment benefit? The prognosis report cannot answer the response report's missing treatment comparison merely because both mention the marker.

Second, which device produced the observation? The source assay and cutoff matter because a different commercial call can select different patients. An expert-defined pathology feature likewise does not automatically become the deployed

scanner/model/reader result. Association, measurement and clinical performance can share observations while retaining distinct conclusions.

Third, which people generated the evidence, and which reports describe them? Disease stage, intended population and user affect applicability; participant identity affects independence. The first response report and its companion analysis remain one cohort. The conflicting report's missing follow-up needs to be understood before the reviewer relies on, or discounts, its result.

Finally, which conclusion survives the same challenge applied to favorable and unfavorable findings? Supported wording, the unresolved predictive question, commercial bridge and update trigger belong together. A changed deployed claim can reopen applicability. It does not silently inherit the old conclusion, and an unchanged, adequately supported part need not lose its credit.

The consultant turns that record into a manufacturer decision. A saved search and screening ledger, cohort map, extracted evidence and reasons for bias judgments make the short scientific-validity report recoverable. The filled claim-boundary table then separates management's options: retain an adequately supported marker role, commission the missing comparative or assay-bridge work, or consider a narrower first claim with a separately evidenced future extension. “Staged” describes a planning option; it guarantees neither certification nor a post-approval date.

For a medicine-linked proposal, diagnostic and medicine partners need the same proposed eligible population, cutoff, treatment-evidence question and wording. A partner's preferred label cannot supply a missing interaction estimate or device result. Management chooses the actual offering, and the manufacturer controls the evidence decision. The consultant explains what work and specialist input each option requires.

Changes after deployment can separate the questions again. A credible new pathology study about a feature's meaning may reopen scientific validity; a changed scanner's extraction can reopen analytical and possibly clinical performance. A new conflicting treatment study or a knowledgebase reinterpretation in a sequencing product may require updated literature appraisal, claim and risk review, and performance evaluation. These are conditional mechanisms, not events asserted in either the opening dossier or the marker library.

### **Changed-case transfer**

A separate pathology product is proposed to mark an image feature for a pathologist. What question would test the feature's association with the stated finding? What question would test reproducible scanner/model extraction under the claimed image conditions? What question would test the complete reader-plus-software result in the intended population and workflow? How would a different scanner or reader setting change those inquiries?

**How the three questions survive the change of modality**

The first concerns why the feature relates to the clinical finding. The second concerns whether the configured scanner and model reliably extract the claimed feature under the proposed image conditions. The third concerns how the actual reader-plus-software result relates to that finding in the intended people and workflow.

A different scanner calls for comparison of image and extraction conditions; a different reader setting calls for comparison of population, use and complete-result evidence. The earlier association may remain relevant, but applicability must be explained for each affected strand. The packet supplies no pathology study result, and the marker library's treatment observations supply none on its behalf.

**SECTION 07****Quick discrimination lab**

These two records are independent of each other, the opening dossier and the four-report library. Each question concerns only the observations supplied with it.

**A new population and clinical role**

A biomarker is strongly associated with advanced cancer in treatment studies. The proposed intended purpose now adds screening adults without symptoms for disease. No screening study or outcome is supplied.

**Decision**

**What remains potentially useful from the treatment-setting evidence? What new association and use questions does screening create, and can the previous scientific-validity conclusion carry over unchanged?**

**How a reviewer follows the claim into the new population**

The advanced-disease evidence keeps its value for its actual people and role. Screening asks whether the marker distinguishes the relevant condition at an earlier or different disease spectrum, and whether the result has an appropriate action in that population. Prevalence and the consequences of incorrect results can differ too.

Evidence applicable to those new questions could change the claim conclusion. The present packet requires an applicability appraisal and identification or generation of the missing support. Treatment-response evidence cannot simply be carried over as a screening or prognosis conclusion. No observation here establishes either the success or failure of a screening test.



## A proposed benchmark

A draft state-of-the-art section summarizes competitor websites and uses their advertised performance as the clinical benchmark. It supplies no recoverable search of clinical practice or underlying study methods.

### Decision

**What can these pages contribute? What evidence action would make the account of current practice and its chosen benchmark defensible?**

### How a reviewer follows a marketed claim to its evidence

The pages may identify products and claims worth tracing. Their summaries alone do not reliably establish the study population, reference, uncertainty or limitations needed for a clinical benchmark. A documented search of current practice, credible guidelines, peer-reviewed evidence and available alternatives would allow those methods and limits to be appraised.

The benchmark then needs a reason connecting the chosen comparator and expected performance to the proposed use. Appraised underlying evidence could support that choice; advertising alone cannot. This packet says nothing that establishes any particular competitor's inadequacy.

## SECTION 08

### Which conclusion can the evidence carry?

Scientific validity concerns why a marker or feature relates to the stated condition or state. A recoverable search and claim-specific appraisal can explain the relationship, its conflicts and its population or role limits. The particular device still needs applicable evidence for its measurement and clinical result. The connected utility question concerns what happens when the result is used; it is not a fourth IVDR pillar.

The closed library makes these limits visible: four reports represent three cohorts, inclusion is not endorsement, and response association is not automatically predictive treatment benefit. It leaves assay C's bridge open. Separately, the opening dossier's high overall agreement leaves its sparse treatment-decision boundary unresolved. Each is a useful bounded conclusion rather than a product verdict.

The seven checks retain the inherited IVDR-CH09 lookup identifiers for Workbook Part A2. Familiar working item states remain A, adequately explained for this item; U,

present but unclear; M, missing or inconsistent; D, deeper analysis needed; C, consultation indicated; and N/A, with a reason. An evidence location or identified absence and a reason make the state useful. None is a device grade or certification decision.

### **Which association does the claim promise? • card IVDR-CH09-A01**

Do the intended-purpose sentence, population and treatment comparison distinguish diagnosis, predisposition, prognosis, predictive treatment benefit or another stated role? The prognosis report cannot become the first response report's missing treatment-effect comparison.

### **Which evidence answers which question? • card IVDR-CH09-A02**

Are marker meaning, measurement and clinical use separated? Trial assay A and commercial assay C need their actual cutoffs and result mapping. The pathology counterpart separates feature meaning, extraction and the complete reader workflow before any finding can close.

### **What challenges the desired conclusion? • card IVDR-CH09-A03**

Can the saved search and screening ledger recover contradictory and population-specific evidence? The conflicting report's missing follow-up needs an explicit weight and reason. Weak or unfavorable evidence cannot disappear because it complicates the desired claim.

### **How many independent observations are represented? • card IVDR-CH09-A04**

Does the cohort map link the first response report and its companion analysis to the same people? A study can legitimately contribute to scientific, analytical or clinical reasoning when its design permits; reuse cannot be counted as independent replication.

### **Can another reader recover the bounded conclusion? • card IVDR-CH09-C01**

Does the scientific-validity report join question, recoverable search, cohort map, appraisal, favorable and unfavorable synthesis, supported wording and unresolved predictive or population questions? The filled library table illustrates that record, not a successful commercial-assay submission.

### **Which scope or evidence option can management consider? • card IVDR-CH09-C02**

Are a supported marker role, the commercial measurement/cutoff bridge and a future population kept distinct? A narrower first claim or later extension needs its own evidence and route. The option comparison cannot guarantee a certificate.

## Do both partners mean the same treatment decision? • card IVDR-CH09-C03

Do the diagnostic and medicine proposals agree on eligible patients, treatment evidence, cutoff and wording? Disagreement and missing evidence belong in the record. Partner preference cannot establish a predictive relationship.

## Further reading

Book 1C Chapters 5 and 11 offer optional practice with reference limits and the path from a finding to clinical meaning. They add no observations to the local four-report library and establish no bridge for the opening assay.

## Source trail and limits

- **Binding law, checked 2026-09-25:** IVDR consolidated 10 January 2025, Article 2(38)–(41), Article 56(1)–(5), Annex XIII Part A §§1.2–1.3. These define scientific validity, analytical and clinical performance; require systematic identification and appraisal of relevant data, including gaps; and place their assessment in the performance evaluation report. They do not define clinical utility as a fourth performance-evaluation pillar or validate the fictional assays by name.
- **Nonbinding context:** Medical Device Coordination Group (MDCG) 2022-2 §§6.3–6.4 informs the claim-specific literature and scientific-validity explanation. The European Medicines Agency anticancer guideline, revision 6, biomarker-enrichment discussion, informs the distinction between prognostic factors and treatment-effect questions; it is not an IVDR device-approval shortcut. Licensed read-only **CLSI EP49-Ed1 (2026)**, §§1.1 and 2.1–2.2, was checked only for its noninterventional clinical-validity scope and explicit limits on interventional and clinical-utility evidence development. No licensed text, study design, cutoff or numerical result is transplanted into the cases.
- **Packet limits:** The opening anonymous CDx trial/assay dossier, separate marker Z four-report library, two microcases and conditional DP transfer are not one product or cohort. Marker Z has four reports from three cohorts; L1 lacks a supplied predictive interaction estimate, L4's unfavourable attrition remains in the appraisal, and assay C has no established cutoff bridge. No source figure or artwork is used.

## CHAPTER 9

# What Can This Measurement System Actually Report?

**Before reading.** Chapter 8 separated a marker's relationship to a clinical state from a commercial assay's ability to identify it. What could a published association establish, and what would it leave open at that assay's chosen cutoff? How would an unfavourable study remain visible in the appraisal?

An association can explain why a finding matters while leaving a practical question: can this system produce the promised result? **Do the tested specimens, counts and product conditions support the analytical claim for the system actually offered?** A useful answer will credit the supported conditions, identify any needed evidence bridge and leave genuinely untested conditions open.

## What lies behind the result the user receives?

**Opening packet.** An anonymous fictional oncology sequencing panel reports **99.3% overall agreement**. Only **three positives** represent a complex variant class highlighted in its labeling. Extensive tissue validation sits beside a proposed liquid-biopsy claim, using body-fluid-derived material, supported by a small contrived set. After the wet-laboratory studies, the company updated its bioinformatics pipeline to improve low-frequency calls. It presents the one percentage as support for the complete claim.

The packet supplies no denominator for that percentage, uncertainty interval for the three complex positives, or demonstrated equivalence for the liquid or updated configuration. It is separate from the prepared-specimen accounting exercise and the later 98% example. Those boundaries matter before we calculate anything: a number from another packet cannot repair this one's missing denominator.

A sequencing result passes through more than an algorithm. A specimen arrives in a particular condition; collection and preparation affect what enters the instrument; its output passes through quality checks, software and report rules. **Analytical performance** asks how well that complete device detects or measures its target under the claimed conditions. An in vitro diagnostic examination takes place outside the body on human-derived material. Its target's clinical relevance is Chapter 8's scientific-validity question. The relationship of this device's result to the clinical condition in intended use is the separate clinical-performance question.

Even a correctly calculated percentage can hide which of those operations it represents. One specimen may be tested several times, several variants may be sought in it, and one run may produce many calls. Specimens, attempts, variants and patients therefore supply different denominators. An invalid or no-call result means the ordinary positive/negative answer was not produced under the system's rules. Excluding it from an agreement calculation does not make it disappear from the user's experience.

For digital pathology, the promised output could be an image feature, a case score or an image-quality decision. Scanner, preparation, model and report rule can each alter that output, and an unprocessable slide remains part of the workflow. The experiments differ from sequencing, but the first questions are the same: which output, counted unit, configuration and unusable cases does the evidence describe?

Common, easily read sequence differences can dominate an average while leaving a rare claimed class poorly represented. That need not make the opening 99.3% false. It makes the observations beneath it necessary. Three complex positives are a small coverage cell to examine; they are not three demonstrated failures.

## What would count as a good measurement?

First, what kind of answer is offered? An amount of target per specimen volume is a quantity. The quantity intended to be measured, together with its specimen and relevant conditions, is the **measurand**. A positive/negative answer instead needs a specified target property and a rule turning instrument information into categories. “Positive” is not itself a concentration. A sequencing variant class needs its reportability rule; a pathology score needs the threshold, if any, that converts it into a review flag.

Now consider a **separate invented quantitative example**, unrelated to the opening panel and the accounting exercise. A prepared material has an assigned reference value of **10 arbitrary units**. Three repeated readings are **9, 10 and 11**. Another configuration gives **7, 10 and 13**. Both average 10, yet the second scatters more. **Precision** concerns that scatter under specified repeat conditions. If the readings were **12, 13 and 14**, they would cluster similarly to the first set while sitting systematically above the assigned value. **Trueness** concerns systematic offset from an appropriate reference; **accuracy** depends on both trueness and precision.

Those three-read examples expose different questions without establishing any product's performance. The assigned reference itself needs a justified value and uncertainty. Where a measurement chain applies, **metrological traceability** records how the reported quantity connects through that chain to an appropriate reference, with uncertainty at each link. A tidy set of repeats cannot justify an unexamined reference value, just as an accurate reference cannot remove scatter in the device's readings.

### Which conditions were held still, and which changed?

Immediate repeats on one instrument under closely held conditions answer a **repeatability** question. If the intended workflow includes storage or transport, that is another comparison: appropriately matched fresh and stored material at the claimed times and temperatures. A shift after storage could reveal a **stability** problem despite tidy immediate repeats. Unchanged readings would support the tested handling conditions, with no automatic extension to other temperatures or times.

Changing operators, lots, instruments or sites asks whether the result holds under those particular changes. One operator or lot swap does not establish **reproducibility** across laboratories. The within- and between-laboratory design must match the claim and identify what varied. In sequencing, low input and handling can change what reaches the instrument. In pathology, stain or scanner can change what reaches the model. The comparison follows the actual result path rather than borrowing the name of a broader experiment.

### Which comparison reveals a different error mechanism?

Keep the target amount constant and add a plausible non-target constituent. If the reading shifts only with that constituent, the next inquiry is **interference** with the target measurement. Now remove the target and present a look-alike substance. Reporting the absent target would suggest **cross-reactivity**. That is a different mechanism from changing the reading of a target that is actually present.

A third comparison changes what ran before the blank. Compare a blank following a high-target specimen with one following another blank. A new signal only after the high specimen would suggest **carryover** from the earlier run. These are conditional teaching outcomes, not findings in the oncology packet. They show why clean repeats of the original 10-unit material cannot answer all three questions. Claimed instruments or scanners and intended software inputs also need their relevant comparisons; development examples alone do not supply them.

### What can be detected, and what can be reported as a number?

For a quantitative result, the **measuring or reportable range** is the span over which numerical results have justified performance. At the low end, detecting presence and estimating amount can separate. A **limit of detection (LoD)** concerns reliable detection at a stated low target level. A **limit of quantitation (LoQ)** concerns the level at which adequate quantitative performance permits an amount to be reported. A specimen can therefore contain detectable target below the level at which its amount can be stated reliably.

An analytical **cutoff** is a rule for converting measurement information into a categorical call. If that call guides treatment, the clinical treatment-decision threshold needs a separate clinical rationale. Stability of a laboratory cutoff cannot supply that rationale.

Changing a specimen condition might alter detection without changing the intended medical action; changing the treatment role might alter the needed clinical justification without changing the instrument's calculation.

For a categorical sequencing report, “reportable range” instead needs the variant classes and regions that can be called, the relevant levels and specimens, and the reporting rule and reference assignment. It should not sound like a concentration interval. A pathology system likewise needs to say whether it offers a continuous score or a case flag, under which image and scanner conditions. Neither scope is established by another assay's numerical measuring range.

This also keeps two uses of sensitivity apart. **Analytical sensitivity** asks whether the assay detects its target at the challenging levels and conditions it claims. **Analytical specificity** asks whether it distinguishes that target from non-target material or signals. Clinical sensitivity and specificity concern classification of a clinical condition in an intended population against a suitable condition reference. A target-level experiment answers the first kind of question; it cannot become a patient-condition estimate just by changing its heading.

The In Vitro Diagnostic Medical Devices Regulation (IVDR) Annex I §9.1(a) requires the relevant analytical characteristics **where applicable**. The examples above explain why precision, trueness, accuracy, detection and quantitation limits, range, cutoff and interference matter differently to different outputs. Annex XIII requires analytical reporting and a justification for a listed parameter judged inapplicable. A qualitative variant call, numerical concentration and image-based output therefore need reasoned, claim-specific experiments. Justifying one omission leaves the other applicable questions intact.

**Reviewer Lens — make the claim testable.** The useful comparison separates the opening tissue **single-nucleotide variant (SNV)** call—a one-letter sequence difference—from the complex class and proposed liquid call. Each needs specimen, target, output, pipeline and conditions. Numerical evidence needs scatter, reference offset and range distinguished; categorical evidence needs valid-detection rules and target assignment. A pathology score needs its image/scanner conditions before one model statistic can be interpreted. The three complex positives and unknown headline denominator should stay visible, without a guessed precision estimate.

**Consultant Lens — cost the consequential comparison.** Product and laboratory owners must agree what output and conditions will appear in labeling and instructions before additional easy repetitions consume the budget. Difficult complex variants or a second specimen may be the cells that decide the claim. A narrower initial offering is an option only when the actual report, configuration, wording and evidence support it. The experiment should answer that commercial choice; a proposed experiment cannot promise a certificate or extension date.



## What authority does a method provide?

A Clinical and Laboratory Standards Institute (CLSI) method can help design an experiment. That technical help does not by itself confer EU conformity. Under IVDR Article 8, a relevant harmonized European standard or part has a scope-limited presumption only when its reference is published in the Official Journal and the covered requirements fit. The dated list under Commission Implementing Decision (EU) 2021/1195 must be checked. An uncited or out-of-scope method may still be useful technically, but cannot supply that particular legal effect.

Article 9 **Common Specifications** have a different threshold: within their applicable scope, the manufacturer must comply unless it duly justifies another solution with at least equivalent safety and performance. Regulation (EU) 2022/1107 concerns specified class D devices, so it cannot be assumed to set every oncology panel's criterion. Regulation (EU) 2026/977 governs notified-body quality-management and assessment procedures; it supplies no product analytical acceptance percentage. The covered device, requirement and source role decide which branch matters. None supplies the opening packet's missing observations.

## Which difficult cell disappears inside the average?

The instrument reads many short pieces of genetic sequence. Software aligns them to a reference, combines their support and decides whether a difference can be reported. A one-letter change and an **insertion or deletion (indel)**, which adds or removes letters, can pose different detection problems. Complex changes can pose another. Abundant easy single-letter examples do not establish all those mechanisms.

The amount of variant signal also matters. **Variant allele fraction (VAF)** is the proportion of relevant sequence observations carrying the variant in the examined material. A low fraction can approach the detection boundary. Class, fraction, sequence neighbourhood, target region, input amount and specimen quality can each affect whether a correct call emerges. These are reasons to examine the difficult conditions, not measured failure rates in this packet.

Suppose, as arithmetic alone, that easy negatives vastly outnumber a rare claimed alteration. Their agreements will dominate the pooled fraction even when the rare cell is poorly estimated. The opening dossier gives no such counts, so its class-specific rate cannot be calculated. Nor can its **confidence interval (CI)**, the uncertainty around an estimated measure given the data and method. Even if three positives were all called correctly, their interval could remain broad. Actual counts, sampling rationale and uncertainty are needed before the cell can carry the label's breadth.

A **coverage matrix** makes that question visible by crossing claimed output with the conditions that might change it. For this panel those include class and level, specimen

type and quality, instrument, reagent lot, operator or site, software version and reference method. The map distinguishes direct data, justified bridging evidence and assumption. It does not demand an independent trial for every possible combination.

For pathology, the corresponding rows cross feature or case flag with scanner, stain and image quality, site, model version and reader conditions. A pooled result cannot establish that low-quality images or another scanner behaved like the represented images. The same question makes both matrices useful: which changed input or condition would make this evidence insufficient for the proposed output?

**Reviewer Lens — find the hidden denominator.** For the complex class, eligible specimens, valid calls, correct and discordant calls, reference assignment, fraction/context and uncertainty expose what the three observations can support. For liquid material, the specimen path and reference relationship matter beyond “same gene.” A scanner/image-quality row needs the equivalent accounting. The reviewer can credit useful pooled evidence while leaving the unsupported cell explicitly open.

**Consultant Lens — make a coverage choice possible.** A direct/bridge-proposed-or-tested/open marker beside each actual study configuration lets management see where targeted material could change the decision. Difficult positives, clinical specimens or controlled pathology images have value when they address a critical claim cell. A narrower report can remove a cell only if it truly removes the offered function. Added specimens and revised words remain proposed work until the resulting evidence and product scope have been assessed.

## How could the same raw data produce another report?

Follow one hypothetical specimen, without adding an observation to the opening dossier. The instrument produces short sequence **reads**. **Alignment** places them against a reference sequence. In a difficult region, ambiguous placement can alter apparent support for a difference. A **variant caller** weighs the aligned observations and proposes a difference. A proposed difference is still several decisions away from the clinical report.

A **quality gate** can reject inadequate specimen data before a call is issued. For passing data, a **filter** checks whether a candidate has enough support under configured rules. A stricter rule might withhold a low-fraction candidate; another might retain it or represent an indel differently. Unchanged raw reads therefore do not guarantee unchanged calls, and an improvement for one class does not prove unchanged performance for its neighbour.

For a retained difference, **annotation** uses a knowledgebase to connect it with a gene or interpretation. Report rules then determine what the user sees. Changed content can alter interpretation even when detection is unchanged. The study's identity consequently includes the wet-lab materials and instrument that made the reads, along

with alignment, caller, quality and filter rules, thresholds, annotation source and final report state. An application number alone does not identify that assembly.

The opening company's later update was intended to improve low-frequency calls. Its intention establishes no general improvement. Paired old and proposed outputs can show which decisions change; targeted challenges can test affected behavior and neighbouring boundaries. A **regression test** asks whether previously supported behavior remains supported after the change. An evidence bridge explains why an earlier study can still inform the new configuration. Some questions may remain directly supported while others require new work. Neither a complete repeat of every study nor automatic transfer follows from a familiar release label.

After deployment, a sequencing manufacturer's outside knowledgebase update can reopen the call or treatment-facing report comparison through its controlled change and monitoring records. A pathology scanner or model update can reopen image acceptance, prioritization or displayed score. These are possible postmarket mechanisms, with no added failure or patient harm in either separate deployed teaching scene.

**Reviewer Lens — compare the decision path.** Release manifest, change description and paired affected outputs show more than filenames. A changed gate needs newly invalid cases and report differences examined alongside agreement among remaining valid cases. For the opening panel, the decisive factual request includes whether the three complex positives ran under the proposed configuration. A shared application label cannot answer it.

**Consultant Lens — order the changed-state work.** Study-to-configuration rows can place low-frequency single-letter gains, indel/complex filtering, liquid material and no-call handling in consequence order. Each row identifies who may authorize a supported claim, who owns a needed bridge and which proposed wording must wait. A release schedule then reflects the unresolved work; it supplies no evidence of its completion.

## Which question can an engineered specimen answer?

Rare variants and very low levels may be difficult to obtain in enough clinical specimens. A laboratory can add or engineer a target into chosen material, creating **contrived material** that deliberately challenges a boundary. A reference material may also have a characterized property. Those are useful contributions, but background, extraction, degradation and target distribution can differ from specimens collected under the proposed conditions.

The relevant comparison has a precise name: **commutability** asks whether the material behaves between the measurement procedures being compared as the relevant clinical specimens do. Looking like tissue or plasma is insufficient. One assay's result on the

material does not establish a second assay's truth either. Fitness for the specific question, reference limitations and any needed clinical-specimen work or bridge must therefore be explained. The opening small engineered liquid set may inform low-level detection without carrying the whole liquid claim from tissue.

Curated digital images can similarly challenge blur or unusual stain, while leaving scanner optics, specimen preparation and the full workflow unrepresented. A reviewer needs to know what behavior the challenge exercises. A consultant can combine targeted challenges with representative intended-use material and versioned evaluation. Neither source of evidence automatically replaces the other.

## What happens to a specimen that never gets an answer?

A quality-gate failure, invalid or indeterminate output, no-call, or exclusion before analysis can remove a specimen from the positive/negative comparison. It does not remove the person waiting for a result. Repeating, obtaining another specimen or acting without the result can have different practical consequences. The ledger therefore begins with entered specimens, then follows first answers, reasons for missing answers, allowed repeats and final reports. A successful repeat leaves the first failed attempt in that history.

Low input or contamination could cause a sequencing no-call; a changed filter could alter calls and invalid rates together. An unreadable slide or image-quality rejection could likewise disappear from a pathology model calculation. These are mechanisms to test, without supplied opening failure counts. The specimen-flow diagram and denominator at each stage show whether a percentage describes agreement among usable results or the offered workflow's ability to deliver one.

**Reviewer Lens — follow the missing result.** The opening dossier needs an entered-specimen ledger for tissue and proposed liquid material under each pipeline, including invalids and repeats. A pathology extension needs quality-rejected images included in entered-case accounting and their fate visible to the reader. Without a supplied count, a rate is unknown. Reporting zero would invent evidence.

**Consultant Lens — make recovery feasible.** A repeat protocol needs to say when extraction, rescan or another specimen is possible, how both attempts remain visible and what the operator does if no answer can be produced. Repeating consumes material and time. Challenges also need to resemble the claimed poor-quality specimens or images; recovery on convenient clean material cannot establish recovery everywhere. Those proposed tests remain separate from observed results.

**Additional exhibit — the opening packet.** The updated pipeline improves low-frequency single-nucleotide calls but changes filtering for indels and complex variants. Its release record still uses the same application version. Counts of gained or

lost calls, class-specific retest results and proof that every old conclusion transfers are not supplied.

### Decision

**Does 99.3% overall agreement support every claimed class and specimen? Which denominator and configuration records come first, and how does an unanswered question differ from a measured failure?**

### How a reviewer bounds the headline result

The pooled result may describe its tested conditions. Three complex positives do not by themselves characterize the whole labeled class, and the small contrived liquid set does not automatically establish the liquid claim. Neither limitation supplies a measured failure rate.

The behavior-changing update also needs its own identity despite the unchanged application label: which version produced each result, what changed in indel and complex filtering, and which direct results or targeted bridges remain applicable? Eligible, valid and no-call denominators, reference assignments and uncertainty belong beside each claim cell. The packet establishes coverage gaps, with no authorization outcome or invented underrepresented-cell rate.

**Consultant Lens — choose the next evidence branch.** Management can compare targeted complex/context work, a supported liquid-specimen bridge or study, focused pipeline regression and a genuinely narrower initial offering. Each option protects a particular report decision and has evidence-dependent timing. The manufacturer chooses the claim and release; the consultant makes the work visible. The reviewer separately decides whether the selected evidence closes the open cell. A plan cannot perform either decision in advance.

## Which denominator answers the question?

**The separate accounting exercise, AP-1,** supplies the counts the opening dossier lacks. It is an invented qualitative-assay exercise at a specified challenge condition and configuration, using **100 distinct prepared specimens**, not patients. Each receives one initial attempt. A prespecified comparator assigns **40 positive** and **60 negative**. It is the reference for this agreement exercise, with no established status as a diagnosis of patient disease. The candidate assay can report positive, negative or no-call. The initial record contains no repeats.

Follow the forty comparator positives first: the assay gives 36 positives, two negatives and two no-calls. They sum to forty. For the sixty comparator negatives, it gives two

positives, 56 negatives and two no-calls. They sum to sixty. Four of the original hundred specimens have no first answer; 96 have a positive or negative result. Those counts establish the denominator before a performance name is attached.

| Initial assay result | Comparator positive: 40 specimens | Comparator negative: 60 specimens |
|----------------------|-----------------------------------|-----------------------------------|
| Positive             | 36                                | 2                                 |
| Negative             | 2                                 | 56                                |
| No-call              | 2                                 | 2                                 |
| <b>Column total</b>  | <b>40</b>                         | <b>60</b>                         |

How often did the first attempt provide a valid result? **First-pass valid-result yield** is  $(36 + 2 + 2 + 56)/100 = 96/100 = 96\%$ . The complementary **no-call rate** is  $4/100 = 4\%$ . Before these results were seen, the local teaching protocol required **at least 95% valid initial results** as an operational point-estimate gate. The observed 96% meets that gate for these prepared specimens and conditions. That is a real, bounded positive conclusion.

The gate says nothing about correctness, detection limit or clinical adequacy. A real product's plan might also need an uncertainty bound or another challenge set, but those are further evidence decisions. Changing this local point-estimate criterion after seeing the outcome would alter the question that was prospectively asked.

Now restrict the comparison to initial valid pairs. Of the forty comparator positives, 38 received a positive or negative candidate result; 36 of them agreed. **Positive percent agreement (PPA)** is  $36/38 = 94.7\%$ . Of the sixty comparator negatives, 58 received a valid candidate result and 56 agreed. **Negative percent agreement (NPA)** is  $56/58 = 96.6\%$ . Together, **valid-pair overall agreement** is  $(36 + 56)/(38 + 58) = 92/96 = 95.8\%$ . These denominators deliberately condition on a valid answer; the four no-calls must remain visible beside them.

What proportion of *all* comparator-positive specimens received a correct positive report on their first attempt? The numerator stays 36, but the denominator is forty, including the two no-calls:  $36/40 = 90\%$ . This is **first-attempt positive-report yield**. It is neither PPA nor clinical sensitivity in patients. The difference between 38 and forty is a change in question, not rounding: agreement among usable pairs versus delivery of a positive report among all comparator positives.

A specimen is one of these hundred prepared materials; an attempt is a run on one specimen. A sequencing variant is a target difference, and several may occur in one specimen. Adding variant-level calls to this table would therefore not create more independent specimens. Patients are not the unit here at all. Comparator assignment

may have error, and prepared materials may differ from clinical specimens. Passing the operational gate removes none of those limitations.

The filled decision record below keeps the narrow pass beside the open questions. Read its first row back into the table: one original specimen per first attempt is why the operational denominator is a hundred.

| AP-1 decision field       | What the record establishes                                                                                                                                                                                                                                                          |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Question and unit         | First-pass validity for 100 distinct prepared specimens, one initial attempt each, at the stated challenge condition and configuration. Specimen identities and the 40/60 comparator split remain in the ledger.                                                                     |
| Criterion and observation | Prespecified local gate $\geq 95\%$ valid initial results; observed 96/100 = 96%, with four no-calls.                                                                                                                                                                                |
| Bounded conclusion        | <b>Operational point-estimate gate met.</b> PPA 36/38, NPA 56/58, valid-pair agreement 92/96 and positive-report yield 36/40 answer separate questions.                                                                                                                              |
| Open evidence and owner   | Analytical lead retains specimen/results/configuration and assesses reference, material, levels/classes and repeat policy. Statistical lead reports uncertainty appropriate to sampling and inference. No clinical prevalence, patient benefit, LoD or broader claim is established. |

The same observations appear in several rows because each row asks a different question. They are not several independent demonstrations of correctness. If the protocol's criterion concerned an uncertainty bound, the point estimate alone would not answer it. Here, however, the stipulated gate concerns the initial-validity point estimate, so its positive finding should not be discarded merely because wider performance remains open.

**Reviewer Lens — defend the counted unit.** The two column totals make every no-call recoverable. Keeping positive-report yield beside PPA exposes how a usable-pair calculation can omit missing first reports. In pathology, the declared unit might be an image, slide, case or reader decision. Several images from one case cannot silently become independent patients. Accounting comes before the larger clinical-performance inference.

**Consultant Lens — a repeat is extra work, not extra people.** A workable repeat policy states when another attempt is allowed, how both attempts remain visible, what the user learns after another failure and whether enough specimen remains. Re-extraction consumes material and time; rescanning may restore an image but still



requires case-level accounting. Operational burden and performance deserve separate estimates. A better final yield would not erase the first invalids.

### Decision

The accounting exercise now permits a second attempt for two original no-calls. One comparator-positive no-call becomes positive; one comparator-negative no-call becomes negative. Every other initial result stays as recorded. What is final specimen validity, how many total attempts occurred, and may the initial 96/100 be overwritten?

### How the two attempts change the final ledger

Two more original specimens now have valid results: **98/100 = 98% final specimen validity**, after **102 total attempts**. Two specimens still have no-call results. The extra attempts did not create extra independent specimens, subjects or patients.

The initial record remains **96/100 = 96%** first-pass validity and **4/100 = 4%** first no-calls. Both belong beside final yield, repeat burden and the preplanned rule. The arithmetic closes neither analytical accuracy nor a clinical claim.

## What happens close to a yes-or-no boundary?

An instrument may generate a continuous signal or score before a cutoff turns it into positive or negative. If repeated readings fluctuate around that boundary, one specimen can fall on different sides on different attempts. Obvious high and low specimens tell little about that transition. A pathology case flag may similarly switch near its threshold when the scan or model run changes. The repeat design must match what actually varies; neither packet supplies such a transition experiment.

CLSI EP12, **third edition (2023)** calls two useful positions **C5** and **C95**. On the relevant response or target scale, they are levels where the binary examination reports positive about **5%** and **95%** of the time under stipulated conditions. The span between them is the positive-response **transition region**. Its width indicates how much repeated-result variation surrounds the cutoff. A wider transition would change the evidence about reliability near the boundary even if obvious positive and negative specimens continued to agree.

The scale matters: concentration in a detection experiment need not be the same as another assay's internal score. The way C5 and C95 are estimated depends on how the device generates its binary result and on its stated precision conditions. They are not universal acceptance levels, percentages of clinical sensitivity or confidence limits on

the accounting exercise's PPA. That exercise supplies no repeated-level series from which either position could be estimated.

A confidence interval asks another question: how uncertain is an estimated quantity from a finite study? An agreement percentage can have one, and an estimated transition position can have one too. The C5-to-C95 span describes the response transition; a confidence interval around an estimated C95 describes uncertainty about that position. The “95” in C95 is not a 95% confidence level. Which transition width and estimation uncertainty are tolerable depends on the intended decision and a plan made before observing the results. EP12 terminology supplies neither packet's analytical cutoff nor a treatment rule.

The detection limit returns us to a particular target level and specified conditions: how reliably does the system detect it, often expressed through a prespecified probability? It is not an absolute line below which detection is impossible. Low-fraction sequencing requires the class, specimen, input, processing and pipeline to be named. Tissue evidence needs a reason to transfer to body-fluid-derived material. The cutoff is the system's reporting rule; the treatment-decision threshold determines use in care and needs clinical justification. A change may affect more than one, but measuring one does not establish the others.

### **What does the comparison procedure know?**

The accounting exercise's reference assigns prepared material positive or negative. It is not described as the best available assessment of a patient's disease state. That premise is why its metrics are PPA and NPA against the comparator. Two procedures can agree while both miss the true condition. Clinical sensitivity or specificity instead needs an appropriate condition reference, relevant patients and a suitable sampling design. Changing the reference and sampled population could change the legitimate inference; renaming the existing fractions could not.

Further testing may explain disagreement, but it cannot silently replace the original comparison. Suppose only discordant specimens receive a third referee test and inconvenient cells are replaced with its answers. The resulting table has selectively verified cases, potentially flattering the candidate. Original results should remain, with investigated cases annotated. Estimating performance against the referee needs a defensible design that also addresses concordant cases.

That distinction is especially useful when the comparator may miss low-level variants. A discrepancy could reflect candidate error, comparator limitation, a threshold difference or specimen handling. Those possibilities require investigation before blame is assigned. The opening packet supplies no discrepancy resolution; the method explanation adds none.

There is another familiar word with a different denominator. Measurement precision concerns repeated values under stated repeatability or reproducibility conditions. In **machine-learning (ML)** classification reports, precision often means **positive predictive value (PPV)**: among the results called positive, what fraction are reference-positive? That ratio depends on the reference and case mix. An enriched pathology image set can describe its own mix without establishing a deployed population's predictive value. Naming numerator, denominator and reference prevents the word “precision” from joining two different claims.

**Reviewer Lens — identify the uncertainty.** Input scale, cutoff and repeat conditions make a C5/C95 statement interpretable. Study-specific confidence intervals on agreement describe another uncertainty. The original discordance table, comparator limits and no-calls must survive both readings. The accounting exercise has neither a transition experiment nor a patient-condition reference to credit, even though its operational gate passed.

**Consultant Lens — protect the boundary before testing.** Repeatability near the cutoff, detection under the claimed specimens and discrepancy investigation require distinct plan entries. A treatment-facing claim also needs clinical ownership of the medical threshold. For a pathology score, how the reader uses a borderline flag and whether scanner or model changes move cases across it determine the needed work. These are testable plans and questions, with no assumed successful validation.

## Could another reviewer reconstruct the offered system?

A useful evidence row begins with the result and unit: specimen call, variant call, slide or case. Conditions identify specimen quality, class and level, site, instrument or scanner and user. Configuration identifies reagents and relevant software or database. Comparison identifies reference assignment and its limits. Accounting follows entered, excluded, invalid, repeated and valid results. Only then can the conclusion state its estimate, uncertainty and supported range.

Direct evidence, a tested bridge and an untested extrapolation belong in different findings. A supplier or pipeline change that can alter the result reopens its affected rows. Copying an old overall percentage would conceal the question that changed.

For the opening panel, the immediate records are the 99.3% denominator and strata, the three complex-positive results, tissue-to-liquid rationale and paired pipeline outputs. For pathology, the parallel questions concern the image/case unit, scanner/model state and quality-rejected images in entered-case flow. Those requests remain precise even before the eventual adequacy is known.

## Which missing cell is worth the next study?

A coverage matrix and configuration-specific bridge plan give each open row an owner, prospective criterion, material strategy and management consequence. Scarce complex-positive specimens and a defensible coverage rationale may answer more than another large set of easy negatives. For a scanner extension, controlled comparisons and representative intended images may be more useful than repeating the original scanner's best images.

Management can broaden evidence, narrow the actual supported claim or develop a separately evidenced future specimen or class extension. The offered report, instructions and configuration must make any narrower scope real. The consultant can explain evidence cost and lost scope, while timing and certification remain conditional on the resulting assessment.

**Transfer — a separate pharmacy test.** A respiratory test is proposed for trained pharmacy staff. Temperature, operator, site, specimen viscosity or quality, lot, interference, invalid results, connectivity loss and repeat availability replace variant classes as relevant conditions. Is the counted unit a cartridge attempt or the person/specimen receiving a final result? How does a missing result reach the user? The setting supplies no pass. A condition-by-result ledger can answer the reviewer while representative challenges and feasible instructions become the consultant's proposed work.

## Practice studio — what action follows from the number?

This studio uses a compatible but separately presented anonymous sequencing dossier from the opening exercise. Its **class C is stipulated**, without being inferred from performance. It adds no observations to the independent prepared-specimen exercise. The useful result of review will be both a precise evidence question and a manufacturer's next decision.

**Practice packet.** The dossier reports **99.3% overall agreement**. Tissue evidence is strong; a proposed liquid claim rests on a small contrived set; complex variants are sparse; no-calls were excluded; and a later pipeline update changed low-frequency filtering. Its performance evaluation report treats the percentage as support for the whole intended purpose. Full stratum denominators, liquid detection estimate, cutoff-region results and a bridge to the current configuration are not supplied.

### Which output and conditions belong in each row?

Begin with what the user is offered: common and complex tissue calls, the proposed liquid result and any treatment-facing call near a decision threshold. Each row needs specimen, variant class and level, report rule and tested pipeline. Input amount and quality, sequence region and context, instrument, lot, site or operator and reference method then identify conditions that could change the answer. Counts, exclusions, invalids and uncertainty show how far the evidence reaches.

An absent relevant experiment makes a row open, without making it a failed experiment. A study on the previous configuration likewise remains a study on that state until its applicability is established. The table below retains these distinctions. Its first row shows what can be credited about tissue evidence while leaving the other claims with their own questions; none of the proposed work is a completed pass.

| Claim cell and counted unit                                                 | Supplied evidence and limit                                                                                                   | What could resolve the next decision?                                                                                                          |
|-----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Represented tissue classes; specimen and variant-call ledgers distinguished | Extensive tissue validation and pooled 99.3% described; full denominators, failures, class detail and uncertainty absent here | Class counts, entered/valid specimen flow, reference assignment and exact tested pipeline; credit demonstrated conditions                      |
| Claimed complex class                                                       | Three positives identified; wider representativeness and uncertainty unknown                                                  | Targeted class/context coverage and sampling rationale, or a genuinely supported narrower claim                                                |
| Proposed liquid result                                                      | Small contrived set; tissue and liquid material can behave differently                                                        | Specimen-specific detection, interference and stability, with clinical-material/commutability bridge where appropriate; no borrowed tissue LoD |
| Low-frequency and near-cutoff calls after update                            | Filtering changed after wet-lab work; complete old/new decision map absent                                                    | Study-version reconciliation, paired regression and targeted boundary analysis before old results transfer                                     |
| Usable first report and repeat path                                         | No-calls excluded from headline; counts and repeat burden absent                                                              | Entered, invalid, repeated and final-result denominators, with actual user consequences                                                        |

For the complex row, the question is what those three specimens represent and what they actually produced. For the liquid row, the question includes whether the material

comparison represents the proposed specimen. One request for “more validation” would hide that difference. In pathology, the first row would concern scanner/model/image-quality/reader output: strong evidence for represented images can remain useful while a second scanner or difficult slide remains open. A deployed manufacturer would compare changed scanner or model states with the approved configuration and actual monitoring, without an invented field incident in this studio.

**Reviewer Lens — make the row answerable.** Source data should reveal which specimens produced the pooled result. Each complex or liquid row needs a disposition grounded in direct evidence, a justified tested bridge or untested extrapolation, with its specific missing numerator, denominator, conditions or reference limitation. A three-positive cell cannot be called precise before its sampling and uncertainty are justified.

**Consultant Lens — order by dependency and consequence.** No-call accounting and configuration identity come first because later percentages cannot be interpreted without them. Complex-positive material, liquid conditions and near-cutoff decisions can then be prioritized by claim consequence and feasible material. A targeted bridge and narrower first offering have different costs and value; both need a review of actual behavior and labeling. Commercial preference alone cannot settle the evidence boundary.

### **Does the physical specimen meet the software study?**

Collection, transport, preparation and extraction determine what reaches reagents and instruments. Their signal or sequence reads then pass through quality checks, alignment and calling or calculation, filters, annotation and report rules. These stages can interact: poor extraction lowers input; a stricter gate might prevent an erroneous call while increasing no-calls; a changed filter might recover low-frequency single-letter calls while changing indel behavior. A manifest must join the physical and digital parts that actually produced each result.

In the practice packet, an unchanged application name cannot establish that filtering code and content still match the wet-lab studies. Appropriate paired processing of stored data, challenges on current specimens and a documented comparison can show which old conclusions survive. Rare-variant material may be useful if its particular challenge and between-method behavior are explained. The small contrived liquid set remains a sensible possible starting point with explicit size and material limits. A pathology system has the corresponding scanner acquisition, preprocessing, model, threshold and display path. Neither device ends at its algorithm.

## What evidence would change each finding?

A finding becomes useful when it connects an observed fact to the decision it affects, then to the smallest plausible assessment or study that could change it. Sometimes the resulting work makes a genuine claim limitation worth considering. The five comparisons below retrieve those different reasons; they do not predict that any future experiment will pass.

**The complex class.** Three positives leave the class's breadth and difficult sequence contexts uncertain. Their actual calls, reference assignments, contexts and uncertainty would show what can already be credited. A targeted class/context plan and coverage rationale address the remaining scope. If representative positives cannot be obtained in time, an initial supported report might exclude the class—but only if report behavior and labeling can truly enforce that boundary. Three observations alone supply no class-level rate without the denominator and sampling argument.

**The liquid specimen.** Target amount, background, extraction and degradation may differ from tissue. A tissue detection limit therefore describes the tissue system under its tested conditions. Liquid-specific low-level detection, interference, stability and representative-material evidence would address the proposed extension. A tissue-only first scope could be examined if the liquid claim is unnecessary, retaining tissue evidence within its limits. No exact liquid detection limit is supplied here.

**The unusable result.** Excluding no-calls prevents first-report performance from being read from the headline. Entered specimens, exclusion reasons, initial invalids, repeat attempts, final reports and user actions are the missing path. If the intended setting cannot repeat, that changes the practical claim or workflow options. The accounting exercise's four no-calls and two successful repeats belong to its own dataset; they supply this dossier no rate.

**The changed pipeline.** Historical studies concern an older computational state. Old/new filtering and paired outputs for low-frequency single-letter changes, indels and complex calls can locate changed behavior. Regression and targeted boundary work then determine whether each historical conclusion remains direct, transfers through a tested bridge or needs new evidence. Improvement in one intended behavior supplies no general class conclusion, and an application label supplies no missing identity record.

**The treatment-facing boundary.** Easy agreements can dominate while decisions near the threshold differ. This packet supplies no near-cutoff counts or effects, so neither a safe nor an unsafe rate can be calculated. Paired results around the relevant boundary, reference limits and the effect of changed calls on the claimed decision would answer the actual question. Separating that role from a narrower reportable claim may be an option for clinical and regulatory owners to assess. An “analytical only” label cannot remove an unchanged treatment function.



Together these support a focused finding: the pooled agreement excludes no-calls and does not supply the needed results for the three-positive complex class, liquid specimens or updated pipeline at the decision boundary. The next evidence is the entered-specimen path, class/specimen counts and uncertainty, full tested configuration and a targeted bridge or supported limitation. Represented tissue evidence remains separately appraised. The finding identifies a gap; it does not turn any untested cell into a failure.

Management needs the corresponding view: which work consumes scarce specimens or time, what is lost by an initial limitation and what a later claim would need to establish. Those dependencies can support a choice. Neither an assessment memo nor a management memo can supply a certificate, date or treatment decision from missing numbers.

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### **Decision**

**Does the practice dossier's pooled 99.3% support every proposed specimen and alteration class? What can the percentage be credited for now, and which evidence and configuration boundaries must be resolved first?**

### **How a reviewer retains the represented evidence**

The result may be useful for tested, valid, represented tissue conditions. It cannot settle the sparse complex class, proposed liquid material, excluded no-calls or changed pipeline. Exact class/specimen denominators and uncertainty, full specimen/repeat flow, reference limitations and old/new configuration decisions are needed before the wider inference.

Targeted evidence or an actually supported limitation can then be assessed. The percentage supplies no unreported cell rate, patient outcome or approval. This is compatible with crediting the tissue work that genuinely applies.

### **Who uses the analytical finding?**

A notified-body reviewer assesses the actual analytical claim and configuration through the applicable device route, distinguishing direct evidence, justified bridge and unsupported extension. If obtaining new evidence involves a regulated performance study, the national competent-authority process is a separate question. A numerical gap alone supplies no country authorization or complete study-route facts.

A medicine-linked claim also needs its operative premise kept visible. A companion diagnostic is essential to a corresponding medicine's safe and effective use, identifying before or during treatment those most likely to benefit or at increased risk of serious adverse reactions. Where that relationship applies, the medicinal-products authority's

scientific consultation can address the treatment-linked implication. It does not replace the device's analytical assessment, and this paragraph supplies no missing medicine relationship to a local packet.

The manufacturer owns purpose, studies, release and postmarket decisions. A consultant can organize the matrix, denominator audit and alternatives. Helpful advice cannot certify a device or make an untested row pass. Changed product, study or corresponding-medicine facts determine which actor and evidence question next become relevant.

**Transfer — the separate pharmacy cartridge.** With one result unit fixed, how would temperature, specimen viscosity or quality, operator, lot, interference and connectivity appear in a condition row? Which can prevent a result, and is another cartridge or specimen available? The reviewer needs first-pass and final-report denominators by condition; the consultant needs representative challenges and feasible instructions. No oncology percentage or variant-class count transfers to this test.

## Quick discrimination lab

These packets are independent of one another, of the prepared-specimen exercise and of the 99.3% dossier. Each has its own evidence boundary.

### A broad report with a different percentage

A different sequencing assay reports **98% overall agreement**. Nearly all tested variants are common single-nucleotide changes. Its proposed purpose also includes **fusions**, joining sequence from different genomic regions, and **copy-number changes**, changing how many copies of a region are present. No class-specific counts or results for those additional classes are supplied.

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### Decision

**Does 98% establish the entire claimed reportable range? Which counted unit and class-specific evidence are needed, without borrowing the earlier three complex positives?**

### How a reviewer finds the unsupported classes

The percentage may describe represented common single-letter changes. Fusions and copy-number changes require different detection and reporting mechanisms, whose coverage is absent from this packet. Specimen and variant-call counts, reference methods by class, relevant region and input conditions, pipeline and uncertainty would show the actual support.

The unsupported cells need a justified bridge, added evidence or a narrower actual claim. The earlier three-positive count belongs to another case. This assay has not supplied a class-specific success or failure rate for the added claims.

### **A detection study without the difficult specimens**

A separate detection-limit study uses high-quality contrived material. Low-tumor-content tissue and degraded clinical specimens named in the proposed label are excluded. No results for those conditions are supplied.

#### **Decision**

**Can its detection limit be generalized to those labeled conditions? Which material relationship and difficult-condition result would make that inference defensible?**

#### **How a reviewer keeps the measured condition intact**

The high-quality material can challenge a target level under its own conditions. Low tumor content and degradation can change the input and detection, so automatic transfer is unsupported. Relevance of the material, a test or justified bridge to the difficult conditions, and a probability-based detection result with conditions and uncertainty would address the gap.

A genuinely narrower specimen claim is another option to assess. The packet supplies neither a numeric detection limit nor success or failure in the excluded conditions. Retaining the measured study's value and limiting its inference are compatible conclusions.

### **Leave a conclusion with its denominator attached**

An analytical conclusion concerns one result from a configured system under stated conditions. Its specimen or image unit, target/class and level, comparator limits, entered/valid/repeated accounting, estimate and uncertainty show what that conclusion means. A useful pooled result can coexist with an open rare class, second specimen, difficult image, near-cutoff region or changed pipeline.

The accounting exercise demonstrates a bounded positive: 96/100 meets its initial-validity point-estimate gate. PPA 36/38, NPA 56/58, valid-pair agreement 92/96, first-attempt positive-report yield 36/40 and final validity 98/100 after 102 attempts answer different questions. None establishes clinical performance or a

treatment threshold. Chapter 10 now asks what the device's result means in the intended people and workflow.

These seven checks attach to evidence items: **A**, adequately explained; **U**, present but unclear; **M**, missing or inconsistent; **D**, deeper analysis needed; **C**, consultation indicated; **N/A**, inapplicable with a reason. The location and bounded inference give the letter meaning. They are not device grades or certification votes. Original identifiers retrieve optional expanded Workbook Part A2 cards; their source-era topic number differs from this chapter number, as the original-to-integrated crosswalk shows.

### **Every analytical claim has a tested state • card IVDR-CH10-A01**

Do purpose, instructions and study manifests identify the specimen, analyte or variant range, platform and version? Tissue, liquid, complex class and pipeline state need their own evidence rows before a pooled result can support them.

### **Every result keeps its denominator • card IVDR-CH10-A02**

Do entered flow, no-calls, exclusions, repeats and class-specific uncertainty remain visible under challenging conditions? Initial validity 96/100 and valid-only PPA 36/38 in the accounting exercise cannot substitute for each other.

### **Direct evidence and transfer are distinguishable • card IVDR-CH10-A03**

Which specimen, class, level and configuration were tested, and which extension has an evidenced comparison? Tissue detection and contrived liquid results each retain their own conditions; neither is universal coverage.

### **A behavior change reopens the affected inference • card IVDR-CH10-A04**

What do old/new manifests and paired affected calls show, including no-calls and threshold crossings? A shared application label does not establish unchanged measurement behavior.

### **The coverage matrix can direct work • card IVDR-CH10-C01**

Does each row identify unit, claim cell, tested conditions, reference, raw counts, criterion, observed result, uncertainty and version, then the next study or limitation? Planned work remains visibly distinct from observed evidence.

### **Management's scope options describe a real product • card IVDR-CH10-C02**

Would broader evidence, supported narrowing or a future extension leave report, instructions and configuration coherent? A later liquid or complex-class plan is neither obtained evidence nor a guaranteed market route.

## Scarce material answers the consequential question • card IVDR-CH10-C03

How do complex-positive, low-input, liquid and near-cutoff needs compare with already represented easy strata? Material choice and sample-size reasoning must explain how the intended decision becomes answerable, rather than merely making a large study larger.

## Further reading

Book 1C Chapters 7–10 offer optional background on measurement, detection boundaries, variation and the complete sequencing process. This chapter is self-contained; those readings add no specimens, variant counts or outcomes to its local packets.

## Source trail and limits

- **Binding law, checked 2026-09-25:** IVDR consolidated 10 January 2025, Article 8, Article 9, Annex I §9.1(a), and Annex XIII Part A §1.2.2, for claim-relevant analytical characteristics, justified nonapplicability and analytical reporting. Commission Implementing Decision (EU) 2021/1195, consolidated 17 June 2026 identifies the current dated cited-standard list, not a blanket presumption for any CLSI method. Regulation (EU) 2022/1107 has covered class-D Common Specifications; Regulation (EU) 2026/977 addresses notified-body quality-management and assessment procedures, not a product analytical cutoff. These sources do not determine a fictional product's outcome.
- **Method, read-only licensed text:** CLSI EP12-Ed3 (2023), definitions and §§3.1.2, 4.2.5.1–4.2.5.2, inform the original explanation of binary response levels, agreement versus target-condition performance, and discrepancy resolution. C5/C95 transition positions and confidence intervals are distinct; EP12 supplies no AP-1 acceptance threshold, oncology cutoff or treatment rule. The local AP-1 exercise is invented accounting data, not a clinical cohort or a claim of EP12 conformity.
- **Packet limits:** The 99.3% oncology packet, AP-1's 100 prepared specimens, the independent 98% SNV-heavy record and the high-quality-material LoD record are not one evidence set. No unprovided stratum count, detection probability, reference truth, patient outcome or regulatory certificate is inferred. No source figure or artwork is used.

## CHAPTER 10

# Does the Result Work in the Hands of Its Intended User?

**Before reading.** Why did Chapter 9's 36/38 positive percent agreement and 36/40 first-attempt positive-report yield answer different questions? What was counted, and where did the no-calls go? Why might high overall sequencing agreement still leave a particular patient decision unsupported?

### How the counted result carries into clinical use

The independent exercise counted prepared specimens, not patients. Among 40 comparator-positive specimens, 38 had valid initial calls and two had no-call results. The 36 agreed positives therefore give 36/38 agreement among usable pairs, but 36/40 positive reports among all initially tested comparator positives. Excluding the two missing reports changes the question. Neither figure establishes a patient-condition reference or the commercial result's usefulness for every clinical decision, specimen or configuration.

The next question is **whether this device's result relates to the claimed clinical condition or state for the intended people, users and workflow**. A study can answer that question well for its tested use while leaving another use open. The reviewer needs to preserve the positive finding and locate the exact boundary; the manufacturer needs to know which next decision that boundary creates.

## SECTION 01

## What result does the user actually reach?

An assay may identify a sequence variant accurately without establishing what its reported call means for the intended clinical or treatment-facing state. A pathology model may recognize a feature in stored images without establishing what a pathologist sees and decides with the offered product. These are reasons to follow the result into use. Under the In Vitro Diagnostic Medical Devices Regulation (IVDR), clinical performance concerns the device's ability to yield results correlated with a clinical condition or physiological or pathological process or state, in accordance with the target population and intended user. Correct software execution and demonstrated patient-outcome benefit are different conclusions.

For an assistive pathology tool, a slide becomes a scanned image, the tool flags or scores a case, and a worklist brings it to a pathologist for interpretation. A model-only experiment examines one part of that path. A study of readers using the tool can examine more, provided its interface, case list and operating conditions represent the offered use. An accurate score on a case that does not reach review promptly leaves a timing question unanswered. Recognition of a finding and time until its review therefore remain separate outcomes.

Even recognition needs a comparison. Before counting correct positive decisions, the study must explain how it decided which cases had the target finding. Its **reference standard** is that specified assignment method or process. Independent experts, another examination, follow-up or a defined combination may supply it. The reference is the benchmark for the actual study question; it is neither infallible nor automatically the device output being tested. Its sources and unresolved cases affect what the comparison can establish.

An anonymous fictional pathology study **meets its primary endpoint**. The model alone detects cases accurately, and average reader sensitivity improves. Most slides come from **two academic centers and one scanner family**. Community pathologists participate only in a **short enriched session**. Queue prioritization and low-confidence suppression are disabled during the study, although the commercial workflow includes them. No total case count, effect size, scanner-specific rate, endpoint definition or patient outcome is supplied. The success is a real positive in this packet; its applicability to the offered workflow remains to be assessed.



## Which task would make the study result relevant?

The claim needs a sentence a user can recognize: who receives which result about which finding, and what happens next? “Identifies cases” and “helps pathologists review possible positive cases in time” promise different work. A stand-alone classification output needs its appropriate comparison. A feature highlight asks what the reader and device achieve together. A triage promise concerns which cases reach review by the intended time. A numerical score, prognosis or treatment-selection result has its own result-to-condition and decision questions. These are illustrative claim forms; the opening packet has not supplied the exact finding or endpoint details.

Once the task is clear, the study can specify what outcome will judge it. That is an **endpoint**. The **primary endpoint** is the main outcome, with population, calculation, criterion and timing set before the result is known. For finding detection, one possible measure is **reader sensitivity**: among cases assigned positive by the reference process, how many does the reader identify as positive? A priority claim may additionally require a time-to-review outcome for reference-positive cases. An improvement in average sensitivity does not tell us whether a low-score positive case was late; a faster list does not show better diagnostic classification.

The connection to clinical benefit has to be explained through that actual use and the possible consequences of wrong or late information. The word “benefit” cannot supply an unreported endpoint result. Here the primary criterion was met and average reader sensitivity improved, but neither the criterion's definition nor its effect and denominator is given. Numerical precision and subgroup estimates cannot be calculated from that description, and one endpoint cannot cover every label claim. The separate sequencing question from Chapters 8–9 retains the same discipline: biological association, measurement and the commercial result's clinical or treatment-state relation each need applicable evidence.

**Reviewer Lens.** For the opening product, the most useful record joins intended purpose, pathologist task, prospective primary and secondary endpoints, reference-positive/negative counts, exclusions and criteria. Recognition and time to review need separate findings. The met endpoint keeps its credit while the reviewer determines whether the commercial queue was part of the tested task. For a sequencing call, the analogous comparison identifies the clinical state and the device/cutoff that actually produced the study result. A result can be adequate for that bounded question without supporting the new one.

**Consultant Lens.** Agreement among clinical, software and product owners on the user action makes the next study a manageable choice. A priority claim needs a case entry point, acceptable review time, fallback and a rule for missed or deferred cases. Interpretation assistance needs a reader comparison and intended settings. Those choices determine whether a targeted simulation, reader study or another comparison

is useful. A request to repeat everything can waste effort; reliance on model success alone can leave the important task unobserved. Management needs the work behind each option.

## SECTION 02

### Which intended people and conditions are represented?

Two centers provide two sources of study material. They do not necessarily provide two acquisition technologies or a routine community workload. The opening slides support their actual academic-center case mix and scanner family. A community laboratory using another scanner or a different mix of findings asks whether that support remains applicable. The comparison begins with what can vary in the cases themselves.

Easy and difficult stages, subtypes and look-alike conditions make up the **disease spectrum**. A set of obvious findings can yield different performance from borderline routine work. **Prevalence** is the frequency of the target condition in the population. An enriched session deliberately selects more of particular cases, often positives or difficult cases, than routine work would contain. Such selection can make a useful challenge test while changing workload and the fraction of flags that are true positives. It does not automatically invalidate sensitivity. Its selection method, actual reference-positive denominator and fit to intended use determine what the estimate can support.

The person working through that set also matters. A reader familiar with the tool and a short selected list may notice a false negative that a community reader in a high-volume queue does not. Training, fatigue and access to a secondary queue can affect whether and when a case is reviewed. These are mechanisms to examine, not supplied training failures. Likewise, another academic site using the same scanner family cannot establish performance on a second family. A multi-scanner claim needs the corresponding acquisition and processing evidence. The original scanner result remains positive for its represented images; no result is supplied for an untested scanner. Missing, uninterpretable and deferred slides stay visible in the case flow.

A sequencing study has its own version of this applicability question. Selected high-quality specimens may not represent intended low-input or degraded material. A trial-to-commercial comparison needs the actual people, specimen availability, reference assignment and cases near a treatment-facing cutoff. A deployed scanner, pipeline or knowledgebase change can reopen an affected part of the evidence comparison. None of these conditional examples supplies a new field failure or patient-harm observation.

**Reviewer Lens.** The route into the denominator is more informative than “two sites” or “many slides.” Enrollment, actual scanners, exclusions, reference-positive counts and the allocation of cases to readers reveal what was observed. The opening community

session needs its duration, enrichment rule, training and reader-group analysis before a subgroup estimate can be accepted. The sequencing counterpart needs the intended specimens and boundary cases that entered the study. Missing numbers remain unknown; they cannot become zero exclusions or a favorable community rate.

**Consultant Lens.** A site/scanner/user matrix turns a broad claim into represented and unrepresented conditions. A realistic community-reader workload exercise might answer a gap that more academic slides on the same scanner would leave untouched. A targeted bridge or genuinely narrower scope can be compared against the actual task, evidence and product behavior. The budget should follow the decision-critical missing condition. Funding an experiment and proposing a narrow label establish neither its result nor its approvability.

### SECTION 03

## What did the comparison and reference actually measure?

For an assistive claim, imagine a pathologist making a decision without the tool and, under a controlled comparison, with it. The comparison concerns the reader's clinical result, rather than the model score alone. A **reader study** identifies intended readers, cases, reference result, task and aided/unaided or other relevant conditions. Decisions, time, missing reads and deferred reads are recorded as the claim requires. A queue-timing question might need a workflow comparison instead of that aided/unaided design. The label “artificial intelligence,” or AI, cannot choose the design.

Reading the same case twice creates another possible explanation for a changed answer: memory. The pathologist may remember a difficult image. Planned order and adequate separation between conditions can reduce that bias, while training should resemble the offered instruction without revealing the answer key. Readers also differ in experience, and cases differ in difficulty. When several readers evaluate several cases, observations share both readers and cases. They cannot be treated as independent patients. A **multi-reader multi-case (MRMC)** design and analysis addresses that structure when it matches the intended inference. The opening packet supplies no reader or case count from which to infer such a design.

The interface belongs to the comparison too. If a study and an offered worklist show the same case in different ways or at different times, identical model weights do not guarantee the same reader task. Quality failures, connectivity loss or low confidence can require a visible and usable fallback. The relevant comparison occurs where the person sees and acts on the case. Final diagnostic responsibility cannot remove an earlier product effect on attention or timing.

**Reviewer Lens.** Reader instructions, training, case order, separation, missing reads, endpoint calculation and exact interface identify the actual experiment. The commercial queue, low-confidence state and fallback then have something concrete to be compared with. A time-to-review claim needs an observable clock and a way to learn whether readers can retrieve deferred cases. A positive endpoint sentence gives none of those details by itself, nor does it identify an MRMC sample size or analysis method.

**Consultant Lens.** If visibility is the open question, a realistic workload and timed recovery task can be scoped before prescribing a new diagnostic reader study. A changed user population instead raises reader selection and training questions. Model-only tests remain useful for their input-to-output behavior; the aided human result needs its own evidence. Restoring a studied interface is a conditional option, requiring comparison with the actual claim and released state. It is not a completed correction merely because it is technically possible.

### **Where did the reference answer come from?**

Sensitivity cannot be interpreted without knowing which cases count as positive. Independent expert slide review may suit a tissue finding; another examination, follow-up or a specified combination may suit a sequencing-related clinical state. The claimed condition and study question determine the appropriate reference. Its reviewers' actual information must be recoverable. If experts assign labels while viewing the model overlay being evaluated, the reference may reproduce a model error. Agreement would then provide less independent reassurance. **Blinding** keeps a reference reviewer unaware of the candidate result where appropriate, interrupting that feedback path.

Experts may still disagree. A prespecified **adjudication** process identifies who reviews a disputed label, which extra evidence may be used and which rule produces the final assignment. Adding a convenient vote after inspecting the candidate's disagreements would change that process. Some cases can remain unresolved; their counts, reasons and handling in exclusion or separate analysis remain part of the result. Removing only difficult cases can improve apparent sensitivity and specificity while making the evidence less representative. Reference uncertainty need not invalidate the whole study, but it limits the confidence the device comparison can bear.

The opening packet supplies neither the label-assignment method nor dispute resolution. That is a request for the reference record, not evidence that the reference was invalid. For new work, the consultant can scope a blinded, recoverable process and explicit unresolved-case accounting. A sequencing comparator can also be imperfect near a cutoff. A preplanned investigation of disagreements should preserve the original comparison rather than selectively replacing inconvenient results.

The adjudication record is therefore a traceable decision process, not a claim of perfect truth. Named reviewers, evidence and rules can improve consistency while leaving an unresolved category. Its effect on the denominator still needs to be shown.

## SECTION 04

### What can close while the queue remains open?

The opening primary endpoint was met and average reader sensitivity improved. An applicability review should keep both findings. Neither supplies routine community-reader performance, a second-scanner result or the commercial queue's effect on review time. The packet has one scanner family; the later studio's three-scanner scope belongs to another dossier. Its short enriched community session is limited evidence, not a supplied failed community study. A useful finding names the represented images, users and interface before the unanswered conditions.

Post-market performance follow-up can observe a suitably supported marketed device and its residual questions or emerging changes. It cannot supply the missing core premarket basis for a materially different queue or claimed scanner path. A narrower purpose, interface rollback or targeted new study remains a choice to assess against the actual product, route and claim. None becomes certified or deployed through its appearance in an options memo.

**Additional exhibit — a later simulation of the opening workflow.** The study used a flat case list; the released product hides low-confidence cases in a secondary queue. In a later simulation, **two positive low-score cases exceeded the intended turnaround time**. No simulation denominator, patient outcome or real-world incident rate is supplied. The timing observations and default-hidden behavior warrant workflow assessment, while workload, other causes and the interface's share of the delay remain to be investigated.

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#### Decision

**Can the manufacturer rely on the successful reader study for the offered queue and low-confidence behavior? What remains positive, what does the simulation add, and which missing comparison comes first?**

#### How a reviewer preserves the study while reopening the affected task

A reader study evaluates intended readers carrying out a defined task under stated conditions. An aided/unaided comparison and a timed worklist comparison can answer different questions. The study name does not identify the tested task or interface.

The met primary endpoint and improved average reader sensitivity remain positive for the study's actual readers, cases and flat list. They do not demonstrate the commercial queue's performance when low-confidence cases are less visible. The simulation adds two observed positive low-score cases beyond the intended turnaround. Their full case paths, the timing denominator, default visibility, recovery and consequences now need examination.

Nothing here establishes patient harm, a field rate or the interface as the sole cause of delay. Claim/workflow applicability, use-related risk, configuration and software evidence, labeling and monitoring reopen as appropriate to the changed path. A focused timed workflow evaluation may answer the gap if it observes the actual task under suitable conditions. Model accuracy alone cannot make a delayed case reach the reader.

**Consultant Lens.** The company has three conditional options with different work behind them. Restoring the visible studied worklist requires an equivalence assessment of the resulting product and claim. Testing the priority/suppression path requires representative workload and prospective timing and recovery criteria. Narrowing a priority claim requires the actual use and evidence to support the resulting boundary. Cost and schedule belong beside clinical consequence. A broader reader population or diagnostic task could require a new reader study, but that choice follows the task and endpoint. The reviewer independently judges whether obtained evidence closes the selected gap.

### What exactly would the next record change?

The first row below starts with the two academic centers and one scanner family. It credits the represented acquisition and cases, then identifies the enrollment/scanner record that could establish broader applicability. Each later row follows that same observation→bounded conclusion→next evidence sequence. Proposed work remains visibly distinct from obtained results.

| Supplied opening observation                                       | Bounded conclusion                                                                                                          | Next evidence that could change the finding                                                                                                     |
|--------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Most slides came from two academic centers and one scanner family. | The represented cases and acquisition retain support. Site count cannot establish community or other-scanner applicability. | Eligibility and reference-positive case flow by site, scanner/acquisition conditions, and rationale/data for the additional claimed conditions. |

| Supplied opening observation                                                                                                                     | Bounded conclusion                                                                                          | Next evidence that could change the finding                                                                                                           |
|--------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Primary endpoint met; model detects cases accurately and average reader sensitivity improves. Community readers joined a short enriched session. | The overall positive stands. Routine intended-reader performance and precision remain unspecified.          | Endpoint protocol, reader/case counts, enrichment and training rules, reader-group estimates and analysis of shared readers/cases.                    |
| Priority and low-confidence suppression were disabled in the study but operate commercially.                                                     | Visibility and review-time tasks differ; the original study supplies no comparative timing result for them. | Configuration/workflow comparison, default queues, timed use, visibility, fallback and recovery.                                                      |
| Later simulation found two positive low-score cases beyond the intended turnaround.                                                              | Timing and interface path require investigation; no denominator, harm or field rate is established.         | Full simulation case/attempt flow, clock rule, workload and each delayed path; separate cause, consequences and proportionate protection assessments. |

The third row shows why more model-only accuracy would not answer the missing task. The fourth shows why the two observed delays still cannot become an incidence estimate. Both need their actual case paths, but they ask different questions about applicability and observation. Reference sources, independence, blinding, adjudication and unresolved labels also remain necessary for the first two rows. Calling a label “ground truth” cannot make that process certain. The table is a claim-to-evidence handoff, not a completed performance evaluation report.

The next design is independent of this opening study. Its numbers teach counting and dependence; they provide no missing sample size or performance result for the product above.

### Decision

**Six intended readers each read 80 distinct patient cases once unaided and once aided, with randomized order and an appropriate separation plan. How many reading observations, distinct cases and readers are there? Could a simple independent-binomial confidence interval across all reads establish precision?**



### How the repeated observations affect the inference

6 readers  $\times$  80 cases  $\times$  2 conditions = 960 reading observations. The design still contains **80 distinct patient cases and six readers**, not 960 independent patients. Each reader contributes many decisions, each case is seen by several readers, and the aided/unaided decisions on each reader–case pair are related.

An interval that treats all 960 as independent overstates the available information for the intended comparison. The analysis and uncertainty plan must account for readers, cases, pairing and the target inference. Appropriate order and separation address memory; they do not create new people or cases. No positive/negative split, effect estimate or sample-size sufficiency result is supplied for this design.

Two independent positive records show what a proportionate closure can look like. In the first, a supplied workflow-bridge report covers intended users and case mix, actual queue/default behavior, prospective performance and delay criteria, and the exact unchanged scanner/model configuration. Its sample size and uncertainty assessment are justified, and all criteria are met. After appraisal, that record can close its specified workflow question. Its adequate evidence adds no community subgroup or scanner-extension result to the opening dossier.

The second record concerns a spelling-only display correction with documented absence of an effect on the clinical task. A reasoned impact review can close that narrow change without a full new reader study. The operative fact is the demonstrated absence of task effect. A change in what a reader sees, retrieves or acts upon would remove that premise and require its own assessment. It cannot inherit the spelling correction's conclusion.

## SECTION 05

### What should the reviewer request and management decide?

A clinical finding becomes reviewable by following a case into its reported decision. What condition did the reference process assign, and what did the reader or device report? Which cases entered, failed, left the analysis, waited or remained indeterminate? The scanner, site, user, training and software state locate each result. The endpoint then has to measure the task promised in the label. For the opening packet, that sequence preserves the primary-endpoint success while keeping community-reader, scanner and released-queue questions separate. The aggregate average cannot describe the two delayed simulation paths.

A proposed new reference procedure or clinical study needs its sampling, blinding, adjudication, unresolved labels and uncertainty examined before a narrow estimate becomes a broad conclusion. A changed product calls for a new evidence decision,

which may retain some direct support, use a justified bridge for another part and leave a third part open. It does not require the same testing for every update. The two independent positive records show why the finding's scope matters in both directions.

The consultant makes this comparison usable for management: claim, task, endpoint and criterion, reference, intended users/cases, studied configuration, offered configuration, observation and open consequence, each with an owner and next record. A timed queue simulation, reader expansion, scanner bridge, interface rollback and true claim limitation answer different questions. Costs and development time should be compared with those questions' clinical consequences and uncertainty. A cheap experiment that never observes deferred positive cases cannot settle a delay concern.

For a separately deployed pathology or sequencing product, a complaint, changed workflow or new study can prompt investigation and postmarket evaluation. The actual field counts and chronology matter; the opening simulation cannot become a field incident. Proportionate protection and timely assessment of applicable reporting duties need not await a complete causal explanation. Evidence sufficient to close the claim still needs its own assessment. The consultant can structure these options while the manufacturer retains product and evidence responsibility.

### **Changed-case transfer**

#### **A sequencing treatment result.**

A separate commercial assay reports a variant near a treatment-facing cutoff. Which clinical state and population should the result distinguish? Which trial and commercial assays, cutoffs, reference process and reference limits matter, and how should discordant decisions be investigated? What changes if specimens near the boundary are unavailable?

### **How the clinical-result question transfers**

The record needs the actual paired decisions, their assay/cutoff identities, the reference and the missing-specimen ledger. Unavailable boundary specimens are an applicability limit, not permission to guess concordance. Pipeline-only agreement cannot replace the commercial result's clinical relationship, just as a pathology model score cannot replace the reader-plus-queue outcome.

The reviewer requests the comparison that can address the stated patient decision. The consultant scopes a justified bridge with diagnostic and medicine owners. This packet supplies neither treatment benefit nor a complete companion-diagnostic relationship, and no such conclusion is inferred.

## SECTION 06

**A marketed product has three changed paths**

The following fictional pathology dossier is separately presented. It does not inherit the opening study's two academic centers, short enriched community session or two simulation delays. Here the comparison is between a specialist/one-scanner/flat-list study and a marketed product supporting community readers, three scanners, priority and suppression. "Marketed" is a supplied fact. Deployment extent, conformity status and field results are not. No case count, effect size, scanner-specific result or patient outcome is supplied either.

**Practice case.** A pathology study meets its primary endpoint using specialist readers, one scanner family and a flat case list. The marketed product supports community pathologists and three scanners, with queue prioritization and low-confidence suppression. The company argues that these differences do not matter because the pathologist still makes the final diagnosis. The study's success is positive for its tested conditions; whether it answers the marketed tasks is open.

**What is the product supposed to change?**

If the promise is assistance in detecting a finding, the relevant result belongs to the pathologist using the tool, compared with an appropriate baseline and reference. A priority promise asks whether reference-positive or otherwise high-priority cases reach review within the intended time. Suppressing low-confidence output creates another task: recognizing uncertainty, retrieving the case and following the manual path. These tasks may share cases without sharing an endpoint. The protocol needs an outcome, criterion, comparison and excluded/deferred-case rule for each consequential promise.

A model-only agreement result remains useful for algorithm scoring on the examined images and scanner. It cannot establish whether either a specialist or community reader notices, follows or overrides the score, or sees a hidden case in time. The product changes the information and order presented before the final diagnosis. A benefit claim about faster review or fewer missed findings therefore needs evidence observing that path and its reference-positive cases. No such outcome is supplied here. Human responsibility does not perform the missing experiment.

**Reviewer Lens.** Finding classification, visibility/recovery and time to review need distinct rows. Reference labels and eligible-case flow make each outcome calculable; comparison of study and marketed interfaces locates the changed task. The primary endpoint can keep its bounded positive status while unreported queue outcomes and additional-scanner performance remain open. The same restraint applies when a study includes a favorable overall average but gives no subgroup denominator.

**Consultant Lens.** Defining the changed decision first lets the team compare a timed simulation, paired reader task or scanner bridge. Feasibility work and conditional budgeting can begin while the design is developed. Execution of a new study still depends on prospective criteria, suitable users/cases and workload, the tested interface and applicable permissions. Narrowing or rolling back a marketed claim requires matched product behavior, instructions, evidence and assessment of its actual released state and obligations. Removing a sentence from marketing cannot by itself control use already offered.

### **Do the studied conditions span the offered use?**

The studio supplies three differences. Reader role changes from specialists to a scope including community pathologists, whose workload and access to consultation may differ. Scanner scope grows from one family to three paths that may introduce different focus, color or artifacts. Worklist behavior changes from a flat list to priority and suppression, altering order and sometimes location. These facts call for applicability analysis. They establish no automatic product failure, and the specialist/one-scanner/flat-list success remains creditable.

Case selection needs the same discipline. If positives are enriched for efficiency, the estimate retains its actual positive and negative denominators. Routine prevalence and workload may change the **positive predictive value (PPV)**—the fraction of positive flags that are truly positive under the reference. Disease stage, subtype, specimen quality, comorbidity and demographics matter where they can alter the claimed clinical relation or use. A rationale can explain why a dimension was not separately sampled; silence cannot. Another site using the same scanner and central preparation may add observations without adding scanner diversity. Missing, invalid, hidden and indeterminate cases remain part of the offered path.

The studio's reference standard is unspecified. Its assignment sources, reviewers, exposure to model output, adjudication and unresolved labels need to be recovered. Specialist consensus is not automatically independent truth; an overlay-assisted label can share the candidate's error. A more elaborate reference procedure is valuable only if it answers the intended condition without selectively removing the difficult cases.

**Reviewer Lens.** Counts by scanner, reader role, spectrum and disposition show where the boundary actually lies. The study's training and reference process need to accompany them. Community use and added scanners are not yet demonstrated by this packet; they are not measured failures. Following one low-confidence case from acquisition to final review under both lists identifies the evidence question that a broad “clinical validation” request might miss.

**Consultant Lens.** A three-axis comparison of reader role, scanner path and list state separates direct study support from marketed-product conditions. Some differences may admit a tested scientific bridge; others change the human task itself. Additional

work can be sequenced, but post-market performance follow-up cannot retroactively supply a missing core foundation for the marketed intended-user claim. The company must assess current scope and applicable obligations while resolving the gap, rather than treating a future study as present support.

### **Which change reaches an observable outcome?**

Six parts of a software-assisted service help locate possible effects. In this studio, priority, suppression, scanner mix and intended reader role are supplied differences. Cloud preprocessing, fallback adequacy and training details are questions to characterize. Their presence, change or failure cannot be invented. Each passage below follows a mechanism into prospective evidence and then into a possible observation from use.

**Queue prioritization.** Moving high-score cases upward can leave a low-score positive waiting. A pathologist may recognize that finding correctly when it is eventually seen while the case has already missed the intended review time. A priority study therefore needs the case stream and workload, defined clock start and stop, and time-to-review analysis by reference state and score. Average sensitivity among completed reads cannot supply the elapsed time for cases still waiting. If deployed, timing patterns by site, version and workload could become monitoring signals; none is a reported outcome in this studio.

**Low-confidence suppression.** Withholding a weak score may prevent unwarranted reliance on it. If the default list or recovery route also obscures the case, however, the user needs to recognize the manual-review task, find the case and complete it in time. Failure-state testing with the actual interface can observe that sequence. Hidden/deferred-case logs, overrides, complaints and recovery time would be relevant in use, but are not supplied here. A pathologist's responsibility cannot make an inaccessible case visible.

**Scanner mix.** One scanner family establishes the study's acquisition path. The three-scanner marketed scope adds two paths that may change focus, color or artifacts before the same model receives the image. Scanner-specific comparison and boundary challenges can test whether acquisition, preprocessing and output preserve the clinical task. Scanner B or the third scanner cannot be called failed without results. If used in practice, stratified image-quality, accuracy and artifact trends could show where further investigation is needed. They remain prospective signals in this packet.

**Cloud preprocessing.** Resizing or normalizing an image before inference can alter model input while the weights remain unchanged. Whether this studio product has such a service, or changed it, is unknown. Actual versions and relevant input/output comparisons would come first if that mechanism exists. The consultant can then scope regression and clinical-boundary checks around the changed behavior. A cloud label

cannot settle applicability. Later input shifts, discordance or version-specific trends could prompt investigation without proving cause.

**Fallback.** If an image cannot be processed or the queue service is unavailable, the clinical task still needs a route. Can the user find the case, recognize that no model result is available and complete appropriate manual review in time? Testing the critical task under the claimed degraded condition must observe both availability and understanding. This packet provides no tested fallback to call effective or failed. Fallback use, failures and turnaround would be records to inspect if the product were used in that state.

**Reader training.** Community users need to understand a score, an absent score, a suppressed case and manual recovery. The studio provides neither the specialists' training duration nor evidence of its effectiveness. It therefore cannot establish the adequacy of the community instruction. A representative evaluation can test the actual instructions and retention required for the critical tasks. Support calls, use errors and retraining needs may then inform review where the product is in use. No adequate program or failure count is supplied here.

The table retrieves those mechanisms after they have been explained. For the first row, the outcome is time to review, the evidence needs a realistic stream and workload, and the final column describes a possible future signal. The columns do not convert planned tests or monitoring into achieved results.

| <b>Workflow element</b>         | <b>Claim-specific question and prospective evidence</b>                                                                                               | <b>Possible signal if deployed</b>                              |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| Priority                        | Do reference-positive/high-priority cases reach review in the intended time? Timed high-volume simulation or workflow comparison with full case flow. | Delay by score, site, version and workload.                     |
| Low-confidence suppression      | Can intended users recognize uncertainty and recover hidden/deferred cases? Actual-interface critical-task and failure-state testing.                 | Hidden/deferred cases, overrides, complaints and recovery time. |
| Three-scanner path              | Does every scanner-to-preprocessing-to-model path preserve the clinical result? Scanner-specific comparison and boundary bridge.                      | Stratified accuracy and artifact trends.                        |
| Cloud preprocessing, if present | Has a relevant transformation changed input distribution, score or report? Versioned input/output comparison and regression.                          | Input shift, discordance and version trend.                     |

| <b>Workflow element</b> | <b>Claim-specific question and prospective evidence</b>                                                                                      | <b>Possible signal if deployed</b>        |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| Fallback                | Is manual recovery available, understood and timely when the service or result fails?<br>Degraded-state critical-task testing.               | Fallback use, failures and turnaround.    |
| Reader training         | Do commercial instructions support community users' score, suppression and recovery tasks? Representative training and retention assessment. | Support calls, retraining and use errors. |

The scanner row can be investigated from a supplied scope difference. The cloud row first needs evidence that the service exists and changed relevant input. That distinction prevents every row from becoming a fictional defect. Across the rows, invalid images, deferred cases and indeterminate results cannot become successful negatives by exclusion. Their entered counts and final clinical-answer status belong in the record. Planned follow-up signals and action thresholds can address residual marketed-use uncertainty; they cannot retroactively support this studio's missing core three-scanner/community/hidden-queue evidence.

### **Decision**

**Does the successful specialist, one-scanner, flat-list study support the marketed community, three-scanner, priority-and-suppression purpose? Which positive remains, which task paths changed, and what proportionate evidence or scope option can the company evaluate?**

### **How a reviewer separates study success from marketed breadth**

The met primary endpoint remains positive for the specialist, one-scanner, flat-list conditions. It cannot establish every marketed extension. Community readers, added scanner paths—including the Scanner B question—and changed queue order or hidden cases need their own applicability analysis with reference-positive flow and task-specific endpoints. Preprocessing, fallback and training need characterization where relevant, not a fabricated failure verdict.

The company could assess a narrower specialist configuration using the studied scanner family and visible list, an interface rollback, targeted scanner/workflow bridges or broader reader evidence. Feasibility depends on actual marketed and released state, claim, behavior, adequate results and applicable obligations. None is an easy first-launch path or an already completed correction. Final diagnosis by a pathologist supplies neither timely case visibility nor safety evidence for an untested path. Follow-up cannot replace the missing core foundation.



## Who owns the remaining decision?

The notified body assesses clinical evidence for the stated purpose and complete human–device configuration through the applicable conformity route. Both the positive study and unsupported extension conditions belong in that assessment. The manufacturer defines the purpose, controls the offered interface, selects and executes proportionate evidence work, and assesses actual marketed/use state while maintaining performance. The consultant prepares the comparison, conditional options and cost/consequence account; it cannot grant a certificate or study permission.

A new clinical performance study also raises applicable national competent-authority and ethics requirements, separately from conformity assessment. No country-specific start authorization is supplied here. A medicinal-products authority may have a role in a genuine companion-diagnostic question elsewhere, but the studio establishes none. The decision record must follow the actual product question to the actor entitled to answer it.

### Changed-case transfer

#### Autonomous negative screening.

Suppose the proposed task changes from assisting review of every case to allowing a negative model result to remove a case from immediate human review. Where has the opportunity to notice a missed positive changed? Which user/oversight, endpoint, delay, case-spectrum, uncertainty, fallback, labeling and risk questions need a new answer?

## How the changed task reopens the evidence

The negative result now controls which cases immediately reach a human. That changes the point at which an unnoticed positive could be recovered. The intended user and oversight, reference-positive missed-case endpoint, tolerable delay, case spectrum, sample-size/uncertainty plan and fallback for absent valid scores all need definition for the new path. Labeling and risk controls must describe that actual use.

An assisted-reader study cannot simply be copied into the new purpose. Clinical and regulatory classification, evidence sufficiency and the postmarket plan require their own assessment. No autonomous-product outcome or class is supplied by this hypothetical change.

**SECTION 07****Quick discrimination lab**

These independent records contain neither the opening simulation's two delays nor the studio's three-scanner facts. Each provides its own evidence boundary.

**A proposed user expansion**

A reader study uses expert pathologists at academic centers. The intended purpose also includes community pathologists with different workload and scanner access. No community performance result is supplied.

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**Decision**

**What does the expert study support? Which user/task and scanner questions need answers before extending its result to community practice?**

**How a reviewer keeps the expert finding bounded**

The expert result remains useful for its represented conditions. Community experience, workload, interface, scanner access and fallback may change the complete human–device result. Direct evidence or a justified tested bridge could support the added purpose. The supplied expert-only study establishes neither community success nor failure. Any limitation of the actual claim must fit the offered configuration and evidence.

**A default worklist**

An AI product leaves final diagnosis to a pathologist, while its default list places low-score cases behind a second screen. The packet supplies no count, delayed-case outcome or patient harm.

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**Decision**

**Does final diagnostic responsibility remove the need to evaluate this list? Which timing and recovery evidence could answer the concern without assuming an observed injury?**

**How a reviewer follows the case before diagnosis**

The interface can affect attention and review time before the pathologist reaches a final diagnosis. The relevant evidence observes full case flow under intended workload, especially reference-positive low-score cases, with prospective timing criteria. Default visibility, user recognition, recovery from the second screen and fallback all belong in that comparison.

The supplied design is potentially consequential. It is not a measured patient injury or incidence rate. Evidence about the actual timing and recovery tasks could change its assessment; the fact that a pathologist signs the diagnosis cannot answer those tasks.

**SECTION 08****Which clinical conclusion remains supported?**

Clinical performance belongs to a defined device result, clinical condition or state, population, intended user and use path. The claim and task give the endpoint and reference their purpose; readers, cases and analysis then have a question to answer. A met primary endpoint can remain a real positive while a new scanner, community use, hidden queue or autonomous role needs additional support.

The opening two low-score positive delays are simulation observations requiring workflow and causal assessment, not harm or a field rate. Separately, 960 related reads still involve 80 distinct cases and six readers. A well-evidenced workflow bridge can close its exact task; a documented spelling-only change can close its narrow impact question. Neither adds support to another product's open claim.

The inherited IVDR-CH11 identifiers retain the lookup into Workbook Part A2. The working item states remain A, adequately explained for the item; U, present but unclear; M, missing or inconsistent; D, deeper analysis needed; C, consultation indicated; and N/A, with a reason. An evidence location or identified absence and a reason support each state. These are bounded work findings, not device grades or certification decisions.

**Does the endpoint measure the device's clinical job? • card IVDR-CH11-A01**

Do the assistive or priority claim, prospective endpoint/criterion and reference assignment agree, including unresolved cases? Model-only scores or agreement with another method need not establish the intended task.

**Did the study test the offered use path? • card IVDR-CH11-A02**

Does the comparison trace the flat list into priority, suppression, recovery and software state? A changed task needs an applicable tested bridge or its own evidence decision.

**Which people and conditions are represented? • card IVDR-CH11-A03**

Do site, scanner, reader, subgroup and spectrum counts support the proposed scope, with selection, training and uncertainty explained? The independent 960-read exercise does not contain 960 independent patients.

**What happened to cases without an ordinary completed result? • card IVDR-CH11-A04**

Does every entered case remain visible through reference assignment, device output, queue and final user action? Missing, invalid, deferred and indeterminate states may be exactly the cases that challenge the intended path.

**Can the next reviewer see the supported boundary? • card IVDR-CH11-C01**

Does the clinical/workflow comparison retain studied conditions, reference, endpoint, uncertainty, product differences, consequences and next records? The opening endpoint keeps its credit while its changed queue remains open.

**Which option answers the missing task? • card IVDR-CH11-C02**

For a targeted bridge, reader expansion, interface rollback or real claim limitation, are the task, configuration and criterion explicit? A proposed option supplies neither its results nor an earlier certificate.

**Are clinical consequence and development cost visible together? • card IVDR-CH11-C03**

Does the account explain how a low-score positive could be delayed, what is observed versus unknown, and the cost/time of resolving the gap? A cheaper path cannot support a task it never tests.

**Further reading**

Book 1C Chapters 12 and 15–16 and Book 1D Chapter 13 offer optional practice with clinical studies, readers and human–AI use. They add no observations to the local pathology dossiers or independent six-reader exercise.

## Source trail and limits

- **Binding law, checked 2026-09-25:** IVDR consolidated 10 January 2025, Article 2(37), (39)–(42), Article 56, Annex I §9.1(b), Annex XIII Part A §§1, 1.2.3 and 2.3.2, Part B §§4–5. These connect clinical performance and clinical evidence to the stated purpose, target population and intended user; study plans identify endpoints, population and device conditions; PMPF concerns a placed-on-market device within its intended purpose. They do not set an endpoint, scanner-specific result or AI Act status for the fictional product.
- **Nonbinding method/context:** Medical Device Coordination Group (MDCG) 2022-2, especially clinical-evidence and intended-use principles, helps explain the IVDR's continuous evaluation; it is not an independent approval route. MDCG 2019-11 rev.1 is earlier software-qualification context, not a Chapter 10 classification finding. The United States Food and Drug Administration (FDA) computer-assisted-detection reader-study guidance and FDA multi-reader multi-case research tool inform the general reader/case-dependence explanation, not an EU legal requirement or a mandated design for this DP product. AI Act interplay or applicability is deliberately not inferred from this case; that source question belongs to Chapter 14 if relevant.
- **Packet limits:** Opening DP study: two academic centers, one scanner family, short enriched community session, met primary endpoint and improved average reader sensitivity. Its two delayed positive low-score cases arise only in a later simulation, with no denominator, harm or field incident rate. The separate specialist/one-scanner/flat-list studio describes a marketed product supporting community readers, three scanners, prioritization and suppression; it does not add details to the opening study or supply field results. The six-reader/80-case design, stipulated favorable workflow bridge, spelling-only change and two microcases are independent teaching records. No source figure or artwork is used.

## CHAPTER 11

# From evidence plan to a defensible performance conclusion

**Retrieval check — before reading.** Chapter 10 kept the reader study's passed endpoint beside its unresolved worklist question. Which user, task, configuration, reference process and endpoint did the study represent? What changes when the offered worklist gives the same cases a different path to review?

**Which parts of the offered claim does the evidence support now, which need a justified limit, and which need more evidence before they can be included?**

A useful conclusion allows the next person to see both its basis and its boundary. The work begins with a planned question, continues through appraisal of what was actually found, and returns to risk, instructions and new observations when the product or its clinical setting changes.

## What the evidence report must decide

Imagine a laboratory result offered to inform treatment. Before judging its evidence, we need to know what it reports, from which specimen, for whom and for which decision. A **biomarker**, or marker, is a measurable sign in the specimen. Its relationship to the clinical condition or treatment state is one question; this device's ability to measure it is another; the reported result's relationship in the intended population and use is a third. In digital pathology, a feature can matter clinically and software can identify it while the pathologist's worklist task still needs evidence. A conclusion about one link cannot silently close the others.

Those questions already have a place in the **performance evaluation plan (PEP)**: the methods and criteria that guide the evaluation. The **performance evaluation report (PER)** takes the next step. It appraises the available evidence and explains what each claim can support, what remains uncertain and what new information would change the conclusion. A particular clinical study keeps its own protocol and results. The device-wide plan and appraisal connect that study to the claim; they cannot stand in for its record.

**Opening record — an impressive-looking report.** A fictional PER has **380 references**, detailed tables and polished summaries. It calls the device state of the art and clinically beneficial. Yet another reviewer cannot reproduce the search,

contradictions are listed without appraisal, comparator performance comes from manufacturer brochures, and the conclusion never joins each intended-purpose element to its evidence and remaining uncertainty. No device type, study denominator or additional hidden result is supplied.

What would turn those references into a conclusion? The sources may contain useful evidence, but the reader needs to see why they form the relevant set and how each bears on a promise. In a separate sequencing illustration, even sound marker-association literature may say nothing about the commercial assay's **decision threshold**—the value separating two reported categories. In a separate pathology illustration, image-classification literature may leave a community pathologist's task with a hidden secondary queue untested. Neither missing connection is repaired by increasing the reference count.

### **Which sources answer the promised question?**

Consider a proposed tissue marker result intended to inform use of Drug A. Searching for “oncology biomarkers” would locate a field, while the decision needs evidence about the particular marker, population, treatment role and specimen. Databases and other sources, exact search terms, dates, eligibility rules and exclusion reasons make the selection reproducible. The screened set, extracted information and review decisions then let a second qualified person retrace how the evidence relied on emerged from the records located.

Selection leads to the next question: what does each report establish? Two publications may describe the same cohort, so they cannot count as independent confirmation. A prognosis study may describe what happens regardless of treatment, while a treatment-selection claim needs evidence about who benefits more from one treatment than an alternative. An unfavourable report with missing follow-up needs appraisal of both its limitations and its possible effect on the conclusion. An appendix entry supplies neither. A dated search also needs an update rule so that later contradictory or better evidence can change the assessment.

For pathology, evidence that a disease feature matters can justify trying to find it. It still leaves the specific scanner, model, reader and worklist to be examined. Tracing source selection and cohort identity preserves the literature's actual contribution while exposing that device question. A consultant can use the same trace to show which conclusion each source changes. Reference counts then describe the library; they do not vote on the claim.



## Ask what current practice makes the result worth using

Once the relationship is understood, what makes the proposed result worth using now? **State of the art** is the current clinical and technical context for that comparison. For the tissue marker, it includes the test or clinical decision that would otherwise be used, how often it yields no answer and what a wrong or late treatment-facing result could do. A pathology aid promising better recognition asks a different benefit question from one promising earlier review. The relevant baseline and acceptance criteria depend on the promised contribution.

Clinical guidelines, independent studies, actual comparator performance, limitations and unmet needs help establish that baseline. A brochure may describe a competitor's claim; by itself it supplies no independently appraised performance estimate. If another method becomes routine after launch, an unchanged device may need its comparator, claimed benefit, risk balance, instructions or follow-up questions reconsidered. Stable code does not freeze the clinical comparison.

**Reviewer Lens — test the comparison.** The opening record's brochures can be credited for describing claims while leaving performance estimates open. Clinical-pathway information, independent comparator evidence and the rationale for each criterion would resolve that difference. For a deployed pathology aid, the useful comparison is the current review pathway against the one assumed in the PER. The model version alone would miss a change in care.

**Consultant Lens — turn context into choices.** A narrower role or different endpoint may change the evidence required because it changes what the product promises to add. That is a concrete management comparison. For a deployed sequencing test, a dated watch for therapy guidance and competing methods can identify which PER and instruction passages to reopen. It is a plan to detect change, with no supplied finding that this fictional device has been displaced.

## Connect the different kinds of evidence

The comparison now needs the evidence connected in the order of the medical task. **Scientific validity** addresses the marker or feature's relationship to a clinical condition or state. **Analytical performance** addresses the device's detection or measurement under the relevant specimen and operating conditions. **Clinical performance** addresses the device result's correlation with the relevant condition or state for the intended population and user. Clinical benefit and safety then concern what that result contributes and what wrong, absent or late information could do. A usable interface supports the path, but does not by itself prove the result's clinical relationship.

One study may contribute to more than one link. Its tissue specimens could support both measurement agreement and correlation with a clinical state. Those are two uses of the same observations, not two independent populations. The PER needs the

reference method, selection, uncertainty and limitations made visible, then a separate explanation of what the study contributes to each claim. Likewise, one pathology reader study can measure diagnosis and timing on the same cases. Each outcome still needs its own definition and analysis. Changing the question does not create additional patients.

**Reviewer Lens — follow a single cohort.** Cohort identity is often the quickest test of apparent corroboration. Which people or specimens underlie the scientific, analytical and clinical reports, and are they reused in another publication? That record lets the reviewer judge the total evidence without treating repeated reporting as independent support. Conflicting observations and missing or invalid results remain part of the conclusion, even when the same data serve several purposes.

**Consultant Lens — connect the claim to its sources.** Putting each study beside the population, specimen, user and decision it supports reveals where an additional connection must be tested. A proposed bridge stays a question and work item until its evidence exists. Management can compare its cost with a genuinely narrower product claim, but neither option comes with an assured passing conclusion.

### **Write the conclusion at the level of the claim**

A product-wide “sufficient” label would conceal these differences. Four plain communication choices can make them visible: proceed on the stated evidence; proceed with a justified limitation and defined follow-up; narrow the proposal to an actually supported configuration or use; or obtain more evidence before putting forward the affected claim. These are **author-created communication dispositions**, not categories under the **In Vitro Diagnostic Medical Devices Regulation (IVDR)**, notified-body decisions or certificate amendments. For a restriction to mean anything, the remaining product, instructions, controls and evidence must actually fit it.

Which choice fits depends on where the uncertainty sits. A bounded residual question within an otherwise supported use may suit focused follow-up. A missing measurement boundary that can reverse treatment information first needs evidence or a justified control that adequately bounds that consequence. Restricting an unrelated part of the claim cannot do this work. For pathology, specialist-reader evidence with a visible list may remain useful while community use and a hidden queue need direct evidence or a tested bridge. The report needs to say which connection is supported and which is still a proposal.

## Let new evidence reopen the conclusion

A supported conclusion still has to survive actual use. **Post-market surveillance (PMS)** is the broader system for collecting and analysing experience, complaints, performance signals and relevant changes. **Post-market performance follow-up (PMPF)** is its planned, proactive performance-evidence work within the supported intended purpose: focused questions about continued performance and justified residual uncertainty. Findings return to the PER, risk, labeling and change control. Until collected and appraised, the planned observations remain a plan.

A sequencing service could track first-attempt invalid results by version and site instead of counting only eventual successful reports. A pathology service could track whether reference-positive cases in a supported queue reach review in the expected time. These are possible monitoring designs, with no new outcome attributed to an earlier case. If observations cross a defined action rule, investigation asks what happened and whether immediate protection, applicable reporting or a changed benefit–risk conclusion is needed. The rule can trigger that work without establishing its outcome.

IVDR Article 56(6) makes continuing appraisal a lifecycle requirement: performance evaluation and its documentation are updated using PMPF and PMS data. For class C and D devices, the PER is updated when necessary and at least annually. A relevant new observation can therefore make an update necessary before the annual date. Scheduled review and event-driven reopening need both a place in the quality process and a versioned link to the product in use. The calendar is a minimum, not evidence that an outdated conclusion remains sound.

**Additional exhibit — same opening PER.** The broad intended purpose includes **community settings**, but almost all clinical evidence comes from **expert centers**. The PMPF plan promises only to collect user feedback. This adds an intended-user applicability question to the search and claim-map defects already seen; it supplies no community result, follow-up metric or finding of harm.

What must the completed appraisal make possible? **Clinical evidence**, under IVDR Article 2(36), is device-related clinical data and performance-evaluation results of sufficient amount and quality to support a qualified assessment of safety and intended clinical benefit in intended use. This is why an untested intended-use condition or evidence that cannot be appraised matters even in a thick file. The decision requires relevant support, not volume alone.

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## Decision

**What contribution can the 380-reference file make despite its irreproducible search and missing appraisal? Which records would resolve the new community-use question before the PER's conclusion can be accepted?**

### How a reviewer turns the library into a claim-level appraisal

The volume cannot compensate for the missing method. A reproducible search, appraisal of favourable and contradictory findings, and identification of overlapping cohorts establish what evidence is actually available. Each claim then needs its direct support, any bridge backed by evidence, any bridge still to be tested, limitation and residual uncertainty made explicit.

For the community purpose, actual user/workflow evidence or a genuinely supported restriction is needed. Generic feedback collection does not define or close the missing core question. A focused PMPF plan needs objective, data source, denominator, review rule, owner and action within a supported use; it cannot retroactively establish the initial basis for an unsupported one.

**Reviewer Lens — hold the conclusion to its evidence.** Missing records, weak methods and unsupported intended-use conditions may call for different repairs. The search protocol and screening log address selection; contradiction appraisal addresses interpretation; the claim table and community-user evidence address applicability. Separating those findings makes the next request answerable. The packet does not justify calling the whole device a failed product.

**Consultant Lens — make the next decision concrete.** Management can compare repairing the review method, generating targeted user evidence or a scientifically justified bridge, and a real scope restriction with product and labeling consequences. Feasibility funding and conditional budgets can prepare that choice. Executing a new study still depends on prospective criteria and any applicable permissions. More references alone cannot fill the user gap, and a proposed restriction supplies no certificate.

### A supported use can still have a focused follow-up question

A different record shows when follow-up is a justified next step. **Assay R** is an independent fictional example with an expressly adequate restricted premarket foundation: its commercial configuration and cutoff are covered, relevant users and conditions are represented, uncertainty is justified, and its benefit–risk assessment supports the restricted purpose. It is neither the opening PER nor the later tissue/Drug A studio. Its remaining question is whether known first-attempt invalid-result behavior stays stable in routine use. No missing high-risk group or new specimen is being hidden in that question.

To observe stability, what should be counted? The initial unit is an **initial specimen attempt**. A specimen that first produces no valid result and later succeeds on repeat has two facts worth keeping: the original interruption and the eventual answer. The plan therefore separates initial invalids among all initial attempts from final specimen results and repeat effort. Treating the repeat as another independent person would distort one

denominator; dropping the initial invalid would erase the event the follow-up is meant to detect.

The table below connects that unit to an action. Its internal rule is stipulated for this fictional program: more than five initial invalid results in a complete rolling window of 100 initial specimen attempts opens investigation. It is neither an IVDR limit nor a ready-made statistical trend criterion. The measure row determines what reaches the trigger row; the owner row determines what happens next.

| <b>Assay R record</b> | <b>Bounded plan and action</b>                                                                                                                                                                                                                        |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Question and capture  | Is known first-attempt invalid behavior stable in routine use at participating sites? Record every initial specimen attempt, eventual result, repeat effort, configuration, site, relevant handling conditions and missing records.                   |
| Measure               | Initial invalid results ÷ all initial specimen attempts, examined by configuration and site. Final specimen yield and repeat attempts remain separate.                                                                                                |
| Internal action rule  | More than five initial invalids in each complete rolling window of 100 initial specimen attempts opens investigation. This does not decide cause or legal reporting.                                                                                  |
| Owner and response    | PMS lead reviews incoming events and each complete window. Quality and clinical leads investigate a trigger, assess proportionate immediate controls and applicable reporting on actual facts, then update risk, labeling, PER and PMPF as indicated. |
| Missing capture       | If unsuccessful attempts are absent from the log, the rate is uninterpretable until ascertainment is restored. No complaints cannot supply missing attempts.                                                                                          |

Follow an initially invalid specimen through the first two rows. Its eventual answer belongs in the final-result record; the initial invalid remains in the initial-attempt measure. That is the value compared with the action rule. If the capture row is incomplete, even an apparently reassuring rate cannot be interpreted. A complete ledger, rather than a complaint count, is the fact needed to repair that branch.

A pathology counterpart would first specify the supported worklist event and where its clock starts and stops. Eventual review cannot make a late case on time. That comparison explains why the original event must remain counted; it supplies no delay result for assay R or the studio.

**Decision**

In assay R's next complete window of 100 initial specimen attempts, six initial attempts are invalid. All six eventually give a valid result after repeat. Does the internal trigger fire, and what can be concluded now?

**How the initial-attempt rule treats successful repeats**

Yes:  $6/100 = 6\%$ , and six is more than five. All six original invalids remain in the measure even though the later valid answers also deserve to be recorded. The stipulated action is investigation, with clinical consequences, immediate protection and applicable reporting assessed from the actual facts.

This result establishes no cause, patient injury, device-wide unsafe state or legal reporting threshold. It also does not remove the example's adequately supported restricted premarket premise. That premise remains assay R's own and cannot close another product's gap.

**Reviewer Lens – four records to open together**

The search/appraisal record, current clinical comparison, purpose map and focused follow-up objectives answer connected but different questions. An unfavourable report needs to be weighed, an extension needs evidence or a defensible limit, and a residual question needs an observable measure and action. For assay R, the initial-attempt ledger is the record behind the trigger; final yield cannot substitute. A precise request can start timely investigation and duty assessment without inventing a cause or reportability finding.

**Consultant Lens – make the report usable for a company decision**

A short executive conclusion becomes useful when each claim retains its exact configuration, strongest support, contrary evidence, uncertainty, medical consequence and next action. Owners, budget assumptions and decision dates can then attach to real evidence or scope choices. Planning funds differ from study-start permission. A specific PMPF question differs from a promise of “more data.” Keeping those distinctions visible lets management act without overstating the report's conclusion.

**Changed-case transfer****Care changes after launch.**

A sequencing assay or pathology aid is technically unchanged, but the clinical standard of care changes. Which scientific relationship, comparator, benefit, consequence, labeling and follow-up questions should be reconsidered?

### How a stable device can acquire a new comparison

The marker or feature may need its claimed relationship reappraised in the new context. A different routine method may change the comparator and value of the promised result; wrong or delayed information may now have a different consequence. Dated practice evidence connects those changes to the PER, risk, labeling and PMPF objectives. The manufacturer needs a conditional update based on that comparison, not an assumption that unchanged software freezes medical practice.

## How far can the strongest evidence carry a broader proposal?

The next fictional company is separate from both the 380-reference record and assay R. It has useful completed studies and a plan written afterward. That timing creates a history to reconstruct, rather than a reason to erase the studies. The harder question is how far their actual scope reaches into the company's broader proposal.

**Practice case — a broader purpose than the studies.** The company writes its PEP after its studies pass and declares no gaps. Its proposed purpose includes a tissue biomarker, a liquid-biopsy specimen, community laboratories and a second therapy relationship. The studies cover tissue, expert laboratories and the first therapy. Hundreds of PER references never become separate findings about supported, conditional or unsupported claim elements. No product name, actual therapy identity, study size, approval or launch is supplied.

Follow what changes between those two descriptions. A body-fluid-derived specimen brings different material from tissue; the exact fluid and processing path remain unspecified. A community laboratory may prepare, repeat and report differently from an expert site. The second therapy creates a new clinical relationship even if the measured target is unchanged. Each boundary can affect a different step from specimen to treatment information. One strong result will therefore need several separate applicability judgments.

### Which decisions must precede the next study?

For the tissue/Drug A proposal, the next study needs a defined commercial result before it needs a contents page. Which marker and treatment-facing role, tissue specimen, users, assay and cutoff, population and reference process are intended? What rule would make near-cutoff discordant results acceptable or require more work? The PEP connects these questions to the evaluation. A clinical study protocol turns its part into actual sampling, methods, endpoints and analysis. Neither document proves that its criteria were met.



The blueprint below is useful before evidence is judged. Its first row fixes the product and promise. Its last concerns only residual questions after a sufficient foundation; the rows between show what would establish that foundation. Reading the ends together prevents “continued evaluation” from becoming a destination for every unfinished claim.

| <b>Planning question</b> | <b>What must become answerable?</b>                                                                           | <b>Record supporting the decision</b>                                        |
|--------------------------|---------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| Product and purpose      | Which configuration, marker/result, tissue or liquid specimen, population, user and treatment role?           | Controlled claim and product-identity map.                                   |
| Clinical contribution    | What decision could improve compared with current practice, and what wrong or absent result matters?          | Benefit and risk questions tied to the clinical pathway.                     |
| Scientific relationship  | Which marker–condition or marker–therapy relationship, including contrary evidence?                           | Search and appraisal method, criteria and update rule.                       |
| Device measurement       | Which specimen, platform, decision boundary and difficult sample conditions must the commercial assay handle? | Analytical study matrix with prospective acceptance criteria.                |
| Clinical result          | Which population and user workflow, and what relation to the intended clinical state or treatment decision?   | Study/literature strategy, reference, endpoints and representativeness plan. |
| Sufficiency              | How will quality, cohort overlap, contradictions, uncertainty and each claim be judged?                       | Prespecified decision and gap-handling rules.                                |
| Continued evaluation     | Which bounded questions remain after support for the use is established?                                      | PMPF objective, data source, measure, review rule, owner and response.       |

Take the tissue cutoff through the middle rows. The clinical contribution identifies why crossing it matters; the measurement row identifies the assay behavior to examine; the clinical-result row connects the reported category to its intended decision. The sufficiency row must explain how much uncertainty remains acceptable. Only then can the final row identify what is genuinely residual. This is a planning path for the studio, with no new completed result implied.

Completed work can enter that path with its original protocol, criteria, dates and result-access history intact. Appraisal asks whether later choices could bias the current

conclusion. A current PEP can guide remaining work prospectively while identifying reconstructed decisions as reconstructed. Feasibility funding and conditional estimates can prepare a new study; execution still waits for the necessary protocol, criteria and applicable permissions. A new date cannot make a cutoff rule prospective for favourable results already seen.

**Reviewer Lens — inspect the chronology.** Original protocols and criteria, dated PEP revisions, data-access records and the gap log distinguish what was decided before results from what was added later. Liquid, community and Drug B questions may have different histories. A polished current table can organize those histories, but cannot establish them merely by containing the questions now.

**Consultant Lens — make the next decision affordable.** Separate tasks and execution gates let management fund feasibility without treating the proposed claim as ready. Useful tissue and expert-laboratory evidence remains in the plan. The budget concerns what must be learned about each unresolved boundary and the consequence of claiming it before that evidence exists.

### **Appraise the clinical relationship itself**

The strongest row begins with the Drug A tissue relationship. Does the marker describe prognosis regardless of treatment, or identify who may benefit from Drug A compared with an alternative? The studio supplies pivotal and replicated evidence for an established Drug A association. That is a real positive, whose reach still depends on population, comparison and contradictory reports. The same measured target does not create an association with Drug B. A second therapy changes the scientific question even if the assay operation stays fixed.

The next question concerns the device producing the result. A pivotal treatment study might have used another assay. Its scientific contribution can remain strong while the commercial assay's assignment of the same patients across a treatment-facing boundary remains unresolved. Naming both methods and appraising their bridge preserves that distinction. Pathology has an analogous separation: a disease-associated image feature does not establish that the released software and worklist convey it to the intended reader in time.

**Reviewer Lens — follow the therapy evidence.** The pivotal and replicated studies need their assays, cutoffs, populations and overlapping cohorts identified. An appraisal of unfavourable findings belongs beside the commercial bridge, especially where methods could put a specimen on different sides of the treatment-facing boundary. General agreement cannot resolve a particular missing comparison merely because the association evidence is strong.

**Consultant Lens — state what transfers.** Management needs the established Drug A association, the commercial measurement question and the Drug B relationship

question kept separate. A shared target may allow reuse of analytical information; it does not supply the second predictive relationship. A proposed bridge is a costed task until evidence allows an appraisal of its result.

### **Build one conclusion for each claim element**

The four claim elements below belong to one product-strategy problem. Their dispositions are communications about evidence, not four approvals. Beginning with tissue tests the strongest case first: what if the remaining gap lies exactly where the output can change a treatment decision? Specimen, user and second-therapy changes then test different parts of the same reasoning.

#### **Tissue marker for Drug A**

For tissue and Drug A, the packet supplies pivotal and replicated scientific evidence, representation of the relevant clinical population, and an adequate commercial-assay bridge away from the cutoff edge. At the edge, discordance data are sparse. The cutoff converts measurement into a reported category. A near-boundary specimen placed on opposite sides by trial and commercial methods can therefore change treatment information even when most specimens agree. No discordance count or patient given the wrong treatment is supplied; the clinical consequence remains insufficiently bounded.

Could a control make that boundary defensible? A **guard band** would define a justified region around the cutoff in which uncertainty leads to a specified confirmation or indeterminate-reporting route, rather than a confident category. Evidence would have to show that the rule controls wrong decisions while addressing the consequences of no result or delay. The packet supplies no validated guard band, reliable exclusion rule or other control that establishes this.

The tissue association and represented population still deserve credit, but the affected treatment-selection conclusion remains open. Near-boundary paired results, discrepancy analysis and justified clinical-consequence assessment could change that finding. A genuinely defensible restriction may also be examined on its own evidence. Limiting users to experts would address a different question; it cannot by itself fix a measurement edge. Neither that restriction nor PMPF supplies a tissue-first go-ahead here.

## Liquid-biopsy specimen

Changing to a body-fluid-derived specimen changes what reaches the assay. Amount and form of target material can differ from tissue. A scarce target near the **limit of detection (LoD)**—the low amount at which detection performance is characterized—may be missed even when a tissue sample would be called positive. Other fluid constituents may interfere. These are mechanisms requiring examination, with no particular fluid or observed failure supplied. The studio has limited low-input, detection-limit and interference data and no direct clinical evidence in the intended liquid use.

The Drug A association may still contribute if the biology and population justify its relevance. Appraising that possibility avoids discarding useful tissue evidence. It also exposes what the association cannot show: the liquid result's own analytical and clinical performance. A **false-negative result**, reporting no target when the relevant target is present, could alter treatment information; no such event is reported here. On this packet, the liquid claim should be excluded from the current certification proposal. A later proposal needs a supported specimen path and actual evidence for its intended use.

## Community-laboratory user

The community-user row changes another link. The association is unchanged and the core assay is described as stable, while clinical evidence comes from expert laboratories. Specimen preparation, recognition of invalids, repeat work and communication of delayed or absent answers may differ in community practice. The packet raises invalid/repeat and delay risks without a field rate or demonstrated community failure. An expert-network restriction could be assessed if product, training, labeling, distribution and evidence truly support it. That would bound user scope; it would still leave the separate tissue cutoff question open.

For pathology, the corresponding comparison concerns intended community readers, scanner paths and worklist behavior, rather than whether specialists recognize the same feature. Chapter 10's marketed studio product is a separate case. Its history and results supply no laboratory-user evidence here.

## Drug B relationship

The Drug B row changes the treatment relationship. Same-target detection may remain useful analytical information, but the scientific work for Drug B is early and the treatment-selection relationship immature. Partner labeling and outcome evidence also remain open. Those facts call for generating and appraising evidence before putting forward the separate therapy claim. No medicinal authority's position or therapeutic outcome is supplied.

The table now collects those four explanations for a claim decision. Read the tissue row across before comparing rows: the positive association and represented population remain visible beside the sparse, consequential cutoff gap. Each later row changes a different link; their missing evidence cannot be supplied by the strongest row.

| <b>Propose<br/>d claim</b>     | <b>Support that remains<br/>useful</b>                                                                        | <b>Unresolved<br/>consequential<br/>question</b>                                                            | <b>Present bounded<br/>conclusion</b>                                                                                                   |
|--------------------------------|---------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Tissue<br>marker /<br>Drug A   | Pivotal and replicated<br>association, represented<br>population, commercial<br>bridge away from the<br>edge. | Sparse near-cutoff<br>discordance; no<br>supplied control<br>bounds the<br>treatment-facing<br>consequence. | Hold the affected conclusion<br>until the edge is resolved or<br>adequately bounded;<br>appraise a real restriction on<br>its evidence. |
| Liquid<br>specimen             | Tissue association may<br>remain relevant after<br>transfer appraisal.                                        | Limited low-input/dete<br>ction/interference<br>work; no direct<br>intended-use clinical<br>evidence.       | Exclude from the current<br>certification proposal; plan<br>specific evidence without<br>automatic transfer or discard.                 |
| Commun<br>ity labora<br>tories | Unchanged association<br>and stable core assay.                                                               | Unrepresented<br>operational workflow;<br>invalid/repeat and<br>delay consequences.                         | Assess an actual<br>expert-network restriction or<br>targeted user evidence;<br>neither closes the tissue<br>cutoff.                    |
| Drug B                         | Same-target<br>measurement may<br>contribute analytically;<br>early scientific evidence<br>exists.            | Immature treatment<br>relationship, partner<br>labeling and outcome<br>evidence.                            | Generate and appraise<br>evidence before the separate<br>therapy claim.                                                                 |

The community row illustrates why narrowing is specific. Evidence and distribution controls might make an expert-only user boundary real. The tissue row would nevertheless retain its measurement question. Likewise, evidence that establishes liquid detection would not automatically establish Drug B's treatment relationship. A new result changes the conclusion for the connection it actually supports.

**Reviewer Lens — challenge the strongest row first.** The cutoff-edge ledger, reference process, paired decisions and clinical-consequence analysis test whether the seemingly small tissue exception is actually bounded. Liquid and community denominators and Drug B scientific sources then answer their separate questions. The expert-user option cannot shortcut the edge inquiry. Strong association evidence should be retained without being made to answer a different question.

**Consultant Lens — make the executive conclusion precise.** Management can credit the established tissue/Drug A association and represented population while seeing that the commercial cutoff still leaves treatment selection open. Liquid development, community-workflow evidence and the Drug B relationship need separate cost and evidence decisions. A staged or restricted scope is conditional on the actual product, adequate evidence and conformity assessment; a tissue-first certificate is not the supplied outcome.

### **Decide what can be followed after a supported claim**

Which of these gaps is genuinely a follow-up question? Calling an unresolved cutoff, liquid use or Drug B relationship “PMPF” does not establish its initial support. Within a supported intended use, PMPF can confirm performance, address justified residual questions and detect emerging risk or changed practice. The plan needs a specific question, user and population, observation capture, counted unit and denominator, analysis and action rule, review interval, owner and return path to PER and risk. Generic feedback alone cannot measure a particular low-prevalence group, scanner problem or near-cutoff discrepancy. The method must observe the question being asked.

The document roles now form a connected sequence. The PEP specifies needed evidence and how it will be judged; the PER appraises what exists and fixes the supported claim boundary. Once marketed, the PMS plan looks for experience and signals, while the PMPF plan proactively examines specified performance questions. A changed question can require a changed method; repeating the document titles does not supply the connection.

What records the follow-up's answer? The **PMPF evaluation report** contains its actual targeted findings and returns their conclusions to the PER and risk file. A different record serves the broader surveillance question. For class C and D devices, IVDR Article 81 requires a **periodic safety update report (PSUR)** summarizing PMS analyses and actions, including the benefit–risk conclusion and main PMPF findings; it is updated at least annually. Its broader account does not replace the claim-specific PER. The class condition determines this PSUR duty; a generic reference to “follow-up” cannot decide the reporting record.

A new observation may change the PER, risk control, labeling or configuration, which can in turn change what future surveillance must watch. The records need to share the same product state and uncertainty. Cross-referencing their titles without that agreement would leave the decision loop incomplete.

Assay R showed a supported-use question and its six-in-100 action trigger. Its adequate restricted foundation does not transfer to the studio. A deployed pathology aid could similarly monitor timing and complaints for a supported worklist, provided its initial reader-plus-product evidence supports the role. New complaints or adverse literature

may require prompt investigation and proportionate protection or reporting assessment while cause remains unresolved. The facts from that work are needed before the evidence conclusion can close.

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### **Decision**

**Can the studio's late PEP be used when it accurately describes strong completed studies? What must the PER disclose about timing and the four proposed claims, and when could a proposed bridge support a current conclusion?**

#### **How useful completed work and an open claim coexist**

The current plan can organize honest appraisal and guide future work. It cannot establish that criteria preceded results or erase the tissue-cutoff, liquid, community and Drug B gaps. Original protocols, dates and criteria remain; reconstructed choices and changes made with knowledge of results need their possible bias assessed.

The PER distinguishes direct support, a bridge backed by appraised evidence, a genuine limitation and justified residual follow-up. A scientifically plausible bridge remains in the work plan until its necessary evidence is obtained and appraised. If it concerns a decision-critical gap, that claim remains open meanwhile. The strong tissue/Drug A association is useful evidence, without making the affected treatment-selection claim ready on its own.

### **Who owns which decision?**

The manufacturer defines the purpose, maintains truthful plans and reports, funds and controls studies, assesses risk and acts on marketed-product signals. Under the applicable route, a notified body assesses the evidence and reasoning; an internal disposition or narrower label is not its certificate decision. Any required national competent-authority or ethics process for a new study remains separate from the PEP and PER. No country, authorization or medicines-agency conclusion is supplied here. A consultant can assemble the claim map, reproducible appraisal, sufficiency table and conditional cost/consequence choices, while those responsibilities remain with their actual owners.

#### **Changed-case transfer**

**Add liquid biopsy.**



For a sequencing treatment-information proposal, which tissue association evidence might remain relevant to a new body-fluid specimen and population? Which extraction, low-input detection, interference, cutoff agreement, false-negative and final-result questions need an explicit bridge or new evidence? Which questions could only become residual after initial support is established?

### How a bridge earns a place in the conclusion

The marker–condition or marker–therapy relationship is appraised for actual relevance to the new specimen and population. It is neither inherited automatically nor discarded merely because the material changes. The liquid pathway needs its own measurement and clinical-result questions answered, including the consequence of an absent or wrong result. Old studies retain their supported scope; a proposed bridge remains work until its evidence is obtained and appraised. Core missing support requires a premarket disposition, while PMPF is reserved for genuinely residual questions within the supported use.

## Quick discrimination lab

The two records below are independent of the opening PER, assay R and the studio. Each keeps its own evidence and unknowns.

**Record A — a broad liquid claim.** A company lacks clinical evidence for a broad liquid-biopsy claim and proposes to collect it after certification through routine PMPF.

### Decision

**Can routine follow-up replace this missing core clinical evidence? What present claim-level option would preserve the evidence boundary?**

### How the present claim determines the next step

No. Justified residual follow-up presupposes an adequate foundation for the intended use. A missing clinical relationship that controls safety or performance needs evidence before the affected claim can be put forward, or a genuinely supported narrower proposal. Either option requires its actual product and evidence to be assessed; neither is automatically accepted.

**Record B — an overlapping library.** A PER counts 18 studies as sufficient. Several reuse the same patients; none represents the community-user workflow.

**Decision**

**What can the 18-report count establish, and what remains necessary before concluding that community use is supported?**

**How a library count differs from independent support**

The count describes reports, while several share patients and none observes the community workflow. It therefore establishes no sufficient basis for that claim. Cohort identity, quality, contrary data, population and user relevance, uncertainty and precise coverage need appraisal. The reports may still answer narrower questions. Counting the same patients again cannot supply independent confirmation or a missing user task.

**Carry the conclusion forward**

A PER says what the total evidence supports now for the identified device, configuration and use. Its positives remain bounded to their evidence, and a consequential gap must be resolved or adequately controlled before the affected claim is put forward. PMS and justified PMPF then examine supported use, returning new findings to risk, labeling, changes and the next appraisal. For class C and D, an annual PER review is a minimum; a consequential signal may make earlier updating necessary.

**Check what you just learned**

The working states describe bounded items beside evidence locations and reasons: **A**, adequately explained; **U**, present but unclear; **M**, missing or inconsistent; **D**, deeper analysis needed; **C**, consultation indicated; **N/A**, inapplicable with a reason. They are neither product grades nor approval decisions. The seven original identifiers below retrieve expanded Workbook Part A2 cards through the original-to-integrated crosswalk.

**Reproducible search and appraisal • card IVDR-CH12-A01**

Can another qualified person follow the dated sources and searches, selection and exclusion decisions, cohort identities, quality judgments and treatment of unfavourable evidence? A reference total cannot supply that method.

**The current clinical comparison • card IVDR-CH12-A02**

Do dated practice and comparator evidence support the claimed benefit and acceptance rationale? What has changed since the last PER, and which risk or follow-up question does that change affect?

**A finding for each purpose element • card IVDR-CH12-A03**

Do tissue cutoff, liquid specimen, intended user and therapy each have their own evidence finding? Direct support, an appraised bridge, a genuine limit and an unsupported proposal have different meanings.

**Residual uncertainty with an action • card IVDR-CH12-A04**

Does the PMPF question have a captured unit and denominator, rule, owner, review date and PER/risk return path? Assay R keeps its six initial invalids in 100 even when all later repeats succeed.

**A usable executive conclusion • card IVDR-CH12-C01**

Does the short conclusion retain exact configuration, supported use, contrary evidence, unresolved clinical consequence and next record? Brevity cannot erase a cutoff gap that can change treatment information.

**Conditional company choices • card IVDR-CH12-C02**

What evidence and product-state changes would support proceeding, a justified limit with follow-up, actual narrowing or further evidence? These communication choices are not legal approval categories.

**A supported-use follow-up question • card IVDR-CH12-C03**

Where is the adequate foundation for the proposed PMPF question, and what will the focused plan observe? Generic feedback cannot replace missing core specimen, user or cutoff support.

**Further reading**

Book 1C Chapters 16–17 offer an optional route into clinical applicability and how evidence travels. Keep the facts and limits of this chapter's separate records.

**Source trail**

- **Binding law, checked 2026-09-25:** IVDR consolidated 10 January 2025, Article 2(36)–(41) for clinical evidence and performance terms; Article 56(1)–(6) for claim-specific clinical evidence and lifecycle updating, including the class C/D annual PER minimum; Articles 78–81 for PMS planning and the applicable PSUR; Annex XIII Part A §§1–1.3 for prospective planning, objective favorable/unfavorable appraisal and PER contents, and Part B §§4–8 for PMPF's marketed-within-purpose role, plan, evaluation and risk/PER feedback. These provisions do not provide a product-specific cutoff criterion or certify any fictional case.

- **Nonbinding context:** Medical Device Coordination Group (MDCG) 2022-2 explains in vitro diagnostic clinical-evidence and continuous-evaluation principles; MDCG 2025-10 addresses PMS-system integration. Neither changes the IVDR or supplies evidence for these fictional devices. The four dispositions and worked assay R threshold are author-created teaching devices; they are not regulatory categories or numerical limits.

## CHAPTER 12

# Route the Study People Will Actually Conduct

**Before reading.** Chapter 11 separated a decision-critical evidence gap from a bounded question that can be followed within a supported use. Which claim would need to wait, or become narrower, while a new study supplies its missing foundation?

A participant may experience a new blood draw even when the assay result stays hidden from the treating team. Those two facts answer different questions. **A defensible study route follows the whole path through people, specimens, results and data, then connects each site's activity to its actual starting conditions.** The protocol and country records must describe the same work.

## What does this study ask a person to undergo?

A clinical performance study examines whether a device result bears its claimed relationship to a clinical condition or state in the intended population and use. The scientific question does not determine how people become involved. One design tests material already left after care. Another requests a new collection. One hides results from clinical decisions; another allows them to influence care. Changing any of those facts can change burden, bias, safety handling and the legal route.

**Opening record — a planned oncology study.** A sponsor proposes to test an investigational oncology assay on leftover specimens from a medicinal-product trial in **three Member States**. Its results will **not guide treatment**. A subset of participants will also provide an **additional blood draw** to investigate disagreement between results. The team assumes that “leftover samples” means notification only everywhere. The record does not yet identify the country with the draw, supply national permissions or establish a companion-diagnostic role.

Follow the two collection paths. Material remaining from a specimen obtained for the medicine trial can be tested without collecting again. The subset's new draw happens because this diagnostic study asks for it. Keeping the assay result out of treatment decisions does not remove that contact, its procedure risk or the need to address consent and data flow. Conversely, a study using no new collection could still raise a different question if its result were allowed to affect patient management.

The **sponsor** initiates and manages the performance study and arranges its financing. The **manufacturer** supplies and controls the device and its performance and conformity work. One organization may hold both roles, but it needs to discharge each set of responsibilities. Here the sponsor must establish the actual study route and local start conditions. Advice from a notified body about eventual device evidence cannot supply a national authority's study-start decision.

### Which changed fact selects another legal branch?

The **In Vitro Diagnostic Medical Devices Regulation (IVDR)** first supplies a baseline. Article 57 requires the study device to meet applicable safety and performance requirements apart from the aspects being studied, with precautions for those aspects. Subject rights, safety, dignity and well-being take priority; the data must be scientifically valid, reliable and robust. Applicable data-protection law also covers leftover material. Thus removing a new draw can reduce one burden without turning the remaining work into an unregulated activity.

Next ask whether Article 58(1) adds its full study framework. It identifies three distinct triggers: surgically invasive sampling performed solely for this study; an **interventional clinical performance study**, in which results may influence patient-management decisions or guide treatment; and additional invasive procedures or other risks for subjects. Where a trigger applies, Article 58, Articles 59–77 and the relevant annex requirements apply alongside the baseline. They govern design, authorization, conduct, recording and reporting.

In the opening, blinded assay results do not establish the treatment-influence trigger. The newly requested blood draw nevertheless makes the additional-procedure trigger relevant. That reasoning does not require calling the draw “surgically invasive.” If the team removed the draw but began using results to guide care, it would have changed which trigger matters rather than eliminated the framework.

Another branch depends on what the device is for. A **companion diagnostic (CDx)** is essential to a corresponding medicine's safe and effective use by identifying, before or during treatment, patients most likely to benefit or those likely to face increased risk of serious adverse reactions. Oncology use or a medicine association alone does not establish that relationship. Under Article 58(2), CDx performance studies normally follow the same full requirements. Its particular exception concerns a CDx study using **only leftover samples**, which must be notified to the competent authority. A separate Article 58(1) trigger still has to be assessed.

That exception cannot be borrowed by any study merely because some specimens are leftovers. A non-CDx leftover study does not inherit the CDx-specific notification provision. A mixed CDx protocol with a newly required draw is not only-leftover work. The opening lacks an established CDx relationship and contains a new collection, so neither missing premise can be supplied by its title.

A different comparison starts with an already **CE-marked device**: the manufacturer's marking of conformity with applicable Union device requirements. What exactly was its assessed intended purpose? Article 70(1) concerns further assessment **within** that purpose. If such a post-market performance follow-up (PMPF) study adds procedures beyond normal use and those procedures are invasive or burdensome, the sponsor must notify the Member States concerned at least **30 days before** commencement and supply the required study documents. Applicable subject, ethics, modification and other protections remain; notification is not an unrestricted permission.

Study outside the CE-marked device's intended purpose instead brings Articles 58–77 into play under Article 70(2). Changing the population or clinical role can therefore change the branch while the device itself remains identical. Conversely, a within-purpose study without additional invasive or burdensome procedures does not acquire Article 70(1)'s additional-procedure notification trigger merely because it collects follow-up information. The actual design and remaining duties still need assessment. The opening investigational assay supplies no CE-marked within-purpose premise for this route.

For a separate deployed **next-generation sequencing (NGS)** test, following its assessed specimen and user conditions differs from requesting a new participant draw. For a separate **digital pathology (DP)** system, observing a normal reader task differs from studying another clinical purpose. These comparisons explain what to inspect; they add neither CE status nor field results to the oncology packet.

The table retrieves those questions after their premises have been explained. Its entries select inquiries, not permissions possessed by the fictional sites.

| <b>Fact established by the actual study</b>                                                                           | <b>Consequence of that fact</b>                                                                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Performance study, including leftover material                                                                        | Article 57 precautions, subject protection, scientific quality and data-protection duties remain.                                                                           |
| Surgically invasive study-only sampling, possible care influence, or additional invasive procedure/other subject risk | Assess the Article 58(1) full framework, including Articles 59–77 and relevant Annexes XIII–XIV. The opening's new draw makes the third trigger relevant.                   |
| True CDx study using only leftovers, with no independent Article 58(1) trigger                                        | Article 58(2)'s specific notification exception may apply. Mixed specimens or an unproved CDx relationship cannot establish it.                                             |
| CE-marked device, within its assessed purpose, plus additional invasive or burdensome procedures                      | Article 70(1) requires notification at least 30 days before commencement and applicable protections. Outside-purpose work instead engages Article 70(2) and Articles 58–77. |



Read the second and third rows together: a site's leftover specimens do not erase another procedure within the same protocol. Read the last row as a conjunction: CE marking alone is insufficient; intended purpose and additional burden determine the stated notification branch. A changed procedure, result use or assessed purpose can change the route even if the project name does not change.

**Reviewer Lens — ask for the branch's evidence.** The full protocol, specimen flow, participant procedures and result-access rules show what the sponsor proposes. They should reveal whether a result can affect enrollment, treatment or other care, and whether the diagnostic study causes a collection or other risk. CDx and CE/intended-purpose records then answer their separate questions. A convenient label is a reason to inspect these records, not a substitute for them.

**Consultant Lens — compare real designs.** The mixed design may answer a question that archived material cannot. Feasibility work can compare its scientific value, burden and cost with a genuine no-new-draw design using lawfully available material. That comparison can be funded before final routing is complete. Executing the chosen design requires its controlled protocol and applicable national, ethics and start conditions. Splitting the current protocol into two spreadsheet rows changes none of its procedures.

### **What would the extra specimen allow the study to compare?**

“Discordance analysis” does not yet identify an endpoint. Which results disagree, from which specimens and times, against which reference or adjudication process? Are comparisons paired by participant, by specimen or another unit? The plan must answer those questions and explain invalid or missing comparisons before results are examined. Otherwise, the same apparent disagreement could reflect a changed assay, reference or specimen path. A direct leftover-versus-new-draw pair is one possible design; it is not supplied in this packet.

The IVDR's **clinical performance study plan (CPSP)** connects the rationale and objectives to participant and specimen selection, device identity, reference, endpoints, prospective criteria, uncertainty, handling, monitoring, data and safety processes. It makes the proposed study executable and its eventual result interpretable. A document written after seeing favorable results cannot make its criteria prospective.

The voluntary **International Organization for Standardization (ISO) 20916:2019** uses CPSP for a *clinical performance study protocol*. Its good-study-practice methods support planning, conduct and reporting. A shared abbreviation does not make the method document an IVDR permission or replace the legally required study plan and country processes.

When an actual file is discussed, its full title and revision make the distinction clear: the IVDR **plan** carries the legal study-document requirement; the ISO **protocol** describes

the method document. Both need to identify the study they govern. An approved internal protocol may be scientifically useful while a local start condition remains unresolved.

Product identity belongs in that same plan. In a separate sequencing example, a cutoff, pipeline or knowledgebase change can alter a reported variant. In a pathology example, a scanner, model, display or worklist change can alter what a reader sees. Linking each observation to its actual configuration prevents a final study report from describing an assembly that was never tested. Neither hypothetical change is an event in the opening study.

**Reviewer Lens — recover the endpoint's denominator.** Participant and specimen flow, discordance definition, reference process, exclusions, missing/invalid rules and the version ledger belong beside the endpoint. Together they show which observations could answer it. An attractive endpoint title without those links can conceal an uninterpretable comparison.

**Consultant Lens — make the next experiment answerable.** A proposed protocol-to-report trace can compare what the extra-sample and leftover-only designs would actually yield. Recruitment and method feasibility can receive a conditional budget while criteria, consent, versions and applicable permissions are being resolved. Those planning decisions do not release the study for execution.

### **When is each site actually ready?**

One scientific protocol may require different local submissions, ethics documents, languages, consent and sample-transfer arrangements. Central readiness does not establish that a country is ready, and progress in one country does not establish another's start basis. The sponsor needs a dated record of the activity each site may conduct, the applicable authority and ethics steps, what remains pending and who verifies it. Translation, supply, data protection and training can determine the next feasible date even after the science is settled.

Under Article 66(7), a study in the additional-invasive-procedure or treatment-influence branches ordinarily waits until the Member State has notified authorization, with no applicable negative ethics opinion. The different, limited branch for surgically invasive sampling solely for the study with no major clinical collection risk can start after application validation, unless national law says otherwise, again subject to the ethics condition. Validation of an application and authorization are therefore different records. The opening's additional draw does not acquire the limited sampling branch simply by being a blood collection.

The packet supplies **three Member States**, with no names or verified procedures. It has not yet located the extra draw. Neither identical participant procedures at every site nor

acceptance of a common notification can be assumed. A coordinated route, if available and chosen, still needs its actual national start conditions satisfied.

### **Can the site identify the device behind each result?**

Imagine one site testing an assay/reagent/software combination while another receives an update. If the update changes calling or interpretation, a pooled estimate may combine unlike outputs. The accountability record therefore connects assay, reagent, instrument, software, algorithm, database and labeling to release and use dates by site and specimen. A plan may prohibit the change, control it prospectively or require a justified bridge. Recording only the newest version leaves earlier observations unexplained.

A sequencing knowledgebase could change treatment-associated interpretation without changing raw reads. A pathology scanner or queue update could change visibility while model weights stay fixed. Those are separate applications of the principle, not supplied study events. The reviewer needs a reconstructable version-to-result map; the consultant needs a change gate defining what evidence permits pooling or requires another analysis.

### **Which problem must be recorded, and which must be reported?**

Sites need to recognize a problem before deciding its route. An **adverse event (AE)** can be an untoward medical occurrence or inappropriate patient-management decision in the performance-study context. It may concern subjects, users or other people without an established causal link to the device. A **serious adverse event (SAE)** depends on specified consequences, such as death, a life-threatening situation or hospitalization; “serious” does not merely mean a symptom feels severe. The planned draw creates a possible procedure risk, with no injury reported here.

A **device deficiency** concerns inadequate identity, quality, reliability, safety, performance, use or supplied information. A hypothetical identifier mismatch intercepted before release can therefore deserve assessment even without harm. A **protocol deviation** instead means actual conduct departed from the controlled plan, such as collection outside its specified time window. One occurrence can raise both questions, but neither term automatically establishes the other.

For applicable studies, Article 76 separates full recording from reporting. The sponsor records plan-defined events critical to evaluating results, all SAEs, deficiencies that could have led to an SAE without intervention or under less fortunate circumstances, and relevant new findings. Recording an SAE does not require first proving that the device caused it.

The reporting premise is more specific. An SAE with an actual or reasonably possible causal relationship to the device, comparator or study procedure is reported without delay to the Member States where the study is conducted. A deficiency that could have

led to an SAE under the stated counterfactual conditions is also reportable, together with relevant new findings. Severity affects timing, and an incomplete initial report followed by fuller information can preserve timely reporting. A caught defect can therefore require reporting, while a documented deviation does not automatically meet that threshold. Sites should protect people and preserve information while the sponsor assesses the facts; perfect causal certainty is not the general prerequisite.

These distinctions establish a response method, not an occurrence in this study. The actual device/study state, event, possible consequence and relationship must be known before assigning a channel or deadline. Study-safety reporting and ordinary market vigilance answer different legal circumstances.

**Additional exhibit — the same planned study.** The extra draw occurs in **only one** of the three countries. Results remain blinded, but a **device deficiency could reveal the medicinal trial arm to the local team**. The record does not say disclosure or harm occurred, or that authority and ethics conditions were satisfied. The one-country location limits where the procedure happens; it does not remove it from the common protocol.

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### Decision

**Is one notification-only strategy justified for the entire protocol? What must the sponsor establish for the draw country and the two other countries before new relevant activity begins?**

#### How a reviewer follows the common protocol into each country

No. The additional invasive procedure belongs to the whole protocol's Article 58 assessment. A locally leftover-only subset cannot make the mixed study an exclusively leftover CDx study. Each country's actual route, ethics and legal start basis still needs verification.

The draw requires its consent and procedure safeguards, while result access and the possible trial-arm disclosure require blinding and deficiency controls. Another country's progress or a notified-body email supplies neither. The supported next decision is to establish the missing basis and hold new relevant activity pending verified conditions. No fact here identifies what an unnamed authority has already decided, or converts a possible disclosure into an observed one.

**Reviewer Lens — compare the proposed paths.** The common protocol and local addenda need to agree with collection/consent records, blinded access and deficiency escalation. The medicinal-trial interface also needs an owner: the IVD sponsor and medicine-trial sponsor, people and systems may differ. A path across two organizations

deserves a documented handover rather than an assumption that both use the same control.

**Consultant Lens — determine what the draw buys.** Its discordance question, the adequacy of archived alternatives, specimen availability, bias and burden determine whether redesign is worthwhile. Country submissions and timing follow the real design. Removing a spreadsheet label adds no scientific or legal evidence.

For an investigational-device study in the Article 58 framework, Article 76 supplies the study-safety route just explained. In an Article 70(1) within-purpose CE-marked PMPF study, Article 76(5) instead directs the applicable events into market vigilance under Articles 82–85. Article 76(6) restores the study-safety article when a causal relationship between an SAE and the preceding performance study **has been established**. That narrower switch must not be confused with Article 76(2)'s reasonably possible causal relationship for reporting an SAE when that study-safety route applies. A change in device/study status or the established event relationship can therefore change the channel; the opening's possible deficiency supplies neither a reportability verdict nor that causal finding.

### Read one country record without losing the common study

The following **A, B and C are fictional placeholders**, not actual Member States. Let B be the country with the extra draw; A and C use leftovers under the same protocol. No national submission, ethics opinion or lawful start is supplied. B needs the procedure, consent and blinding inquiry; A and C need the whole-protocol assessment and their local readiness evidence. A different local collection path does not make another study.

| Placeholder and supplied activity                 | Evidence currently established                                                                     | Sponsor's next decision                                                                                                                |
|---------------------------------------------------|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| A: leftovers; no local added draw stated          | No verified route, submission, ethics or activation basis                                          | Obtain the local route/ethics determination and readiness record before new activity; no exception follows from this sample row alone. |
| B: common protocol plus draw for discordance      | Article 58(1)(c) procedure question is live; actual consent, sampling and blinding controls absent | Establish authority/ethics basis, safeguards and start decision before a new draw or other study activity.                             |
| C: leftovers; handling and transfer not finalized | Route, specimen/data arrangements and readiness unresolved                                         | Obtain local route, ethics, transfer and product-accountability records before new activity.                                           |

Read B's middle cell carefully: identifying a legal trigger is a positive route finding, but it is not a permission. Read A and C against it: their local tasks differ without erasing the common protocol. A verified country decision and readiness record could close a start question for its actual scope; no row contains one now.

The sponsor owns the integrated record. A **responsible-accountable-consulted-informed (RACI)** chart can identify who prepares each entry, who approves it internally, which specialist advises and which investigator receives it. It cannot assign a competent authority's statutory decision to an employee, or make a notified body an ethics committee. Internal accountability makes the external decisions easier to trace; it does not replace them.

If the sponsor is not established in the Union, Article 58(4) ordinarily requires a Union **legal representative** responsible for ensuring the sponsor's obligations are met and receiving its regulatory communications. The stated exception is narrower than “appoint a local contact”: a Member State may waive that representative for studies conducted solely on its territory, or on its territory and a third country's territory, provided at least a contact person is established on its territory. A three-Member-State protocol does not meet that territorial premise merely because it has a contact at each site.

A non-Union manufacturer's **authorized representative** instead acts under the manufacturer's written mandate for specified device-regulation tasks. The same service provider might receive both appointments, but the mandates and duties remain separate. Neither appointment confers study authorization. Changing who the sponsor is or where the actual study runs could change the representation analysis; simply renaming the manufacturer contact would not.

The opening remains a **planned** study. If evidence later showed a site had already conducted activity before its basis was established, the response would also require reconstruction: what happened, when, to whom, under which records; what protection, corrective action or authority/ethics steps now follow? The separate studio later explicitly has opened sites. Its history cannot be rewritten as a prospective hold already observed.

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### Decision

The sponsor actually removes every new draw from a CDx protocol, uses only lawfully available leftovers, keeps results blinded and has no other Article 58(1) trigger. Does a new study title make it immediately ready to start?

### How a real redesign changes a branch without supplying a permission

No. The stipulated substantive changes can place the revised CDx study in Article 58(2)'s only-leftover notification branch. The sponsor still needs a controlled revised protocol, required notification, national ethics and data-protection checks, and applicable starting conditions.

An already authorized or active site also raises change duties before implementation. The revised study's premise does not retrospectively remove the new collection from the original protocol. A title alone could accomplish neither change.

### Could the final report reconstruct what the plan required?

The eventual report must explain whether sites collected and tested the planned specimens, with the identified device, under the applicable basis, and analyzed the entered observations as intended. Recruitment, collection, testing, permission and activation each need their own dates. A main endpoint can pass while missing specimens, changed cutoffs or undisclosed deviations limit its meaning. Entered, excluded, invalid, repeated, missing and analyzed subjects/specimens must therefore reconcile.

This filled trace records supplied purposes and missing records. It invents no comparison, event or result. The first row cannot be completed until the plan defines what the draw is meant to compare; direct leftover-versus-draw pairing remains only one possible design.

| Planned question or control                                                                                      | What the report must reconcile                                                                                                           | Present packet status                                                        |
|------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| Draw for discordance: actual comparison units, pairing if any, reference/adjudication, criterion and uncertainty | Eligible participants, specimen/time paths, all planned comparisons, missing/invalid results, exclusions and near-boundary disagreements | Draw purpose known; design, endpoint, counts, criterion and results unknown. |
| Investigational assay identity and change rule                                                                   | Reagent, instrument, software/database and label per specimen; amendments and any bridge before pooling                                  | Product/version ledger absent.                                               |
| Site permission and participant procedure                                                                        | Submission/ethics, activation, consent, draw dates and deviations, apart from mere opening                                               | Permissions and actual activity unknown in this opening packet.              |



| Planned question or control | What the report must reconcile                                                                    | Present packet status                                                     |
|-----------------------------|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| Blinding, safety and access | Who could see results or trial arm; deficiency/event logs, investigations and reporting decisions | Possible arm-disclosure path only; no occurrence or consequence supplied. |

Following a specimen across these rows connects the scientific comparison to its version, lawful conduct and visibility. A missing comparison definition prevents denominator reconstruction; a missing date prevents checking which permission or amendment applied. Completed records could resolve those particular questions. A passed percentage alone could not.

ISO 20916's voluntary methods can strengthen the conduct-to-report trace, including device changes, accountability, deviations and limitations. **Clinical and Laboratory Standards Institute (CLSI) EP49, Edition 1** supplies a framework for noninterventional clinical-validity studies. It is not a complete interventional CDx protocol or a clinical-utility-evidence plan. Changing results from blinded measurement to care-guiding use therefore changes the question that the method must serve. Binding IVDR and applicable country processes retain their separate permission role.

**Reviewer field aid — four records together.** The patient/specimen/result path, country authority-and-ethics status, version ledger and safety/access record should all describe the same protocol. In the opening, the added draw and possible arm-disclosure route remain relevant despite treatment blinding. An absent record should become a precise request; it should not become an invented approval or reportable event.

**Consultant field aid — make a conditional choice executable.** A route memo, country matrix, CPSP gap list and report trace let management compare keeping the scientifically justified draw, removing it through adequate redesign, or staging genuinely separable work. Local specialists, sequence, feasibility cost and stop points make each option concrete. None guarantees an earlier start or a certificate.

**Transfer — a separate CE-marked near-patient test.** Its study proposes another population and results that guide immediate care. The first comparison is with the actual assessed intended purpose. If the population/use is outside it, Article 70(2) brings Articles 58–77 into the study; the treatment influence also makes the interventional premise relevant. If facts instead established use within the assessed purpose, the additional-procedure question would need its own assessment. Labeling, complete protocol and national records decide the route; this example supplies no outcome or authorization.

## Practice studio — what did the opened sites actually do?

The next dossier is separately presented. It is **not an established continuation of the opening sponsor or study**. Here sites have already opened before a country matrix exists. The route method transfers, but the opening case's possible deficiency mechanism and country record do not. “Opened” establishes a chronology question; by itself it establishes neither participant contact nor permission.

**Practice case.** A sponsor plans a three-country oncology performance study using medicinal-trial leftovers. Assay results are blinded and will not guide treatment. One country has a new blood draw for discordance analysis. Treating leftovers as notification-only everywhere, the team **opens sites before making a country matrix**. No actual national route, ethics opinion, participant activity, trial-arm disclosure or device outcome is supplied.

The first useful record is a dated activity ledger. Was opening limited to setup and training? Did it include recruitment, consent, specimen access, a draw, testing or data release? For any activity, the sponsor needs its site/date, participant and specimen path, device state, result access and the authority/ethics basis then in force. Records should be preserved and proportionate immediate protection or suspension and applicable duties assessed while those facts are resolved. No unlawful start or completed correction follows from the word “opened.” A consultant can organize the reconstruction; the sponsor owns the response.

### Does a separate row describe a genuinely separate study?

A leftover row and a draw row can make activities visible without separating the protocol that governs them. This studio's planned new collection therefore enters the **whole common protocol's** Article 58(1) assessment, including the countries contributing only leftovers. The only-leftover CDx exception cannot describe a mixed study; moreover, this assay's CDx role has not been established. The intended blind should be compared with actual access, but no deficiency-to-disclosure mechanism from the opening may be assumed here.

A genuine redesign could change those facts. Removing the draw may remove its additional-procedure trigger while losing needed discordance information. An independent second study could have its own objective, specimen access, consent, analysis and country basis. Those are substantive changes to the work. Calling the original draw an “optional sub-study” while the same participants, common protocol and data path remain does not establish independence.

**Reviewer Lens — inspect the implemented boundary.** Country addenda, consent and analysis records show whether the draw shares the sponsor's common design. The site ledger then shows which version of that design was actually used. A newly divided table is weaker evidence of separation than those records.

**Consultant Lens — price what would be lost.** A no-draw alternative must still answer the actual discordance question with lawful access, appropriate quality and sufficient archived material. A revised design also needs its applicable modification and ethics steps. Feasibility work can begin now; attractive projected savings do not authorize changed procedures.

### **Which country record could change the next action?**

The earlier A/B/C table was an illustration, not this studio's authorization history. A fresh dated matrix needs seven fields. Each connects an actual activity to a different decision maker or constraint, so an apparently complete row cannot borrow another field's conclusion.

### **Authority route and effective date**

The country regulatory lead establishes what the actual protocol requires: application, notification or another applicable process; what was submitted; which decision or acknowledgement exists; and when it took effect. The extra draw, possible CDx role and intended blind are route inputs. They are not an authority record. In this opened-site dossier, comparison with the first actual activity matters: a later decision cannot silently become an earlier start basis. A verified record could close that chronological question within its scope; absence requires investigation rather than a presumed violation or approval.

### **Ethics and participant information**

An authority submission does not create an ethics opinion. The local record must identify which protocol and consent revision were reviewed, in which language, and what participants were told about the draw, reuse of data and potential effects on the blind. The relevant opinion and participant-facing activity need dates. If opening meant only training, that is a different chronology from enrolling or drawing before the relevant conditions were met. Neither history is supplied.

### **The procedure and its actual occurrence**

The new collection is a physical intervention for the diagnostic objective, unlike testing a remainder from an earlier collection. Its record identifies who orders and performs it, when and for whom, its scientific value, consent and complication management. A participant may be free to decline an optional draw while the protocol still offers an additional invasive procedure. If a draw occurred, its participant/specimen record must survive; if it remains planned, it must not be described as an event. Actual occurrence changes retrospective protection and accountability questions without changing what the proposed protocol contains.

## Specimen and data movement

Blinding depends on access, not only an instruction that assay results will not guide treatment. Coded specimens may connect through databases, reconciliation files or queries to people and trial information. The studio supplies **no deficiency mechanism or event**; the opening's possible arm-disclosure path is separate. Here the record needs lawful specimen/data arrangements, locations and transfers, who can link codes to participants or trial arms, access controls and incident handling. An unexplained flow can compromise both protection and endpoint reconstruction. Verified access and transfer records could resolve that gap without inventing an unblinding incident.

## The device released and used

A shipping date cannot identify what generated a result. Study supply needs the released reagent, instrument, software, database, labeling and training state; receipt and use dates; and unused, returned or destroyed material. A mid-study change would require a controlled impact assessment before unlike states were pooled. No import, release, version or return history is supplied here. A reconstructed ledger could show unchanged conditions, an assessed change or an unresolved mixture; the current packet chooses none of those findings.

## Safety and deficiency response

Sites need a path for recognizing and escalating serious events, potentially serious deficiencies and possible loss of blinding while facts are still emerging. The sponsor's safety lead then applies the device/study-status distinction and the event predicates taught earlier. Article 76 reporting can be required for reasonably possible causal SAEs or potentially serious deficiencies without waiting for final proof. For within-purpose CE-marked PMPF, Article 76(5) points to market vigilance; the Article 76(6) switch requires an established SAE-to-study causal relationship. The studio supplies no deficiency mechanism, event, harm or reportability outcome. A real event and its relationship would change the assessment; a placeholder in the matrix cannot.

## A change after opening

Removing the draw, changing a cutoff or software rule, or changing access to trial information can affect subjects or the reliability of the data. Article 71's substantial-modification premise is a **likely substantial impact on subject safety, health or rights, or on the robustness or reliability of study data**. That consequence, assessed against the current protocol, matters more than whether the edited file is short. A spelling correction without those effects raises a different assessment from a cutoff change that could change endpoint assignments.

Where the Article 71 premise applies, the sponsor notifies the Member States concerned within one week, explaining the nature and reasons and supplying

identifiable document changes. Implementation is no earlier than **38 days after notification**, provided that neither the Member State concerned has notified the sponsor of its refusal nor an ethics committee in that State has issued a negative opinion on it that is valid for the entire Member State under national law. The Member State may extend the period by **seven days** for expert consultation. The applicable route and country records still need checking. An internal version approval is insufficient, and no modification status or completed waiting period is supplied for this studio. Effective dates then let the final report assign participants and specimens to the design actually used.

The compact record below retrieves those seven jobs. Read the procedure and authority rows together: a draw can establish a route question while the authority basis stays unverified. Each entry is next work, not an invented permission.

| <b>Country-matrix field</b> | <b>What the opened-site audit needs</b>                                    | <b>Working owner and possible consequence</b>                                                        |
|-----------------------------|----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Authority route             | Actual protocol assessment, submission, decision/acknowledgement and dates | Country regulatory lead prepares; sponsor confirms. An incorrect basis may require immediate action. |
| Ethics and consent          | Opinion, document versions, language and first participant activity        | Clinical operations/local ethics specialist; participant-facing work may be affected.                |
| Additional procedure        | Country/site, planned or actual draw, consent, risk and specimen record    | Medical lead; a local draw stays in the whole-protocol analysis.                                     |
| Data/specimen transfer      | Lawful flow, linkage, trial-arm visibility and any actual access event     | Privacy/trial-data leads; protection, blinding and endpoint integrity may be affected.               |
| Device supply               | Import, release, version, training, use and return/destruction             | Supply team and manufacturer; pooling may need version analysis.                                     |
| Safety and deficiency       | Logs, escalation, causal assessment and actual reporting route             | Sponsor safety lead; timely duties depend on the supplied facts.                                     |
| Modifications               | Proposed/implemented changes, consequence, dates and country/ethics steps  | Sponsor regulatory owner; changed conduct may alter route or interpretability.                       |

The sponsor's RACI chart can assign preparation, internal accountability, specialist consultation and information flow across these rows. It cannot change statutory responsibility or who issues a national decision. Manufacturer authorized representation and sponsor study representation also remain separate functions, even if one provider supplies both services. The matrix is useful because it shows which decision is missing and whose record could establish it.

### **Can the planned analysis account for what each site used?**

The discordance question first needs its comparison defined. The plan then follows each unit to original collection of leftover material, the added draw's time/site where relevant, assay and comparator states, invalid or missing results, adjudication and endpoint calculation. It might pair leftover and new specimens directly, but that design is not given. A precise trace can support several scientifically different comparisons without silently choosing one for the sponsor.

If a site changed its variant-calling software or reference database during enrollment, the same assay name could cover different decision rules. The record must establish whether change was barred, permitted under a controlled amendment or supported by an adequate bridge before pooling. No such update is asserted for this studio. The condition explains why a version ledger is needed, not what it will reveal.

The CPSP therefore keeps sample availability, cutoff, device state, analysis population, safety monitoring and stop rules visible. The report reconciles amendments and actual conduct, including deviations, exclusions, deficiencies and limitations, even if its main endpoint passes. ISO 20916 methods support that trace without deciding country permission. A separate pathology study would similarly need scanner, model, display and reader-task identity; its example contributes a method, not another oncology incident.

**Reviewer Lens — test the pooled estimate's meaning.** Subject/specimen disposition and per-site versions show which observations can legitimately be combined. Resolved discordances, persistent indeterminate comparisons and actual study changes then determine whether reanalysis or modification assessment is needed. An unexplained pooled denominator is a focused missing record, not proof that every observation is unusable.

**Consultant Lens — define the amendment gate.** A proposed reagent, software or country-procedure change should be linked to its endpoint and subject-safety consequences. Bridging, re-consent or revised analysis can be scoped where appropriate, with owners and decision dates. Their feasibility is different from evidence that a bridge already works or a modification has been permitted.

### **Which design reduces burden without losing the answer?**

The optional draw may improve discordance evidence while adding burden, route work and time. Its value depends on the exact comparison, which is still unknown. Archived material can remove new collection only if lawful access, specimen quality, timing and available relevant material support that comparison. A genuinely separate study or staged country can reduce operational dependence while shrinking the represented population. These trade-offs belong beside cost rather than being hidden by a simpler diagram.

A common United States/Union protocol is useful only if it preserves the configuration and endpoints needed for each claim. One document does not guarantee simpler or sufficient evidence. Likewise, review or stop points need prospective criteria for inadequate recruitment, lost reference material, version changes, important safety signals, missing truth assignment or serious deviations. Such conditions can threaten welfare or make a costly endpoint uninterpretable. The studio supplies no numerical threshold, sample count or actual stop event; budgets and feasibility conversations can proceed while the final design and required permissions are resolved.

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### **Decision**

**Because most specimens are leftovers and results are blinded, may this opened-site study use one notification-only strategy? What facts must be established before judging a proposed redesign?**

### **How the sponsor reconstructs conduct before changing its plan**

No. The actual common protocol includes the one-country draw. Its device status, possible CDx role, result influence and other subject risks must be assessed with the applicable national and ethics processes. True only-leftover CDx work has a specific notification exception; mixed CDx work and independent Article 58(1) triggers require their full-framework assessment. CE-marked within-purpose and outside-purpose studies have the separate Article 70 conditions.

A draw can be removed only by actual redesign or a genuinely independent study, with the lost evidence assessed and change duties satisfied. Here sites already opened, so first establish what each did and which permissions, if any, were in force. Setup alone and participant testing are different facts. The packet supports neither a blanket compliance conclusion nor a finding that an unlawful start occurred.



## Which actor can close each question?

The sponsor owns initiation, management, financing arrangements and study safety reporting. The manufacturer controls the supplied device and performance-relevant changes. The competent authority and ethics process address their actual national start and modification questions. A notified body can later assess whether the evidence supports device conformity; it does not authorize a site or issue an ethics opinion.

A medicinal-trial authority may also be involved where the studies interact, but this packet supplies no such decision. The consultant's route memo, matrix, plan gap list and design comparison make each actor's question answerable. They do not transfer statutory power to the consultant or convert an unverified row into an authorization.

**Transfer — another near-patient proposal.** A separate CE-marked test is proposed for a pediatric population outside its intended purpose, with results guiding immediate treatment. The changed population is explicitly outside the assessed scope, so Article 70(2) brings Articles 58–77 into the analysis; care influence also satisfies the interventional premise. The reviewer needs the actual labeling and whole protocol; the consultant needs the evidence and country path before execution. Neither an existing mark nor this example supplies pediatric performance or site permission.

## Quick discrimination lab

These two records are independent of each other, the opening and the studio. Each supplies a route question without adding national facts to another case.

**Record A — another collection.** A study mainly uses leftover specimens but adds an optional blood draw at two sites to resolve discordance.

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### Decision

**Can the whole common study be treated as exclusively leftover-sample work?  
Which actual procedure fact determines the answer?**

#### How an optional collection remains part of a common protocol

No. The planned additional draw changes the procedure path even if a participant can decline it. The whole protocol needs its Article 58(1) assessment and the applicable country, ethics and participant safeguards. An administrative division into leftover and draw rows does not establish separate studies.

The record does not say a draw occurred or an authority decided. Removing every new draw through real redesign could change the procedure premise, but the revised protocol would still need its remaining route and start conditions assessed.

**Record B — another population.** A CE-marked near-patient test is studied in a pediatric population outside its assessed intended use. Its results guide immediate treatment.

### **Decision**

**Does the existing CE mark eliminate the study-route question? What difference would actual use within the assessed purpose make to the next inquiry?**

#### **How the assessed purpose limits what the mark establishes**

No. The supplied population is outside the assessed purpose. Article 70(2), with Articles 58–77, therefore governs that outside-purpose inquiry, and the immediate-treatment role also makes the interventional condition important. Subject protection and actual country and ethics requirements remain to be established.

Within-purpose use would require the separate Article 70(1) examination, including whether additional invasive or burdensome procedures trigger notification. That changed premise is absent here. No pediatric outcome or permission follows from either the mark or a route label.

### **Carry the real protocol into the next decision**

The route follows what the study asks people to undergo and what it does with specimens, results and data. Article 57 supplies its baseline; Article 58 responds to specified procedures, care influence and actual CDx status; Article 70 distinguishes further assessment within a marked device's purpose from study outside it. Those premises explain why one local leftover subset cannot settle a mixed protocol.

The country record then connects those duties to actual decisions and dates. At opened sites it also reconstructs conduct. Finally, the plan-to-report trace preserves the scientific meaning of the observations and their device versions. Neither table supplies a permission, an endpoint result or a completed bridge merely because its fields are filled. Chapter 13 follows the corresponding medicine relationship and the evidence needed when diagnostic and treatment decisions meet.

**Check what you just learned**

Each item belongs beside an evidence location and reason. The working states are **A**, adequately explained for this item; **U**, present but unclear; **M**, missing or inconsistent; **D**, deeper analysis needed; **C**, consultation indicated; and **N/A**, inapplicable with a reason. They are neither study authorizations nor certificates. The original IVDR-CH13 identifiers locate expanded Workbook Part A2 cards through the topic crosswalk; their source-era number differs from this chapter's number.

**Route from the actual study • card IVDR-CH13-A01**

What do device status, objective, specimen origin, participant contact, result use and other risks establish? An extra draw in the common protocol defeats a blanket only-leftover conclusion; the next route still needs its complete premises.

**Country process and timing • card IVDR-CH13-A02**

Which actual application or notification, ethics opinion, language, consent and activation record supports each site's activity and date? A/B/C are placeholders, not verified jurisdictions or transferable permissions.

**The device behind every observation • card IVDR-CH13-A03**

Can each specimen and result be connected to reagent, instrument, software/database, label and protocol versions? An amendment needs its impact and evidence assessed before unlike states are pooled.

**Safety facts and their proper route • card IVDR-CH13-A04**

What do event/deficiency logs, causal assessment, device/study status, escalation and recipients establish? Article 76, the within-purpose PMPF market-vigilance route and the narrower established-study-causality switch require different premises.

**One protocol across the decision records • card IVDR-CH13-C01**

Do route memo, country matrix, CPSP gaps and subject/specimen-to-report trace describe the same work and unknowns? No missing national decision can be supplied by a more polished table.

**Real design and sequence options • card IVDR-CH13-C02**

Which evidence does the draw add, what would removing it lose, and is proposed separate work actually independent? The changed design must meet its applicable modification and start conditions.

## Expertise with accountable roles • card IVDR-CH13-C03

Who prepares country, consent, supply and safety evidence; who advises; and who makes each statutory decision? RACI helps organize the work while sponsor duties and legal representation remain identifiable.

## Further reading

Book 1C Chapter 12 offers optional practice in clinical-performance-study design. Actual conduct still depends on the real protocol's applicable IVDR, country and ethics requirements; the separate study packets here retain their own facts.

## Source trail

- **Binding law, checked 2026-09-25:** IVDR consolidated 10 January 2025, Article 2(7), (25), (35), (46), (57), (60)–(62); Articles 57–58 for baseline, full-framework triggers, CDx leftover exception and sponsor/ethics roles; Articles 66, 70–71 and 76 for country commencement, CE-marked within/outside-purpose studies, modifications and the study-safety versus market-vigilance branches; Annex XIII Part A §2 and Annex XIV for study planning and documentation. The law does not supply fictional national permissions, sample counts or study results.
- **Nonbinding guidance:** Medical Device Coordination Group (MDCG) 2025-5 explains performance-study pathways and combined-study context; the Commission guidance register lists MDCG 2024-4 on study safety and MDCG 2022-19/20 on submissions and modifications. Guidance helps plan questions but does not decide a country outcome.
- **Method scope, not a legal route:** International Organization for Standardization (ISO) 20916:2019 describes good-study-practice planning, conduct and reporting for IVD clinical performance studies and names its Clinical Performance Study Protocol; it does not replace the IVDR's study plan or authorizations. Clinical and Laboratory Standards Institute (CLSI) EP49, Edition 1 provides a clinical-validity framework whose noninterventional study scope does not supply an interventional CDx protocol. Neither standard is claimed to set a country permission or an outcome in these fictional cases.

## CHAPTER 13

# Connect the Trial Result to the Commercial Treatment Decision

**Before reading.** Chapter 12's study combined leftover trial material with an added blood draw. Why did its route depend on the whole protocol? What would change for a genuinely leftover-only companion-diagnostic study, and why would that study-start decision still leave device certification to be addressed?

A medicine trial uses one assay, but patients would later receive another. **What connects the treatment decision made from the commercial result to the evidence generated with the trial assay?** The connection needs both a scientific argument and a clear account of who judges it.

The work begins with the specimen and the trial result's actual use. Those facts determine what the commercial comparison must preserve, what scarce or missing material can leave unanswered, and which questions belong to the diagnostic company, medicine partner and regulatory bodies.

## SECTION 01

### What decision did the trial assay actually support?

Imagine tissue entering a trial laboratory. Its test produces information that could affect enrollment, treatment analysis or another planned trial task. Which task it actually performs matters: a result used only in a later analysis has a different evidential history from one used to select participants. Neither the oncology setting nor proximity to a medicine establishes a **companion diagnostic (CDx)**. Article 2(7) of the **In Vitro Diagnostic Medical Devices Regulation (IVDR)** defines that device by its essential role in the safe and effective use of a corresponding medicine: identifying, **before and/or during treatment**, patients most likely to benefit, or patients likely to face increased risk of serious adverse reactions from that treatment. The actual medicine relationship must meet that premise. If it does, the device and medicine still have separate regulatory decisions, joined by the question of which patients the result identifies.

**Opening record.** A pharmaceutical **pivotal trial**, a main study intended to support medicine authorization, is enrolling with a **central laboratory assay**. The diagnostic

partner proposes a commercial kit with a different extraction method, a broader variant list and use in distributed laboratories. Tissue is scarce and the pharmaceutical labeling deadline is fixed. Both partners call the trial assay a “prototype” of the kit. The record supplies no medicine or biomarker identity, exact specimen preservation, treatment cutoff, assay release state or account of how the trial used its result.

The word “prototype” supplies none of those missing links. The central laboratory generated its result through its own preparation, interpretation and quality rules. The proposed kit would use different extraction and more laboratories, and would offer results for additional **variants**—differences in deoxyribonucleic acid (DNA) sequence relative to a reference. To connect the two, the trial protocol must first identify the result's role. Only then can the partners ask whether the kit's actual process and intended population support the corresponding treatment classification. A shared marker name cannot answer what a changed process reports.

**Reviewer Lens.** The first useful comparison joins the trial protocol and central-assay identity to the proposed kit purpose, report and draft medicine-label relationship. The protocol can establish whether the central result selected patients, supported an analysis or had another role. If that use remains unstated, a request for “equivalence data” has no precise clinical decision to test. Clarifying this input can be more valuable than immediately commissioning another agreement study.

**Consultant Lens.** A dated specimen-to-treatment map lets the diagnostic partner, medicine team and trial laboratory identify the same dependency. They can reserve scarce tissue and fund feasibility while refining the comparison. Those early steps preserve options; they do not require a claim that the bridge, device evidence or conformity decision is complete. The useful management estimate says which missing decision controls the next expenditure.

### **Which evidence answers which part of that decision?**

First, why should the marker identify a clinically relevant group for this medicine? That is the **scientific-validity** question: the association must fit the clinical state and proposed medicine use. Finding a genetic difference does not establish treatment benefit. The medicine evidence has to show what its assay-defined groups support, while preserving the trial's actual use of that assay. The opening supplies no such analysis, so the relationship remains a question to appraise rather than a conclusion from the study's pivotal status.

Given a relevant relationship, can the proposed kit produce its claimed result? Its extraction, chemistry and interpretation need **analytical-performance** evidence for the tissue, variant classes and laboratories claimed, including difficult low-content material. Central-assay evidence concerns the conditions under which that assay was studied. The next question, **clinical performance**, concerns how the kit's result relates to the relevant clinical state and medicine-selection use in its intended population. Successful

detection or agreement with another assay can contribute, but neither alone establishes that clinical relationship. Changed measurement conditions and changed clinical use therefore reopen different questions.

An **evidence bridge** explains why evidence from one defined product state applies to another despite specified differences. Here its task is to connect the trial and kit treatment decisions: when both assays apply their own prespecified rules to relevant material, do they classify medicine eligibility consistently, and what do differences mean for the treatment evidence? This is an argument linking the three performance questions, not a fourth legal pillar. The eventual device assessment and medicines consultation examine the corresponding claims for their distinct purposes.

Consider the direction of a disagreement. A trial-positive result paired with a kit-negative result could exclude someone the trial rule would have considered for the medicine. The reverse could include someone outside the trial-defined positive group. These are possible consequences of the comparison, with no affected patient or outcome supplied here. Knowing only that the assays differ leaves both the direction and clinical meaning unresolved. Knowing the classifications and treatment evidence for the affected groups could change that conclusion.

Those questions also locate responsibility. The diagnostic manufacturer controls device design, validation, purpose and changes; the pharmaceutical partner controls its medicine program and label submissions. The trial sponsor, laboratory and data holders may hold specimen or source-data access under their actual agreements. The notified body assesses diagnostic conformity and seeks the required medicines opinion. A partner's deadline can organize their work, but it cannot provide missing material or a performance result.

## SECTION 02

### Which specimens and decisions does the comparison represent?

The opening supplies three reasons to examine transfer. **Extraction** makes target material available from the specimen; changing it can alter the usable amount reaching measurement, especially with scarce or low-tumor-content tissue. A broader variant list offers more genetic differences for detection or interpretation, so evidence for the old list may leave new classes unanswered. Distributed use adds possible differences in handling, instruments, operators, software versions and repeat paths beyond the central laboratory. These mechanisms explain why a comparison is needed; the record supplies no size or performance effect for any of them.

Before an agreement percentage can be interpreted, the partners need to know which material entered it. The inventory begins with trial specimens, lawful access for the



proposed use, remaining amount, handling and retest quality. Depleted or unavailable specimens could differ from those retested in marker status, tumor content or clinical outcome. That possibility requires investigation; none of those causes is established here. If missingness selects a clinically different group, even excellent agreement in the remaining material may poorly describe the intended patients. An adequate explanation of availability and representation could justify a wider inference.

Next comes the result itself. Clear positives and negatives can dominate an overall agreement figure while observations near a treatment cutoff remain few. Because crossing that cutoff can change eligibility, the comparison needs classifications and uncertainty at the decision boundary as well as the average. A failed or uncalled test is another state: the person has no reportable result, not a negative result. Eligible, ineligible, indeterminate and failed outcomes, relevant variant classes and low-content conditions therefore need explicit accounting. The medicine evidence must then be examined for the patients whose decisions could change.

**Reviewer Lens.** Three counts answer different questions: trial specimens inventoried, material usable for commercial retesting, and valid matched results. Following a missing or failed specimen through that flow can reveal why the reported denominator became smaller. A treatment-classification cross-tabulation then shows which pairs changed category. Without it, a high agreement fraction may be arithmetically correct while leaving the clinical question open.

**Consultant Lens.** Rights and inventory work belongs before destructive retesting because the next experiment can consume the only material able to answer a critical question. A useful sample plan identifies needed strata, repeat and no-call rules, and the alternative if the archive is inadequate. Prospective collection or other evidence may be worth investigating. An archive becomes a feasible bridge resource only when access, quality and representation support that particular comparison.

## SECTION 03

## Who judges the diagnostic–medicine relationship?

For a device that meets the full CDx definition, the treatment relationship creates an additional question within conformity assessment. Under **Annex IX §5.2**, the notified body assesses the device documentation and seeks a medicines scientific opinion **before issuing its European Union (EU) technical-documentation assessment certificate**. The alternative **Annex X type-examination** route, paired with the applicable Annex XI production-assurance requirements, also requires that opinion through **§3(k)**. Type examination concerns the defined device type; changing to that route does not remove the medicine question. Both consultation provisions use the draft **summary of safety and performance (SSP)** and draft **instructions for use (IFU)** to examine suitability for the corresponding medicine. The notified body gives the opinion due consideration and retains its own device decision. A favorable opinion can support the relationship it examined without supplying every device requirement or issuing a certificate.

The medicine's actual authorization path determines which institution receives that question. **Annex IX §5.2(c)** and **Annex X §3(k)** allow a Member State's designated medicinal-products competent authority or the **European Medicines Agency (EMA)**, with specific constraints. A medicine falling exclusively within the mandatory centralized-procedure scope requires EMA. If the medicine is already authorized, or its authorization application has been submitted, the consulted authority or EMA must be the one responsible for that authorization. Thus an existing or submitted authorization ties the opinion to its responsible regulator; where neither constraint selects the body, the legally permitted authority/EMA choice remains. The opening supplies no medicine identity or regulatory path, so no branch is selected. Establishing those facts could turn this open routing question into a definite institutional choice.

The partners can nevertheless prepare a coherent question. It needs the intended diagnostic use, medicine and population, both assay versions, difficult and failed comparisons, and aligned draft device and medicine wording. A later cutoff or specimen change can alter both the treatment population being evidenced and the documents sent for consultation. Early coordination is useful because each reviewer needs a stable question and has a different decision to make; an informal discussion is not a completed bridge or opinion.

**Reviewer Lens.** A consultation plan becomes assessable when the proposed conformity route, medicine authorization path, draft labels and bridge identify the same product relationship. The record should show where the opinion enters the notified body's final device decision. If one document describes an earlier kit or narrower population, the existence of an opinion does not close that mismatch. The request is for the applicable version and decision trace, not simply evidence of a meeting.

**Consultant Lens.** Evidence freeze, partner review, device assessment, consultation questions and possible revision form a dependency chain. Conditional dates can guide staffing and funding before all evidence exists. The fixed pharmaceutical deadline remains a business constraint; it does not determine how long a substantive question will take or when a certificate will issue. A realistic schedule exposes which answer would permit the next step.

### **When can two regions reuse the same evidence?**

Shared evidence is useful when it answers the same question for both proposed uses. Agreement on patient groups, assay versions, specimen rules, cutoffs, reference decisions and endpoints can avoid needless consumption of tissue. But United States (U.S.) and EU claims and submission procedures remain separate. A broader European purpose or changed kit may need additional work even when part of a study remains relevant. A defensible evidence map distinguishes direct relevance, a difference needing a bridge and a study that does not answer the claim, with the actual reason for each finding.

For a **next-generation sequencing (NGS)** kit, a controlled tumor-specimen set and known alterations may support shared work while new extraction, variant classes and treatment interpretation still need evaluation. In a separate **digital pathology** example, an image score studied with one scanner at an expert center may leave a distributed reader workflow unanswered. That comparison transfers the method, not a product identity: the opening kit is not thereby image-based, and an image biomarker is not thereby a CDx. Its actual purpose and medicine-essential role would have to establish those conclusions.

### **What changes would reopen that relationship?**

The opening kit has no supplied approval or commercial release. The partners can still decide who will identify a new medicine indication, specimen, laboratory, software rule, knowledgebase or cutoff, who can obtain samples, and who assesses evidence and labels. For an approved CDx under **Annex IX §5.2(f)**, changes affecting performance, intended use or suitability for the medicine require the manufacturer to inform the notified body before making them. The body decides between a new conformity assessment and a certificate supplement; the supplement path includes assessment and a further opinion from the consulted medicines authority. Under **Annex X §5.5**, changes affecting those same aspects of an approved type trigger medicines consultation. These premises explain why actual impact and the existing route matter. A change that does not affect those aspects does not acquire that particular trigger merely because it is called an update; other applicable change duties still need assessment.

For a future marketed sequencing CDx, changed interpretation could switch a variant's therapy-positive status while measurement remains similar. For a future pathology treatment-selection tool, a scanner or queue update could change which image a reader assesses or when a case is seen. These are possible future mechanisms, with no incident or deployed state added to the opening. A reviewer would need the actual change and field evidence to judge its impact; a consultant would give both partners a shared gate for evidence, labels and any required notice or renewed consultation.

**Additional exhibit — opening program.** Most observed discordance lies near the treatment cutoff and in the variant class defining the therapy-positive population. The proposed commercial assay also fails more often in low-tumor-content samples. No counts, effect sizes, causes, patient outcomes or final eligibility classifications are supplied.

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### Decision

**Could a high overall agreement figure alone connect this kit to the trial assay's treatment evidence? Which observations and conditions would need to be resolved before judging the treatment-selection claim?**

### How the decision boundary changes the weight of the average

The supplied disagreements concern places where a result can change treatment classification. More failed low-content samples can also leave an important group outside valid-pair analysis. Overall agreement alone supplies neither the affected eligibility decisions nor the missing specimens' implications.

The relevant evidence includes the full inventory and failure flow, cutoff-region and therapy-positive-class comparisons, kit configuration and distributed-use evidence, and an account of how changed classifications relate to the medicine data. The packet supports holding a broad equivalence conclusion open. It does not quantify harm, establish a failure cause or make every part of the trial evidence unusable.

**Reviewer Lens.** The central trial can retain credit for its own assay and represented conditions while the commercial bridge remains incomplete. A narrow finding identifies which therapy-positive classes, boundary decisions or failed low-content specimens are unsupported. A proposed narrower claim must then have its own complete scientific, analytical and clinical basis; narrowing the wording alone does not create the missing overlap.

**Consultant Lens.** The real choices include reducing kit differences, targeted comparison with accessible archived material, a justified cutoff or claim change, and a staged laboratory footprint. Each has a material-rights and evidence dependency, an effect on treatment classification, label and consultation work, and a conditional

schedule. Explaining those dependencies lets management compare options without promising that any one will meet the medicine deadline.

SECTION 04

What can the current bridge record actually establish?

A filled record can be useful even while its evidence gates remain open. In the table, material must first exist, be usable and be accessible for this purpose. The two configurations then define the comparison, and the resulting classifications determine what medicine evidence is needed. The final rows carry that question into device assessment. “Tissue” still gives no preservation method—fixed, frozen or otherwise—and a planned kit is not an identified released configuration.

| Opening-program dependency       | Supplied boundary and next decision                                                                                                                                                                                              |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Specimen access and availability | Selection, handling, remaining amount, permitted access and depletion need records before retesting is feasible. No preservation method or count is given.                                                                       |
| Actual trial and kit states      | Central assay versus changed extraction, broader variants and distributed kit sites. Each difference and combination may affect signal, failure, classification or use; “prototype” establishes no equivalence.                  |
| Treatment-decision comparison    | Eligible/ineligible, no-call and unavailable specimens need accounting, with cutoff, therapy-positive class, low-content and uncertainty analysis. The exhibit supplies patterns of discordance/failure, without counts.         |
| Consultation question and actor  | The manufacturer prepares device evidence, draft summary and instructions; the notified body assesses and seeks the medicines opinion under IX §5.2 or X §3(k). The medicine path remains unknown.                               |
| Present disposition              | Broad trial-to-commercial equivalence is unsupported by this packet. Additional evidence or a genuinely supported narrower purpose is needed before a particular kit claim can be judged. No opinion or certificate is supplied. |

Follow the first row into the third. Even a sound comparison plan cannot retest unavailable tissue; even a complete retest inventory cannot say whether classification changed until results are examined. This separates **availability**, the fraction of inventoried specimens retested, from **observed agreement**, the fraction of retested

comparisons that match. A wider claim about the intended population or treatment decision additionally needs representation, failures and the clinical meaning of discordance. Better access could resolve the first gate without resolving the latter ones.

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### **Decision**

**A separate exercise inventories 120 distinct trial specimens. Twenty are unavailable for commercial retesting. Of the 100 retested, 96 agree with the trial assay and four disagree. Which fractions can be reported, and which conclusion do they leave open?**

#### **What each denominator permits**

Availability is  $100/120 = 83.3\%$  of the inventory. Observed agreement is  $96/100 = 96\%$  among retested specimens, with  $4/100$  disagreements in that group. The twenty unavailable specimens are neither agreements nor disagreements. The inventory is not identified as the entire intended population.

The record does not explain whether availability depends on marker state, cutoff region, quality or clinical outcome, or whether the four differences alter treatment classification. Positive/negative and failed-result accounting and those missingness questions therefore remain necessary. A cause cannot be inferred from absence, and 96% cannot by itself establish population or CDx treatment-decision equivalence. The later 95% microcase is independent and supplies no denominator or result to this exercise.

### **Reviewer Lens – give the evidence an order**

Exact central and kit identities tell the reviewer what to compare. Inventory, access and failure records tell which observations are possible. A cross-tabulation then shows the paired eligibility calls by relevant class and cutoff region; intended-user/site evidence establishes the setting to which they apply. Finally, the medicine relationship and consultation package show what decision is being requested. This sequence preserves the positive trial contribution while locating each commercial difference that remains unbridged. An overall percentage cannot stand in for an absent cross-tab.

### Consultant Lens – make each next step conditional on its input

An integrated diagnostic–medicine timeline joins the versioned comparison, rights-checked sample plan and draft-label alignment. It gives sample access, analysis, change notice, publication and authority selection named owners. Management can fund planning and feasibility now and compare convergence, targeted work, narrower purpose or staged sites. Study execution still depends on an adequate protocol and applicable approvals; a claim or launch still depends on actual evidence and the relevant decisions. Separating those commitments keeps an estimate usable as conditions change.

**Changed-case transfer — a proposed body-fluid extension.** A different approved tissue CDx proposes testing body-fluid-derived material. The exact fluid and analyte path are unspecified.

#### Note

Which marker–medicine evidence might remain relevant, and which collection, abundance, detection, population and treatment-classification questions would reopen? What would the reviewer and consultant need before extending the label?

### How a specimen change changes the bridge

The scientific relationship may remain useful if it fits the proposed use. It does not establish the new collection path, target abundance, analytical detection or clinical interpretation. Those differences need an applicability argument and evidence, including their effect on eligibility and labeling. The reviewer locates the remaining gaps; the consultant scopes specimen-specific work and the actual notified-body/medicines consultation consequences. A new specimen word is not itself an approved extension.

## SECTION 05

### How do two partners make the next comparison possible?

The next studio is an independent dossier, with no established identity or release link to the opening manufacturer. It adds a new interpretation component: a software knowledgebase. Stored reference information and rules can change a reported meaning after a variant has been detected, even when raw sequence reads remain unchanged. The studio therefore uses the bridge method to organize joint access, version and decision work. It inherits no opening result.



**Practice record.** A pharmaceutical pivotal trial uses a central laboratory assay. Its diagnostic partner plans a kit with different extraction, a broader variant list, distributed laboratories and a new software knowledgebase. Tissue is scarce, the medicine-label deadline is fixed, and both partners call the trial assay a prototype. This packet supplies no assay comparison, inventory, cutoff, medicine authorization path or commercial release.

### **Which partner controls the missing input?**

The first dependency is the trial result's job. What did the assay measure, and how did that result enter the medicine study? The proposed kit's specimen requirements, classification rule, laboratories and patient group then define what needs comparing. If the kit puts different patients in the therapy-positive group, its engineering team cannot resolve the clinical implication alone. The medicine partner needs to establish what treatment evidence applies to that changed group.

A joint configuration ledger makes that comparison recoverable. For the trial it identifies assay, extraction, targets, software/database, cutoff, reporting rule, laboratory and dates; beside those it records the intended kit state. In this studio the new knowledgebase is an explicit difference, with possible interpretive effects after measurement. The ledger needs its actual content and version relationship, not a conclusion that a shared family name makes the states identical. A documented match or justified bridge could close a particular difference.

The next dependency is access. Someone must inventory tissue, check consent and data-use rights, release material and preserve its custody history. The partners also need owners for analysis approval, assay/medicine-label change notice, publication and regulatory responses. If a specimen can be consumed only once, its most decision-relevant test matters before it is used. Missing rights or material holds the proposed archive comparison open while lawful, scientifically credible alternatives are investigated; it does not establish failed assay performance.

**Reviewer Lens.** The configuration comparison, access basis, trial result-use rule and written change/escalation agreement show whether the partners can execute the stated plan. A signed collaboration intention is useful only within its actual commitments. It cannot substitute for specimen permission, and an access problem should not be reported as an adverse experimental result. Each missing input calls for its own evidence request.

**Consultant Lens.** Inventory and method finalization precede bridge execution; the resulting evidence feeds the device file, notified-body questions, medicines opinion and label alignment. This sequence supports conditional budgets even when a critical input is unresolved. Funding protocol work may be sensible while study execution remains blocked by inaccessible material. Keeping those states separate prevents the fixed medicine deadline from becoming an invented certificate date.

## What can change between tissue and the reported eligibility call?

Begin with what arrives. The studio says tissue but gives no preservation method. Preparation and extraction could leave different amounts of usable target for the two assays. A low-content specimen might then fail before producing either a positive or a negative result. This is a mechanism to examine, not a demonstrated studio failure or cause. Restricting the comparison to easy retests could remove the very material needed to evaluate that intended condition.

For specimens that produce results, the broader variant list can change what is found; the new knowledgebase can change what the finding means. A cutoff can place the same sample on a different side of medicine eligibility. Distributed laboratories add possible handling, instrument and operator differences absent from the single central setting. The chain therefore asks four connected questions: whether a result is available, whether the target is found, how it is interpreted, and whether intended users can apply it reliably. Resolving one question does not silently resolve the next.

The original inventory is the starting denominator for that chain. Material not accessed, material tested without a valid result and valid disagreeing pairs have different explanations and must remain distinguishable. Treatment classifications can then be compared for evaluable pairs, with cutoff-near, therapy-defining and low-content strata visible. The final clinical question is whether the medicine evidence represents patients shifted between categories. Without the absent cross-tab and flow, neither a failure rate nor an adequate bridge can be calculated for this studio.

The table retrieves each link after its mechanism has been considered. Its rows describe the next evidence decision, with no completed study result added. Reading the specimen and missingness rows together prevents an average among available material from becoming a conclusion about people missing from it.

| Comparison layer      | Difference or condition to establish                                                                 | Decision it can support                                                   |
|-----------------------|------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| Specimen and handling | Collection, preservation if known, extraction, input, stability, remaining tissue and permitted use. | Who can receive a valid result, and who is absent before paired analysis? |
| Measurement           | Targets, chemistry, instrument, quality control (QC) and no-call rules.                              | Whether claimed variant classes are found or missed at relevant levels.   |
| Interpretation        | Calling pipeline, cutoff, new knowledgebase, report rules and versions.                              | Whether the reported information changes medicine eligibility.            |

| Comparison layer            | Difference or condition to establish                                                                              | Decision it can support                                                               |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Intended workflow           | Central versus distributed laboratories, operators, training, turnaround and repeats.                             | Whether the intended commercial process reliably delivers that decision.              |
| Discordance and missingness | Inventory, unavailable, failed, valid and disagreeing specimens; cutoff, low-content and therapy-defining strata. | Whether agreement and its uncertainty represent the treatment decision being claimed. |
| Two labels                  | Diagnostic purpose and limitations compared with medicine population and use.                                     | Whether the claims are coherent and which relationship belongs in consultation.       |

**Reviewer Lens.** Following one specimen through the table can establish its trial result, kit result or reason for absence, versions and eligibility calls. The inventory-to-analysis flow then asks whether that record represents the relevant group. A persuasive matched pair answers an individual trace question; it cannot supply population performance. For example, matching target detection in the measurement row still leaves a changed knowledgebase's eligibility effect in the next row to examine.

**Consultant Lens.** The comparison protocol turns these rows into prespecified inclusion, access, missing/no-call categories, treatment-classification analysis and uncertainty rules. Distributed-site evidence needs its own plan alongside paired testing. If the archive cannot answer a critical class or cutoff question, targeted new data or a genuinely supported narrower claim become options to assess. A larger count of irrelevant easy specimens would not resolve that specific gap.

### Which version of the question goes to consultation?

The manufacturer's submission needs to identify the candidate kit, intended purpose, performance evaluation and trial-to-kit bridge. The notified body assesses that device and seeks the medicines opinion under the applicable Annex IX or X route taught earlier. The opinion concerns suitability for the corresponding medicine; it does not repeat the entire device assessment. The notified body must give it due consideration in its own final decision. A different kit state or claim can therefore remain unsupported even when an opinion on an earlier relationship is favorable.

This studio's medicine path is unknown. Its joint ledger cannot establish “EMA consultation booked” merely by writing it. The medicine's actual authorization route must first resolve the authority/EMA branch, including the mandatory-centralized and already-authorized/submitted conditions. Draft instructions and the summary of safety and performance must then align with the medicine's proposed population and

biomarker wording. Added variants or a changed cutoff require an account of the people those choices identify and the treatment evidence supporting them.

After a future authorization, a new medicine indication, specimen, laboratory model or knowledgebase could change the same relationship. The actual impact and existing route determine which evidence, label, notified-body assessment and renewed-consultation duties apply. The Annex IX/X change premises taught above make this a joint change-control question. **Post-market performance follow-up (PMPF)** can address a justified residual question in supported use; it cannot create a missing initial trial-to-kit bridge. Nor can a partner email provide the evidence that the intended claim requires.

**Reviewer Lens.** The technical assessment, consultation request and opinion, final notified-body decision and two labels should describe a recoverable chain. A favorable opinion deserves bounded positive credit for the relation it addresses. The critical comparison is whether the submitted kit, population and versions still match that relation. An additional device requirement or changed claim remains a separate finding, not an implied rejection of the opinion.

**Consultant Lens.** A joint question cycle identifies which evidence and label versions both partners endorse, who answers each query and who resolves diverging device and medicine language. Contingency for new evidence is part of a useful estimate. Neither that estimate nor a fixed pharmaceutical labeling date is a legal review timetable or assurance of launch.

### **What would make a staged option defensible?**

Reducing the kit's differences is one option if it can technically and operationally use the trial method; a renamed prototype does not accomplish it. Archived-sample comparison is another if adequate, relevant material exists, can lawfully be used and supports a prespecified analysis. Changing a cutoff creates a new treatment-classification rule, so any improvement in agreement must be weighed against analytical, clinical and medicine-link evidence for that rule. Restricting initial specimen, variant or laboratory scope can reduce the work only when the actual offered product fits the boundary and has adequate support there. In each branch, a real change in those facts changes what can proceed; none guarantees a certificate or deadline.

A useful option record therefore compares evidence preserved and lost, tissue consumed, changes to purpose and labels, consultation questions, cost, timing and work that can occur in parallel. A global evidence plan can share scientifically applicable specimens and analyses while maintaining each region's claims and submissions. The consultant makes these constraints visible early enough for management to choose. A fixed medicine deadline does not resolve an unsupported device decision.

## Decision

The partners offer a high overall concordance, or agreement, figure for the studio kit. Is it sufficient to connect the central trial assay to distributed commercial use? What must be resolved before a staged option can be selected?

### What must support the partners' next choice

The figure needs its denominator, unavailable and failed specimens, cutoff-near and therapy-defining comparisons, knowledgebase/cutoff versions and intended distributed workflow. The medicine evidence must also apply to any group whose treatment classification changes. These inputs determine whether convergence, targeted comparison, a narrower claim or staged sites can solve the actual gap.

The studio supplies no percentage, completed layer-specific evidence or authority outcome. The offered aggregate is a result to appraise within its described comparisons, not sufficient support for the entire bridge. A proposed option remains conditional until its material, scientific, product and legal dependencies are satisfied.

## Who owns the decision that remains?

The diagnostic manufacturer owns the kit's purpose, evidence and changes; the pharmaceutical partner owns its medicine program and label. The notified body evaluates technical and performance evidence and makes its conformity decision. The medicines authority or EMA supplies the required scientific opinion under the actual selection branch. A consultant can integrate the sample/version ledgers, comparison, schedule and consultation package, but cannot acquire those statutory powers. Keeping the missing input beside its owner shows exactly which next decision is unsupported.

**Changed-case transfer — governing a later liquid claim.** A separate marketed tissue CDx proposes testing body-fluid-derived material. Its marker–medicine rationale may remain useful, while the sample path and patients yielding a result may change. No liquid extension or medicine-label update is supplied.

### Note

Which specimen-specific evidence and field information would the reviewer need? Which study, access, label, follow-up and consultation responsibilities should the partners settle before extending the claim?

### How the partners govern an extension

Analytical and clinical applicability, eligibility impact and actual relevant field information determine what the old evidence can support. The notified-body change assessment and applicable renewed-consultation question follow the real effect and route; the proposed specimen word alone supplies no approval. The consultant identifies who owns each study, sample right, label decision and PMPF task across the two partners. A useful scientific rationale can remain while the liquid claim is still unestablished.

## SECTION 06

### Quick discrimination lab

Each packet below is independent of the opening program, studio and 120-specimen exercise. Its numbers and history apply only to its own question.

**Record A — a reported average.** A separate trial-to-commercial comparison reports 95% overall concordance. Most disagreement occurs near the treatment cutoff and among low-tumor-content samples that determine eligibility.

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#### Decision

**Does the high overall figure establish an adequate bridge? What comparison would change the judgment?**

### How the average and the clinical boundary coexist

The 95% is a positive observation for the comparisons producing it. Yet disagreement concentrated where eligibility is determined needs its own assessment. The denominator, unavailable and failed specimens, eligibility cross-tab, cutoff behavior, variant claims and actual commercial configuration would show its reach. Those records could support a bounded bridge or identify further work; the aggregate alone cannot decide.

This is not the earlier 96/100 result and supplies no patient-harm count.

**Record B — a changed medicine population.** A pharmaceutical partner expands its label to a new population. The diagnostic purpose and evidence plan stay unchanged.

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#### Decision

**Can the existing CDx opinion simply be carried over to that population? Which changed relationship must be assessed?**

### Why a new medicine population reopens the device question

An opinion addresses the relationship and scope actually examined. The expanded population therefore needs appraisal of the marker–medicine relationship, specimen and assay performance, diagnostic purpose and both labels, followed by the applicable notified-body and consultation assessment. Relevant post-market surveillance (PMS) or PMPF evidence may contribute; an unchanged plan does not establish the new claim.

The packet gives no new result, opinion or device extension. Evidence showing applicability to the new population could change the support finding, but that conclusion cannot be inferred from the medicine-label change.

## SECTION 07

### Keep the treatment boundary attached to its evidence

The trial's central assay can provide valuable scientific and clinical context without establishing the proposed kit's entire claim. The bridge must account for extraction, variant scope, laboratories, unavailable and failed material, and the treatment decisions that actually change. The medicine's opinion informs the notified body's separate device judgment. Management's narrower or staged option becomes defensible only when the product, evidence and applicable process support that boundary. Neither a shared assay name nor a medicine deadline can supply the missing link.

### Check what you just learned

The working item states remain **A**, adequately explained; **U**, present but unclear; **M**, missing or inconsistent; **D**, deeper analysis needed; **C**, consultation indicated; and **N/A**, inapplicable with a reason. An evidence location and rationale explain the state and what would change it. These are not product grades or approvals. The seven source-era IVDR-CH14 identifiers retrieve expanded Workbook Part A2 cards through the original-to-integrated crosswalk.

### Every decision-changing layer • card IVDR-CH14-A01

Does the trial/kit comparison identify extraction, target and variant scope, pipeline, knowledgebase where relevant, cutoff, laboratory and report rules? The studio's new knowledgebase is its own fact; “prototype” establishes equivalence in neither main record.



**Eligibility before the average • card IVDR-CH14-A02**

Do inventory, availability, no-calls, cutoff and class-specific comparisons, low-content strata and uncertainty identify the actual treatment boundary? Every fraction needs its denominator; the independent 83.3% availability and 96% retest agreement answer different questions.

**Two purposes, two decisions • card IVDR-CH14-A03**

Do draft device and medicine labels, actual trial-result use, consultation input/opinion and the notified body's decision describe the same relationship and versions? A favorable opinion on that relationship does not issue the device certificate.

**A later change and its consequence • card IVDR-CH14-A04**

Who examines a new medicine population, specimen, site, cutoff or software interpretation, and what evidence, label, surveillance or consultation step follows from its actual effect? A change name alone cannot settle the branch.

**An integrated evidence timetable • card IVDR-CH14-C01**

Do specimen rights and inventory, prespecified comparison, configuration freeze, device assessment and consultation have explicit dependencies and conditional dates? Feasibility funding and permission to execute are different decisions.

**Conditional product and scope choices • card IVDR-CH14-C02**

How do convergence, targeted evidence, narrower purpose and staged sites compare in technical feasibility, treatment impact, tissue use, new work, cost and timing? Which actual evidence would make the chosen boundary supportable?

**Recorded partner commitments • card IVDR-CH14-C03**

Are sample/data access, change notice, evidence ownership, publication and escalation responsibilities established? An intention to collaborate cannot grant an absent right to consume tissue.

**Further reading**

Book 1C Chapter 13 offers optional practice with oncology CDx and bridge reasoning. Its case results do not fill the specimen, medicine or consultation gaps in these independent records.

## Source trail

- **Binding law, checked 2026-09-25:** IVDR consolidated 10 January 2025, Article 2(7), Article 48(3)–(4), (7)–(8), Article 56 and Annex XIII for the CDx definition, device performance and applicable conformity/consultation work; Annex IX §5.2 and Annex X §3(k) for the notified body's medicinal-products opinion, authority/EMA selection, and distinct final device decision; Annex IX §5.2(f) and Annex X §5.5 for relevant later change assessment. The law does not choose a medicine authority or establish a certificate for this fictional packet.
- **Official procedural context:** EMA CDx page identifies the current consultation guidance and Q&A and the mandatory EMA branch for medicines exclusively within the centralized scope; EMA consultation guidance, revision 1, effective 17 December 2024, §§1–3, describes the notified-body applicant, initial/follow-up consultation and source documents. Guidance is not a device certificate or a substitute for the binding text.
- **Method and packet boundary:** Trial/commercial comparison layers, specimen inventory, missingness and eligibility cross-tabs are original explanatory work applied to the live fictional source. No assay, medicine, patient outcome, sample-preservation method, release or bridge result beyond the stated independent exercises is inferred.

## CHAPTER 14

# Control the whole software release

**Retrieval check — before reading.** Chapter 13 followed a trial result into a changed commercial kit. Which specimen, method, cutoff and treatment-decision connections did “prototype” leave unexplained? Why did the medicine opinion and device certificate answer different questions?

**When images, decision rules, security dependencies or the user's view change, what evidence supports offering the complete new configuration?**

The release decision begins with the medical information a user will act on. It follows that information through every changed component, then asks what was tested, what scope may be offered and what each hospital actually installed. The same path makes device, AI and security questions concrete without turning one answer into all three.

## A release is more than its model file

A pathologist cannot act on a model file alone. A scanner produces an image; preparation changes what reaches the model; a rule turns its score into a category or priority; a worklist makes the case available for review. Each step can affect the meaning or availability of the final information. Before judging a release, we therefore need its medical job and the exact combination that delivers it.

**Opening record.** A cloud pathology platform is **certified for Scanner A and locked model M3.2**. The manufacturer proposes one quarterly release containing **Scanner B**, a **normalization upgrade**, a **threshold adjustment**, and a **user-interface redesign**. The software team calls it one release; the regulatory team sees four changes with potentially different evidence and certificate consequences. The packet does not give a class, certificate wording, measured new-scanner performance or permission to deploy the proposed combination.

Follow one image through the certified combination. **Normalization** prepares incoming image data for the model, perhaps handling color or intensity differences; this packet gives no exact method. Locked M3.2 produces a score. A threshold turns scores into displayed categories or priorities, and the interface determines what the pathologist sees and how. **Model weights** are numerical parameters learned during training. Keeping those parameters fixed leaves the scanner, preparation, decision rule and user task open to change. The quarterly name identifies a package, while this path identifies what needs evidence.

**Reviewer Lens — locate the existing support.** The actual Scanner A/M3.2 certificate, assessed configuration and labeled medical task identify the positive starting point. A manifest can then expose what the proposed scanner, processing, decision rule and interface change. The supplied certification deserves credit within its scope; its exact wording and configuration are still needed before deciding what else it covers.

**Consultant Lens — make the release a set of answerable decisions.** Each change needs an owner, affected clinical question, evidence and possible notified-body communication. Those dependencies allow time and budget to be planned now. Release and hospital deployment remain conditional on support for the assembled configuration.

### **What medical job does the software actually perform?**

Copying an image into storage leaves its clinical interpretation to another process. Selecting suspicious regions, assigning a diagnostic category or ordering cases by disease likelihood can change the specimen-derived information reaching a clinician and when it arrives. Which of those jobs is offered? “Workflow,” “cloud” and “assistive” describe a product without answering that question. The intended output and its use supply the starting facts.

For **medical-device software (MDSW)**, an independent medical purpose and a supply arrangement are separate questions. An integrated function can have its own medical purpose. Another function can drive or influence a device, or be intended specifically to enable its use or directly assist its medical function. Merely being used nearby does not establish that accessory relationship. A bounded archive that copies and retrieves can lack an independent medical purpose; a content-selecting score needs a different analysis. The opening does not supply enough purpose and relationship detail to requalify the certified platform from its name.

Which device law fits that job? Under Article 2(2) of the **In Vitro Diagnostic Medical Devices Regulation (IVDR)**, the device is intended for examining specimens taken from the human body, solely or principally to provide the specified medical information. That includes information about physiological or pathological states, predisposition, predicted treatment response and therapeutic measures. A digitized tissue path can meet that specimen-based purpose; merely receiving a health-related file does not establish it.

For a different medical job, the **Medical Devices Regulation (MDR)** definition requires the manufacturer's specified medical purpose and a principal intended action that is not achieved by pharmacological, immunological or metabolic means, although those means may assist it. Its purposes include diagnosis, prediction, monitoring and treatment of disease. MDR Article 1(6)(a) excludes IVDs covered by the IVDR, so a purpose fitting the IVD definition is not assigned to MDR simply because software is involved. A function outside both definitions stays outside that device-qualification

conclusion unless another applicable relationship is established. Changed purpose, output or device relationship can therefore change the branch. Class, route, certificate scope and claim support follow as separate judgments; none is supplied for the proposed release by the Scanner A certificate alone.

A separate sequencing example makes the distinction tangible. Archiving raw sequence read files preserves data. Accepting or rejecting a patient's variant result before reporting can pass an unsupported call or suppress a reportable one. Those possible errors explain why the classifier's real purpose and device relationship matter. They supply neither a class for that separate function nor missing purpose facts for the pathology platform.

**Reviewer Lens — compare the promise with the data path.** Labeling, sales claims, screens and example outputs should describe the same medical job. New specimen-derived information and an output that changes another device's result raise different relationship questions. Comparing them with the actual assessed claims helps locate precisely where the qualification rationale or scope is missing.

**Consultant Lens — make a purpose memo useful to engineering.** Input, transformation, output and user decision connect legal questions to a function the team can inspect. Own medical purpose, device relationship and supply arrangement can then be explained before class, AI status or release route. A changed feature reopens that reasoning even if the product adjective stays the same.

### **Can the evidence follow the same result?**

Once the job is known, the evidence needs to follow its path. The certified record identifies Scanner A, normalization, locked M3.2, threshold, interface and supported cloud environment. The proposed record identifies every changed version and combination. A requirement specifies a needed behavior; risk analysis explains a possible failure and consequence; verification tests the specified behavior; intended-use validation examines the assembled task with its intended users. Release, supplier and field records then identify what was built, assessed and installed. Each record answers the next person's question about the same configuration.

Suppose a worklist requirement is to keep low-confidence cases reachable. Queue behavior, representative user tasks, recovery and instructions would need to support it. A passing score test alone would leave case access unanswered. If the view changes, those task and recovery conditions need re-examination. A separate sequencing pipeline similarly needs a trace from raw reads and knowledgebase through the report, user action and later update. These are possible evidence paths, with no passing or failing result added to the opening.

The IVDR makes that product path consequential. Annex I §16 requires software repeatability, reliability and intended-use performance, with risks or impaired

performance from a single fault reduced as far as possible. Section 16.2 requires state-of-the-art development that takes account of lifecycle, risk management including information security, verification and validation. Section 16.4 requires the minimum hardware, network and security conditions needed for intended operation to be specified. A changed environment can therefore reopen the evidence even when the model is fixed. Annexes II–III require the relevant technical and postmarket documentation.

The **quality management system (QMS)** controls how that work is done; product records show what it established. **IEC 81001-5-1:2021**, from the International Electrotechnical Commission and read with Interpretation Sheet 1:2025, offers a voluntary selected method for the security lifecycle. Selecting the method supplies neither a product passing result nor a claim of harmonized presumption under the IVDR. Its particular requirements matter when the manufacturer uses that method; the binding device duties remain their own obligations.

**Reviewer Lens — find the missing connection.** A visibility or specimen-identity requirement can be followed through architecture, risk, tests, user evidence, open anomalies, release manifest and installed state. An absent link is a specific finding. Counting all the document titles would conceal which question remains unanswered.

**Consultant Lens — give teams a shared product identity.** Engineering, quality, security and regulatory work can use one configuration and traceability ledger. Their gates still have different owners: code freeze and deployment need evidence, while the selected notified body's procedure and actual certificate conditions determine the regulatory communication.

### **When does the AI Act add another question?**

Before asking whether artificial intelligence is high-risk, does the offered function meet the Act's **AI-system** definition? Article 3(1) describes a machine-based system designed to operate with varying autonomy which, for explicit or implicit objectives, infers from its input how to produce predictions, content, recommendations or decisions that can influence physical or virtual environments. It **may** adapt after deployment; continuing adaptation is not required. The practical inquiry is how input becomes an output and what that output can affect. Fixed learned weights do not by themselves establish or defeat the definition. M3.2's name leaves the actual system characteristics to be documented.

If the Act applies, its duties join the device obligations. Versioned risk, data, test and change records may serve both where they contain each required output. Sharing a file does not remove a distinct duty or prove that either framework's requirements were met.

The product-law route in **AI Act Article 6(1)** next joins two conditions. First, the AI system is itself a product covered by legislation listed in Annex I, or is intended as a safety component of such a product. Second, that product or system requires third-party conformity assessment under the listed law. Both must be established. Changing the intended role or the legal assessment requirement can change this route's conclusion; an “AI” feature name establishes neither.

The 2026 European Union (EU) amendment explains what makes a **safety component**. Under amended Article 3(14), it either performs a safety function for the product or AI system, or its failure or malfunction endangers the health and safety of persons or property. A safety function means that its intended purpose is to prevent or mitigate those risks. Being installed inside a regulated product alone does not establish that purpose.

Article 6(1a) excludes AI systems used **solely for non-safety aspects** of user assistance, performance optimization, service efficiency, automation, convenience or quality control (QC) from **safety-component status**. The alternative in which the AI system is itself a listed product still needs analysis on its own facts. But Article 6(1b) overrides that exclusion when failure or malfunction would endanger health and safety: such a system qualifies as a safety component. The failure path therefore changes the branch even if the feature keeps its friendly name. Article 6(1c) separately excludes from the second condition a product needing third-party assessment **solely** for risks unrelated to health and safety. A safety-component finding still leaves that assessment condition to establish.

For the opening, a worklist or result function that could affect timely clinical review warrants examining its safety role. That mechanism supplies no finding that M3.2 meets the AI definition, Article 6(1), or any harm outcome. A “quality control” label cannot establish a solely non-safety role if failure can affect diagnosis. The useful next record explains the actual function, failure path and legal route, so case-specific Union legal review can reach a conclusion on those facts.

If a system is ultimately classified as high-risk **under Article 6(1) and Annex I**, amended Article 113 sets **2 August 2028** for Chapter III Sections 1–3, except Article 6(5). That exception concerns the Commission's duty, after consulting the European Artificial Intelligence Board, to provide practical classification guidelines and examples of high-risk and non-high-risk systems. The postponement does not postpone that guidance duty. Nor does the 2028 date apply to every AI Act provision or every AI system; a different classification route or obligation requires its own date analysis.

The June 2025 joint Medical Device Coordination Group (MDCG)/Artificial Intelligence Board (AIB) guidance predates the 2026 amendment and is nonbinding. The Commission's Article 6 classification guidelines were still labeled **draft** on its page



checked 25 September 2026. Those sources help frame the opening's missing function and route questions; they cannot overrule amended law or certify this product.

**Reviewer Lens — preserve the joined conditions.** The definition rationale, actual function and foreseeable failure, Annex I product-law coverage and basis for third-party assessment belong beside the obligation/date map. A certificate for one configuration is relevant evidence, but it does not state every AI-definition or amended classification premise.

**Consultant Lens — identify what a shared record really covers.** Risk, data, oversight, transparency, logging and monitoring work can be coordinated across IVDR/MDR and any applicable AI Act duties. Each distinct required output and date still needs an identified place. An open qualification question remains open in the release plan until actual function and route facts resolve it.

### What does each change do to the next stage?

Scanner B supplies a different incoming image. New normalization then changes its preparation before M3.2 receives it. Acceptable isolated changes would not establish that the **new scanner and new normalization together** produce a supported input. Next, a threshold changes where scores become positive, negative or priority categories. **Sensitivity** is the fraction of true target cases detected; **specificity** is the fraction of non-target cases correctly excluded under the chosen reference. A changed threshold can affect both, with no measured effect yet supplied. Finally, the interface changes a user's route to a case, including possible low-confidence retrieval and recovery. Accuracy alone cannot show whether that task works.

The matrix collects those questions. Its rows need different evidence because they change different links. Read scanner and normalization together first, then follow the resulting score into the threshold and user view. The last row asks whether that entire proposed path is supported; it does not assign one legal step to every modification.

| Proposed part of the quarterly release | Mechanism to test                                                   | Decision record to open or create                                                                     |
|----------------------------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Scanner B                              | Different image acquisition enters the same model path.             | Scanner-specific analytic/clinical applicability, supported inputs, labeling and certificate scope.   |
| Normalization upgrade                  | Preprocessing changes the pixels or features passed to locked M3.2. | Old/new transform, image quality, paired compatibility with both scanners, and risk of new artifacts. |

| Proposed part of the quarterly release | Mechanism to test                                           | Decision record to open or create                                                                                                          |
|----------------------------------------|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Threshold adjustment                   | Same score may cross a different decision boundary.         | Prespecified operating point, reference/uncertainty, sensitivity and specificity, clinical and label effect.                               |
| Interface redesign                     | Case visibility, ordering or recovery may change.           | Intended-user task tests, hidden/low-confidence access, delay risk, instructions and monitoring.                                           |
| <b>Combined release</b>                | Input, score-to-category and worklist changes can interact. | Integrated performance/usability/security validation, supported configuration, change assessment, notified-body decision and release gate. |

For example, evidence for Scanner B under the old preparation would leave the new paired input open. Even a supported input would not settle a changed threshold or prove that the intended reader can retrieve every necessary case. The table therefore ends at the assembled medical task. New evidence changes the row and combination it actually addresses.

**Reviewer Lens — retain the old positive while testing the new path.** Scanner A/M3.2 remains the stipulated certified baseline. Per-change evidence identifies local effects, and an assembled test examines their interaction. The certificate's conditions then determine whether the new inputs, decision rule or user view enter its change procedure. A favorable isolated test cannot answer an untested integrated task.

**Consultant Lens — make stages technically real.** One candidate stage might pair Scanner B with new normalization while retaining the existing threshold and interface. Another combination may be more defensible. Either needs compatible components, its own evidence and labeling, and the applicable certificate/change process. Staging changes the question being assessed; it supplies no earlier lawful launch by itself.

## Separate release, permission and installation

What turns a packaged build into an offerable medical release? Its exact combination needs verification and intended-use evidence, addressed or acceptably bounded risk, security review, labeling and any applicable notified-body decision. Under IVDR Annex IX §4.11, where that route applies, planned changes to an approved device that could affect safety, performance or prescribed use conditions must be notified to the issuing body and require its approval. That body determines whether a new conformity assessment or a certificate supplement is needed. This is a conditional rule about the actual approved device and change; the packet does not establish its exact route or make every software update the same branch.

Installation answers a later question. A hospital may still run an older supported combination after a new release is approved. Its record needs to identify installed versions and whether a partial update or rollback remains supported. The change-impact and certificate decision establish the permitted release; confirmation from the hospital establishes its deployment state.

The same separation matters after deployment in other products. A sequencing library or knowledgebase update could affect result integrity or treatment interpretation without changing the assay's name. A pathology cloud or queue change could affect case availability without changing weights. These are conditional mechanisms in separate examples. Complaints, post-market surveillance (PMS) and relevant post-market performance follow-up (PMPF) may reopen their risk and product records. A future observation plan still cannot replace evidence needed before a proposed release.

**Additional exhibit — same Scanner A/M3.2 record.** Scanner B together with normalization **shifts model inputs**. The threshold **changes sensitivity and specificity**. The redesigned interface **hides low-confidence cases**. The manufacturer's cybersecurity reassessment is triggered **only by application-code changes**. No number of shifted images, delayed cases, harmed patients, attack or new certificate is supplied.

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### Decision

**Do unchanged M3.2 weights make the newly described combination one minor change? Which performance, user, security and certificate questions remain before release?**

### How the changed path determines the release inquiry

No. Input shift, the sensitivity/specificity effect and hidden cases reopen different evidence questions, and the assembled path may add interactions. A code-only security trigger also misses scanner, preprocessing, interface, service and supplier changes that can alter access, misuse paths or clinical availability. Each modification and the complete configuration need impact assessment, controls, tests, labeling and deployment conditions, followed by the actual certificate and notified-body steps.

The exhibit establishes these changed behaviors. It supplies no harmed patient, breach or permission to offer Scanner B. The existing Scanner A/M3.2 certification remains its own bounded positive.

**Reviewer Lens — connect each finding to a decision.** Baseline versions, old/new images and scores, cutoff analysis, low-confidence task tests, threat changes, anomalies and certificate conditions answer different parts of the review. Keeping their combination visible prevents one passing test from closing a failed or untested medical path.

**Consultant Lens — preserve a supported state during new work.** A sequence may keep Scanner A available while candidate combinations are tested. Its supplier participation, user communication, rollback and notified-body questions must fit each real stage. Fixed weights or a postponed interface do not guarantee that a stage is acceptable.

### What could change a message before the right person uses it?

Consider an accounting office. A clerk receives an invoice for a customer, records it and sends an approved statement. The office needs to establish which customer the invoice belongs to, who may change the ledger and which statement was approved. It also needs usable copies and recovery after an interruption. The final statement can be perfectly formatted while carrying the wrong customer's invoice. Checking the final file alone would miss that earlier mistake.

A **digital signature** adds a specific check. A private key signs particular electronic content; a receiver uses the corresponding trusted public key to verify it. Successful verification supports the integrity and claimed origin of those signed bytes under that key trust. It does not establish the customer match, freshness of the statement or absence of compromise of the signer or private key. Those are different propositions requiring different evidence.

Now follow a diagnostic result. A hospital service sends specimen and result identifiers to an application, which connects them to the examination and signs a report. The report crosses into a clinical record. At each crossing, someone needs authority to send or change information, a means of checking identity, an authoritative copy and a

recovery path after rejection or interruption. A crossing between components with different control or assumptions is a **trust boundary**. The invoice analogy exposes mistaken identity and unauthorized change; only the device's own risk and performance evidence can establish diagnostic consequences and a safe clinical response.

What if the sender is genuine but the message is old? **Authentication** checks a claimed identity; **authorization** checks whether it may perform the particular action. A **replay** resends a previously valid request. Identity verification may pass even though stale data should not be attached to the current specimen. Binding identifiers to the permitted sequence or time addresses that different question. A modified update package, by contrast, changes the signed contents: correctly trusted signature verification should detect that alteration, within the key limits already explained. Neither mechanism is an observed event in the Scanner A platform.

### **Which property and dependency need protection?**

A result may be disclosed to the wrong person, attached to the wrong specimen or unavailable when needed. These are different security questions. The **confidentiality, integrity and availability (CIA) triad** names them: limit who may see information; protect its content and identity against unauthorized or accidental change; keep the function reachable when needed. A **threat model** connects valuable information and functions to data/update paths, trust boundaries, plausible misuse and controls. It must therefore consider both an old authenticated request and unauthorized replacement of a component. A failed-login test cannot cover both.

Which products contain the affected component? A **software bill of materials (SBOM)** records components in a versioned product and helps answer that question when a vulnerability is reported. Inventory is a starting point for assessment, not evidence of security. A cloud service or external identity provider may sit outside the packaged components. A complementary service/supplier inventory needs its interfaces, responsibilities, supported versions and change contacts. Otherwise a dependency affecting the same medical result could disappear from the assessment because it was delivered differently.

## How does a security question become release evidence?

The selected IEC 81001-5-1 method, with its 2025 interpretation sheet, begins by assigning competent owners and protecting development and delivery systems. A threat model then supports requirements: for example, reject an identifier mismatch or verify an update's origin. Architecture identifies where those controls act. Tests examine expected operation and plausible wrong paths, and release records identify exact versions, issue dispositions, update methods and conditions of use. Later supplier changes, vulnerability reports and retired installations return to that same record. Application-code change is only one possible reason to reopen it. The method complements product risk and clinical performance work without replacing binding IVDR requirements.

One test may deliberately attempt an attack. **Penetration testing** is an authorized, scoped attempt to discover and exploit weaknesses in the specified environment. Under the selected method's §5.7, security testing includes requirements, threat mitigations and vulnerability work as well as penetration testing, with documented means to keep the effort objective. A clean scoped result supports what was actually challenged. A new interface or configuration creates a new coverage question; it cannot inherit universal security from the old test.

Follow the specimen identifier into the lifecycle table. The boundary sketch explains why a binding rule is needed. The replay challenge tests that rule while a valid transaction tests continued operation. The release manifest then identifies the configuration that earned those results. A later library notice sends the team back to its versions and risk assessment. The rows are successive decisions about that path.

| Lifecycle question                                         | Evidence and decision it supports                                                                                                                                   |
|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Who controls development and delivery? (Clauses 4 and 5.1) | Roles, competence, supplier controls and protection of build, signing and delivery systems.                                                                         |
| What must resist misuse? (5.2–5.4 and 7)                   | Threat model, security requirements, architecture, interface assumptions and potential effects of a control on clinical use.                                        |
| Do those controls work? (5.7)                              | Positive and adverse-path tests, vulnerability assessment and applicable penetration testing, with the tested configuration, objective review and remaining limits. |
| What can be released or updated? (5.8, 6 and 8)            | Exact components, authenticity and integrity checks, compatibility, issue disposition, instructions, recovery and supported environments.                           |
| What changes the judgment later? (7 and 9)                 | Vulnerability intake, affected-version analysis, possible safety/performance consequences, response, communication and decommissioning.                             |

The release row needs the earlier test results and issue dispositions. Under §5.8, findings go through problem resolution and security issues are addressed and tracked to closure before release; an unresolved title in a log is not a completed assessment. The relevant records include remaining-risk treatment and conditions for use. If a later vulnerability affects a listed version, the last row returns to those conditions and can change the release judgment without any new model weights.

**Reviewer Lens — inspect rejection and recovery together.** The hospital request, signed report and clinical record show whether the right examination survives every boundary. The supplier-update path adds authenticity, compatibility and installation evidence. An old request or altered package should be challenged, but rejection also needs to leave a usable clinical recovery route. A component list cannot establish that combined result.

**Consultant Lens — connect supplier delivery to product use.** Supplier versions, notifications, compatible combinations and installation confirmation need named responsibilities in the release plan. A supplier can provide a fix while the manufacturer still has to assess the medical product that uses it. The same ledger connects those two jobs without transferring away product-level regulatory responsibility.

### **What work has the manufacturer undertaken for each item?**

The interpretation sheet makes three declared responsibilities explicit. For **maintained software**, the manufacturer assumes the security risk and provides updates whose integrity users can verify. For **supported software**, it notifies users of known security risks and update compatibility, including mitigations when an update is not approved. For **required software**, it considers known security risks before the health-software release, even where another party maintains the item. These declarations tell the operator which security work and information to expect; they are not names for particular technologies.

The categories are nested: maintained is also supported and required; supported is also required. The nesting applies when a clause names a category. Thus §5.2.3's required-software risk work covers all three; §6.3.1's supported-update communication also covers maintained items; §§6.3.2–6.3.3's update delivery and verifiable integrity apply to maintained items. It does not add unspecified duties to every clause. Changing the declared legal category needs its risk transfer and communications assessed, and does not waive legal responsibilities.

Annex F concerns a different question: a **whole health-software product** developed without all the method's lifecycle activities. Its alternative requires the specified gap assessment and security work, including full system security testing and an evaluation of overall remaining security risk to decide fitness for continued use. The version and continued-use rationale must be documented, with a migration plan and the required



postrelease activities under Clauses 6–9. That is not a fourth component category or a shortcut obtained by calling an old library “transitional.” No such conformance is established for either fictional product.

A signing library, operating system and hospital identity service can therefore have different declared responsibilities. If the manufacturer maintains the library, its provided update needs assessment and verifiable delivery. If it supports the operating system, operators need known-risk and compatible-update information. If it requires the identity service, known risks and the application's dependence still need consideration. Changed declarations alter the promised work and transfer of risk; the component's technology name does not choose the category. An integration failure can affect the clinical report in every category.

### **Can the proposed controls support this connected application?**

**Independent security exercise (SC-1).** This diagnostic application is separate from the Scanner A platform. Its manufacturer declares the report-signing library **maintained**, the third-party operating system **supported**, and hospital identity-provider software **required**. Required status includes pre-release consideration of known security risks. It does not remove the interface from the threat model. These are stipulated responsibilities for this exercise.

An authenticated hospital service sends specimen/result identifiers across a network boundary to this application. A signed report then crosses into the clinical record. A replay could attach stale data to a current report; stolen update credentials could permit substitution of the signing component. Neither possibility is an observed replay, substitution, wrong report or injury. Where would the system block each path, and what evidence would show that it did?

The proposed controls bind request, specimen, result and allowed sequence; reject duplicates and invalid identifiers; authenticate updates and detect package alteration; log the decision and preserve recovery. Their proposed challenges include a correct request, replay, altered identifier, expired credential and modified package. Rejecting an attack is only part of the result needed. Legitimate reports must remain recoverable, and a mismatched result must not be accepted later by the clinical system. No passing result has yet been supplied.

Now the company proposes a library fix and operating-system update together. Exact versions and supplier notices identify the candidate pair, but cannot show whether the assembled application signs the correct result and remains available. Paired testing needs to address partial installation and rollback because either can leave an unsupported combination. The required identity-provider interface supplies another testing premise. Under the selected method's §5.7.1(d), each software service used for intended functionality needs testing in the context of the provider, manufacturer and operator responsibility agreements. Here the application relies on that service, so

testing must challenge the connected boundary within those agreed responsibilities. A changed service or interface reopens that coverage question even if application code is fixed.

The record below follows the same message from possible error to release decision. Read its flow and control rows together: a proposed binding rule addresses a replay mechanism, while only executed tests could establish effectiveness. The supplier row then identifies the combination those tests would need to cover.

| SC-1 record             | Filled entry                                                                                                                                                               | Current decision                                                                             |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Flow and possible error | Hospital identifier request □ diagnostic application □ signed report □ clinical record. Replay could attach stale data; unauthorized update could change signing behavior. | Threats identified, not observed failures.                                                   |
| Control and challenge   | Bind identity/sequence; authenticate updates; test replay, mismatches, expired credentials, package alteration, valid operation and recovery.                              | Requirements and tests proposed; effectiveness not yet shown.                                |
| Supplier combination    | Maintained signing library plus supported operating-system update, with required identity-provider interface.                                                              | Obtain exact versions, supplier information and combined compatibility/performance evidence. |
| Release and fleet       | Quality and qualified security/clinical reviewers decide on residual risk and applicable change procedure; operations records installations.                               | Hold the combined release pending evidence; no installation state established.               |

After the last row, there is still no release or installed-fleet result. The missing facts are combined-configuration evidence, qualified review of remaining risk and the applicable change decision. Installation confirmation would answer a further question even after those gates close.

**Reviewer Lens — preserve the clinical purpose of rejection.** Operator visibility, a recovered report's specimen match, supplier versions and adverse-path tests make the medical task inspectable. They also keep an imagined report error in the risk analysis from becoming an alleged field event.

**Consultant Lens — assign the work behind the gate.** Replay, tampering, compatibility, result-integrity and recovery tests need the combined versions. Supplier responses, qualified risk acceptance, applicable change assessment and hospital

installation confirmation have different owners. A vendor's “fixed” notice can start this work without completing its product-level decisions.

### **What can reopen the decision without new application code?**

A reported vulnerability can affect an unchanged component already in hospitals. Clause 6 therefore requires an update policy that considers technical and safety impact, public knowledge, available exploits, the affected fleet and effective external controls. Those circumstances govern a timely response; the method supplies no universal number of days for every patch. A **Common Vulnerability Scoring System (CVSS)** score rates technical severity and characteristics. The medical path and actual use are still needed to assess patient-harm probability and severity.

Clause 7 requires periodic threat review, at least annually for released products, and updating when new threats require it even without a design change. Clause 9 receives and evaluates vulnerability information, affected versions and safety/effectiveness implications. A newly exposed service or supplier issue can therefore change the accepted risk before another code release. At retirement, the record also needs who withdraws access and credentials, handles retained clinical records under applicable rules, and confirms treatment of interfaces and copies. Disabling one login cannot establish that the product was safely removed from use.

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### **Decision**

**All prespecified replay/tampering, compatibility, result-integrity and recovery tests now pass on the intended configuration. Qualified reviewers accept documented residual risk, and the applicable change assessment is complete. What release conclusion can this support? What remains unknown if one hospital has not confirmed installation?**

### **How a passing release and an unknown installation coexist**

The stipulated evidence can support controlled release of that tested configuration and approved scope. Its requirements, exact versions, results, issue dispositions and acceptance record preserve the basis of the decision. The unconfirmed hospital has an **unknown deployment state** requiring follow-up. That missing confirmation is neither a failed product test nor evidence that the whole fleet has updated.

New vulnerability information that challenges accepted risk can reopen the decision without a code change. This bounded positive belongs to the independent security application; it does not close the Scanner B proposal.

## Reviewer Lens – inspect the offered and installed states

The opening's medical function and applicable device law lead to its actual Scanner A/M3.2 certificate and configuration. Each modification then needs its own clinical, security and user evidence plus an assembled-release assessment. Any AI Act duty follows the actual legal trigger and date. That leaves assessed release, lawful scope, labeling and hospital installations visible as distinct findings. The independent security application needs the same state discipline without importing its hypothetical threats into the Scanner A record.

## Consultant Lens – make evidence part of release planning

A purpose memo and change matrix become useful when each changed function or interaction has an owner, risk path, evidence question and closure condition. Placing those gates inside software planning exposes supplier dependencies, supported combinations, notified-body questions and rollback conditions before the end of a sprint. Planning and provisional budgets can proceed while evidence and decisions are obtained. A staged label supplies neither a supported configuration nor a promised market date.

### Changed-case transfer

#### Move inference into the hospital.

Suppose the same model moves from central cloud inference to local hospital deployment. Which environment, access, logging, installation and clinical-performance assumptions need to be revisited before the new combination is offered?

## How a familiar model acquires a different operating context

The hospital now controls parts of computing, access, logging and update installation. The supported architecture, scanner/preprocessing compatibility, trust boundaries, supplier responsibilities, clinical performance, monitoring and certificate-change questions need to match that actual environment. A familiar image and model do not establish those connections. This conditional transfer supplies neither authorization nor a finding that either deployment works.

## Quick discrimination lab

**Micro-case 1 — a cloud service's output.** A service called “workflow software” combines specimen-derived results and patient factors to generate a treatment-response category.

**Decision**

**Does “workflow software” establish an administrative role for this output? Which purpose and device-relationship facts must precede a class or AI Act conclusion?**

**How the output directs the qualification inquiry**

No. A treatment-response category may supply medical information regardless of its administrative name. Exact inputs, transformation, output, users and intended use need comparison with the IVDR/MDR definitions and the software's device relationship. Class, claim support, certificate route and any AI Act analysis then need their own premises. This record supplies no accuracy or authorization finding.

**Micro-case 2 — several changed parts.** An AI model update is validated and signed. The preprocessing library and scanner calibration profile also change, but are called infrastructure maintenance.

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**Decision**

**Does the validated and signed model update settle this release? What happens to its evidence when the preprocessing library and scanner profile change as well?**

**How the assembled input path changes the test question**

The controlled model remains a useful positive, while changed preprocessing and calibration can alter what it receives. The assembled versions need impact assessment and suitable paired performance tests. Actual scope can also reopen security, risk, documentation, user information and certificate communication.

A signature's presence alone proves nothing about verification. Successful verification with a trusted key supports integrity and claimed origin of the signed bytes within the key assumptions. It does not establish clinical correctness, freshness, performance of all combinations or permission for a new deployment.

## Bring the release decision together

A release conclusion concerns the complete configuration delivering the offered medical function. Scanner A with locked M3.2 retains its certified baseline. The proposed scanner, normalization, threshold and interface need their own support; the later exhibit establishes changed input, performance and visibility without supplying harm or an approved new certificate. The independent security application has a different progression: proposed controls first, later stipulated passing tests and qualified acceptance, then a hospital installation still unknown. Model files, signatures, supplier notices, certificate conditions and fleet records each support a particular part of that reasoning.

### Check what you just learned

The working states describe bounded findings beside evidence and reasons: **A**, adequately explained; **U**, present but unclear; **M**, missing or inconsistent; **D**, deeper analysis required; **C**, consultation indicated; **N/A**, inapplicable with a reason. They are neither regulatory classes nor approvals. The eight original identifiers retrieve expanded Workbook Part A2 cards through the original-to-integrated crosswalk.

### The actual medical output • card IVDR-CH15-A01

Do the function/purpose memo, data origins and device-law rationale explain what the user receives and why it fits the relevant definition? What output or relationship fact would change qualification before class or AI status is considered?

### The complete result-producing configuration • card IVDR-CH15-A02

Which scanner, preprocessing, model, threshold, interface, services and versions produced the assessed result? Does the supported-combination manifest identify the proposed path as precisely?

### Shared records and distinct legal duties • card IVDR-CH15-A03

Does the amended-law map identify each applicability condition, obligation and date? Which implemented quality records serve both device and AI duties, and which required output remains separate?

### Individual and combined modifications • card IVDR-CH15-A04

Do per-change performance, use, risk and security records connect to tests of the intended assembled release? Which interaction would remain invisible if only isolated changes were examined?

### **Release support and installed state • card IVDR-CH15-A05**

What evidence, certificate/change decision, labeling and monitoring support the offered configuration? What does the deployment log establish about the combinations actually installed?

### **A usable qualification memo and change matrix • card IVDR-CH15-C01**

Does each changed medical function have a decision owner, evidence and closure condition? A quarterly name cannot replace the four individual analyses or their interactions.

### **Conditional release stages • card IVDR-CH15-C02**

Are proposed stages compatible, evidenced and supported under the selected body's procedure? Which supplier, certificate, user or rollback dependency makes the sequence contingent?

### **Security evidence inside release planning • card IVDR-CH15-C03**

Can the selected IEC 81001-5-1 baseline, threat-to-requirement-to-test trace, qualified risk review and release record reconstruct the product decision? Which new field information would reopen it?

### **Further reading**

Book 1B Chapters 2–7 and Book 1D Chapters 11–15 offer optional deeper work on security, integrated verification, human use and controlled change. Keep this chapter's local cases and evidence limits distinct.

### **Source trail**

Binding sources checked: IVDR Regulation (EU) 2017/746, consolidated 10 January 2025, Annex I §16 and Annex IX §4.11, with Article 10 and Annexes II–III for the product documentation and change record; EUR-Lex. AI Act Regulation (EU) 2024/1689 as amended by Regulation (EU) 2026/1744, Articles 6 and 113 (official amended text checked 25 September 2026). The Commission's Article 6 classification-guidance page still labels the guidance draft on that check date. MDCG 2019-11 rev.1 and MDCG 2019-16 rev.1 are nonbinding guidance, not certificates.

Selected licensed method checked read-only: IEC 81001-5-1:2021, definitions 3.21, 3.25 and 3.36; Clauses 4–9 (particularly 5.7, 5.8, 6.1–6.3, 7 and 9); Annex F, read with Interpretation Sheet 1:2025, printed pp. 2–3 on category nesting, category-named duties and whole-product transitional use. These are methods, not evidence that this product passed them or that the standard is harmonized under the IVDR. The lifecycle table, SC-1 walkthrough and filled decision thread are original teaching syntheses; neither fictional case supplies a patient harm or a new regulatory authorization.



## CHAPTER 15

# Hear the field, find the configuration, act

**Before reading.** Chapter 14 followed a proposed software change through the other components and user tasks on which it depended. What evidence and applicable certificate decision would gate that release? If installation stops halfway, what does a sent package leave unknown?

A field observation becomes useful when the manufacturer can find the product state and people exposed, assess what could happen clinically, and show what happened next. The difficult part is often the connection between records that already exist.

## SECTION 01

## Why did a complete procedure miss the question?

Six months after launch, an anonymous pathology device has a complete **post-market surveillance (PMS)** procedure and an annual safety-update-report calendar. Customers nevertheless describe delayed low-score cases, scanner-specific color artifacts and difficulty installing a security update. Each complaint type sits in a different system, below its own escalation threshold. The device is described as using artificial intelligence (AI). Its class, complaint counts, root cause and actual patient harm are not supplied, and nothing identifies it as Chapter 14's Scanner A platform.

Real use brings users, scanner combinations, specimen mixes, workloads, suppliers and update paths that a controlled premarket study could not fully show. Why might three quiet complaint systems leave an important question unanswered? Follow the information toward the pathologist. A color artifact could change the image delivered to the scoring function. A low score could change where a case appears in the **worklist**, the queue awaiting review. An interrupted update could leave old and new components working together. Those are possible connections to investigate, not findings that these complaints share a cause.

To examine them, the original observation needs its site, scanner, component versions, dates and clinical workflow. That allows the investigator to ask whether a reported image problem and a late case actually passed through the same **configuration**. Keeping the original reports matters: combining records for analysis must not turn

three descriptions into one invented event. The immediate clinical question is what could happen while the technical cause remains unsettled, including whether protection or reporting is needed now.

**Reviewer Lens.** A complete procedure deserves credit for the process it establishes. Its operational reach becomes visible in one complaint-to-result trace: can the complaint record, configuration manifest, contact log and exposure data identify the relevant users and cases across systems? A narrow request for that connection is more revealing than another copy of the procedure. It can uncover a low-score case that never triggered its complaint category without presuming a harmed patient.

**Consultant Lens.** Common product, version, site and result keys are cheap to establish before several teams need them urgently. An owner for reviewing signals across systems makes those keys useful. Clinical and security assessment can proceed while engineering examines possible mechanisms; a yearly meeting can summarize those decisions. If it is the first occasion on which the records meet, the calendar has become a delay in the response.

The company's annual calendar points to a familiar document, the **periodic safety update report (PSUR)**. For the IVDR classes to which it applies, this draws together surveillance findings, benefit–risk conclusions and preventive or corrective actions over a reporting period. Having a calendar is a positive planning fact. It does not establish the device's class or show that any of these complaints has been appraised.

### **What was exposed to the possible failure?**

Suppose the team wants to know whether delayed review is becoming more common. The number of delayed cases needs the number of cases that could have been delayed. This **denominator** gives the count its meaning. Ten installations, their processed results and repeated attempts at one specimen are different units; an installation can process many specimens, and a repeat cannot become another independent specimen.

The observation window and unit therefore come before the rate. Initial specimen accessions, individual results, sequence reads, reports and installations can each answer a useful question, provided the numerator describes that same unit and period. Scanner, model, preprocessing, interface, site, user and update state then show who was exposed to which conditions. A pooled rate can hide a problem concentrated in one state even when its arithmetic is correct.

Counting also depends on what the system hears. Complaints, service calls, incident reports and supplier notices arrive because someone notices something. Scheduled performance sampling, user-task checks, literature, external quality assessment, registries, cybersecurity intelligence and suitable lawful telemetry actively seek information. External quality assessment brings an outside performance comparison into that picture. PMS joins and analyzes these sources against expected performance,

with a specified review trigger and owner. A count threshold can prioritize work; it cannot safely close a possible serious clinical consequence merely because the count is small.

This is work to design before launch and use after deployment. A sequencing team needs to know which supported variant and **knowledgebase** states can affect interpretation. If a later content update changes a report, the specimen count may stay the same while the affected report versions and entries change. A digital-pathology team needs scanner and worklist-state recording for a different reason: a service change could delay visibility with model weights unchanged. Its relevant exposure is cases processed through that service state. Neither illustration supplies an observation or cause for the opening company.

**Reviewer Lens.** “Below threshold” becomes assessable when the file shows the exposed population, exact version, period, missing records and complaint-to-result link. Comparing the pooled rate with scanner and version rates then tests the scope of that conclusion. An untraceable denominator is an evidence gap even if the percentage looks reassuring.

**Consultant Lens.** A useful signal specification begins with a clinical failure path and carries its source, counted unit, configuration fields, expected rate or limit, trigger, owner and immediate action question. Preserving the initial count beside dated corrections lets the company explain what it knew when. Overwriting it with the latest number would lose the history needed to judge the response.

## SECTION 02

### What evidence would answer the remaining question?

If the unresolved question concerns low-confidence cases in a real worklist, a collection of satisfaction comments cannot show how often those cases were hidden, delayed or recovered. The required observation follows from the question: the actual queue state, intended operators, relevant cases and a suitable reference for the claimed result. This is why **post-market performance follow-up (PMPF)** needs a targeted method rather than a general promise to collect feedback.

There is a boundary to that reasoning. Premarket evidence may adequately support a bounded claim while leaving a justified uncertainty about a scanner, population or real-use workflow. Follow-up can address that residual question within the assessed purpose. If evidence essential to the initial marketed claim is missing, calling it PMPF cannot supply the missing basis or justify waiting. If PMPF is inappropriate for a particular device, the **performance evaluation report (PER)** must explain why.

A useful objective connects the residual question with the actual version, population, method, result measure, collection frequency and action rule. For the separate

sequencing illustration, eligible results under each controlled knowledgebase version might show whether a supported variant class remains reportable. For the pathology illustration, observation of intended operators handling low-confidence cases on supported scanners might address the user path. These are possible objectives; neither is an executed result, and neither closes the opening complaints.

Once results exist, the PER appraises whether scientific validity, analytical performance and clinical performance still support the **intended purpose**. The PSUR has a wider period view, bringing surveillance, follow-up, benefit–risk and actions together. Thus a useful PMPF plan can coexist with an unanswered performance question, and a completed period report can coexist with an investigation still underway. The records are connected by evidence rather than made interchangeable by their titles.

The legal schedule follows the device class under the **In Vitro Diagnostic Medical Devices Regulation (IVDR)**, the regulation governing examinations of human-derived specimens for the specified medical information. For class A/B devices, Article 80 requires a PMS report updated when necessary, without a fixed annual minimum. For class C/D devices, Article 56(6) requires the PER to be updated when necessary and at least annually; Article 81 requires the PSUR at least annually. Knowing the class selects the relevant periodic record. A new safety or performance concern can require action before that record's next scheduled date. The opening packet leaves the class open.

**Reviewer Lens.** The most useful comparison puts the PER's residual question beside the PMPF data source. If low-confidence worklist handling is the concern, the source must include the users, states and cases through which that concern could occur. “Collect feedback” may be a feasible activity yet still be the wrong method. Enough means the method can answer the stated question and its result reaches a claim-specific appraisal.

**Consultant Lens.** Feasibility work can begin while the conclusion remains open: access and privacy arrangements, version capture, analysis, review points and conditional budgets all need owners. The temptation is to fund whatever data are easiest to collect and call the resulting volume follow-up. Agreeing the decision and action threshold first makes that mismatch visible before collection becomes expensive.

### **If a scanner effect is established, where does it lead?**

Suppose investigation establishes a scanner-bound image effect. The complaint records the affected observations; the next question is how altered image information could reach a clinical decision and what harm could follow. That brings the finding into risk management. Engineering then examines the image path and a possible control, while the PER asks whether the claim remains supported for this scanner. A changed performance question may need a revised PMPF method, and the applicable period report carries the resulting conclusion and action.

The same finding can therefore require linked revisions to labeling, technical documentation, supplier controls and change assessment. Where an actual nonconformity is found, a **correction** addresses affected instances; **corrective action** removes an established cause and checks that it does not recur. Fixing an instance and demonstrating that the cause has been addressed require different evidence. Neither conclusion follows from closing the complaint's administrative status.

A vulnerability can reopen the file by another path, even when application code is unchanged. Loss of confidentiality, integrity or availability may affect clinical use. A technical severity score helps describe the security problem; patient consequences require their own product and workflow analysis. Likewise, a supplier's fix notice leads to affected-version analysis, proportionate protection, product security and performance tests, and installation verification. It does not close those steps on the manufacturer's behalf.

**Reviewer Lens.** A field result should lead forward to the complaint, configuration, risk question, performance appraisal, action and customer communication. Reading backward from a changed requirement should locate the fielded versions it affects. These two directions expose different gaps: a well-investigated complaint may have incomplete customer reach, while a distributed patch may lack a demonstrated answer to the original clinical problem.

**Consultant Lens.** A linked action record gives clinical, quality, engineering, regulatory and operations teams their own decisions, evidence and dates. Cause may stay open while a proportionate protective measure or timely report is needed. Making those tracks visible prevents a request for perfect root-cause certainty from becoming an unexamined reason for waiting.

### Which product does that action concern?

A correction has to reach an identifiable product. Its regulatory identifier locates the represented device or model; its **certificate** gives the assessed scope and conditions; its component manifest identifies the assembly. The internal fleet log answers a further question: what is actually installed and working at each customer? A registered and assessed product can still have an uncertain installation state.

The European database on medical devices, **EUDAMED**, holds European Union (EU) device information. On the Commission page checked 25 September 2026, four modules had been mandatory since **28 May 2026**: Actor registration, Unique Device Identification (UDI) and Device registration, Notified Bodies and Certificates, and Market Surveillance. Market Surveillance supports competent authorities and the Commission. It is not the manufacturer's fleet ledger. The separate Vigilance and Post-Market Surveillance module and Clinical Investigations and Performance Studies module were not yet mandatory.

That status determines where particular regulatory information belongs; it supplies no evidence that a customer installed a correction. Nor does one start date impose the same migration deadline on every historical device and certificate entry. Actual records follow the applicable module instructions and transitional provisions. Later in the chapter, the result-to-identity chain will show how the software change itself can affect UDI.

**Reviewer Lens.** One sampled customer's label, listed actor and device, applicable certificate, release manifest and installed version offer a practical reconciliation. If the identity changes after a correction, the same customer and affected results should remain traceable. A correct registration entry can receive credit while the installation question remains open.

**Consultant Lens.** Data stewardship means assigning both an owner and the change that requires that owner to act. Validation before submission and reconciliation after release serve different purposes. Regulatory staff may close an identity entry while operations still owes a deployment confirmation; presenting both states prevents a registration milestone being sold as completed field reach.

### **What belongs on the calendar, and what interrupts it?**

A booked audit cannot establish that a proposed scanner addition is supported. Notified-body surveillance, certificate conditions and relevant technical-documentation sampling are continuing work under the actual route. A release plan joins their dependencies with supplier changes, literature, risk, performance evaluation, PMS/PMPF, scheduled reports, labeling, security, registration and any notified-body change decision.

For a three-year plan, the useful connection is between the work, its owner and the evidence that allows the next decision. An audit date can reserve capacity; a serious possible incident still requires prompt assessment when it appears. Certificate maintenance and competent-authority vigilance retain separate purposes even when one observation raises both.

## **SECTION 03**

### **A shared exposure becomes visible**

The opening record now supplies more information. **All three complaint types concern Scanner B sites and releases after the normalization update. Delayed queues occur mainly after update failures leave a mixed configuration.** There is still no event count, root-cause test, actual patient outcome or field-action determination. These facts do not establish that this marketed device is Chapter 14's proposed Scanner B release.

**Decision**

**May the company keep the categories separate because none crosses its individual threshold? Which shared exposure and clinical path should it examine first?**

**How the new exposure changes the investigation**

The newly shared scanner, release period and mixed state make a product/configuration investigation necessary. The complaint codes no longer describe the whole pattern. Image color could affect scoring; scoring or queue behavior could affect visibility; failed installation could change the combination through which the case passes. Reconstructing that path for affected cases and supported configurations tests the possible connection.

This remains an investigation signal. The exhibit does not prove that normalization or failed updating caused every delay. Clinical continuity and customer communication may need proportionate action while seriousness, reporting, certificate impact and a tested correction are assessed. Those decisions need their actual facts; none is resolved by the low count in a separate category.

**Reviewer Lens.** The revealing comparison joins old and new scanner images, version-specific queue behavior, update-failure logs and service status with case-level clinical review. It also asks when the common exposure first became visible and what the threshold rationale allowed the company to see then. That inquiry can identify a response gap without inventing a patient injury.

**Consultant Lens.** A joined response connects affected configurations and customers to containment options, investigation and later evidence gates. Business trust and service continuity belong in that planning because customers need usable instructions while cause remains open. The resulting plan can connect corrective action, performance evidence, notified-body communication and future release gates without prematurely describing the event as reportable.



## SECTION 04

### How can 1.5% conceal 20%?

A separate fictional record, the **thirty-day field-accounting example**, supplies counts the opening company did not. It concerns the first **30 days** after an update rollout. Each result unit is one specimen accession with its **initial** processing attempt. Repeats have their own log and do not create independent specimens. A result is delayed when it exceeds the device's prespecified turnaround limit; this exercise's product limit is not a statutory reporting threshold. This packet is independent of the opening, practice device and any named product.

Begin with the previous configuration: nine delayed initial accessions among 900 gives  $9/900 = 1\%$ . The complete update has  $2/80 = 2.5\%$ . The incomplete or mixed state has  $4/20 = 20\%$ . Adding the numerators gives 15 delays; adding the matching denominators gives 1000 initial accessions. Thus  $15/1000 = 1.5\%$  is correct. Why does it seem so different from the mixed state's rate? Most observations—900 of the 1000—belong to the previous configuration, so they dominate the pooled result.

The table keeps the denominator beside the state when the rates are compared.

| State during initial processing | Initial specimen accessions | Above the turnaround limit | Rate within that state |
|---------------------------------|-----------------------------|----------------------------|------------------------|
| Previous configuration          | 900                         | 9                          | 1%                     |
| Complete update                 | 80                          | 2                          | 2.5%                   |
| Incomplete or mixed             | 20                          | 4                          | 20%                    |
| All three pooled                | 1000                        | 15                         | 1.5%                   |

Reading the mixed row means four delayed initial results among the twenty exposed to that state. It does not mean four harmed patients, or twenty percent of the fleet. The supplied counts close the descriptive arithmetic. They leave sites, workload, specimen mix, observation time and missing logs to compare before the pattern could support a causal conclusion. Both delayed and non-delayed accessions need their actual configurations checked. No patient outcomes, harmed-patient count, validated new threshold or proof that the update caused delays is supplied.

Now change the unit. The fleet record describes **ten installations: eight confirmed complete, one confirmed mixed and one unreachable**. Complete deployment is confirmed for  $8/10 = 80\%$  at this cutoff. The unreachable installation is unknown, not a confirmed success or failure. Those ten installation records and the 1000 initial

accessions answer different questions; the latter do not establish every installation's state at every later time.

That distinction determines what a first response needs to do. Quality can open a linked investigation while preserving specimen/result identifiers, versions, retries, dates, contacts and original counts. A clinical lead can review delays and possible consequences. Operations needs to identify mixed states and check the continuity arrangement before further affected processing; if no adequate arrangement exists, the relevant clinical-operating decision is still needed. These are proposed actions, not a supplied correction or completed field response.

This short record connects each established fact to the evidence still needed for its next decision.

| Question                         | What this packet establishes                                         | What would advance the decision?                                                                                                 |
|----------------------------------|----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Where do delays concentrate?     | Previous 9/900; complete 2/80; mixed 4/20; pooled 15/1000.           | Reconcile configuration strata, workload and missingness before a causal claim.                                                  |
| Which installations are known?   | Eight complete, one mixed, one unreachable; 8/10 confirmed complete. | Contact and verify the unknown; control the mixed state; preserve dated records.                                                 |
| What could happen clinically?    | The product turnaround limit is exceeded; outcomes are absent.       | Review affected results and potential consequences; decide protection and applicable reporting.                                  |
| What would close the correction? | No tested correction or complete fleet deployment is supplied.       | Verify compatibility, affected-result reconciliation, needed clinical follow-up, effectiveness and updated risk/PER/PMS records. |

The final row explains why a patch file alone could not close this response: its compatible behavior, the affected results and its effectiveness in the field are different parts of closure. Conversely, the first row's arithmetic need not stay “unknown” until the cause is found. Positive descriptive evidence and unresolved causal evidence can coexist.

**Reviewer Lens.** Recalculating the rates is only the start. The ten-installation log raises the operational question of whether the mixed installation continues processing and how a corrected state would be confirmed. Keeping the two ledgers visible prevents a striking result rate being mistaken for fleet coverage or a patient-harm estimate.

**Consultant Lens.** A dated contact and verification plan gives quality, clinical and operations staff distinct work. Interim routing depends on the actual service

arrangements; it cannot be promised from a generic suggestion to use an alternative. Threshold investigation, a possible report and a later validated correction can then progress in parallel, with each waiting only for the evidence its own decision requires.

### Does the observed rate choose a reporting deadline?

It does not. Under **IVDR Article 82**, the relevant route here concerns devices made available on the EU market other than devices for performance study. The first questions are what occurred and when the manufacturer became aware of the relevant facts. A **serious incident** directly or indirectly led, might have led or might lead to death, temporary or permanent serious health deterioration in a patient, user or other person, or a serious public-health threat. Thus the absence of observed injury does not settle seriousness when that consequence might have occurred or might occur.

A **serious public-health threat** concerns possible imminent death, serious deterioration or serious illness that may need prompt action and may cause significant illness or death in people, or is unusual or unexpected for its place and time. That urgency explains a shorter reporting branch. A product turnaround exceedance alone does not establish those conditions.

Once the relevant incident branch is established, the table distinguishes immediate reporting from its outer limit. Its awareness and causal conditions are deliberately different.

| Article 82 branch                                | Reporting condition and maximum period                                                                                                   |
|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Serious incident generally, §3                   | Immediately after device causality is established or reasonably possible; no later than <b>15 days after awareness of the incident</b> . |
| Serious public-health threat, §4                 | Immediately; no later than <b>2 days after awareness of the threat</b> .                                                                 |
| Death or unanticipated serious deterioration, §5 | Immediately once device causality is established or suspected; no later than <b>10 days after awareness of the serious incident</b> .    |

For the general branch, a reasonably possible device relationship makes waiting for final causal proof inappropriate. If the actual facts instead establish a serious public-health threat, the threat-awareness branch supplies the two-day outer limit. The field-accounting packet supplies neither the clinical consequence nor device relationship needed to select among these branches. Its thirty-day observation window and turnaround limit cannot serve as reporting clocks.

Uncertainty does not create an extra waiting period. Article 82(6) allows an incomplete initial report followed by a complete one when necessary for timeliness. Under Article

82(7), uncertainty about whether a potentially reportable incident is reportable does not suspend the applicable timeframe. Expected erroneous results have the Article 82(1)(a) exception only when clearly documented and quantified in product information and technical documentation and subject to Article 83 trend reporting; merely calling an error “expected” cannot establish that exception.

The protective action also has its own route. A manufacturer's action for technical or medical reasons to prevent or reduce serious-incident risk in a marketed device is a **field safety corrective action (FSCA)** when those legal conditions fit. Article 82(1),(8) normally requires its report in advance, without undue delay; urgency can require immediate action. Clinical protection therefore need not wait for the final cause finding. This packet supplies no particular field action or authority report.

**Article 83 trend reporting** asks another question. For non-serious incidents, the increase in frequency or severity must be statistically significant, potentially significantly affect benefit–risk and have led or potentially lead to unacceptable health or safety risks. The other branch concerns a significant increase in expected erroneous results compared with stated performance. The PMS plan supplies the method and observation period for this analysis. The small descriptive table alone gives no significance test or legal trend determination. While the EUDAMED vigilance module is not mandatory, the applicable national reporting arrangements still matter.

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### Decision

**Operations sends the update to the unreachable installation. The delivery service says “request accepted,” but no installation confirmation returns. Can the fleet become 9/10 confirmed complete? Does the message change the earlier 1000-specimen denominator?**

### What the acknowledgment actually changes

It adds evidence of a request, not a verified installation, compatible configuration or return to clinical service. The dated fleet remains eight confirmed complete, one confirmed mixed and one unknown until verification changes the record. Calling it 9/10 would replace an unknown with an invented success.

The earlier denominator also remains 1000 initial specimen accessions in its thirty-day window. A later communication is neither a new accession nor a retrospective change to those observations. New observations belong to a separately dated analysis, with the request and follow-up attempts retained as part of the response history.

## SECTION 05

## Which identity changes with the software?

Imagine a sequencing reporting application issuing a result under a registered device identity. Investigators still need the internal build and rules that produced that particular result. Chapter 1's **unique device identification (UDI)** system connects related regulatory identities; it does not replace that finer component trace.

The **Basic UDI device identifier (Basic UDI-DI)** connects an eligible device family with its documentation and database records. It is not the production code in the UDI carrier. The **UDI device identifier (UDI-DI)** identifies the relevant device model or version. Its companion **UDI production identifier (UDI-PI)** carries production information, including software identification where applicable. In the hypothetical application, these locate the regulated model and applicable software production identity while internal records locate the exact pipeline, model, threshold, database, scanner and site state.

A knowledgebase revision can now raise two connected questions: did interpretation change, and does the regulatory identity need to change? IVDR Annex VI Part C §6.2.1 applies its software-system assignment rule to software commercially available on its own and software constituting a device itself. It does not assign that specific rule to every embedded program. The first gate is therefore whether this software falls within that scope.

Within that scope, §6.2.2 requires a new UDI-DI for a modification changing original performance, safety or intended use, or data interpretation. New or modified algorithms, database structures, platforms, architecture, interfaces or interoperability channels can have such effects. The hypothetical knowledgebase revision needs its actual interpretation effect examined; an internal release label saying “minor” supplies no answer. For a genuinely minor revision, §6.2.3 instead requires a new UDI-PI, not a new UDI-DI. The distinction turns on the change and its effect, not whether the team calls it maintenance.

The field-accounting example gives previous and complete exposure states, with a mixed state between them. It gives insufficient product-change facts to select either identifier branch. A Basic UDI-DI and certificate link can help locate the regulated device, but neither confirms whether its unreachable installation updated. The four mandatory EUDAMED modules require controlled ownership and data quality; historical and new records still follow their actual transitional and module-specific instructions. Registration establishes neither conformity nor completed deployment.

**Reviewer Lens.** A reported result should remain traceable through internal components, the applicable UDI, registered device and assessed certificate scope. The decisive request for a “minor” change is its effect rationale. Names for old and new

builds cannot establish either that a new identifier is required or that the existing one can remain.

**Consultant Lens.** A pre-release identity decision connects UDI, label, database and certificate implications before distribution. A later operating check confirms the installed configuration. Separate regulatory and operations sign-offs make these dependencies practical: correct identity can close its own task without silently closing customer deployment.

A useful field record therefore carries the observation window, denominator and version into the clinical assessment, follow-up method, PER conclusion and applicable period report. The linked operating map gives each decision an owner and connects it to customer reach. A rolling three-year calendar can fund that work, while the more informative management measure follows time from a credible signal to affected-fleet understanding and a reasoned decision, then to verified reach and effectiveness. A count of closed complaints says little about those intervals.

### **Changed-case transfer**

#### **An independent near-patient test.**

A seasonal pathogen change could alter the target mix. A reagent lot may affect invalid results, a connectivity failure may prevent a result reaching the user, and unclear instructions can change how an operator handles an invalid. Which exposures need joining before these observations can be treated as separate nuisances or a shared performance risk?

### **How the exposure changes outside the central laboratory**

Lot, season, operator, site and result denominators connect the different observations to the conditions in which a result is produced and used. Each mechanism remains a possibility to investigate. The transfer supplies no particular device class, patient event or reportable trend, and inherits no counts from the thirty-day packet.

## **SECTION 06**

### **What must the practice team's file connect?**

This practice packet restates the Scanner B/mixed-update complaint pattern. It supplies no field-accounting numbers and does not identify the device as Chapter 14's certified Scanner A/locked M3.2 platform.

An AI pathology device has a PMS procedure and annual PSUR calendar. Complaints describe delayed low-score cases, scanner-specific artifacts and failed security updates. Each category stays below its threshold. All occur on Scanner B after a normalization

release that sometimes leaves mixed configurations. Counts, actual injury, a causation test and the device class remain absent.

### **What happened before and after one delayed review?**

Working backward from a delayed review identifies the specimen and case, scanner, preprocessing, model, threshold, queue, service/update state, user and site. Working forward asks whether the case was eventually seen, whether the delay exceeded the product's intended time, and whether that timing could affect a clinical decision. The complaint code “delay” contains neither account. Artifact and failed-update records become useful when shared configuration and timing keys connect them, while their original observations remain distinguishable.

Reported information needs actively sought evidence alongside it: service logs, suitable telemetry, user follow-up, literature, external quality assessment, supplier alerts and planned sampling. Missing records are themselves a finding. Without installed versions and supported scanner combinations, the company cannot reliably locate the affected population or verify a later correction. A possible common supplier or configuration cause can guide a test; it has not been established by the packet.

**Reviewer Lens.** A single result-to-release trace plus the known Scanner B mixed-state installations can expose whether the search covers the right population. The risk analysis then needs the actual low-score visibility and clinical-timing path. Three individually green complaint categories are not a substitute for that path.

**Consultant Lens.** Common identifiers need agreed handoffs among complaint intake, service, engineering, clinical and regulatory teams. A trigger for the joined pattern gives the cross-system owner something actionable. A separate route for one potentially serious event prevents the aggregate threshold from becoming a condition for clinical attention.

### **Which remaining question belongs in follow-up?**

The practice packet leaves two different performance questions: whether the supported scanner/image configuration continues to meet its claimed performance, and whether intended users reach low-score cases in time under the deployed queue. Either may support a targeted PMPF objective if it is a justified residual question within the marketed purpose. Its method needs the actual version, population, endpoint, observation window and action rule. Generic comments cannot measure either boundary. If the finding instead challenges the basis of a core marketed claim, risk, evidence, claims and protection need reassessment now.

The PER then appraises the resulting scientific, analytical and clinical evidence. The applicable PSUR draws its period conclusion from surveillance, vigilance, follow-up, exposure or sales information, benefit–risk and preventive/corrective actions. Class A/B instead invokes Article 80's report updated when necessary. Because this packet gives



no class, the class C/D annual PER and PSUR floors remain conditional. That uncertainty need not prevent useful staffing and cost planning.

The table shows how a rolling three-year plan accommodates both scheduled work and an event that changes it.

| Planning slot | Recurring work and dependencies                                                                                                                                 | What can interrupt the schedule?                                                                                            |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Year 1        | Assign surveillance, literature/risk, follow-up, supplier and fleet owners; schedule class-specific reports and body surveillance under the actual certificate. | The Scanner B/mixed-state cluster needs prompt investigation, protection and reporting assessment.                          |
| Year 2        | Reassess exposed configurations, follow-up results, risk, PER and applicable report/certificate records on their actual cycles.                                 | A validated fix, supplier-version change or changed unresolved installation can reopen evidence, labeling or customer work. |
| Year 3        | Continue scheduled review and plan the next certificate, version and support decisions from current evidence.                                                   | A new threat, performance change or scope proposal reopens the relevant route before its calendar slot.                     |

In Year 1, allocating staff for a periodic report cannot defer the current cluster's clinical assessment. In Year 2, a validated fix would change the work because its evidence and customer reach would need reconciliation; the table does not assert that such a fix exists. These are planning slots, not a three-year certificate term or proof that class C/D reports apply. Conditional planning is useful to management while the reviewer still needs the urgent dated response.

### **If the software identity changes, which records move with it?**

Suppose investigation leads to a change in the practice software's identity. The first practical question is how an affected customer or authority reaches the responsible party. From there the trace needs to identify the device, assessed scope and actual installation. The five records below answer successive questions in that trace; they do not certify one another.

The actor record identifies the manufacturer and, where applicable, authorized representative and importer, with current roles, mandates and contacts. If an entity, mandate or contact changes, this record needs attention even without a code release. Incorrect contact information can delay reaching the accountable party. The practice packet supplies no actual actor change.

Once the party is identified, the UDI/Device record locates the regulated model and intended purpose represented by the label. A new interface or threshold can matter if

its effect meets the software UDI change rule taught above. “Same model weights” does not answer that effect question. The packet leaves the new-identifier decision open.

The corresponding certificate record identifies the precise device and assessed scope, status, limits and supplements. Its relevance depends on the offered configuration. A current certificate covering one combination cannot establish that a later scanner/software combination was assessed. That is the notified body's conformity question, distinct from a national authority's vigilance work.

If an authority is investigating, EUDAMED's Market Surveillance module supports competent-authority and Commission coordination. The manufacturer must still perform its own applicable analysis and communication and retain its response record. The separate, not-yet-mandatory Vigilance and Post-Market Surveillance module does not become a live manufacturer reporting substitute because Market Surveillance is mandatory.

Finally, a customer-specific action needs the internal fleet record: actual software, model, content, scanner, update and service state, with confirmation and uncertainty. It is not a fifth EUDAMED module. The separate thirty-day example's eight/one/one installation states illustrate that distinction; the practice device supplies no fleet counts, and a request acknowledgment cannot establish the actual state that a registration record also cannot show.

The compact map helps decide which owner must reopen a record after a particular change.

| Record                           | Question it answers                              | Change that requires attention                                 |
|----------------------------------|--------------------------------------------------|----------------------------------------------------------------|
| Actor registration               | Who is accountable and contactable?              | Entity, mandate or contact change.                             |
| UDI/Device registration          | Which regulated model and label are represented? | Purpose, model or UDI-relevant software change.                |
| Notified Bodies and Certificates | What scope and status were assessed?             | Issuance, supplement, limitation or applicable product change. |
| Market Surveillance              | What authority oversight needs reconciliation?   | Authority request, regulatory signal or action.                |
| Internal fleet/configuration     | What is installed and processing results?        | Release, service, failure, rollback or customer change.        |

For an assessed device whose registration and certificate are correct but whose customer remains in a mixed state, the immediate installation problem sits in the last row. It does not make the other records false. If the proposed correction changes the legal software identity or assessed scope, the second or third row also reopens on those

facts. The Commission's four-module start on 28 May 2026 and the two remaining modules' nonmandatory status at the 25 September check leave module-specific transition and national arrangements relevant; there is no universal historical-record migration date to substitute for them.

## SECTION 07

### Does the AI label change the response?

A worklist that could hide a low-score case raises a functional question before it raises an AI label. As Chapter 14 explains, an **AI system** is a machine-based system designed for varying autonomy that infers from inputs how to generate outputs for explicit or implicit objectives, with outputs capable of influencing physical or virtual environments. Adaptiveness after deployment is possible, not mandatory. The practice packet's use of “AI” does not itself establish this definition or a high-risk classification.

For the Artificial Intelligence Act's Article 6(1)/Annex I product route, two conditions must be met together: the AI system is itself a listed product or intended safety component of one, and the product/system requires the relevant third-party conformity assessment. The 2026 amendment excludes an assessment required solely for risks unrelated to health and safety from the second condition. A safety component fulfils an intended function of preventing or mitigating health and safety risks to persons or property, or its failure or malfunction endangers their health and safety. For the component question, solely non-safety user assistance, performance optimization, service efficiency, automation, convenience or quality control is excluded; a function whose failure or malfunction would endanger health and safety qualifies as a safety component notwithstanding that exclusion.

That is why the actual consequence of lost worklist visibility matters. Evidence that the function only supports a non-safety convenience would lead to a different component analysis from evidence that its failure endangers health and safety. The source gives neither a completed AI-definition analysis nor all the product, safety-function and third-party-route facts, so it supplies no Article 6 verdict. The relevant amended Article 6 conditions require a functional rationale, not an exemption inferred from the word “workflow.”

If Article 6(1)/Annex I high-risk status is established, amended Article 113 places Chapter III Sections 1–3, except Article 6(5), from **2 August 2028**. That exception concerns the Commission's practical classification-guidance duty discussed in Chapter 14; it does not turn this date into the start of every AI Act duty. Other provisions have other dates. The joint 2025 Medical Device Coordination Group (MDCG)/Artificial Intelligence Board (AIB) guidance is nonbinding and predates the amendment. The Commission classification-guidance page remained labeled draft at the 25 September

2026 check. Neither that earlier guidance nor the scoped date establishes this device's compliance or exemption.

Operationally, the same scanner, normalization, model, threshold, interface, cloud, supplier or label change can affect medical evidence and any applicable AI duty. A common **quality management system (QMS)** can carry shared risk, data, oversight, logging and monitoring records while making each legal output identifiable. Assessing both each modification and their combined effect keeps that shared file useful. An annual product report cannot perform the change assessment or customer protection for a changed configuration.

**Reviewer Lens.** The useful evidence joins the actual AI-system definition, safety-function rationale, conformity route and obligation/date map to the deployed version. Scanner and mixed-state observations then inform the same risk and performance file. A completed legal gate has to explain why it fits the real function; the case label alone cannot do that work.

**Consultant Lens.** Two parallel paper systems invite the same version change to acquire different assumptions. Shared evidence can reduce that risk where it genuinely serves both duties, while genuinely additional outputs remain explicit. Each release still needs its clinical, security and identity decisions. The consultant can organize the map without deciding an unknown legal classification by analogy.

### **What would allow a response to close?**

The practice device's affected-fleet map and clinical assessment determine which interim measures are proportionate. Depending on those facts, options may include warning users, preventing an unsupported mixed combination or arranging an alternative workflow. Testing scanner, normalization, queue and update hypotheses can then establish what a correction needs to address. Correcting affected instances and eliminating a cause remain different achievements; the changed whole configuration and the user's recovery path need testing before effectiveness can be concluded.

Reporting and certification work proceed alongside the investigation. The competent authority receives applicable vigilance or field-safety communication; the notified body considers the certificate and continuing conformity under the actual route. Customers also need appropriate communication and support decisions. The packet supplies no particular patch, field action, authority report or restored customer. Closure needs evidence of the relevant action, its reach and effectiveness rather than an assumption that a closed complaint means a recovered installation.

## Decision

**Can the three practice complaint categories remain closed because none crosses its separate count threshold? What must be joined before choosing containment, reporting or a product change?**

### How the practice record reaches an owned decision

The common Scanner B/normalization period and mixed update states need a function- and configuration-level investigation. Results, clinical timing, site exposure and installed versions make that pattern assessable. They support deciding what protection is needed and applying the timely reporting rules to the actual seriousness and causal facts; they do not themselves prove harm, reportability or one established cause.

A correction then needs its own test and verification of field reach. The outcome may alter risk, PER/PMPF, applicable period reporting, labeling, notified-body communication or support scope. This is why joining the records changes the work even when the three original category counts stay below threshold.

The **manufacturer** owns investigation, risk protection, reporting assessment, tested correction and customer reach. The **notified body** examines assessed scope, performance evidence and the continuing quality system; an applicable change may need its involvement. The **national competent authority** handles the public-law vigilance and market-surveillance duties. Sending information to one does not perform the other's process. A medicines authority has no automatic role in these pathology complaints merely because other chapters concern companion diagnostics. A consultant can build the signal map, evidence architecture, responsibility matrix and three-year options; relabeling a complaint cannot transfer the statutory decisions.

### Changed-case transfer

#### Central cloud to local hospital.

If the model moves to local inference, who now controls computing, scanner compatibility, credentials, logs, update delivery and recovery? Which hospital installation actually changed, and which cases were exposed to its supported configuration?

### How local deployment changes the evidence route

The actual design determines the new performance, cybersecurity, supplier and economic-operator responsibilities, plus UDI/registration and certificate questions. The exposed-result denominator needs the local supported states. A cloud result cannot automatically bridge local performance, and a sent installer cannot establish local deployment. The transfer supplies no successful local installation or performance finding.

## SECTION 08

### Two independent field decisions

**A pooled complaint rate.** The global delay rate remains below the company's threshold. Nearly all delayed reviews occur on Scanner B after a normalization update. No count or patient outcome is supplied.

#### Decision

**May the aggregate trend stay closed? What denominator and comparison would you open?**

### How a reviewer tests the reassuring average

The aggregate cannot close the concentrated pattern. Scanner, version, site and workflow comparisons need the number of cases exposed in each state. That analysis can show where the complaint concentration lies and guide investigation of intended performance, risk, field protection, PER/PMPF, labeling, certificate scope and vigilance. The global rate alone has answered a broader and less discriminating question.

No count, patient outcome, causal result or reportability determination can be borrowed from the thirty-day example. The next work is to establish those facts for this packet and assess the applicable decisions.

**One communication route.** A manufacturer identifies a serious software-related incident and tells its notified body. It has not analyzed competent-authority vigilance duties.

#### Decision

**Is that communication complete? Which decisions belong to the authority, body and manufacturer?**

### Why the recipient matters

The notified body's certificate and conformity role does not replace competent-authority vigilance or field-safety reporting. The manufacturer must promptly assess the Article 82 route and clock while addressing customer protection, correction and evidence updates. Notified-body change and surveillance work remains a separate question rather than a reason to discard the communication already made.

The packet supplies no incident date or branch facts from which a specific deadline can be calculated. The useful finding is the unperformed vigilance assessment, not an invented number of overdue days.

## SECTION 09

### Can another person follow the response?

The thirty-day packet now supports a precise distinction: its 15/1000 pooled result and 4/20 mixed-state result are both correct, while their clinical meaning and cause remain open. Its 8/10 confirmed installations cannot become 9/10 from an acknowledgment. The opening and practice complaints need their own consequence and cause assessment. A traceable response carries such facts and unknowns from observation through exposure, clinical assessment, protection, applicable reporting, tested correction and verified reach.

The following checks retrieve the evidence behind those decisions. **A** means adequately explained for this bounded item; **U**, present but unclear; **M**, missing or inconsistent; **D**, deeper analysis required; **C**, consultation indicated; and **N/A**, inapplicable with a reason. An evidence location and explanation give the letter meaning. These are working states, not classes or approvals. The trailing historical IDs locate expanded Workbook Part A2 cards; their numbers refer to source-era topics, with the crosswalk preserving lookup.

#### Signal, exposure and owned action • card IVDR-CH16-A01

Can each rate be tied to a counted unit, period, exposed configuration and decision? The PMS plan and actual signal/fleet records show sources, denominators, versions, thresholds, owners and actions together.

#### A follow-up that answers the residual question • card IVDR-CH16-A02

Does the PMPF population, method, endpoint and action rule answer the specific uncertainty in the PER? Generic comments cannot establish a result for a user task they do not observe.



**One result across several systems • card IVDR-CH16-A03**

What concentration would a pooled report conceal? Case-level links across clinical, service, security and installed-version records let function and configuration be examined beyond complaint codes.

**Identity, scope and installation • card IVDR-CH16-A04**

Which actor, UDI/device, certificate and fleet records establish accountable identity, regulatory identity, assessed scope and actual installation? Each needs its own change trigger and evidence.

**A calendar that can respond to events • card IVDR-CH16-C01**

Does the PMS/PMPF operating map connect sources, owners and actions with the rolling three-year calendar? Class-specific report cycles remain conditional on class; urgent work needs its own event trigger.

**Protection through effective correction • card IVDR-CH16-C02**

What is known about potential consequences and causality, and what remains unknown? Interim protection, reporting assessment, a validated fix, communication, evidence and support decisions need verification of reach and effectiveness.

**Time to an understood and corrected fleet • card IVDR-CH16-C03**

Do dated investigation, contact, installed-version and confirmation logs show time from signal to affected-fleet understanding and decision? The request, installation and effectiveness milestones must remain distinguishable.

**Further reading**

Book 1B Chapter 7 and Book 1D Chapters 14–15 offer optional deeper work on deployment, monitoring, incidents and retirement. Keep the local cases and their evidence limits distinct.

**Source trail**

Binding provisions checked 25 September 2026 in the IVDR consolidated 10 January 2025 text: Article 2(68),(69),(71), Article 56(6), Articles 78–83, Article 82(1),(3)–(8), Annex III, Annex VI Part C §§3.9, 5.5 and 6.2.1–6.2.3, and Annex XIII Part B §§4–8. The 15/2/10-day branches have different awareness/causality conditions; no FS-1 event is assigned to one. The European Commission EUDAMED overview, checked 25 September 2026, confirms the four modules mandatory since 28 May 2026, the authority/Commission role of Market Surveillance and the two remaining modules' nonmandatory status; follow current module-specific transition instructions for actual records. MDCG 2025-10 is nonbinding PMS context, not a reporting trigger.

For the conditional AI discussion, the binding text is AI Act Regulation (EU) 2024/1689 as amended by Regulation (EU) 2026/1744, Articles 6 and 113; the Commission classification-guidance page remained draft on the check date. Joint MDCG 2025-6/AIB 2025-1 predates the amendment and is nonbinding. The FS-1 arithmetic and operating tables are original teaching syntheses of the preserved fictional packet; they are not a product conformity, vigilance or patient-outcome finding.

## CHAPTER 16

# What can travel, and what is worth funding?

**Before reading.** For a Chapter 15 field signal, what unit was exposed, over what period, and in which configuration? Which records connect a delayed result to its installed version? Why can an annual report never be the reason to postpone an urgent clinical or reporting decision?

That same attention to the actual product matters before entering another market. A study may cross the Atlantic unchanged while the claim it is supposed to support changes substantially. The investment question is therefore quite concrete: which work would support the offered product in the United States (U.S.) and European Union (EU), and which sequence can the company fund and sustain?

## SECTION 01

### What exactly would the board fund?

A board wants one launch recommendation for two anonymous programs. The sequencing program has strong analytical evidence for the U.S. Food and Drug Administration (FDA) and a U.S. companion-diagnostic partnership. Its EU performance evaluation plan and report are immature, as is its access to notified-body capacity. The digital-pathology program has European pilot sites and a broad scanner claim; its U.S. reader study and cybersecurity change strategy still need work. Capital can support **one major additional study this year**. The packet supplies no study budget, authority review times, market returns or authorization for either proposed regional claim.

Which study should get the money? “U.S. first” cannot yet answer that question, because the two programs have different strengths and open questions. The sequencing evidence positively supports analytical work already done. The European pilots are also real evidence from their tested settings. A partnership and pilot access may make future work feasible. None of those facts identifies the whole product that could now be offered, or supplies approval, certification or a completed scanner bridge.

The missing connection starts at the clinical result. For sequencing, which specimen enters, which treatment information comes out, and which patient and treatment

decision is it for? For pathology, which scanner and image processing feed the software, and what does the reader actually see and do? Those details connect the **intended purpose**—the promised medical job—to the **configuration** that produced the evidence. A regional product name is too broad to do that work.

A scanner illustrates the problem. Even with the same model file, a different scanner can present different image information to it. A sequencing report has an analogous dependency: changing the specimen or interpretation database can change what reaches the clinician even if the gene list stays fixed. A study can remain excellent evidence about its tested state while a changed state needs further support. The first choice is thus which claim and configuration to consider in each market; the study choice follows from the distance between that proposition and the evidence already available.

**Reviewer Lens.** The useful first request joins the exact regional purpose and label to the studied configurations and evidence index. That lets strong U.S. analytical work and European pilots receive their proper credit without turning either into a regional authorization. The gap may be a missing comparison rather than a wholly missing study. A reviewer who rejects everything because the broad claim is open loses the same boundary that an optimistic launch slide obscures.

**Consultant Lens.** Two separate product-and-claim sheets make the one-study limit actionable. Before choosing which study to buy, management needs its cost, recruitment or image-access feasibility, applicability to each claim, specialist capacity and commercial value. Those inputs are missing here. Conditional sequences are still useful: they show which unanswered question each investment would close and which opportunity it would defer, without pretending that a financial winner has already been calculated.

## SECTION 02

### Can one observation do two jobs?

Suppose a pathology reader study used one controlled scanner, while the proposed EU claim names that scanner and another. Can the study be reused? The first-scanner part may match the observation directly. The second-scanner part asks whether the changed input still supports the claimed reader performance. Calling the whole report “reusable” hides that difference.

The comparison therefore follows both ends of the evidence. At the claim end are the condition or treatment decision, population, specimen or image, user, setting, output, clinical role and limitations. At the study end are specimen handling, reference or comparator, hardware, software and knowledgebase versions, user task and exclusions. A match on the scanner's name alone would leave the processing and reader task

unresolved. A justified **bridge** would explain the relevant difference and provide the evidence needed to carry a bounded conclusion across it; proposing that bridge is not a successful result.

The same issue appears in a conditional sequencing example. A study of tissue taken during a procedure describes its tested tissue, operators, analysis pipeline and clinical decision. A proposed use of material recovered from blood may change collection, extraction, detectable material and borderline-result behavior. The same genes do not settle those differences. Conversely, if the controlled assay, population and medical question genuinely match, the observation may answer a scientific question in both regions. Neither example adds an undisclosed specimen or scanner to the opening board packet.

Even that scientific match leaves a second task. The EU performance evaluation report must appraise the observation for the EU purpose and connect it to the wider evidence. The study report might be usable as it stands after the required regional review, might need adaptation or a cross-reference, or might require a new regional record. Those possibilities concern the record's formal job. They do not determine whether an untested scanner performs adequately.

This table separates the two decisions when the word “reuse” is too coarse. Its judgments are **author-created planning aids**, not FDA or In Vitro Diagnostic Medical Devices Regulation (IVDR) acceptance categories.

| Question to resolve                                           | Possible conclusion for planning                                                                                         | What would support it?                                                                                             |
|---------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Does the observation answer the proposed scientific question? | Directly applicable; applicable through a defined, tested bridge; or not applicable to this question.                    | Matching claim, population, input, configuration, endpoint and limits, or evidence across the relevant difference. |
| Can the record serve this region's formal decision?           | Candidate for direct reuse after regional review; adaptation or cross-reference needed; or a new regional record needed. | Required appraisal, format, traceability, procedures and decision maker.                                           |

For the added-scanner example, the first row leaves the second scanner's support unresolved until the relevant bridge or new evidence exists. The second row can still require an EU appraisal of the original report even for the first scanner. Better document formatting would change neither scanner's observed performance. Evidence establishing a bridge could change the scientific judgment while leaving formal regional work to complete.

**Reviewer Lens.** Following one reported result back to its image or specimen, users, versions and reference decision usually exposes where a broad reuse claim exceeds the

observation. Following it forward to each proposed label then identifies the exact boundary to examine. The request can remain narrow: what supports this scanner, specimen or clinical-role difference? A matching dossier heading is easy to inspect but answers a different question.

**Consultant Lens.** A useful evidence register gives scientific match, bridge work and documentary work separate entries, owners and decision gates. Then the team can price the work actually needed. A shared protocol becomes a proposal whose applicability can be tested; it does not enter the budget as an assumed saving before the two claims have been compared.

### SECTION 03

## What would that claim commit the company to support?

Suppose the pathology company supports two scanner families instead of one. The added family may need premarket evidence, but the commitment lasts longer than the study. Compatible software combinations, customer training, updates, complaints and identification of affected installations now have to cover that family too. A cheap first dossier can therefore create a costly product fleet.

Chapter 15 showed why the installed state matters to later investigation. The same mechanism belongs in this budget. If an interface changes, the reader may encounter or recover a case differently. If sequencing knowledgebase content or a supplier component changes, the treatment interpretation may change after the original study. The **variant**—a difference from the reference sequence—has to be turned into clinical information by maintained knowledge and report rules. Security, data access and partner contracts matter because a changed or unavailable component can interrupt that path. The opening packet provides no figures from which to calculate either program's lifetime cost.

Some support can share a controlled foundation. The **quality management system** can govern design changes, supplier checks, risk records, software configuration and corrective action for both markets. Common testing can inform both arguments when the actual claims match. The regional outputs nevertheless have distinct jobs: EU technical documentation, appraisal of scientific validity and analytical and clinical performance, the applicable notified-body route, performance follow-up, surveillance, vigilance, registration and labeling. U.S. submission, labeling, change and reporting decisions also remain necessary under their applicable routes. A common spreadsheet is useful when it leads to those records and their owners.

A change strategy needs that same distinction between planned work and a decision already obtained. A **predetermined change control plan (PCCP)** in a U.S. device submission describes specified planned modifications, how they will be developed and

validated, and their impact assessment. Under the FDA's August 2025 final guidance, a **proposed or submitted plan is not an authorized plan**. For an authorized plan, the particular modification must be within the described changes and implemented according to its specified method. A change outside those terms needs its applicable U.S. change assessment; the plan is not permission to alter any configuration. The guidance itself is nonbinding. (FDA guidance, Sections IV–V)

This matters directly to the board's cybersecurity change strategy. Neither opening program has a supplied authorized PCCP. Obtaining authorization for a plan and establishing that a particular change follows its terms would change the U.S. premise; it would not supply an EU certificate or replace an applicable notified-body change assessment. Evidence and decisions must still be traced by region.

**Reviewer Lens.** A strategy slide can make proposed control look like an existing permission. The decisive records are the actual authorization and covered modification, or the actual European certificate and change procedure, followed back to the supported product variant. Common quality records should let risk, performance, security and field observations reach that same variant. Their shared location cannot enlarge the scope of either regional decision.

**Consultant Lens.** Service, update validation, regulatory maintenance, registration, surveillance and partner commitments need costs by supported configuration. Separating shared infrastructure from per-market obligations shows what remains payable after the first dossier. It also makes a proposed scope reduction concrete: which support commitment would actually disappear, and which clinically influential function would still be supplied?

## SECTION 04

### Which dependency controls the date?

If regional analysis needs the completed study, it cannot finish while that study is still producing its results. Their durations add. If two independent regional adaptations can both start after the study, and both have money and staff, they may overlap. The slower adaptation then determines when **both** dossiers are ready. This chain of prerequisites is the **critical path** behind the date; the number of busy teams is not a substitute for it.

The end of that internal path is **dossier readiness**: the required study and document work for the specified dossier is complete. Applicable authority review, notified-body scheduling, medicines consultation, certification, registration and commercial deployment have their own gates, after or alongside the relevant work. A dossier date contains none of those durations merely by being called a launch milestone. For an IVDR device, Article 48 requires the applicable conformity assessment before placement on the market, and before putting a device into service without placing it on the



market, subject to the Article 5(5) in-house exception. Finishing the file does not perform that assessment.

The opening board can now compare sequences on their premises. U.S. first could use stronger U.S. readiness while EU evidence develops, provided that the first controlled configuration does not force later EU rework. EU first could be justified by a stronger exact European purpose, evidence, real body access and partners. Parallel work could shorten a calendar if one stable product and scientifically adequate protocol serve both claims and the regional tasks are funded. A **portfolio sequence** allows different products or claim increments to follow different paths. A weak broad claim need not become a prerequisite for every narrower proposition. None is established as fastest by the opening facts.

**Reviewer Lens.** The first challenge to an advertised date is whether each predecessor task is actually finished. A proposed bridge and an unconfirmed body slot can look like completed milestones in a diagram. The evidence should distinguish completed work, internal dependencies and external decisions with only estimated availability. The purpose is to identify the unsupported date assumption, not to dismiss a valid shared task.

**Consultant Lens.** Management needs the earliest internal date beside the external steps still to follow, the minimum supported scope on which it depends and the study competing for the available capital. A credible alternative and a date or evidence event for reopening the plan make uncertainty manageable. An attractive date without those conditions leaves the board unable to change its mind rationally.

### **How does the evidence reach the board decision?**

The recommendation needs to say what can proceed for which claim, what waits, what the funded study would buy and what would trigger a switch. A readiness score cannot compensate for a missing clinical result, an ineligible body or an unaffordable study. Even the earliest feasible dossier may have less commercial value than another option. The unchosen scope and its deferred value therefore belong beside the proposed choice.

A one-page identity for **each** candidate product supplies the connection: exact regional purpose, specimen or image, user, setting, output, clinical role, version and limitations. From that identity, linked records address EU qualification, classification and route; applicable general safety and performance requirements in the technical file; prospective performance planning and appraisal of actual gaps; needed study strategy; software, artificial intelligence and cybersecurity change questions; and surveillance and performance follow-up. The U.S.–EU evidence comparison can then support a two-page executive decision. An absent input remains “unknown,” with the owner who will obtain it and the gate it controls. These are linked decisions, rather than headings that prove one another.

An illustrative work row makes the link visible: “Proposed pathology scanner addition; study on existing scanner only; scientific bridge unresolved; regional record needs claim-specific appraisal; clinical performance lead owns the evidence question; expanded claim waits for adequate evidence and applicable change assessment.” This is a method example, not an observation about the opening product. Its corresponding board line would explain that funding a scanner study serves both markets only if both claims and protocols actually match.

For the opening programs, a bounded first sequencing claim, later specimen expansion and pathology scanner or preprocessing work are conditional options to compare. The board owns scope and capital; regulatory, clinical, statistics, cybersecurity and commercial specialists contribute bounded analyses. The manufacturer retains its statutory responsibilities. A local expert can resolve a national study procedure or examine a body's stated scope, but cannot issue that body's certificate.

### **A new exhibit changes the comparison**

The opening sequencing program's analytical protocol can serve both markets, but its EU liquid-biopsy claim is immature. The pathology scanner expansion needs additional data in both markets; its interface change is commercially optional. The exhibit supplies no results for the liquid claim, no designed or funded scanner study, and no finding that the optional interface change is harmless to the validated configuration.

**Liquid biopsy** means testing material derived from a body fluid. Moving from tissue to such material can change collection, detectable material, failure patterns and clinical meaning. That is a conditional comparison here: the opening packet still does not identify the specimen behind its FDA evidence, or the exact fluid and processing path proposed for Europe.

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### **Decision**

**Both markets matter. Does that make a parallel program the fastest option? Which dependency could make it slower, and which item could still be shared?**

### **How a reviewer separates shared work from shared delay**

The analytical protocol is a genuine candidate for shared work. That positive fact does not turn the immature liquid claim into a supported one. Likewise, both regions need pathology scanner data, but no completed study is supplied. If every specimen, scanner and optional interface must enter one indispensable global package, both routes may wait for its weakest claim.

A narrower first proposition with later expansion, or different sequences for the two products, can therefore be costed and tested. The actual supplied product must respect that boundary: deleting wording cannot remove a function still offered. A U.S.-first

tissue companion-diagnostic path would be an alternative to investigate **if its tissue scope and evidence were established**; tissue behind the opening FDA evidence is not given. A shared pathology scanner/preprocessing study is another option to design against actual claims. Partner commitments, lawful routes, one-study capital and feasible timing control the comparison. Neither regulator accepts a global planning label in place of its own product-and-evidence decision. There is no unique fastest or highest-value route on these facts.

**Reviewer Lens.** The shared protocol deserves credit for its designed question. Expanded claims need the EU specimen and clinical-role comparison, actual scanner-study design, interface configuration and body scope. Keeping those requests separate protects both the positive reuse finding and the unresolved expansions.

**Consultant Lens.** The exhibit lets management consider deferring an unsupported liquid claim or commercially optional interface expansion while developing a genuinely bounded product. It does not select the investment. The next useful purchase may be protocol feasibility and a conditional budget before the major study itself; otherwise the only available study could be committed before anyone knows which regional decisions its results can answer.

## SECTION 05

### Which objective does the arithmetic answer?

A separate resource-and-sequence exercise now supplies money and months. These figures are **invented for arithmetic teaching** and price neither opening program. The board has **\$1 million** for new study and dossier work; operations has a separate reserve. Its objective is the **earliest first completed regional dossier**, paying only for work needed for that dossier. Authority review time, revenue, probability of acceptance and a second-market plan are absent.

The **shared-study option** costs **\$0.8 million** and takes **six months** for its study. Afterward, U.S. dossier adaptation takes **one month** and **\$0.1 million**; EU adaptation takes **two months** and **\$0.2 million**. Both adaptations may run concurrently if funded. The study is designed as shared, but its applicability to both scientific questions is not yet established. The **regional-study option** costs **\$0.6 million** and takes **five months**, followed by a **one-month, \$0.1 million** dossier task. It does not answer the other region's clinical question; that region's later cost and time are unknown.

Start with the regional option's dependency. Its one-month dossier task follows five months of study, giving  $5 + 1 = \text{month } 6$ . Its study and dossier together cost  $\$0.6\text{M} + \$0.1\text{M} = \$0.7\text{M}$ , leaving  $\$0.3\text{M}$ . The dossier cannot finish at month 1 because it needs the study result. The shared option's U.S. path similarly takes  $6 + 1 = \text{month } 7$  and

costs  $\$0.8\text{M} + \$0.1\text{M} = \$0.9\text{M}$ . Its EU path takes  $6 + 2 = \text{month } 8$  and costs  $\$0.8\text{M} + \$0.2\text{M} = \$1.0\text{M}$ .

The table is useful for comparing what the original budget can actually fund. All amounts are millions of dollars.

| Funded path                     | Tasks in order                                                               | Earliest dossier-ready month | Cost and balance of the \$1M budget |
|---------------------------------|------------------------------------------------------------------------------|------------------------------|-------------------------------------|
| Shared study, U.S. dossier only | Six-month study, then one-month adaptation.                                  | 7                            | \$0.9M; \$0.1M left.                |
| Shared study, EU dossier only   | Six-month study, then two-month adaptation.                                  | 8                            | \$1.0M; nothing left.               |
| Shared study, both dossiers     | Six-month study, then one- and two-month adaptations concurrently if funded. | 8 if funded                  | \$1.1M; <b>\$0.1M over budget.</b>  |
| Regional study, one dossier     | Five-month study, then one-month dossier task.                               | 6                            | <b>\$0.7M; \$0.3M left.</b>         |

The third row shows why parallel execution does not solve affordability. Its adaptations can overlap in time, but both still cost money:  $\$0.8\text{M} + \$0.1\text{M} + \$0.2\text{M} = \$1.1\text{M}$  exceeds the original budget. Under the stated first-dossier objective, the **regional-study option is the bounded positive answer**, ready at month 6 for \$0.7M. Its \$0.3M remainder has no supplied purchasing power for an unknown second-region study.

This closes the specified scheduling choice, not the business case. Dossier readiness is neither submission acceptance nor completed authority review, EU certification, U.S. authorization or commercial launch. The regional option's second-region route remains unpriced and unsequenced; the shared option's scientific transfer still needs checking. A different objective, budget or evidence-applicability premise could change the recommendation without making this arithmetic wrong.

**Reviewer Lens.** The audit is reproducible: adaptations begin after month 6, an unfunded pair stays out of the feasible choices, and an unexplained second-region cost stays unknown. The most misleading error would be crediting a shared study with both dossiers while charging only for the first. The result is useful precisely because its objective and omissions remain visible.

**Consultant Lens.** The board can accept the first-dossier result while asking what a second region is worth, whether the shared study really covers both claims and what extra funding would buy. Commercial and regulatory inputs remain separate from the

arithmetic. Those are the levers for reconsidering the choice; “month 6” alone cannot tell management when revenue or lawful access begins.

### Decision

The board now wants both regional dossiers ready, raises the study/dossier budget to \$1.2 million, and stipulates that the shared study answers both regions' scientific questions. With the same durations and funded adaptations allowed to overlap after the study, when are both shared-option dossiers ready, and how much money remains? Does this establish commercial optimality?

### How the changed objective changes the calculation

Three premises changed: the objective, the budget and scientific applicability. The shared option now has  $\$1.2\text{M} - (\$0.8\text{M} + \$0.1\text{M} + \$0.2\text{M}) = \$0.1\text{M}$  left. Both adaptations start after the six-month study. The one-month U.S. task finishes first, but both dossiers are ready only when the two-month EU task finishes, at **month 8**. The total is \$1.1M.

That is the earliest date under these supplied shared-option dependencies. It is not an eight-month authority decision. Commercial optimality still lacks authority review, market access, service cost, revenue, partner value and the regional option's second-region evidence cost. Nothing in this changed scenario repairs the original \$1M budget or establishes its original scientific premise retrospectively.

### A recommendation another person can reopen

The review record works backward from proposed result to tested claim, input, users and versions, then forward to each region's scientific question, formal assessment and decision maker. For a numerical recommendation it also exposes dependencies, excluded authority time and lifecycle costs. Its conclusion names the observation that could change it. The same state discipline keeps “proposed,” “submitted” and “authorized” distinct when a U.S. change plan appears in the file; neither opening program has a supplied authorized plan.

The management version connects a controlled scope sheet, shared and regional evidence budget, critical path, alternatives, owners, stop rules and a reopening date. The consultant integrates that choice and commissions bounded local work, while the manufacturer owns scope, investment and statutory submissions. An informal specialist response remains advice, however helpful it is to planning.

### Transfer

The opening packet's body capacity moves nine months.

The selected notified body's capacity is delayed by nine months. Which opening-program work remains scientifically useful, which EU dependencies move, and which U.S. or bounded commercial preparation could proceed? Would a narrower EU purpose or another designated body change the choice?

### How a capacity delay differs from evidence failure

Useful shared scientific work may retain its value while EU capital timing and the critical path change. A supportable narrower purpose needs evidence; another body needs the exact designation and actual capacity to take the application. The delay itself establishes neither a right to switch, a new certificate date nor sufficient EU evidence. The revised options and opportunity cost belong before the board, with verified body and claim assumptions. These are the opening programs: the resource exercise's dollars and months cannot price them.

## SECTION 06

### What changes when the claim becomes specific?

The practice studio supplies a more specific board packet. Its sequencing product has strong **U.S. tissue evidence** and a companion-diagnostic partner, while the EU purpose adds liquid biopsy and the performance plan and report are immature. The pathology product has European pilot sites and a broad scanner claim; its U.S. reader study used a **narrower workflow**, and its interface change is outside the validated configuration. The board can fund one major study. There are no dollar amounts or durations, and neither product's authorization, certificate, outcome or exact investment return is established. The tissue and narrower-workflow facts belong to this studio; the independent resource exercise supplies none of its economics.

### What would the next study actually resolve?

The tissue evidence remains valuable for the tissue question it examined. The proposed liquid purpose changes how genetic material is obtained and may change its amount, failure patterns and the clinical meaning of a result. Sharing the gene list does not establish transfer across that specimen path. Before selecting a study, the team therefore needs the specimen-specific analytical and clinical questions, the proposed bridge or new work and the intended initial scope.

The medicine relationship needs equal precision. As earlier chapters established, an **IVDR companion diagnostic** is essential to safe and effective use of a corresponding medicine, identifying before or during treatment patients most likely to benefit or at increased risk of serious adverse reactions. That relationship makes the

medicine-specific suitability and consultation question relevant. A partner's presence does not supply a missing performance appraisal; a changed medicine or clinical decision role can reopen the relationship. The **performance evaluation plan** states the intended questions and criteria, while the **performance evaluation report** appraises the evidence actually available, favorable and unfavorable. Neither document title creates results for the liquid claim.

For pathology, the result reaches a human task. European pilots may establish feasibility at their tested sites; the narrower U.S. reader workflow may supply useful observations about that workflow. A broad scanner and user claim also needs to account for the actual images, preprocessing, users, tasks, worklist and interface version. Changing what reaches the reader can change the question even if the model is unchanged. Commercial attraction does not bring an interface outside the validated state inside it.

The comparison below identifies what a study purchase must answer before its results could be used in either region.

| <b>Proposed use of the studio's evidence</b>                      | <b>Scientific question still to resolve</b>                                                                   | <b>Regional work that remains</b>                                                      |
|-------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| U.S. tissue results for an EU tissue claim                        | Do device, population, treatment role and methods match? The extent is not supplied.                          | EU plan and appraisal, with applicable conformity and medicine-consultation questions. |
| U.S. tissue results for an EU liquid claim                        | Does the changed specimen path have adequate support? Transfer is not established.                            | Specimen-specific questions, bridge or new study, and bounded initial claim.           |
| Pilots and narrower reader workflow for the broad pathology claim | Do scanner, processing, user, task and interface differences have support? Broad transfer is not established. | Evidence design for the actual claims in each region and their required appraisals.    |

In the first row, a true match could allow the tissue observations to contribute to the EU argument without a duplicate scientific question. In the second, the same report cannot establish the changed specimen merely by moving to the EU file. Evidence for that difference would change the scientific judgment; the regional appraisal would still have to use it. The pathology row applies the same reasoning to an image-to-reader path. These rows preserve the tissue work and pilots' value without inventing favorable bridge results.

**Reviewer Lens.** The crucial comparison lies at the changed boundary: collection, processing, failure and treatment role for sequencing; image path and reader action for pathology. A global protocol is credible when those comparisons show why its



observations can answer both claims. The request need not become an indiscriminate demand to repeat everything already demonstrated.

**Consultant Lens.** The study specification should explain which unresolved decision it buys, which regions could use the answer and which records would still be regional. Protocol feasibility and conditional costing can begin now. Treating a proposal as study-ready or approved would hide the very facts needed to decide whether it deserves the only major-study allocation.

### Which claim has to wait for which other claim?

The studio's tissue evidence makes a narrower **U.S.-first** sequencing option worth testing. Its possible calendar advantage comes from allowing the tissue question to proceed without making immature liquid evidence its prerequisite. That still requires the actual U.S. route, release state and supported tissue scope; neither a partner nor strong evidence supplies a launch date or revenue. The EU bridge needs separate costing, and a controlled global core can help prevent the first configuration from forcing an awkward European redesign. If tissue support or partner scope fails, the reason for this sequence changes.

Could the pathology pilots instead justify **EU first**? Only if the exact European purpose, performance evidence, body designation and real capacity, partnerships and lifecycle economics support that bounded route. Pilots alone do not establish those conditions, and the sequencing program's immature plan and report do not establish them either. A future EU certificate would still leave a U.S. reader-task or change question to its own process. An actual stronger EU file and usable body access would change the comparison; the geographic location of pilots does not close it.

The scanner evidence need raises a third option: could one controlled product and protocol answer both markets in **parallel**? A scanner/preprocessing study could do so if the proposed tasks and endpoints match, with money and people for simultaneous regional analysis. No such completed study is supplied. If an expanded specimen or scanner claim becomes indispensable to both submissions, its unresolved result can delay both. A shared-study gate and separate regional stop rules let the team distinguish useful overlap from a shared bottleneck.

Because the two products differ, the company need not give both the same market order. A **portfolio sequence** can separate sequencing tissue and later liquid claims, or the pathology validated interface and proposed expansion. A bounded product in one region, shared evidence for another product and deferred optional scope can have different gates. The benefit depends on which work unlocks useful value under the one-study limit. Cost, expected demand and study feasibility are missing, so the portfolio idea cannot itself identify a winner.

The table retrieves the premise and failure condition for each sequence when management compares alternatives.

| Sequence                             | Premise that could justify it                                                     | Fact that would reopen it                                                              |
|--------------------------------------|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| U.S. first                           | Supported U.S. scope and actual route are more ready.                             | Failed evidence or partner assumption, or a configuration that undermines EU transfer. |
| EU first                             | Supported EU scope, performance file and usable body access are stronger.         | Pilot-to-claim gap or body capacity invalidates the assumed path.                      |
| Parallel                             | Controlled product and protocol answer both questions; regional teams are funded. | Shared endpoint, claim or resource becomes a bottleneck for both.                      |
| Different product or claim sequences | Readiness and value differ across products or increments.                         | Budget, partnership or bridge results change which work unlocks value.                 |

For the parallel row, a study answering both scanner questions would support overlap only if both regional tasks can actually be funded and staffed. If the shared design cannot answer one reader task, concurrency no longer supplies the assumed advantage. The table is not a scorecard: a cheap date cannot compensate for missing clinical evidence, and no row supplies an authority timetable. Continuing quality, security, change, labeling, surveillance, performance follow-up, vigilance, registration, partner and service costs remain in every credible sequence.

**Reviewer Lens.** The diagram's shared milestone deserves a close read of protocol, configuration and evidence. A treatment-specific or scanner-specific result may have been smuggled into a broad word such as “global.” An external slot may likewise have been assumed. Locating those premises makes a focused hold possible while preserving genuinely completed shared work.

**Consultant Lens.** The alternative not funded deserves visible treatment. A narrow first claim may remain commercially useful or may lose the very value management wants; that requires commercial input alongside regulatory readiness. Comparing the one-study purchase with later expansion and support prevents an earlier dossier from silently becoming the only objective.

### What would make the memo actionable?

One **option to test** is U.S.-first sequencing tissue work while the EU performance architecture is completed, with liquid biopsy deferred until specimen and clinical support are adequate. For pathology, another is keeping the validated interface controlled while designing scanner/preprocessing evidence that could address both markets. The optional interface expansion would wait unless its benefit justified added verification, clinical and change work. These options are neither funded conclusions nor findings that a claim is authorized or commercially best. Different cost, value, capacity or route facts could favor another investment.

The first executive page connects the proposed initial claim to positive evidence, gaps, critical dependency, owner, capital envelope and stop rule. The second compares the unchosen alternatives and identifies the switching trigger. “Fund scanner study” becomes actionable only when it names the scanner/use claim, the result that would matter and the other work displaced by the one-study limit.

The table shows three studio decision lines whose evidence and management consequences remain connected.

| <b>Decision for management</b> | <b>Basis supplied by the studio</b>                                                                    | <b>Owner and reason to reconsider</b>                                                                                                                        |
|--------------------------------|--------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tissue-first sequencing option | Strong U.S. tissue evidence and partner; EU plan/report immature; no specific authorization or return. | Regulatory and clinical owners confirm exact scope, route and appraisal; reconsider failed tissue/partner assumptions or material new liquid evidence.       |
| Shared pathology study option  | European pilots and narrower U.S. reader workflow; broader scanner support needs data in both markets. | Clinical, statistics and software owners define scanner, processing, reader task and endpoints; reconsider if a shared design is infeasible or unaffordable. |
| Optional interface expansion   | Commercially optional and outside the validated configuration.                                         | Maintain control of the initial configuration; reconsider with a defined benefit, evidence and applicable change route.                                      |

The last row shows why an optional feature can have a real funding consequence. Deferring it can preserve a controlled initial boundary only if the supplied product actually stays within that boundary. A demonstrated benefit and completed relevant evidence could change the choice; commercial preference alone cannot remove the change question. The shared-study row also remains a proposed purchase, rather than proof of either scanner claim.

A decision owner can call on body-scope specialists, national authority or study specialists, medicinal-product experts for an actual companion-diagnostic relationship, statisticians, clinicians, cybersecurity specialists and legal advisers. Their answers resolve particular gates. They do not transfer the manufacturer's statutory decisions to the consultant.

### Decision

**Does the strategic importance of both markets establish that parallel work is fastest? Which facts would let the board test that claim for these products?**

### How a board tests the proposed overlap

A stable configuration and genuinely shared scientific protocol can shorten the path when regional people and funding are available. Here the liquid specimen and broad scanner/workflow claims may become prerequisites that hold both sides. Actual study dependencies, bounded purposes, body or authority queues, medicines consultation and capital determine the comparison. Counting simultaneous project streams does not.

A portfolio sequence may preserve a bounded earlier option while retaining later expansion. Its value still requires commercial inputs. The result is a conditional choice with a reason to switch, not a declaration that parallel work always loses or always wins.

### Who actually owns the remaining gate?

The **notified body** assesses the EU product and evidence under the applicable conformity route within its designation and procedures. A national **competent authority** has separate study, vigilance or market-surveillance responsibilities; it is not an alternative product certifier. Where the defined companion-diagnostic relationship requires consultation, a medicinal-products authority or the European Medicines Agency contributes its scientific opinion on suitability with the medicine. That opinion is not the device certificate.

The FDA applies the relevant U.S. device route to the offered U.S. claim. The **manufacturer** owns product architecture, capital and statutory submissions; the consultant integrates the dependencies and obtains bounded local expertise. An informal favorable reply about capacity cannot become the other actors' decisions. Changing the product or medicine relationship can change which process is needed, while a different meeting attendee cannot change the process's legal job.

### Transfer

The studio's body capacity moves nine months.

The selected body's capacity is delayed by nine months. How does that alter the studio's EU critical path and opportunity cost? What genuine U.S. milestone or bounded commercial preparation could proceed, and what event would restore or change the original sequence?

### **How the evidence can retain value while the sequence moves**

Tissue, scanner and software evidence can remain scientifically relevant through a capacity delay. EU timing still changes, and the board needs the revised dependency and capital picture. A narrower EU purpose must have adequate evidence. A different body is a credible application option only with the actual designation and usable capacity. No replacement body, new certificate date or earlier launch is supplied, and there are no studio durations from which to calculate a complete new calendar. The recommendation can still identify the opportunity cost and the verified event that would trigger reconsideration.

## **SECTION 07**

### **Can the method survive a different packet?**

#### **A combined global plan**

An independent global plan bundles tissue, liquid biopsy, three scanners, community users and two drug relationships into one parallel U.S.–EU program. One weak specimen claim holds every study and submission. No completed study, cost or authorized subset is identified.

#### **Decision**

**Is parallel still the fastest strategy on these facts? What would have to be separated before a different sequence could be recommended?**

### **How a reviewer examines whether a claim can stand alone**

The weak shared prerequisite defeats an unqualified fastest claim. The comparison needs product and claim increments that can stand on their own evidence and route, together with the work that genuinely shares a controlled configuration and study. Only then can critical paths and capital be compared. A portfolio sequence may unlock earlier bounded value while preserving expansion. Calling a subset “first” cannot make it lawful, and this packet supplies no authorized subset to assume.

## A remapped dossier

A separate team maps a complete U.S. tissue dossier into EU Annex II headings. It assumes that a liquid-biopsy purpose is covered because the same genes are reported.

### Decision

**Is the remaining task mainly formatting? Which two evidence questions need separate answers?**

### How a reviewer separates the specimen question from the file question

First comes scientific applicability: whether the tissue observations support the liquid specimen, extraction, variant detection and clinical or treatment role. A relevant difference needs a defined, tested bridge or new work. Separately, the actual EU route needs its technical documentation, scientific-validity and analytical and clinical appraisal, applicable consultation and lifecycle records. Correct headings can help locate those arguments. They cannot create either the missing specimen support or the required regional assessment.

## A choice that remains open to evidence

The resource exercise supports a precise first-dossier recommendation under its original objective, while its later disclosure shows how changed premises alter the answer. The opening and studio still need their own evidence and economics. Chapter 17 carries this method into an unfamiliar independent assessment: the offered product and claim, the decision each evidence item supports and the unresolved dependencies. These fictional programs and numerical conclusions are not facts to carry into that assessment.

The inherited cards below retain their Workbook Part A2 retrieval identifiers. The item's evidence location and reasoning precede its working state: **A**, adequately explained for that bounded item; **U**, present but unclear; **M**, missing or inconsistent; **D**, deeper analysis required; **C**, consultation indicated; or **N/A**, inapplicable with a reason. These states are neither scores nor approvals or device classes. Historical card numbers identify source topics, not this edition's chapter numbers.

## The same product and claim? • card IVDR-CH17-A01

Which regional purpose, population, specimen or image, user, clinical role, limits and controlled versions match the evidence? The factual comparison gives document headings something meaningful to organize.

**Two separate reuse judgments • card IVDR-CH17-A02**

Does the protocol and report answer the proposed scientific question? What distinct appraisal, traceability or regional record is still needed? An adequate answer to either leaves the other inspectable.

**The cost after the dossier • card IVDR-CH17-A03**

Which development and continuing quality, software, security, field, partner and regional obligations survive the first dossier? Supported configurations make those costs identifiable.

**A premise that could change the answer • card IVDR-CH17-A04**

Which objective, budget, dependency or applicability fact controls the recommendation? The changed-scenario record distinguishes a legitimate recalculation from silently borrowing a later premise; absent authority and commercial inputs still prevent a launch conclusion.

**An evidence budget management can use • card IVDR-CH17-C01**

Do the claim-specific crosswalk, critical path, capital envelope, authority steps and alternatives lead to the executive choice? Its dates distinguish dossier readiness from market permission.

**The study and value being deferred • card IVDR-CH17-C02**

Which supported first scope fits the one-study constraint, and what value waits? An explicit objective and stop rule make the product, market and capital choice reviewable.

**The decision and its owner • card IVDR-CH17-C03**

Which Member State, notified-body, medicine, clinical or technical work needs local expertise? The owner map connects that work to its gate while retaining the manufacturer's statutory decisions.

**Further reading**

Book 2 Chapters 7 and 11 provide optional depth on prospective evidence planning and actual sufficiency appraisal. Their methods can inform a regional budget; their fictional case facts do not become evidence for these products.



**Source trail**

Binding EU context checked 25 September 2026 in the IVDR consolidated 10 January 2025 text: Article 48(1)–(2) on applicable conformity assessment before placing a device on the market or putting it into service, Article 56 and Annex XIII Part A §1 on intended-purpose performance evidence and prospective planning, and Annexes I–III on applicable requirements and documentation. The European Commission's notified-body/NANDO page describes designation and scope; it does not establish a fictional body's availability or certificate. The FDA's final AI-enabled device PCCP guidance, August 2025 and its guidance text distinguish a planned/submitted PCCP from modifications implemented under an authorized PCCP; the guidance is nonbinding and does not confer EU change permission. FDA pathway and product-specific decisions still govern any actual U.S. submission. The two-axis reuse labels, board matrix and LS-1 arithmetic are original teaching methods from the preserved fictional source packet, not legal classifications or findings of market access. No AI Act status, commercial return or authority review time is inferred for either case.

## CHAPTER 17

# The assessor's flight check

## Before reading

For one study in Chapter 16, would its observations answer another region's scientific question? Could its record serve that region's formal decision as written? Which assumption and new event could change the market sequence while leaving a dossier-ready date distinct from authorization?

An unfamiliar packet can contain good evidence and still leave the requested decision unsupported. The assessor's task is to find that boundary without losing what the evidence already establishes. Following a result back to its promised medical job makes the next question smaller: what is missing, who can supply or decide it, and what would change the conclusion?

## SECTION 01

## What was the evidence supposed to establish?

An assessor may have hundreds of pages, yet the object of review begins with an ordinary event. Someone takes a specimen or image; a controlled device produces information; a stated user acts on it in a stated setting. The **intended purpose** connects those events into the manufacturer's promised medical job. If that promise changes across the file, a polished performance report cannot tell the assessor which proposition to decide.

A useful orientation can therefore fit in one sentence: *this version examines this input for this result, for these people and users, to inform this decision*. The sentence joins the label to the actual configuration, the combination that produces the result. It also exposes the next distinction. Is the manufacturer asking about a proposed claim, a product change, a performance study, a field signal or a market plan?

That question determines whose decision is needed. The manufacturer owns design and evidence; a notified body performs applicable conformity assessment within its scope. A national competent authority performs its assigned public-law tasks, and a medicinal-products authority contributes the prescribed opinion where a companion-diagnostic consultation applies. Management chooses investment within

those constraints. A favorable answer from one actor cannot silently become the others' decisions.

Nor is this a one-way journey through the file. If a pathology worklist update changes which cases a pathologist can see, the earlier claim of the same workflow needs reconsideration even when the model-performance table is stable. A sequencing label that adds a specimen or treatment role can similarly change the scope of earlier evidence and classification reasoning. The useful return point is the first affected premise; evidence whose premise still holds keeps its value.

**Reviewer Lens.** Purpose, label, configuration manifest and one representative result are often most revealing when read together. A contradictory specimen, user, version or decision role shows where a conclusion has lost its object. The narrow request is for the relationship that would resolve that contradiction, with the actor and source appropriate to it. This avoids turning an orientation problem into an unexplained demand for every study to be repeated.

**Consultant Lens.** A one-page account of claim, configuration, market state, actors and unknowns makes specialist work easier to commission. The expensive mistake is to schedule a broad evidence program before the company has chosen which medical job it is offering. An open gate can have an owner and a next event without a confident completion date; management can fund that work while retaining responsibility for the scope choice.

**An evidence critique.** An assessor completes a detailed performance evaluation report (PER) critique, then finds two medical roles alternating in the purpose: a *prognostic* role, giving information about the likely course of a condition, and a *predictive treatment-selection* role, giving information about who may benefit from a particular treatment. Nothing in the packet establishes that they are the same claim.

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## Decision

**Was the evidence review wasted? What is the earliest premise to reopen, and which later conclusions depend on it?**

### How a reviewer preserves the observation while reopening its use

The critique still describes evidence for the question the studies actually addressed. What it cannot do is attach that conclusion to either of two unstable medical purposes. Once the manufacturer settles the role, the assessor can revisit qualification, applicable class, performance planning and appraisal, and labeling against that role. This repairs the sequence without declaring the observations worthless.

The medicine relationship needs its own premise. A companion diagnostic is essential to the safe and effective use of a corresponding medicine and identifies, before or

during treatment, patients most likely to benefit or those at increased risk of serious adverse reactions. A stated predictive role alone does not establish that complete relationship. If it applies, its classification and consultation work belongs in the reopened route; if it does not, that particular consultation cannot be inferred merely from treatment-related wording. The revised purpose and evidence comparison, rather than the report's completeness, determine which later conclusions can stand.

## SECTION 02

### Can an older device use the later date?

A company presents an older device and points to a transition date several years away. Before that date can support a market plan, what must be true of the device? The In Vitro Diagnostic Medical Devices Regulation (IVDR) governs qualifying devices examining human-derived specimens for the specified medical information. An older device made under the former In Vitro Diagnostic Medical Devices Directive (IVDD) may have a conditional transition route. Calling a new device similar to an older model cannot give it that history.

Article 110 offers two relevant doorways. The certificate branch, **3a**, requires an IVDD certificate valid by virtue of paragraph **2**. The formerly self-declared branch, **3b**, requires three facts together: the IVDD procedure did not require a notified body; an IVDD declaration of conformity was drawn up **before 26 May 2022**; and the IVDR procedure does require a notified body. If neither doorway fits, the later end dates do not extend a new product's route or that of an unrelated class A non-sterile device.

For the certificate doorway, “there was a certificate” is too weak. Under Article 110(2), certificates issued **from 25 May 2017**, still valid on **26 May 2022**, and not withdrawn afterward continue beyond their printed expiry to **31 December 2027**. Certificates in that group that expired **before 9 July 2024** need a further basis: before expiry, the manufacturer and a notified body had signed the specified written conformity-assessment agreement for that device or its intended substitute; or a Member State competent authority had granted an Article 54(1) derogation from the procedure, or required the manufacturer under Article 92(1) to carry it out. These are actual agreement or authority records. An inquiry about body capacity supplies none of them. The adopted 2024 amendment supplies these operative conditions.

The older, **pre-25 May 2017** certificates have a different rule: validity lasts to the period stated on the certificate, with IVDD Annex VI certificates becoming void by **27 May 2025** at the latest. They do not inherit the later-issued group's extension simply by being old. This distinction follows the dated consolidated Article 110(2). Certificate

issue, validity, withdrawal and expiry history therefore precede the calendar conclusion.

Even an eligible history leaves the continuing conditions in **3c** to establish. The device must remain IVDD-compliant, have no significant change in design or intended purpose, and present no unacceptable risk to health, safety or public health. An Article 10(8) quality management system had to be in place by **26 May 2025**. The manufacturer or its authorized representative must also have lodged the formal notified-body application for the device or its intended substitute, followed by the required manufacturer–body written agreement, by the applicable deadlines. A capacity email is neither event. A significant change, unacceptable risk or missing prerequisite changes the conclusion even if the end date has not arrived.

The table retrieves those deadlines once eligibility and continuing conditions have been established. The final column is the conditional period for placing on the market or putting into service.

| <b>Eligible route</b>                                            | <b>Formal application by</b> | <b>Written agreement by</b> | <b>Conditional period ends</b> |
|------------------------------------------------------------------|------------------------------|-----------------------------|--------------------------------|
| 3a: device with qualifying IVDD certificate                      | 26 May 2025                  | 26 September 2025           | 31 December 2027               |
| 3b: formerly self-declared device now class D                    | 26 May 2025                  | 26 September 2025           | 31 December 2027               |
| 3b: formerly self-declared device now class C                    | 26 May 2026                  | 26 September 2026           | 31 December 2028               |
| 3b: formerly self-declared device now class B or class A sterile | 26 May 2027                  | 26 September 2027           | 31 December 2029               |

For the class C row, the pre-2022 declaration and old/new body-involvement facts first establish which branch is being considered. The 2026 application and agreement records then answer separate conditions; pointing to the 2028 end date cannot repair either missed deadline. The same reasoning applies across the table, with the dates changing by branch. No fictional device in this discussion is declared eligible.

These are the Article 110(3a)–(3c) dates in the consolidated IVDR checked **25 September 2026**. Under **3d**, eligible legacy devices follow IVDR post-market surveillance, market-surveillance, vigilance and economic-operator/device registration requirements in place of the corresponding IVDD requirements. Continuing through the transition does not defer the response to field signals until full IVDR certification.

**Reviewer Lens.** An old certificate or declaration is the beginning of the evidence chain. The old and new routes, change history, quality-system record, formal application and

agreement show whether the claimed date follows from it. An unknown condition stays visible as an unknown; a distant end date adds no evidence for that missing condition. For an expired certificate, the precise earlier agreement or authority record is more useful than a broad assurance that transition has been discussed.

**Consultant Lens.** A dated eligibility sheet prevents a commercial forecast from outrunning the device's history. Significant-change and risk questions need qualified owners, while ongoing legacy obligations and the new assessment project need separate work plans. The company can price and schedule those tasks before promising a launch based on a 2027–2029 date. Later receipt of a document changes the assessment only if the document establishes the required event and timing.

## SECTION 03

### Where does a changed fact send the review?

Return to the pathology worklist. A model may calculate the same values while a changed display alters which cases reach the pathologist. Where should the assessor reopen the file? The nine passes introduced earlier provide a route to that answer, beginning with **Pass 0**: the exact product, claim, version, market and requested decision.

Once those are visible, authority and product purpose can be checked together. Binding IVDR text, nonbinding Medical Device Coordination Group guidance, a standard used as a method, an actual notified-body designation and a Member State procedure have different jobs. The source must answer this question for this actor. The product's medical function then permits qualification, followed by the applicable Annex VIII class and Article 48 conformity-assessment route. Software can have its own medical purpose while integrated into another system; its packaging does not decide the class. If function or purpose changes, those premises need another look.

The route then needs evidence about one identifiable product. An applicable general safety and performance requirement leads to what the design must achieve, a possible risk and control, the relevant verification or validation, the label, and a way to learn after marketing. Those links become traversable technical documentation only when their versions can be related. A stable model file leaves the changed display and user task somewhere else in that chain.

The performance questions locate that task more precisely. **Scientific validity** concerns the association of the measured marker or image feature with the clinical question. **Analytical performance** asks whether this system detects or measures it under claimed conditions. **Clinical performance** concerns the result's relationship to the condition or state in the intended population and use. For a changed worklist, evidence about the calculation alone does not describe the pathologist's actual encounter with the case.

Changed users, specimens or configurations can similarly change which strand needs further support.

The performance evaluation plan states questions and acceptance grounds prospectively; the performance evaluation report appraises what was actually obtained, including unfavorable findings. Post-market performance follow-up can address justified residual questions within an assessed purpose after marketing. It cannot turn missing core premarket evidence into a supported first claim by promising to collect it later.

Making that evidence may itself require decisions about the actual study. Specimen source, added procedures, result use, protocol, countries, ethics, version and safety responsibilities determine the study questions. A medicine-linked claim adds consultation only when the companion diagnostic relationship holds: essentiality for safe/effective medicine use and the specified benefit or serious-adverse-reaction patient identification before or during treatment. Changed protocol or medicine role can therefore reopen a route that was sensible for the earlier question.

Keeping the evidence true extends the inquiry to software, artificial intelligence, cybersecurity and suppliers. A scanner, cloud service, interface or controlled interpretation source can change the result path without changing a model file. Field information then feeds back into risk, performance appraisal, certification and product identity through surveillance and applicable vigilance work. A signal can reopen an earlier premise; it need not erase unrelated positive findings.

The table is useful when a later observation needs a return address. Its passes organize review, rather than creating nine independent approvals.

| Pass                          | Question to resolve                                | First records to open                                                       | Return when...                                                             |
|-------------------------------|----------------------------------------------------|-----------------------------------------------------------------------------|----------------------------------------------------------------------------|
| 0 — orientation               | Which exact product and decision?                  | Purpose, label, configuration and market state.                             | Later facts change the product or proposed scope.                          |
| 1 — actor and source          | Who can decide, under which authority?             | Actor map, legal text, source status and designation or national procedure. | The question becomes one of studies, vigilance, medicines or national law. |
| 2 — purpose and qualification | What information, for whom, for which medical job? | Actual function, workflow, purpose and presentation.                        | Behavior or label contradicts the purpose.                                 |



| Pass                       | Question to resolve                                               | First records to open                                                                                  | Return when...                                                     |
|----------------------------|-------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| 3 — class and route        | Which Annex VIII rule and Article 48 procedure fit?               | Reasoned class, scope and body-route records.                                                          | Specimen, user, medical role or medicine relationship changes.     |
| 4 — requirement and file   | Does the evidence concern one controlled product?                 | Requirement, design, risk, test and label links.                                                       | Product or supplier baselines disagree.                            |
| 5 — performance            | What scientific, analytical and clinical support fits each claim? | Plan, report, studies, limits and follow-up rationale.                                                 | Population, specimen, user or configuration misses the claim.      |
| 6 — study and consultation | What extra routes follow from actual activities and medicine use? | Protocol, country/ethics, safety and medicine-bridge records.                                          | Procedure, sample source, patient impact or medicine role changes. |
| 7 — software and change    | Is the evidenced system still the one supplied?                   | Architecture, versions, tests, user task, security and change records.                                 | Scanner, model, cloud, interface or supplier state changes.        |
| 8 — field and conclusion   | Can field information reach an owned product decision?            | Surveillance/follow-up, vigilance, risk/report updates, certificate, registration and installed state. | A signal or strategic change alters an earlier premise.            |

Reading the software/change row for the worklist takes the review back to the user task at pass 2, its evidence at pass 5 and its field detection at pass 8. For a sequencing specimen addition, the return may include purpose, class, analytical and clinical performance, and applicable study or medicine questions. Earlier tissue results remain observations about their own bounded claim. These are generic routes through a change, with no case outcome supplied.

**Reviewer Lens.** Following one sampled requirement from label to test and field record can expose a contradiction hidden by complete-looking headings. The finding becomes useful when it identifies the earliest inconsistent premise and the decisions that depend on it. A positive test that still concerns its original configuration should retain that explicit credit, even while the proposed wider conclusion remains open.

**Consultant Lens.** One changed premise may create several dependent decisions, but it need not create nine disconnected projects. An owner and closure event for each

dependency allow product, clinical and software work to meet at the same controlled state. Preliminary design work can be budgeted now; its proposed study cannot be booked as evidence already obtained.

## SECTION 04

### What does an adequate item mean tomorrow?

A pathology scanner-version record can adequately connect tested images and a supported release to a bounded claim. A proposed different scanner raises another question. It does not make the former images cease to exist or necessarily invalidate the former relationship. This is why a working state describes one item at a particular time.

**A** means adequately explained for that item. **U** means present but unclear; **M**, missing or inconsistent; **D**, deeper technical or regulatory analysis needed; **C**, relevant specialist or institutional consultation indicated; and **N/A**, inapplicable with a reason. The letters cannot be averaged into a product grade or approval. A sound class memo can coexist with an unresolved user-evidence question, and new facts may reopen either one according to its own premise.

For sequencing, an adequate analytical method for a tissue variant class can remain adequate within that scope while a liquid specimen or new treatment interpretation is missing or needs deeper analysis. The evidence location and date explain the boundary. The next action says what additional fact or work could change the state and what management can reasonably do meanwhile.

**A changed user claim.** A classification memo was adequate for professional use. Later labeling adds self-testing by lay users. No new rule analysis or use evidence is supplied.

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### Decision

**Should the classification item remain closed because it was once A? Which other decisions need to reopen?**

#### How a reviewer keeps the old judgment without extending it

The former professional-use reasoning and its date remain part of the record. They do not establish the expanded claim. Intended user and purpose now need to connect to the applicable classification rules, user risk and performance evidence, labeling, conformity route and postmarket assumptions.

Lay people using the test activate the self-testing analysis; “professional use” no longer describes the whole proposition. The packet supplies no resulting rule analysis or evidence, so the new state describes an unresolved question and owner rather than an automatic failure. A justified analysis and applicable use evidence could change that

state. They would be a new basis for closure, not proof that the original memo had covered the later claim all along.

## SECTION 05

### How small can a useful request be?

A performance plan may name a specimen absent from the label. A study described as leftover-sample research may add a blood draw in its protocol. A surveillance log may identify the app but omit the knowledgebase that changed a sequencing treatment report. In each case, several pages can look complete while describing different propositions. The first question is which product or activity those pages actually concern.

For pathology, a scanner profile describes how scanner image data are transformed before scoring. Its identity therefore belongs to the path between specimen image and reported performance, not merely to document administration. The next packet makes that connection the subject of a finding.

**Scanner baselines.** The technical file, performance evaluation report, software evidence, instructions for use and surveillance plan each identify a different scanner-profile baseline. The packet supplies no canonical released version.

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#### Decision

**Are these five disconnected document defects? What smallest evidence request would expose the root issue?**

#### How a reviewer connects five discrepancies to one blocked conclusion

The root finding is that the packet cannot connect the claimed and monitored product to a controlled tested and released scanner/profile combination. That uncertainty can affect risk, reader evidence, certificate scope and customer support as well as the five documents. Changing their headers would leave the underlying relationship unproved.

The manufacturer can supply the canonical design and released baseline, the version used in each study, and a controlled mapping into evidence, risk, labeling, certificate/change and surveillance records. Such a mapping may reconcile several records efficiently; each valid observation remains evidence about the version it examined. If the mapping instead exposes an untested supplied state, performance applicability still needs a bridge or new work. The manufacturer owns the correction, while the notified body considers the applicable assessed-scope or change question.

**Reviewer Lens.** “Five documents disagree” states an observation. The useful finding explains why those disagreements prevent establishing that the marketed scanner/profile combination is the one assessed for performance and risk. Exact conflicting identifiers and versions make the request answerable. Evidence that resolves identity can close that finding without automatically resolving a separate performance difference it reveals.

**Consultant Lens.** Product, clinical, software, quality and regulatory owners can often meet around one baseline rather than repair their pages separately. Their minimum closure record must connect what was designed, studied and released. That is also where any additional testing or change assessment becomes visible. Promising that document alignment proves an untested scanner's performance would hide the very decision the repair is meant to expose.

### **Which answer would change the decision?**

A focused request contains an observed fact, the decision it prevents, an owner and a closure condition. Consider an expressly hypothetical illustration, adding no facts to the scanner packet: the reader study names one profile and the proposed label names another. “Which controlled image path generated the result, and what tested comparison supports the proposed profile?” gives the clinical and software leads a bounded task. The version trace and any bridge evidence—the reasoned support for using earlier results across a difference—would let the claim be reappraised.

Some closures require an institution rather than another experiment. Actual notified-body designation and application scope determine eligibility; a Member State study question needs the country's process; a true companion-diagnostic medicine relationship may need the prescribed consultation. The reviewer can identify why the current file cannot settle the question. The consultant can connect the exact question to a qualified specialist, its timing and the manufacturer or board decision that follows. An informal opinion retains its advisory value without becoming permission.

A decision memorandum can then hold four short columns: **supported now, not yet supported, next evidence or institutional decision**, and **owner/trigger**. An actual tissue analytical observation belongs in the first column only for its tested question; a broader liquid claim may belong in the second. For a deployed pathology queue change, proportionate interim protection and applicable reporting assessment may proceed while cause is unresolved. “Unknown cause” does not mean “wait to protect users.” The assessor states the evidence determination, the consultant explains scoped options, the manufacturer owns compliance and release, and management chooses a funded lawful sequence.

## SECTION 06

**Which part of the earlier evidence survives?**

The following sequencing and pathology records are separate fictional mini-cases. Neither inherits trial assays, scanners, drug relationships, incidents or approvals from earlier chapters. Each supplies its own bounded evidence and unresolved scope.

**An expanded sequencing purpose.** Tissue analytical and clinical evidence is strong. The proposed European purpose also includes liquid biopsy—testing body-fluid-derived material—and Drug B, while the performance plan and commercial configuration cover only tissue and Drug A. There are no supplied liquid or Drug B results or exact medicines consultation outcome.

**Decision**

**Which passes must reopen? What remains positive for the bounded tissue/Drug A question, and what has yet to be established for the expansion?**

**How a reviewer gives credit without widening the result**

The strong tissue evidence remains useful for the tissue question. It cannot supply observations about liquid specimens, a Drug B treatment relationship or implementation of those added claims by the tissue/Drug A commercial configuration. Changing the specimen path changes what the analytical and clinical evidence must represent; changing the medicine role changes the clinical question and, where the complete companion-diagnostic premise holds, the consultation work.

That brings the review back to pass **0**, the offered product; **2**, purpose; **3**, class and actual route; **5**, performance evaluation; **6**, study and any true companion-diagnostic consultation; **7**, commercial configuration; and **8**, field and claim maintenance. Pass **4**’s file links must be reconciled once the supported scope is settled. These returns follow the changed claim; they do not declare a new class or consultation outcome.

The manufacturer may assess a genuinely narrower purpose or fund the expanded program. Either needs its own support and applicable institutional decisions. A proposed bridge is not the missing result, and neither option supplies a certificate or launch date.

**Reviewer Lens.** Specimen, treatment-selection role and commercial pipeline belong beside each evidence set. Crediting tissue/Drug A where the claimed relation is supported makes the liquid/Drug B and configuration gaps more precise. A broad adverse label would conceal useful evidence; a broad favorable label would conceal the extension that evidence never examined.

**Consultant Lens.** The practical choice is between work programs tied to defensible scopes. Statistical, clinical and medicines input can help estimate the specimen and medicine-relationship work for expansion and compare it with a bounded initial scope. A new plan or partnership may make work feasible, but it has not completed the bridge or delivered the institutional outcome.

**A changed pathology workflow.** A certified AI-assisted triage product changes default queue visibility and adds Scanner B under a supplier firmware update. Overall model metrics remain stable. No scanner-specific reader result, field event or completed change assessment is supplied.

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### Decision

**Can this close as routine maintenance because the aggregate model metric is stable? Which review questions need the actual changed configuration and user task?**

#### How a reviewer follows the metric through the worklist

The stable aggregate metric remains a positive observation for its tested mix. The changed claim concerns more than that aggregate. A pathologist has to encounter and recover relevant cases in the worklist, and Scanner B's image path includes the supplier firmware. The metric does not establish low-priority-case visibility or intended-user recovery on that combination.

The comparison therefore starts with the certified baseline and the changed scanner/firmware/queue state. Its effects lead into applicable requirements and risk, human–AI clinical performance in the actual worklist, software/security and supplier change control, labeling, applicable notified-body communication and postmarket surveillance/follow-up. A controlled configuration plus targeted reader/recovery, compatibility and security evidence could change the open finding; an unchanged summary number cannot perform those comparisons.

Nothing in the packet establishes harm, deployment of the new state or a certificate supplement. Release work and any field action must keep those unknowns visible rather than inventing a history.

**Reviewer Lens.** The useful performance question reaches from scanner-specific image processing to low-score handling and the reader's task. Firmware compatibility and applicable certificate/change records help establish which combination can be assessed. Keeping the algorithm summary in its tested scope permits positive credit without allowing it to answer a different workflow question.

**Consultant Lens.** A proposed release needs a controlled combined configuration, targeted reader/recovery evidence and security/compatibility verification. If a version

is already deployed, actual fleet reconstruction and proportionate protection become a separate workstream; this packet does not establish deployment. Separating those gates prevents a prospective release plan from being mistaken for evidence that an installed population has been identified or protected.

## SECTION 07

### What changes when the intended operator is a nurse?

**A separate professional-use packet.** A near-patient human immunodeficiency virus type 1 and type 2 (HIV-1/2) antibody test uses capillary whole blood in health clinics. Its intended users are trained nurses, who are health professionals. Near-patient testing occurs outside a laboratory by a health professional, generally near the person tested; it is distinct from lay self-testing. The manufacturer supplies an Annex VIII **Rule 1 class D** rationale for detecting a life-threatening transmissible agent with high propagation risk—risk of spread. The notified body's designation covers the device. Analytical evidence covers the **exact test system, capillary specimen and read time**. The use evaluation employed only central-laboratory staff. No nurse-use study result or self-testing claim is supplied.

#### Decision

**Which bounded items are adequately explained, which evidence question remains open, and who owns the next action? Does near-patient use itself make the product class C?**

#### How a reviewer keeps the class and user-evidence questions separate

The supplied Rule 1 rationale addresses detection of a transmissible agent causing a life-threatening disease with high propagation risk. On these stipulated facts, that class D rationale and the body's designation can each be A for their narrow question. Rule 4(b) requires near-patient devices to be classified in their own right: the setting does not assign a universal C class or override Rule 1. A changed medical purpose would require new rule reasoning; the word “clinic” alone would not supply it.

The exact-system, specimen and read-time analytical evidence is also a positive component. The remaining connection is between laboratory operation and the nurses' intended clinic task. Collecting a small capillary sample, timing the read, interpreting a borderline display and recovering from error are possible use mechanisms to examine, not failures observed in this packet. Under Annex I §19, design and information must support the intended users' skills, available means and reasonably anticipated variation in technique and environment. Near-patient instructions must make the required training, qualifications or experience clear. Safe, accurate use and reduction of



handling and interpretation errors therefore need support applicable to those users, with feasible checks of intended operation and warnings of invalid results considered under §19.3.

The manufacturer can provide representative handling, timing, interpretation and recovery evidence or a justified applicable bridge. That could change the open use finding. A training label or laboratory result alone cannot establish the absent relationship, and calling it documentation-only would assume that it already exists.

The notified body performs the applicable class D Article 48 assessment. If the manufacturer uses the Article 48(3) Annex IX route, §5.1 examines the technical documentation for near-patient use, including intended-user study results, suitability evidence, labeling and instructions. If the manufacturer instead uses the Article 48(4) route combining Annex X with Annex XI, Annex X §2 requires the corresponding intended-user test reports, handling-suitability data, labeling and instructions in the type-examination application. Under either route, the user-evidence question remains part of the procedure as well as scientific appraisal. Relevant Common Specifications require compliance unless a duly justified solution ensures at least equivalent safety and performance. Where an EU reference laboratory is designated for the device, the notified body requests laboratory verification of claimed performance and applicable specification compliance or the permitted equivalent solution. Laboratory verification does not replace the whole conformity assessment or the intended-user evidence relationship.

These bounded positive findings establish neither whole-product conformity nor clinical performance in clinic use or an authorized launch. The manufacturer owns the evidence work; the body assesses the applicable route. The scope of each conclusion remains visible alongside the work still required.

**Reviewer Lens.** Purpose/class, body designation, exact-system analytical evidence and intended-user performance answer four different questions. Keeping them on separate lines avoids reopening the already supported class merely because nurse-use evidence is absent. It also makes the request narrow enough to inspect: the actual clinic task and interpretation path, the proposed evidence or bridge, and the conclusion it could support.

**Consultant Lens.** Existing applicable nurse-use evidence, a representative evaluation or a justified bridge are possible ways to address the clinic question; clinical/human-factors and regulatory owners can determine what is feasible. Preserving the positive analytical and class/body findings avoids rebudgeting work as though it were absent. The remaining class D procedure, including applicable specification and reference-laboratory work, still needs its own plan and actual outcomes.

**SECTION 08****Keep the return route visible**

When the file becomes large, the following questions locate a record, a bounded judgment or an owned next question. A heading with a tick beside it cannot do that work.

1. What exact European product, version, medical purpose and decision are under review?
2. Which actor owns each decision, and which binding or nonbinding source supports it?
3. Is the purpose stable, and does the actual function qualify under the assumed framework?
4. Which Annex VIII rules apply, and which distinct Article 48 conformity procedure follows?
5. Can each important requirement and file claim reach one controlled design, risk control, test and label?
6. Do scientific validity, analytical and clinical performance, the appraisal report and justified follow-up support every claim element?
7. Do study, companion-diagnostic, software, AI, cybersecurity and change routes follow from actual protocol and product facts?
8. Can the postmarket system detect and respond to field information while preserving product, certificate and installed identities?
9. What is the smallest sound assessor action and most realistic company decision, with owner and reopening trigger?

The route works in both directions. A premarket sequencing review can start at specimen/result and follow its evidence scope; a deployed interpretation change can begin with a field report and return to that scope. A premarket pathology scanner review follows image to reader; a deployed queue issue starts with a user who may not see a case and returns to the original task evidence. These examples supply no facts or outcome to the independent assessment that follows.

**Independent assessment — a decision before feedback**

This unfamiliar fictional packet stands alone. The six-part assessment uses its stated facts and the source rules already taught; missing information remains a question, rather than an invented answer. Instructor feedback is held separately until all six responses are complete. The case is not a sequel to any earlier sequencing, pathology, pregnancy, near-patient or method record.

**Product and proposed change.** HarborCheck is a **new** self-testing urine pregnancy IVD intended **solely** to detect pregnancy in adult lay users at home. It makes no fertility-monitoring, fetal-health or treatment-selection claim. Its cassette and dedicated **non-AI optical reader** are supplied as one system. **H1** is the tested candidate. The intended release **H2** changes the rule used to present a reading obtained outside the allowed reading window. The manufacturer says “the chemistry is unchanged.” No H2 verification, use evaluation or clinical/performance bridge is supplied.

**Analytical comparison.** On H1, laboratory personnel tested **200 distinct urine specimens from 200 adults**, one initial test per specimen, against a prespecified independent laboratory reference. **Eighty** specimens were reference positive and **120** reference negative. Among reference-positive specimens, H1 reported **70 positive, two negative and eight invalid**. Among reference-negative specimens, it reported **three positive, 115 negative and two invalid**. No specimen was repeated. The report removes invalids from its agreement tables and states that agreement above 97% supports home use. The packet provides no prespecified acceptance criterion or uncertainty analysis.

**Lay-use report.** A separate report covers **24 adult lay users** performing specified tasks on H1. **Twenty-three** completed all specified tasks; **one** needed moderator assistance to recover after using the wrong read time. The report calls this a recovered difficulty. It does not supply the critical-task definition, whether assistance was permitted, the cause investigation or closure record. These users are **not** the 200 analytical specimen donors.

**Application and commercial request.** The manufacturer's memo cites the pregnancy self-test exception in Annex VIII Rule 4(a), then says, “class B means declaration only; the kit can launch when the company signs.” No notified-body application, conformity outcome, unique device identification (UDI) record or market history is supplied. A prospective distributor offers to begin sales next month. The company asks you to close the file and write the launch recommendation.

The assessment has **six outputs**, in the following order:

1. **Product and decision.** What statement distinguishes H1, H2 and the commercial state, and separates the manufacturer's, notified body's and management's decisions?
2. **Purpose and route.** What qualification, class and conformity procedure follow from the stated purpose? Which parts of the application memo can stand, and which cannot?
3. **Numerical evidence.** What are positive and negative agreement **among valid pairs**, the initial invalid-result rate and positive-result yield **among all reference-positive specimens**? For each, what are the numerator, denominator and scope of support?

4. **Applicability.** How do the analytical comparison, lay-user report and H2 relate, keeping the specimen donors separate from the 24 intended users? What distinguishes the observed assisted recovery from an inferred critical-task or acceptability verdict?
5. **Next evidence.** Which **three focused evidence requests**, each with an owner, source/evidence basis and closure condition, would resolve the decision while preserving conclusions already supportable?
6. **Recommendation, 150–250 words.** What can proceed now, what prevents the requested launch conclusion and what event would change it, using only the supplied product facts?

The complete initial record precedes feedback. An instructor may later provide a changed-premise supplement for a revised recommendation. Both dated decisions remain available so that a changed conclusion can be traced to changed evidence and legal state rather than silently replacing the first answer.

### Further reading

After committing the independent assessment, the cross-volume route map offers optional method depth. It is not additional evidence for HarborCheck, and no earlier case's answer should be transferred into this packet.

### Source trail

Binding source checked 25 September 2026: Regulation (EU) 2017/746, consolidated 10 January 2025, especially Articles 2, 10, 48, 56 and 110(2)–(3e), Annex I §19, Annex VIII Rules 1 and 4, Annex IX §5.1, and Annex XIII. Article 110's branch, condition and date wording was checked against the current official indexed text and the Commission transitional-provisions page; the latter is explanatory, not a substitute for the Regulation. The Commission notified-body/NANDO page supports designation checks, not an assumed product certificate. Medical Device Coordination Group guidance and standards used in earlier chapters remain nonbinding or method sources at their applicable scope. The mini-cases and HarborCheck data are fictional teaching premises, not real device classifications, acceptance criteria, clinical outcomes or authority decisions.

WORKING TERMINOLOGY

# Working terminology and distinctions

Use this as a retrieval aid **after** the chapter's explanation. A short entry helps you find the right question; it is not an acceptance rule, legal opinion or substitute for the relevant clause, method or product record. If a term has two meanings, name the context. Abbreviations expands short forms; source navigation explains what each authority can establish.

The recurring practical question is: *what object is being judged?* An NGS report may be technically correct for a specimen but unsupported for a treatment decision. A DP model may score images reproducibly while a changed scanner or hidden worklist queue alters the reader's actual task. The distinctions below keep those different observations attached to the right product state and decision.

## Product, purpose and decision maker

These entries help locate the next decision when a product description, class and responsible person appear to disagree.

| Term                            | Working meaning and limit                                                                                                                                                                                  |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Intended purpose / intended use | The manufacturer's claimed medical job, population, user, input, result and context. EU purpose is anchored in the manufacturer's specified information; wording must match the product actually supplied. |
| Device function                 | What a component <i>does</i> to information or the clinical path, not what its marketing label calls it. Moving, selecting, interpreting and reporting are different verbs.                                |
| Qualification                   | Whether that function and purpose belong under the IVDR, MDR, another framework or outside device law. Ask before applying a class rule.                                                                   |
| Medical device software (MDSW)  | Software with its own medical purpose under device law. It may be supplied alone <b>or</b> as an integral part of a system; supply arrangement and independent medical purpose are different axes.         |

| Term                                                       | Working meaning and limit                                                                                                                                                                                                                |
|------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Classification</b>                                      | Risk class under the rules applying to the exact purpose. A class guides, but does not itself complete, a conformity procedure or validate a claim.                                                                                      |
| <b>Conformity assessment</b>                               | The applicable process by which conformity is assessed for the device and route, including notified-body tasks where required. A favorable test, QMS certificate or body conversation is not every product claim's completed assessment. |
| <b>Manufacturer</b>                                        | The actor responsible for defining the offered product, establishing conformity and maintaining it, even when suppliers or advisers do work.                                                                                             |
| <b>Authorised representative</b>                           | A <b>natural or legal person</b> established in the EU who has received and accepted a written mandate from a manufacturer outside the EU for specified IVDR duties (Article 2(25)). It is not simply any "EU-based organization."       |
| <b>Economic operator</b>                                   | A role such as manufacturer, authorised representative, importer or distributor with its own statutory duties at a supply-chain handoff; it does not erase the manufacturer's underlying responsibility.                                 |
| <b>Notified body</b>                                       | An independent conformity assessor designated for a defined scope. Designation eligibility, available capacity, accepted application, technical assessment, certificate and surveillance are separate states.                            |
| <b>Competent authority</b>                                 | A Member State public authority for assigned study, vigilance, market-surveillance and related functions. It is not the notified body.                                                                                                   |
| <b>Sponsor</b>                                             | The person or organization responsible for <b>initiating, managing and setting up financing</b> of a performance study. Sponsor and manufacturer may coincide, but their roles must be assigned from the actual protocol.                |
| <b>Person responsible for regulatory compliance (PRRC)</b> | A qualified person with specified regulatory-compliance responsibilities; naming one does not transfer manufacturer or institutional decisions to that person.                                                                           |
| <b>Deliverable</b>                                         | The defined work product—such as a claim memo or evidence map—that a consultant provides. Its existence is not a regulator's conclusion.                                                                                                 |

| Term                              | Working meaning and limit                                                                                                                                                                                                                                                       |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Companion diagnostic (CDx)</b> | An IVD essential for safe and effective use of a corresponding medicine to identify, before or during treatment, people likely to benefit or at increased risk of serious adverse reactions from it. A named medicine or marker association alone is not the full relationship. |

A software product can be supplied inside a larger system while still having its own medical purpose. The medical-device-software row keeps those two questions separate; the supply arrangement alone cannot settle qualification.

## Evidence, reference and measurement

First locate the result in its evidence chain. Scientific validity asks whether the claimed relationship makes sense; analytical performance asks whether the offered system detects or measures it; clinical performance asks whether this result relates to the condition or state in the intended people and use. The plan sets the question before observation; the report appraises what observation actually supports.

| Term                          | Working meaning and limit                                                                                                                                                                                   |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Scientific validity</b>    | Evidence for the association between the analyte, marker or claimed medical information and a clinical condition or physiological/pathological state. Association does not prove this assay measures it.    |
| <b>Analytical performance</b> | How the exact examination system detects or measures its target under defined conditions. State specimen, hardware, software, variant class, errors and invalid outputs.                                    |
| <b>Clinical performance</b>   | Relationship of this device's result to the relevant clinical condition or state in its intended population and use. Successful operation or user comprehension alone is not that relationship.             |
| <b>Clinical evidence</b>      | Clinical data and performance-evaluation results of sufficient amount and quality for a qualified safety and intended-benefit assessment; an evidence index is not that assessment.                         |
| <b>Performance evaluation</b> | Lifecycle appraisal joining scientific validity, analytical performance and clinical performance for each intended-purpose element. A strong result for one element does not support an untested expansion. |



| Term                                                     | Working meaning and limit                                                                                                                                                                                                |
|----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Performance evaluation plan (PEP)</b>                 | Prospective claim-to-question, method, criteria and decision plan. It is not a retrospective list of studies or proof they passed.                                                                                       |
| <b>Performance evaluation report (PER)</b>               | Actual, objective appraisal of favorable and unfavorable evidence against the purpose, benefit–risk and limits. It can close a bounded claim and leave another open.                                                     |
| <b>Performance study</b>                                 | A defined investigation to establish or confirm an IVD's analytical or clinical performance. Study permissions, execution, report and product conformity are distinct.                                                   |
| <b>Clinical performance study plan / protocol (CPSP)</b> | In the IVDR, the clinical performance study <b>plan</b> has its required legal content; ISO 20916 uses the term <b>protocol</b> for its study-conduct method. They may align but are not interchangeable by title alone. |
| <b>Acceptance criterion</b>                              | A justified decision rule specified before the result is known. A post-hoc percentage chosen because it looks favorable is not a prespecified pass.                                                                      |
| <b>Reference standard / reference process</b>            | The specified method and people assigning comparison results, with limitations and unresolved cases recorded. A reference result is not automatically error-free clinical truth.                                         |
| <b>Ground truth</b>                                      | Informal shorthand for that specified reference, not an assertion of perfect truth. Name its method whenever drawing an inference.                                                                                       |
| <b>Adjudication</b>                                      | A prespecified way to resolve disputed reference labels using defined reviewers, evidence and rules; retain cases not resolved.                                                                                          |
| <b>Concordance</b>                                       | Agreement between paired results <b>or</b> consistency among configuration identifiers. State which kind and denominator; one does not prove the other.                                                                  |
| <b>Precision — measurement</b>                           | Scatter or repeat consistency under specified conditions. A tightly clustered measurement may still be systematically offset from a reference.                                                                           |
| <b>Precision — AI metric</b>                             | Positive predictive value in a classification analysis: reference-positive predictions divided by all positive predictions in the stated case mix. It is not measurement repeatability.                                  |

| Term                                                       | Working meaning and limit                                                                                                                                                                                                                                     |
|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Limit of detection (LoD)</b>                            | A low target level detected at a <b>specified probability under specified conditions</b> , not an absolute guarantee that every specimen at or above one concentration will be detected.                                                                      |
| <b>Operating point</b>                                     | The chosen score threshold or rule turning a continuous result into a category or action; changing it changes the claim's error balance.                                                                                                                      |
| <b>Reportable range / scope</b>                            | For a quantitative result, the validated measuring/reporting interval is a numerical interval. For an NGS claim, supported variant classes, conditions and report types form a <b>scope</b> ; do not call a list of variant classes a concentration interval. |
| <b>Generalizability</b>                                    | Whether evidence remains applicable across the users, sites, specimens, instruments and conditions in the claim. It requires a defined comparison, not optimism about transfer.                                                                               |
| <b>Subgroup analysis</b>                                   | Examination within a defined population or configuration stratum; keep its denominator, missingness and purpose visible rather than hiding it in a pooled average.                                                                                            |
| <b>Reader study / multi-reader multi-case (MRMC) study</b> | Observation of specified users reading specified cases under specified conditions. Repeated readings of one case by several readers are not new independent patients.                                                                                         |
| <b>State of the art</b>                                    | Current generally accepted clinical and technical practice used to judge comparators, benefit, risk and performance; neither “latest technology” nor a permanent snapshot.                                                                                    |
| <b>Bridging</b>                                            | A reasoned, evidenced transfer from one assay, specimen, version, site or study to another. Specify the difference, failure mechanism and data needed; a proposed bridge is not an accepted result.                                                           |

The concordance row illustrates why the object matters. Matching database identifiers answers a configuration question; matching patient results answers an agreement question with a denominator. Neither observation can silently stand in for the other.

## Product realization, risk and change

Think of a diagnostic result as travelling through specimen or image preparation, software and interpretation into a report and user action. The file, build and risk record must describe that same path. A changed component can reopen one link without invalidating every older observation.

| Term                            | Working meaning and limit                                                                                                                                                                                                |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Architecture</b>             | Major components, interfaces and flows of information, control and responsibility. Use it to find what a changed scanner, service or database can affect.                                                                |
| <b>Interface</b>                | Point where people, software, hardware or services exchange data or commands; a screen and a supplier application programming interface (API) can both be consequential.                                                 |
| <b>Workflow</b>                 | The actual sequence of users, equipment, defaults, handoffs, timing and decisions through which a result reaches clinical use. A path that worked in a tidy study may differ from the path a busy intended user follows. |
| <b>Pipeline</b>                 | Ordered input-to-output processing steps. For NGS this includes read alignment, calling, filtering, annotation and reporting, not merely a sequencer.                                                                    |
| <b>Preprocessing</b>            | Input preparation before an algorithm or model acts, such as image normalization or sequence-data filtering. An unchanged model can still receive changed inputs.                                                        |
| <b>Model weights</b>            | Learned numerical parameters affecting a model's output. Unchanged weights do not prove an unchanged product, threshold, scanner or workflow.                                                                            |
| <b>Configuration / baseline</b> | The exact supported combination of hardware, software, model, knowledgebase, settings, services and user path; a <b>baseline</b> is the controlled reference state used to compare later changes.                        |
| <b>Build</b>                    | Actual software package made from code, dependencies, tools and settings. A source-code label alone does not identify every installed component.                                                                         |
| <b>Requirement</b>              | Testable statement of what the product or component must do in a specified context. It should lead to design, risk, verification and user evidence.                                                                      |
| <b>Verification</b>             | Evidence a design output meets its specified input or requirement. An implementation test may pass while the intended user's recovery still fails.                                                                       |
| <b>Validation</b>               | Evidence the complete product serves intended users and purpose in representative conditions. Neither term is a synonym for all product performance evidence.                                                            |

| Term                                            | Working meaning and limit                                                                                                                                                                                                                       |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Risk control / residual risk</b>             | A control interrupts a specified path from potential harmful information or source to exposure and possible harm; residual risk is what remains after controls, needing its own assessment.                                                     |
| <b>Benefit-risk</b>                             | Judgment whether expected benefit justifies remaining risks for the actual purpose and state; it is not a blanket label that waives a missing performance result.                                                                               |
| <b>Correction</b>                               | Action on a detected nonconformity or affected instance, such as a justified report replacement. Investigation identifies what happened and cause; it is not the correction itself.                                                             |
| <b>Corrective and preventive action (CAPA)</b>  | Corrective action removes the cause of a <b>detected</b> nonconformity to prevent recurrence; preventive action addresses a <b>potential</b> nonconformity's cause to prevent occurrence. Neither is proven effective by a task being assigned. |
| <b>Release</b>                                  | Manufacturer acceptance of a particular configuration for a particular purpose. It is separate from completed conformity, lawful market state and confirmed installation in the field.                                                          |
| <b>Rollback</b>                                 | Controlled return to a previous <b>compatible and adequately assessed</b> state. An old installer is not automatically a safe recovery target.                                                                                                  |
| <b>Drift</b>                                    | Change over time in inputs, behavior or performance relative to the assessed conditions; a signal to investigate, not a diagnosis of cause.                                                                                                     |
| <b>Predetermined change control plan (PCCP)</b> | U.S. mechanism specifying proposed changes, development/validation method and impact assessment. A proposed/submitted plan differs from an <b>authorized</b> one; it grants no automatic EU change permission.                                  |
| <b>Software bill of materials (SBOM)</b>        | Inventory of identified software components. Complement it with external-service and supplier-dependency records; an inventory alone does not establish security.                                                                               |
| <b>Vulnerability / exploit</b>                  | A vulnerability is a weakness; an exploit is a means of using it to cause an unauthorized effect. A possible path is not proof an attack occurred.                                                                                              |

| Term                                         | Working meaning and limit                                                                                                                                                                                                                           |
|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Authentication / access authorization</b> | Authentication checks the claimed identity of a person or service; access authorization decides which actions that authenticated identity may perform. This <b>security permission</b> is not device-market authorization or a conformity decision. |

The verification and validation rows allow a passing implementation test to remain a success while a user-recovery problem remains open. The file needs both conclusions at their actual scope.

## Legal requirements and continuing operation

These entries connect the product record to the legal requirement or continuing activity it serves.

| Term                                                     | Working meaning and limit                                                                                                                                                                                                       |
|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General safety and performance requirement (GSPR)</b> | Applicable IVDR Annex I obligation. Follow it to a product-specific design, risk, test, label or reasoned non-applicability rather than checking a heading.                                                                     |
| <b>Technical documentation</b>                           | Inspectable Annex II/III account of the device, design, manufacturing, performance, risk, label and postmarket system for one controlled product state.                                                                         |
| <b>Common specifications (CS)</b>                        | Commission-adopted, scope-defined technical or clinical requirements under IVDR Article 9; unlike ordinary voluntary guidance, they have the legal effect stated there.                                                         |
| <b>Harmonized standard / presumption of conformity</b>   | A cited European standard may provide a presumption only for the legal requirements, edition and parts its Official Journal citation covers. An ISO title by itself supplies neither harmonization nor whole-device conformity. |
| <b>Quality management system (QMS)</b>                   | Roles, processes, controls and records by which the manufacturer repeatedly develops, releases, monitors and improves the device; a procedure's existence does not prove an effective transaction.                              |
| <b>Instructions for use (IFU)</b>                        | Manufacturer's directions, intended purpose, proper use and precautions; part of the claimed user task and evidence boundary, not merely packaging prose.                                                                       |
| <b>Post-market surveillance (PMS)</b>                    | Planned gathering, analysis and action on experience with marketed devices; identify the exposed version, denominator and owner.                                                                                                |

| Term                                                  | Working meaning and limit                                                                                                                                                                                                                             |
|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Post-market performance follow-up (PMPF)</b>       | Targeted performance learning after a supported device is marketed within its assessed purpose. It does not defer evidence essential to the first claim.                                                                                              |
| <b>Periodic safety update report (PSUR)</b>           | Applicable periodic synthesis of PMS results, benefit–risk and actions. A calendar does not delay a needed field investigation or timely report.                                                                                                      |
| <b>Vigilance</b>                                      | Formal assessment and reporting/response route for serious incidents and field safety corrective actions under the applicable facts and clocks; not every technical signal is automatically reportable.                                               |
| <b>European database on medical devices (EUDAMED)</b> | EU regulatory information modules for actors, devices, certificates and other specified functions. Registration, certificate scope and the manufacturer's actual installed-fleet record have different jobs.                                          |
| <b>Unique device identification (UDI)</b>             | Traceability system: <b>Basic UDI-DI</b> groups a device family in documentation; <b>UDI-DI</b> identifies a device model/version as required; <b>UDI-PI</b> identifies production information. None is a certificate or proof of field installation. |

The unique-device-identification row is a useful check on overreach: a device identifier supports traceability, while a confirmed installation needs its own field record. Registration cannot fill that gap.

The working letters **A/U/M/D/C/N/A** retrieve a reviewer item's current evidence posture: adequately explained, unclear, missing/inconsistent, deeper analysis, consultation indicated, or not applicable with reason. They are never a product grade or vote. Return to Chapter 17 for the route that uses them.

## ABBREVIATIONS

# Abbreviations at the point of use

The chapters expand a term when it first does work. Return here when a short form reappears. Historical case labels **AP-1**, **FS-1** and **LS-1** refer respectively to the prepared-specimen analytical exercise, the thirty-day field-accounting exercise and the resource-and-sequence exercise. These remain separate fictional records. An **IVDR-CH** code is a stable exercise/retrieval identifier, not a clause citation.

| Short form      | Full term and what to remember                                                                                                                                                                                                                                 |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>AE / SAE</b> | Adverse event / serious adverse event in a performance-study safety assessment; seriousness concerns specified consequences, not simply symptom intensity.                                                                                                     |
| <b>AI / ML</b>  | Artificial intelligence / machine learning. A model is only one part of the supplied diagnostic system and user workflow.                                                                                                                                      |
| <b>API</b>      | Application programming interface: a specified exchange point through which software or services pass data or commands. Its technical access permission is not clinical suitability or device-market authorization.                                            |
| <b>CAPA</b>     | Corrective and preventive action. Keep correction of an affected instance separate from removing an actual or potential cause.                                                                                                                                 |
| <b>C5 / C95</b> | Values on the specified relevant response or target scale at about 5% and 95% positive-response probability under the study's stated conditions. Concentration is one possible scale; these positions are not confidence limits or universal clinical cutoffs. |
| <b>CE</b>       | European conformity marking following the applicable conformity-assessment route; the mark is not an agency's endorsement of every proposed claim.                                                                                                             |
| <b>CDx</b>      | Companion diagnostic; the corresponding medicine and the test's essential treatment-decision role must be specified.                                                                                                                                           |
| <b>CI</b>       | Confidence interval, an uncertainty interval for an estimated quantity under the stated method; not a performance gate by itself.                                                                                                                              |



| Short form         | Full term and what to remember                                                                                                                 |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CIA</b>         | Confidentiality, integrity and availability: different security properties, each needing a mechanism and clinical consequence.                 |
| <b>CLSI</b>        | Clinical and Laboratory Standards Institute; identify the particular document and edition, not merely the organization.                        |
| <b>CPSP</b>        | Clinical performance study plan under the IVDR; compare the actual study-plan content with a standard's <i>protocol</i> requirements.          |
| <b>CS</b>          | Common specifications adopted under the IVDR for a defined scope; not a shorthand for any standard or guideline.                               |
| <b>CVSS</b>        | Common Vulnerability Scoring System; a technical severity score is not a patient-harm finding.                                                 |
| <b>DNA / cfDNA</b> | Deoxyribonucleic acid / cell-free DNA. A fluid-derived cfDNA pathway is not automatically equivalent to a tissue assay.                        |
| <b>DP / WSI</b>    | Digital pathology / whole-slide image. A scanned image, software score and pathologist worklist are different links in one clinical path.      |
| <b>EMA</b>         | European Medicines Agency; its medicine-related consultation role is not device certification.                                                 |
| <b>EQA / QC</b>    | External quality assessment / quality control. A favorable comparison or control result answers its defined question, not every product claim. |
| <b>EU / OJ</b>     | European Union / its Official Journal; an OJ citation's date and scope matter for harmonized-standard effects.                                 |
| <b>EUDAMED</b>     | European database on medical devices; a regulatory registration is not a record of confirmed local installation.                               |
| <b>EURL</b>        | European Union reference laboratory; its class-D performance-verification job is distinct from notified-body assessment.                       |
| <b>FDA</b>         | United States Food and Drug Administration; a U.S. marketing submission or decision does not establish EU claim scope.                         |
| <b>FN / FP</b>     | False negative / false positive against the specified reference and denominator. Include invalid or missing outcomes separately.               |
| <b>FSCA</b>        | Field safety corrective action, a field-risk reduction measure with its own assessment and communication route.                                |

| Short form              | Full term and what to remember                                                                                                                                                                                                 |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GSPR                    | General safety and performance requirement in IVDR Annex I; trace each applicable obligation into product-specific work.                                                                                                       |
| IFU / SSP               | Instructions for use / summary of safety and performance; check the exact intended-purpose and evidence boundaries represented.                                                                                                |
| IVD / IVDD / IVDR / MDR | In vitro diagnostic device / former In Vitro Diagnostic Devices Directive / current In Vitro Diagnostic Medical Devices Regulation / Medical Devices Regulation. The applicable instrument depends on product and legal state. |
| ISO / IEC               | International Organization for Standardization / International Electrotechnical Commission. Their standards need an exact edition and a separate applicability and harmonization check.                                        |
| LoD / LoQ               | Limit of detection / limit of quantitation. Detection requires a stated probability and conditions; reliable numerical measurement asks a further question.                                                                    |
| MDCG                    | Medical Device Coordination Group; its guidance helps interpret, but does not replace, binding law.                                                                                                                            |
| MDSW                    | Medical device software. Own medical purpose and manner of supply are separate axes.                                                                                                                                           |
| MRMC                    | Multi-reader multi-case study; multiple readings of one case are not multiple independent patients.                                                                                                                            |
| NANDO / NB              | New Approach Notified and Designated Organisations information system / notified body. Verify the <b>actual designation scope</b> , not only a body name.                                                                      |
| NGS / SNV / VAF         | Next-generation sequencing / single-nucleotide variant / variant allele fraction. Read-to-report processing and supported variant classes matter alongside the instrument.                                                     |
| NPA / PPA / PPV         | Negative percent agreement / positive percent agreement / positive predictive value. Each needs its stated reference, exclusions and denominator; PPV also depends on case mix.                                                |
| PCCP                    | Predetermined change control plan in the U.S. context. Proposed, submitted and authorized plans are not the same state.                                                                                                        |
| PEP / PER               | Performance evaluation plan / performance evaluation report: prospective evidence questions versus objective appraisal of the evidence obtained.                                                                               |

| Short form                                  | Full term and what to remember                                                                                                                                                                                             |
|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PMPF / PMS / PSUR</b>                    | Post-market performance follow-up / post-market surveillance / periodic safety update report. Targeted follow-up feeds the broader monitoring and applicable periodic synthesis; none replaces timely action.              |
| <b>PRRC</b>                                 | Person responsible for regulatory compliance; a named person does not absorb all manufacturer duties.                                                                                                                      |
| <b>QMS</b>                                  | Quality management system; controlled processes and their actual records, not merely a certificate label.                                                                                                                  |
| <b>RACI</b>                                 | Responsible, accountable, consulted, informed: a task-assignment aid, not a substitute for statutory roles.                                                                                                                |
| <b>RUO</b>                                  | Research use only; the label does not override the actual medical purpose and supplied use.                                                                                                                                |
| <b>SBOM</b>                                 | Software bill of materials; component inventory complements supplier and external-service records.                                                                                                                         |
| <b>TFEU</b>                                 | Treaty on the Functioning of the European Union; Articles 290 and 291 distinguish delegated from implementing acts.                                                                                                        |
| <b>UDI / Basic UDI-DI / UDI-DI / UDI-PI</b> | Unique device identification / basic device identifier for a grouping / device identifier / production identifier. These answer different grouping, device and production-trace questions; none proves field installation. |

For example, PEP and PER name different moments in the same evidence work: the plan states what would answer the question, and the report judges what the obtained evidence supports. Expanding the letters does not establish that either document is adequate.

For a fuller explanation and limits, use working terminology. For the authority behind a term, follow the chapter's source trail or source navigation.

## SOURCE NAVIGATION

# Source authority and edition navigation

Start with the job of the source. The **IVDR** supplies the binding European Union (EU) device requirements. A Commission act can make or amend a binding rule, adopt common specifications (CS), or publish a list of cited standards; these are different legal effects. A cited harmonized standard can give a presumption of conformity **only for the requirements and parts covered by its Official Journal citation**. An uncited technical standard remains a useful voluntary method, not a legal shortcut. Medical Device Coordination Group (MDCG) guidance explains an interpretation but does not replace the regulation. A notified body's designation scope is an eligibility fact, while its local application procedure, capacity and eventual certificate are separate facts. The book's examples and working tools are author synthesis, not any of these authorities.

A usable source record names the source governing the question, its exact edition or consolidation, date and relevant provision, together with the product/configuration and country. A link alone does not establish applicability. The source manuscript's ledger reported selected checks on **6 September 2026**; the chapter-local checks and the two integration updates noted below were scoped to their claims through **25 September 2026**. Neither date is a claim that every external page, Member State procedure, standard status or product-specific decision was freshly verified on that day.

## The 25 source-ledger entries

The numbers preserve the live source ledger for cross-reference. “Official” identifies the issuer or legal publisher, not an approval of a fictional case.

### Binding EU framework and official operating sources

- **Source 1.** Regulation (EU) 2017/746, IVDR — binding framework; the linked consolidation is dated **10 January 2025**. Check amendments and the applicable provision when using it.
- **Source 2.** Regulation (EU) 2017/745, MDR — binding medical-device framework where a function falls on the MDR side; verify the current consolidated text for a live decision.

- **Source 9.** Regulation (EU) 2026/1744 — binding July 2026 amendment to the Artificial Intelligence Act; read amended Articles 6 and 113 with the base act, not a fictional product's assumed classification.
- **Source 10.** EUDAMED overview — official operating information on staged module use; the first four modules became mandatory **28 May 2026**, not every module at once. Registration is not proof of a site's installed configuration.
- **Source 11.** NANDO / Single Market Compliance Space — official notified-body designation source. Verify current **actual scope** before treating a body as eligible; expertise, fee or an available date cannot replace designation.
- **Source 18.** Commission Implementing Decision (EU) 2021/1195, consolidated 17 June 2026 — current list of EN harmonized standards for the IVDR. The integration check confirmed this consolidation and the scope-limited Article 8 point; it did not turn every international standard named here into a harmonized one.
- **Source 19.** Commission Implementing Regulation (EU) 2022/1107 — binding CS for covered requirements of certain class-D in vitro diagnostic devices, subject to its scope and IVDR Article 9.
- **Source 20.** Commission Implementing Regulation (EU) 2026/977 — binding requirements for notified-body quality-system and conformity-assessment procedures. It is **not** a product analytical-performance acceptance criterion or a certificate.
- **Source 21.** Regulation (EU) 2024/1689, Artificial Intelligence Act — binding base act; read as amended by entry 9. Neither an “AI” label nor a source date settles applicability to a given device.

## Interpretive and draft guidance

- **Source 3.** MDCG endorsed-guidance register — official way to find the current nonbinding document and revision.
- **Source 4.** MDCG 2020-16 rev.5, classification rules for IVDs — nonbinding guidance published **9 September 2026**. The imported 6 September ledger listed **rev.4 (March 2025)**; this entry intentionally updates that edition after the official rev.5 publication. Annex VIII remains the legal rule.
- **Source 5.** MDCG 2019-11 rev.1, software qualification and classification — nonbinding June 2025 guidance; function and intended purpose still govern the product-specific analysis.
- **Source 6.** MDCG 2025-5, performance-study questions and answers — nonbinding June 2025 guidance to read with the IVDR and actual protocol/country route.
- **Source 7.** MDCG 2025-10, post-market surveillance — nonbinding December 2025 guidance; a field signal still requires product- and event-specific assessment.

- **Source 8.** MDCG 2025-6 / AIB 2025-1, MDR/IVDR–AI Act interplay — jointly endorsed, nonbinding June 2025 guidance; it predates the binding 2026 amendment in entry 9.
- **Source 22.** Commission draft guidelines on high-risk AI classification — a **draft**, not binding final guidance; the source ledger recorded its draft status on 6 September 2026. Recheck status before live reliance.
- **Source 25.** MDCG 2019-16 rev.1, cybersecurity guidance — nonbinding July 2020 guidance; combine with current legal duties and an appropriately selected technical baseline.

### Other-jurisdiction guidance and voluntary methods

- **Source 12.** U.S. FDA, Content of Premarket Submissions for Device Software Functions — final guidance, 14 June 2023. It does not expand the European claim supported by a U.S. package.
- **Source 13.** U.S. FDA, Cybersecurity in Medical Devices — final guidance, 3 February 2026, for the stated U.S. submission/QMS context.
- **Source 14.** U.S. FDA, Predetermined Change Control Plan recommendations for AI-enabled device software — final guidance, 18 August 2025; a proposed plan is not an authorized plan, and a U.S. plan does not authorize EU changes.
- **Source 15. Selected technical standards and methods.** ISO 14971:2019 and ISO/TR 24971:2020 address risk methods; IEC 81001-5-1:2021 with Interpretation Sheet 1:2025 addresses health-software security lifecycle processes; ISO 20916:2019 addresses IVD clinical performance-study practice. Additional lifecycle methods used here are ISO 13485:2016; IEC 62304:2006+AMD1:2015 CSV; IEC 82304-1:2016; and IEC 62366-1:2015+AMD1:2020 CSV. Check edition, applicability and any EU harmonization or U.S. recognition separately. These methods do not, by themselves, prove product performance.
- **Source 23.** CLSI EP49, edition 1, March 2026 — voluntary framework for clinical-validity evidence and noninterventional clinical performance studies; not a ready interventional companion-diagnostic protocol or clinical-utility evidence method.
- **Source 24.** CLSI EP12, edition 3, March 2023 — voluntary guideline for qualitative **binary** examinations, including response-probability positions C5/C95, comparator choice, agreement, uncertainty and discrepancy resolution. Do not generalize it unexamined to every multicategory or ordinal result.

**Local procedures that must be checked for the actual route**

- **Source 16. Member State procedures** — verify the country-specific performance-study, ethics, data-protection, import, language, vigilance and administrative requirements for the *whole actual protocol*; the fictional Country A/B/C placeholders establish no national permission.
- **Source 17. Selected notified-body procedures** — verify the body's current application, contract, technical-documentation, change-notification and certificate processes and scope. An informal conversation is not national study authorization or a completed certificate.

**Return to the chapter's exact source trail**

The ledger is a starting point. Each chapter's source trail records the narrower provision, guidance passage or method actually used for its teaching claim:

1 · 2 · 3 · 4 · 5 · 6 · 7 · 8 · 9 · 10 · 11 · 12 · 13 · 14 · 15 · 16 · 17.

This is source navigation, not a permission, certification or product verdict. When a live decision changes a claim, user, specimen, version, country or calendar, retrieve the controlling current source again.



**CROSSWALK**

# Original-to-integrated chapter crosswalk

This is a navigation aid for readers of the earlier 28-topic Book 2 source edition. The current edition teaches those topics through 17 connected chapters. A topic's new location is not a claim that two old fictional dossiers were one product; use the premises inside the linked chapter. For the present reading order, return to Contents.

| Original topic | Earlier title                                                                    | Current home |
|----------------|----------------------------------------------------------------------------------|--------------|
| 1              | The Same Product, a Different Regulatory Story                                   | Chapter 1    |
| 2              | Intended Purpose, Scope, and Qualification                                       | Chapter 2    |
| 3              | Classification: The Claim Sets the Risk Context                                  | Chapter 3    |
| 4              | Conformity Assessment and Notified-Body Strategy                                 | Chapter 4    |
| 5              | GSPRs: From Legal Requirement to Evidence Thread                                 | Chapter 5    |
| 6              | Annex II and III: Build a Technical File That Can Be Traversed                   | Chapter 6    |
| 7              | QMS, Risk, Design, Suppliers, and Change                                         | Chapter 6    |
| 8              | The Performance Evaluation Plan: Write the Evidence Strategy Before the Evidence | Chapter 7    |
| 9              | Scientific Validity: The Association Is Not the Assay                            | Chapter 8    |
| 10             | Analytical Performance: Prove the Measurement System You Actually Sell           | Chapter 9    |
| 11             | Clinical Performance and Human-AI Workflow                                       | Chapter 10   |
| 12             | Literature, State of the Art, PER, and Evidence Sufficiency                      | Chapter 11   |

| Original topic | Earlier title                                                      | Current home |
|----------------|--------------------------------------------------------------------|--------------|
| 13             | Performance Studies: Route, Design, and Multi-Country Reality      | Chapter 12   |
| 14             | Companion Diagnostics: One Development Program, Two Products       | Chapter 13   |
| 15             | MDSW, AI Act, Cybersecurity, and Controlled Change                 | Chapter 14   |
| 16             | PMS, PMPF, PSUR, Vigilance, EUDAMED, and Certification Maintenance | Chapter 15   |
| 17             | U.S.-EU Evidence Reuse and the Consultant-Grade Decision           | Chapter 16   |
| 18             | Actor Map, Source Hierarchy, and Route Control                     | Chapter 1    |
| 19             | Intended-Purpose and Classification Workshop                       | Chapter 3    |
| 20             | Conformity Assessment and Notified-Body Engagement                 | Chapter 4    |
| 21             | Technical-File and GSPR Evidence Architecture                      | Chapter 6    |
| 22             | A Worked PEP and PER: From Evidence Plan to Honest Conclusion      | Chapter 11   |
| 23             | Analytical Performance for Molecular and Computational Diagnostics | Chapter 9    |
| 24             | Clinical Performance and Human-AI Workflow under IVDR              | Chapter 10   |
| 25             | Performance-Study Route and Multi-Country Execution                | Chapter 12   |
| 26             | Companion Diagnostics: Co-Development, Consultation, and Bridging  | Chapter 13   |
| 27             | PMS, PMPF, PSUR, Vigilance, EUDAMED, AI, and Controlled Change     | Chapter 15   |
| 28             | Board-Level U.S.-EU Launch Capstone                                | Chapter 16   |

The original technical-file and QMS chapters converge in current Chapter 6 because the product configuration, records, process owners and changes must describe the same system. The earlier stand-alone workshops and studios appear within their present chapters. Chapter 17 is the new integrated review route and independent learner assessment, not a disguised twenty-ninth historical topic.

SERIES ROUTE

# Routes through the series

Book 2 asks for an EU decision about a **specified product, purpose, evidence state and time**. The other books deepen a method used to reach that decision; they do not supply missing Book 2 facts. For instance, a software security method from Book 1B can help test a diagnostic update, but its fictional device cannot become the identity, field history or certificate of a Book 2 case. Return to the applicable source and actual country or notified-body record for a legal conclusion.

| When this Book 2 task needs more method depth                                                  | Continue with                                                                                                                              | What to bring back                                                                                                                                                           |
|------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Defining the medical job, tracing risk and assembling the evidence file in Chapters 2, 5 and 6 | Book 1A Chapters 2–6, beginning with the clinical promise and continuing through risk, architecture, controlled development and evaluation | A clearer claim □ requirement □ control □ test chain, then reapply it to <b>this</b> Book 2 product state. The sequence begins at Chapter 2 and continues through Chapter 6. |
| Establishing system boundaries and managing security or updates in Chapters 6, 14 and 15       | Book 1B Chapters 2–7: boundary, threats, controls, supply chain, testing and field support                                                 | Threat □ requirement □ control reasoning and evidence for operational recovery, with the Book 2 user's diagnostic consequence kept explicit.                                 |
| Planning and appraising performance, references and specimen transfer in Chapters 7–13         | Book 1C Chapters 4–9 and 12–17: diagnostic questions, specimens, measurement, studies and applicability                                    | Denominator, uncertainty and evidence-transfer methods; compare the actual Book 2 assay, population and configuration rather than borrowing a result.                        |
| Assessing artificial-intelligence product evidence and human use in Chapters 10, 14 and 15     | Book 1D Chapters 3–4 and 9–15: product boundary, metrics, robustness, verification, integration, human use, release and change             | The manufacturer's development and release decisions behind the evidence a reviewer evaluates; a model score alone is not the full workflow.                                 |

| When this Book 2 task needs more method depth                  | Continue with                                                                     | What to bring back                                                                                                                 |
|----------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Comparing regional sequence and advising choices in Chapter 16 | Book 3 contents, especially its consulting-practice and company-decision chapters | An engagement and business-decision method after technical, regulatory, budget and dependency assumptions have been made explicit. |

For example, the security row points to a method for following an update into clinical work; it supplies no missing configuration or field result for this book's product. Each route is optional: use the local Book 2 explanation first, then return with the particular method you need.